TUNING

PX4 autopilot

< Exit Vehicle Configuration

- Summary
- Airframe
- Sensors
- Radio
- Flight Modes
- Power
- Actuators
- Safety
- PID Tuning
- Flight Behavior
- Parameters
- Firmware

PID Tuning Config

Tuning Setup is used to tune the flight controllers.

Rate Controller | Attitude Controller | Velocity Controller | Position Controller

Airmode (disable during tuning) ? Disabled Thrust curve ? 0.0

Roll Rate

225.0 180.5 136.0 91.5 47.0 2.5 -42.0 -86.5 -131.0 -175.5 -220.0

deg/s

19.6 19.9 20.2 20.5 20.8 21.1 21.4 21.7 22.0 22.3 22.6

sec

Response Setpoint

Clear Stop

Automatic Flight Mode Switching

Use auto-tuning Use manual tuning

Select Tuning: Roll Pitch Yaw

Overall Multiplier (MC_ROLLRATE_K)

0.3 3 1.5

Multiplier for P, I and D gains: increase for more responsiveness, reduce if the rates overshoot (and increasing D does not help).

Differential Gain (MC_ROLLRATE_D)

4e-04 0.003 0.01

Damping: increase to reduce overshoots and oscillations, but not higher than really needed.



PDF GUIDE

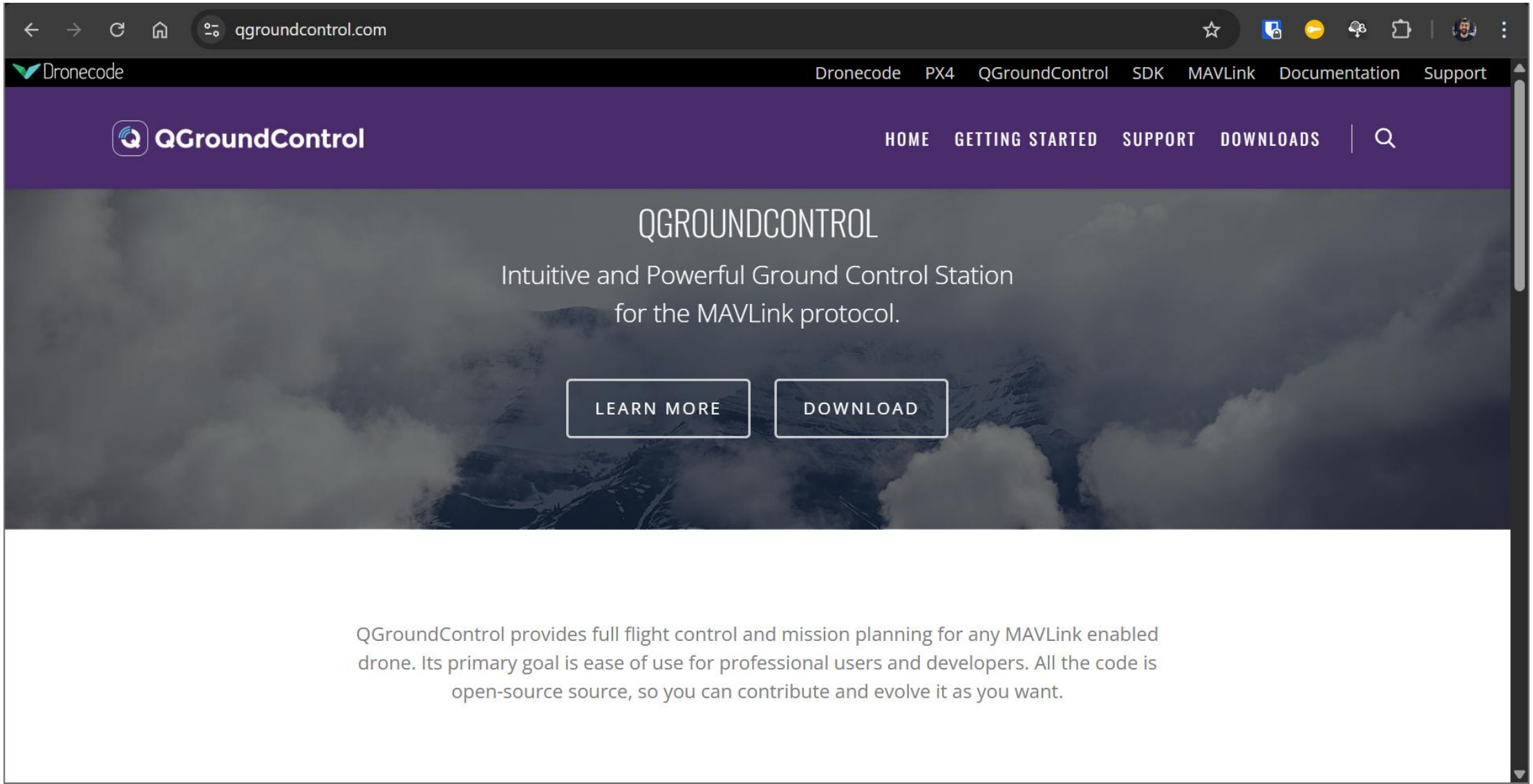
PX4 or Ardupilot? Hobbyist

- As a hobbyist there isn’t a huge difference between PX4 and Ardupilot
- PX4 is very slightly easier to use in my opinion
- Ardupilot is slightly more complicated but has a few more features, especially for non-standard drone configurations
- There is hardware that supports both Firmware options so you don’t necessarily have to choose, you can try both!

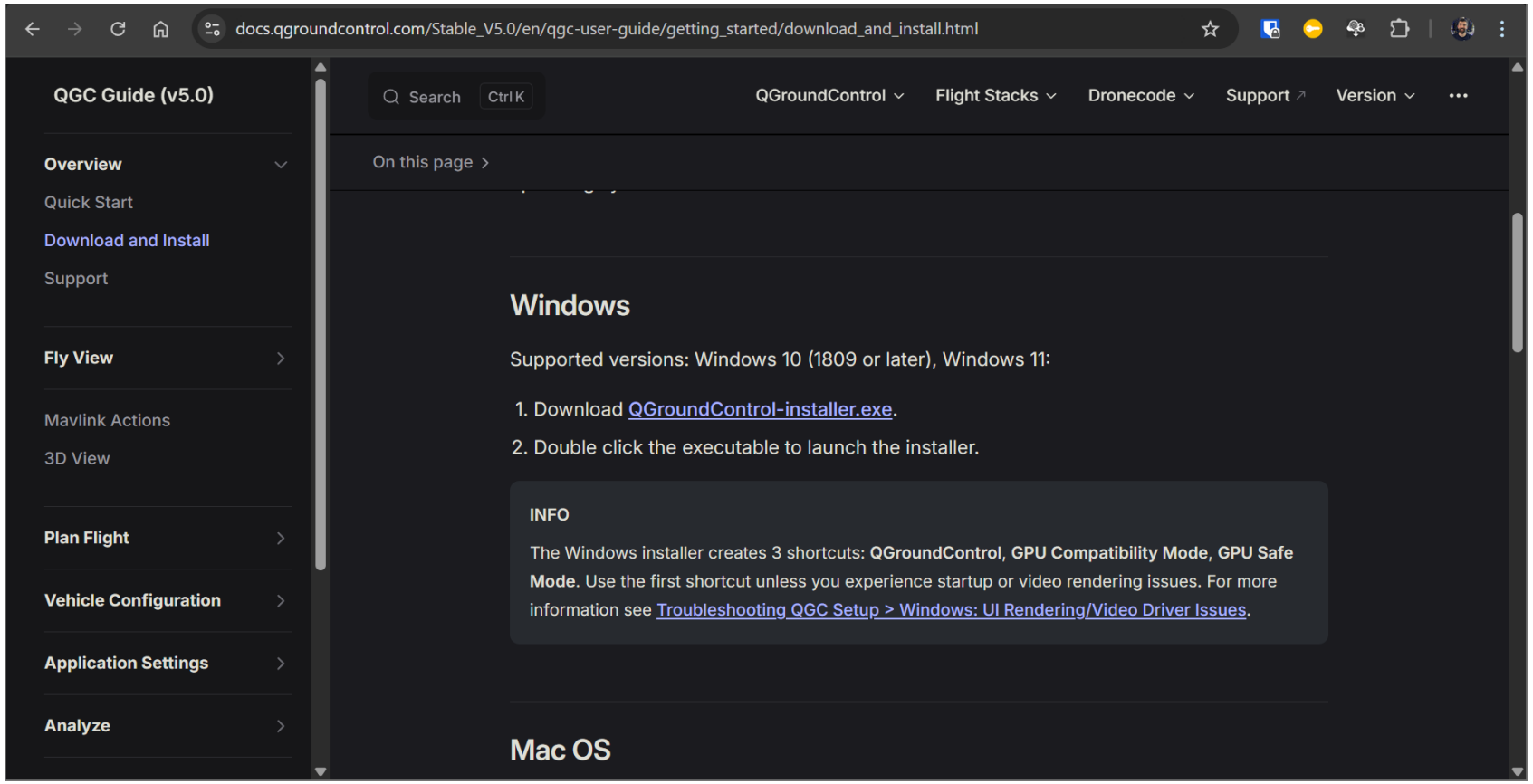
PX4 or Ardupilot? Commercial

- When deciding whether to use Ardupilot or PX4 commercially one of the main considerations is the license:
- Ardupilot is **GPLv3** which means that any further commercial development of the open source code (bug-fixes or enhancements) **must** be open-sourced and contributed back to the project
- PX4 has a **BSD 3-Clause** license which is more permissive and allows any modifications or enhancements to remain closed-source
- This makes PX4 slightly more attractive from a commercial perspective as companies can make custom versions with proprietary enhancements for their customers

Install QGroundControl Software



This guide will assume **Windows 11** (QGC is also available for Android, MacOS and Linux)

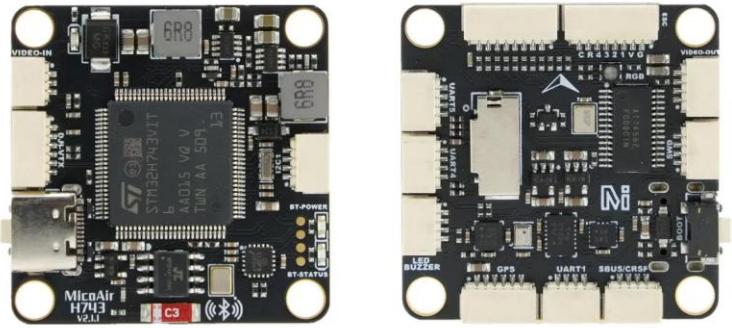


PX4 on the Cheap

- Most PX4 Hardware is expensive and bulky (and out of stock!)
- However the MicoAir743 V2 Flight controller supports PX4, Betaflight, INAV, and Ardupilot out of the box
- It is only $45 USD and also available with a 55A ESC for under $75USD
- I will be using this FC for this guide, but the tuning approach is valid for all PX4 hardware (and drones of any size)



~$700

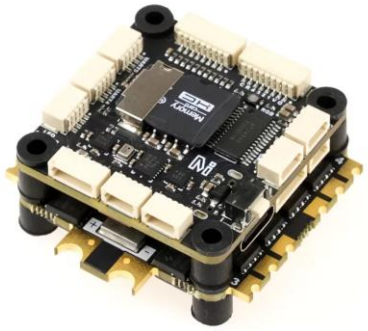


~$45

Hardware used

- AOS UL8 Frame
- MicoAir743 V2 Flight Controller (or any FC with a PX4 target)
- MicoAir743 55A AM32 4in1 ESC
- M10 GPS and Compass module
- ELRS 2.4GHz Receiver
- DJI O3 Air Unit (also works with O4 Pro)











Optional Hardware

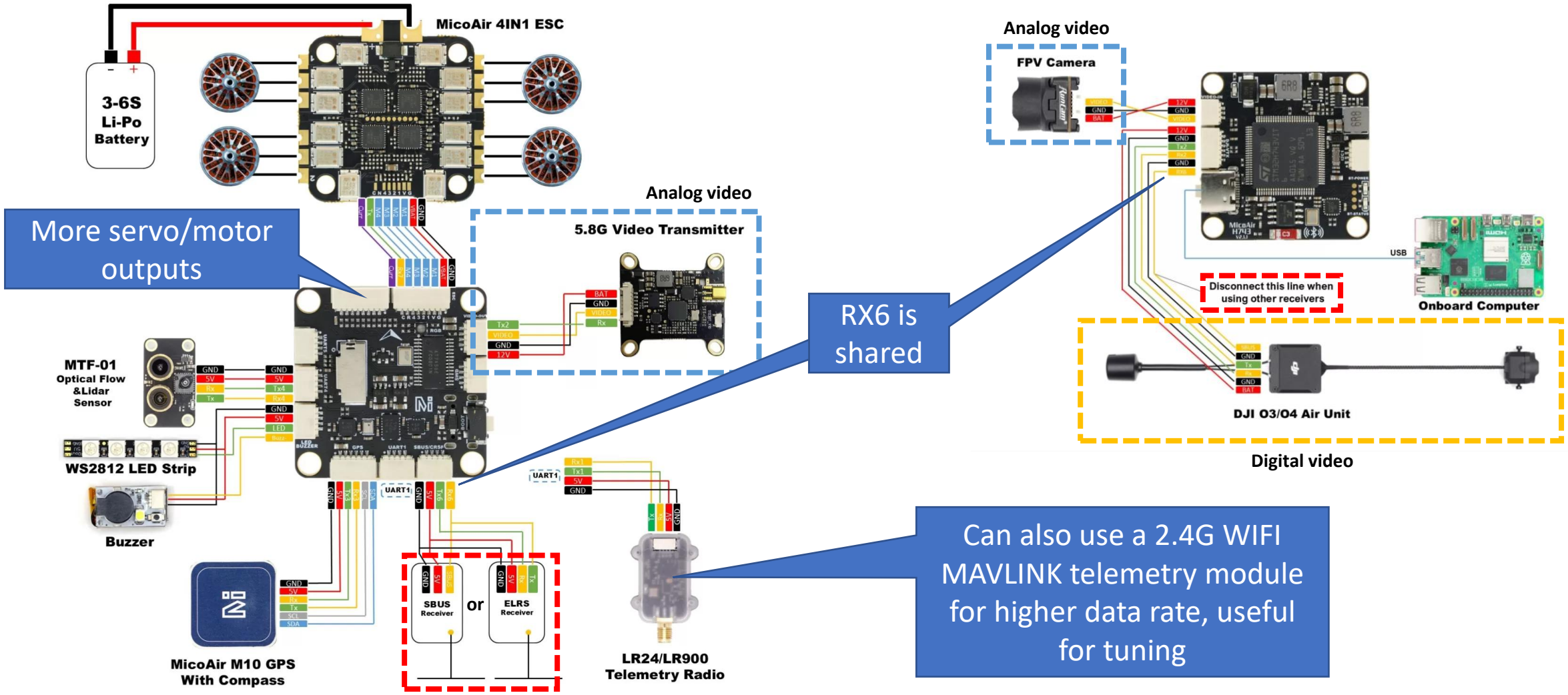
- Telemetry link (433 Mhz / 900 Mhz /2.4 GHz)
- Optical Flow Sensor
- Proximity Sensor
- Not covered in this video series but covered in the tuning course







Wiring the MicoAir743 V2

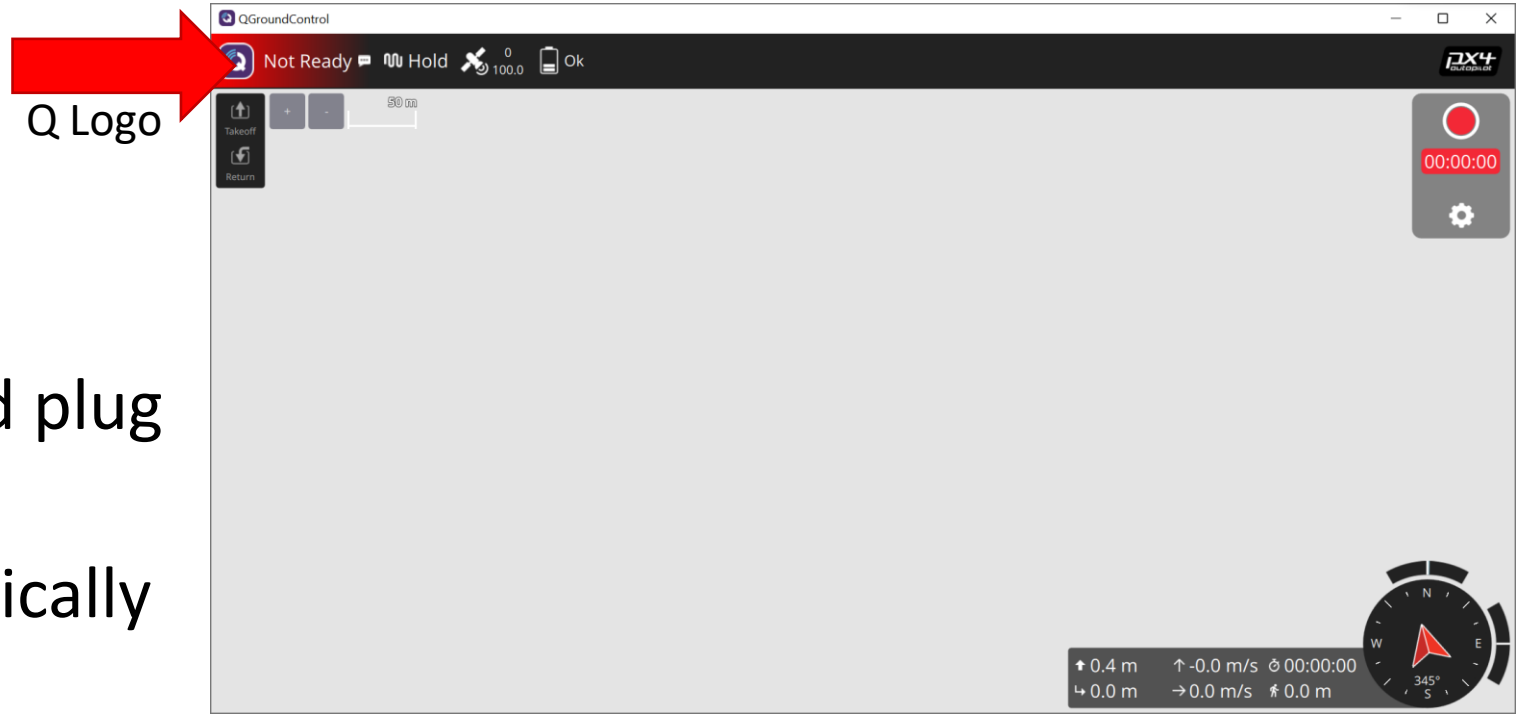


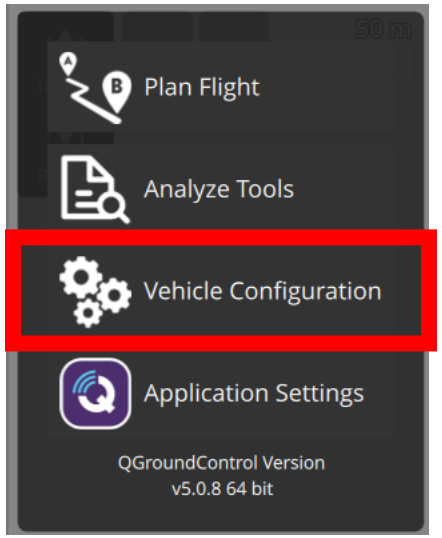
Wiring the MicoAirH743 V2

- PX4/Ardupilot generally expects certain devices to be on certain UARTs so it is very helpful to make sure that each device is connected to the correct UART as shown in the wiring diagram
- This is somewhat different from Betaflight/INAV where UARTs can easily be assigned in any order

Connecting

- Open QGroundControl and plug in your Flight Controller
- It should connect automatically
- Click the **Q Logo > Vehicle Configuration**
- This will be the main page we use to set up our vehicle

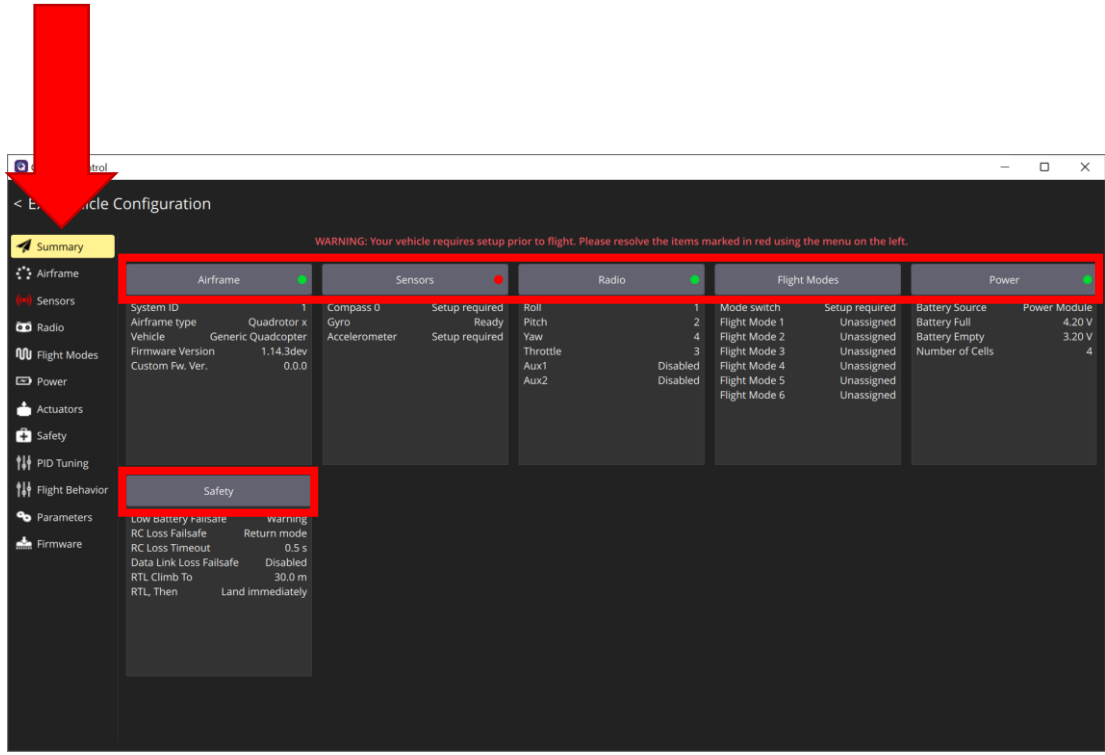




First Setup

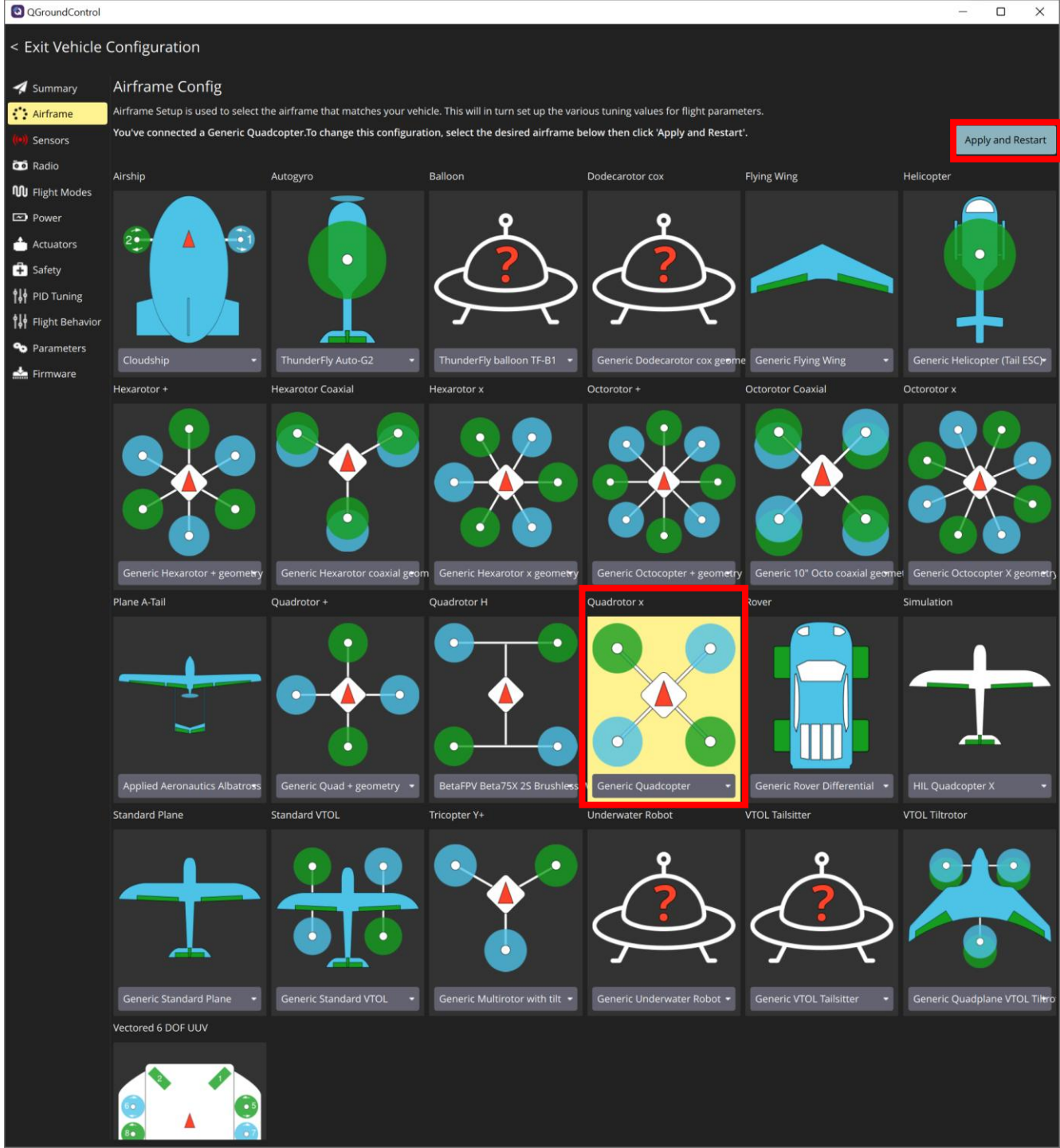
- When you first connect you should see a summary tab with warnings
- We are going to go through each section to get your drone ready for it’s first flight
- You can navigate using the tabs on the left or the headings of the boxes

1. Airframe
2. Sensors
3. Radio
4. Flight Modes
5. Power
6. Safety



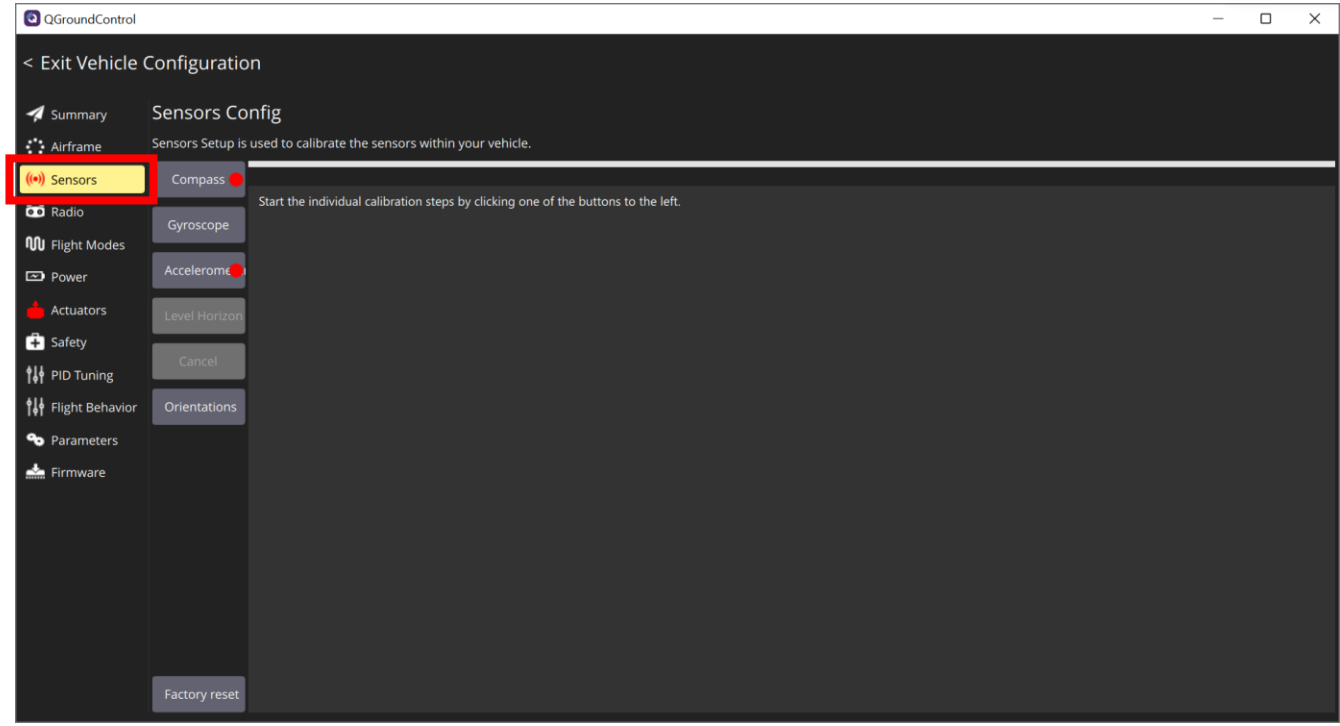
Airframe

- There are a lot of Airframe options in PX4
- For each option there is a dropdown menu for pre-set configurations for particular platforms
- We are going to be using **QuadrotorX > Generic Quadcopter** for this guide but the same tuning approach applies for any multi-rotor / VTOL
- Select your airframe option then click **Apply and Restart** in the top left



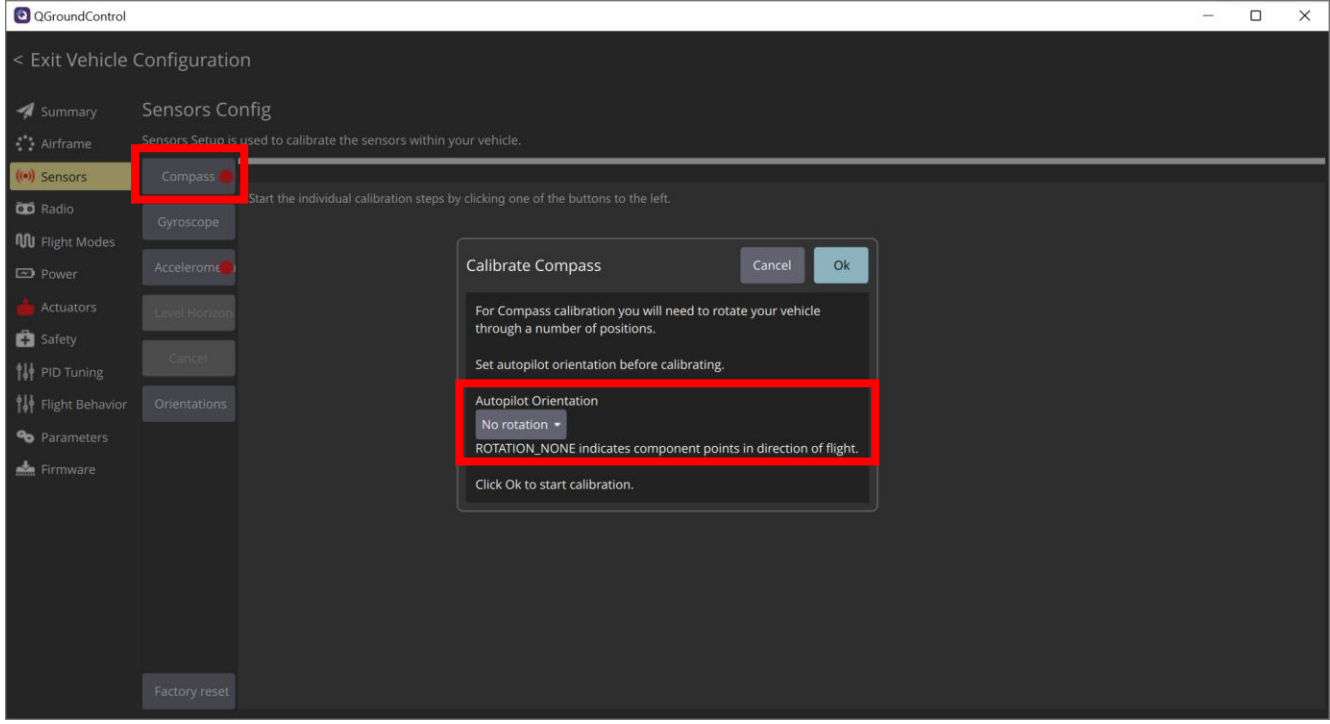
Sensor Calibration

- Click into the sensors tab
- Uncalibrated sensors have a red dot next to them
- For a new build we are going to work through calibrating the sensors in order

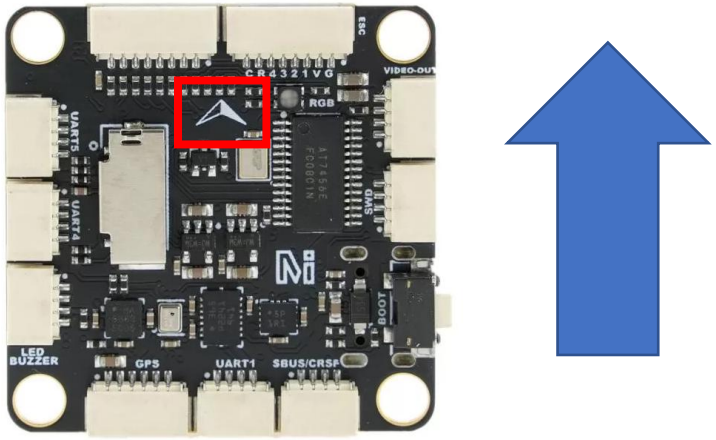


Compass Calibration

- Click into the Compass section
- Set the correct Autopilot rotation
- If you have the flight controller upright and pointing forward then **No rotation** is correct
- If you change the autopilot rotation make sure to power cycle the flight controller after calibrating

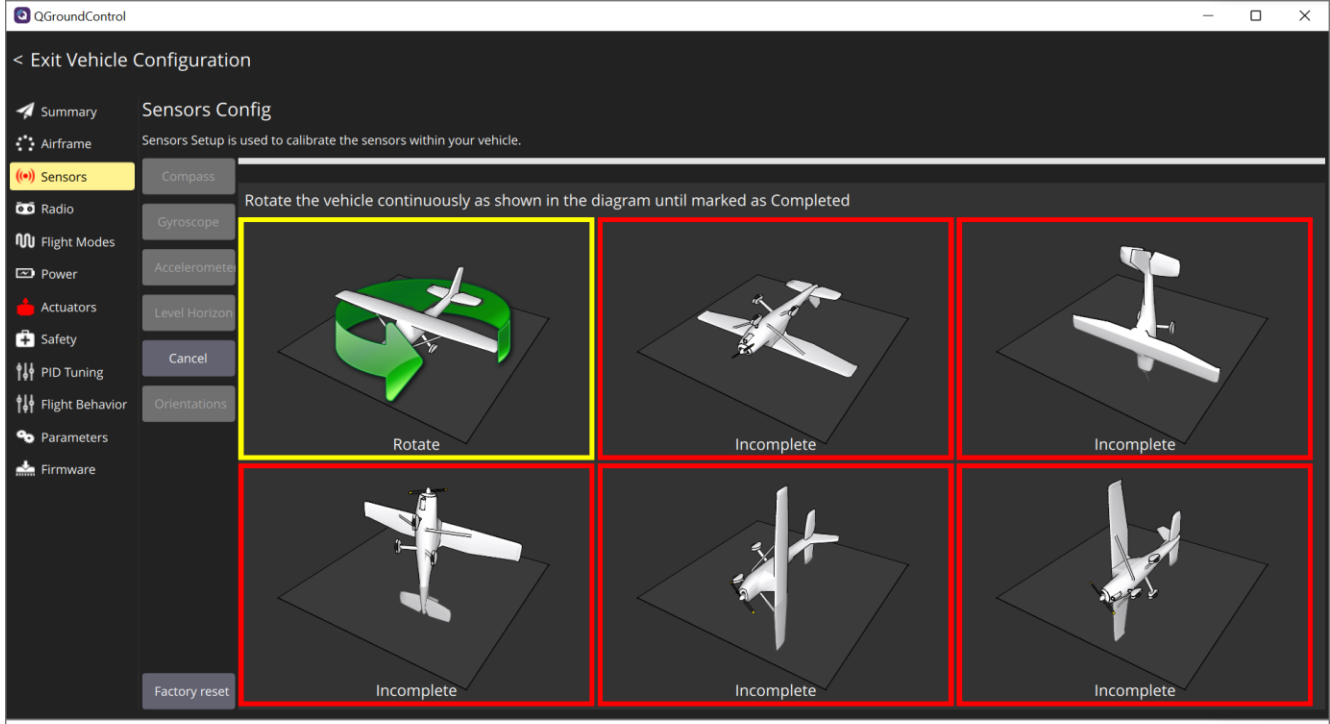


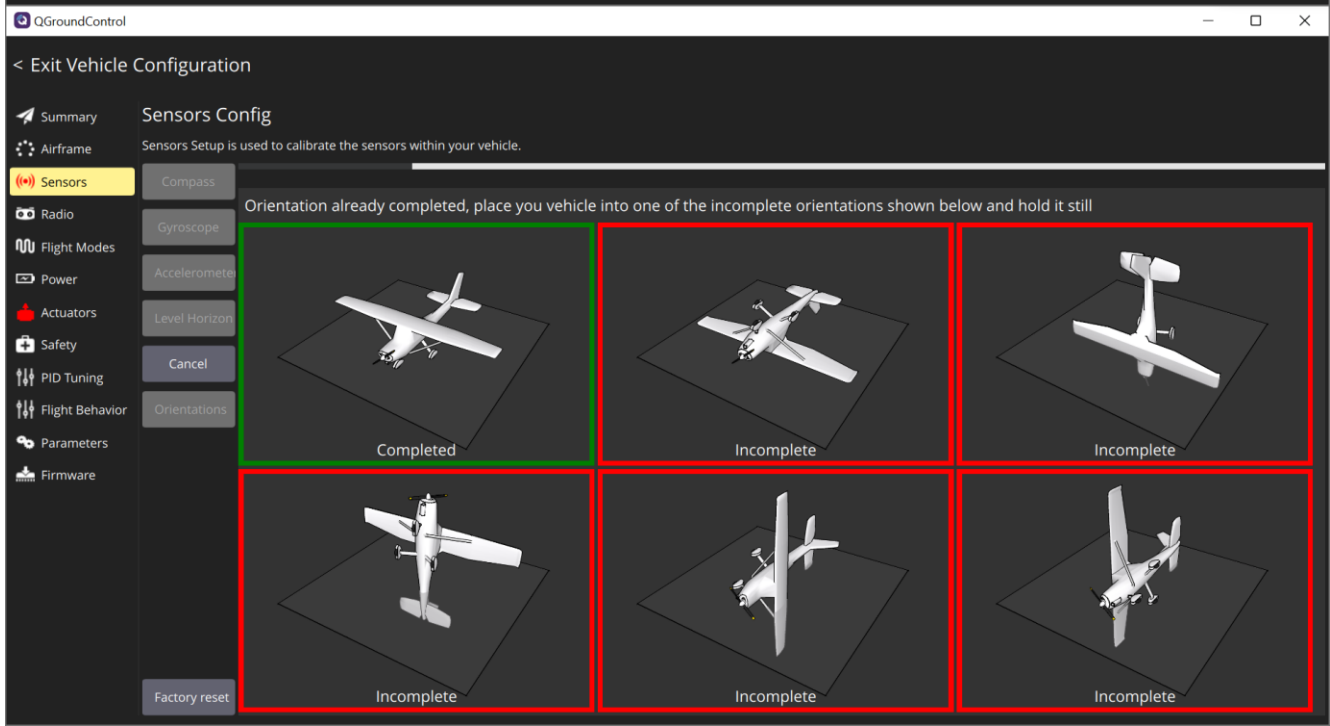
Top Side



Compass Calibration

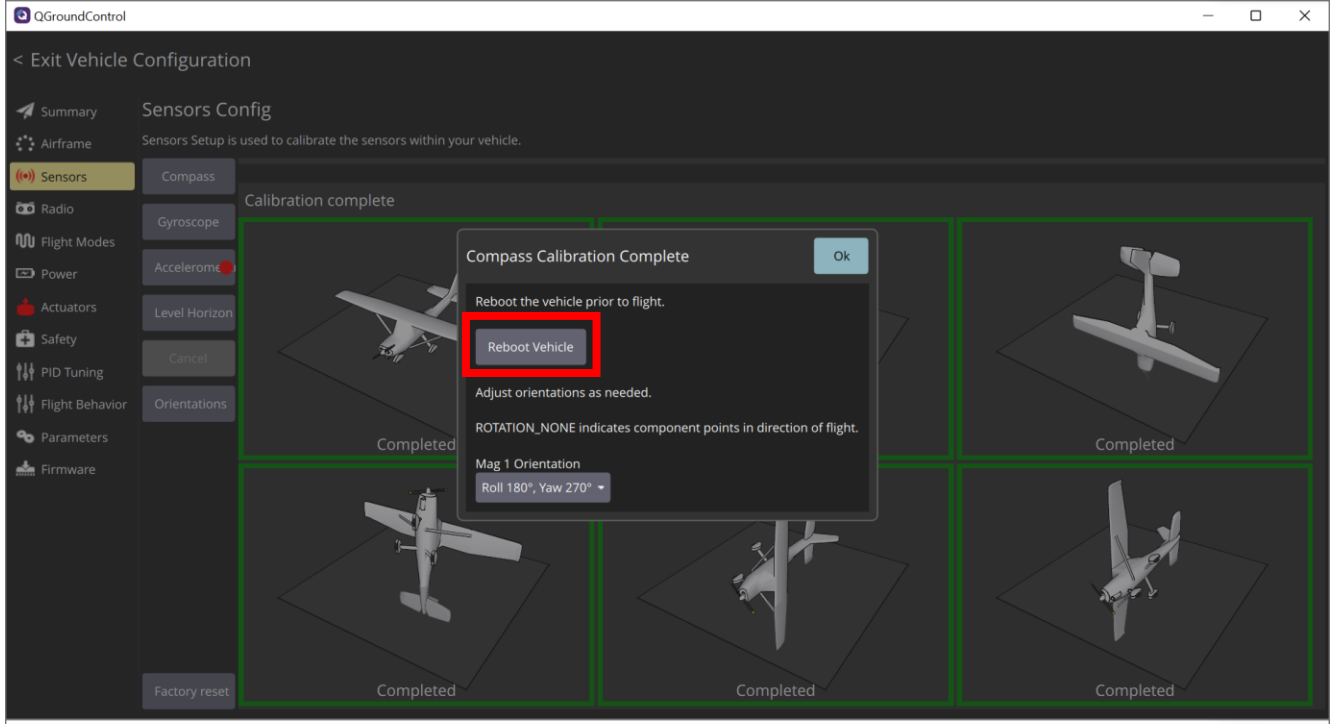
- Hold the drone in each orientation for a few seconds until the calibration box goes yellow
- Then rotate the drone about the vertical axis until the box is marked completed
- Repeat for all 6 orientations

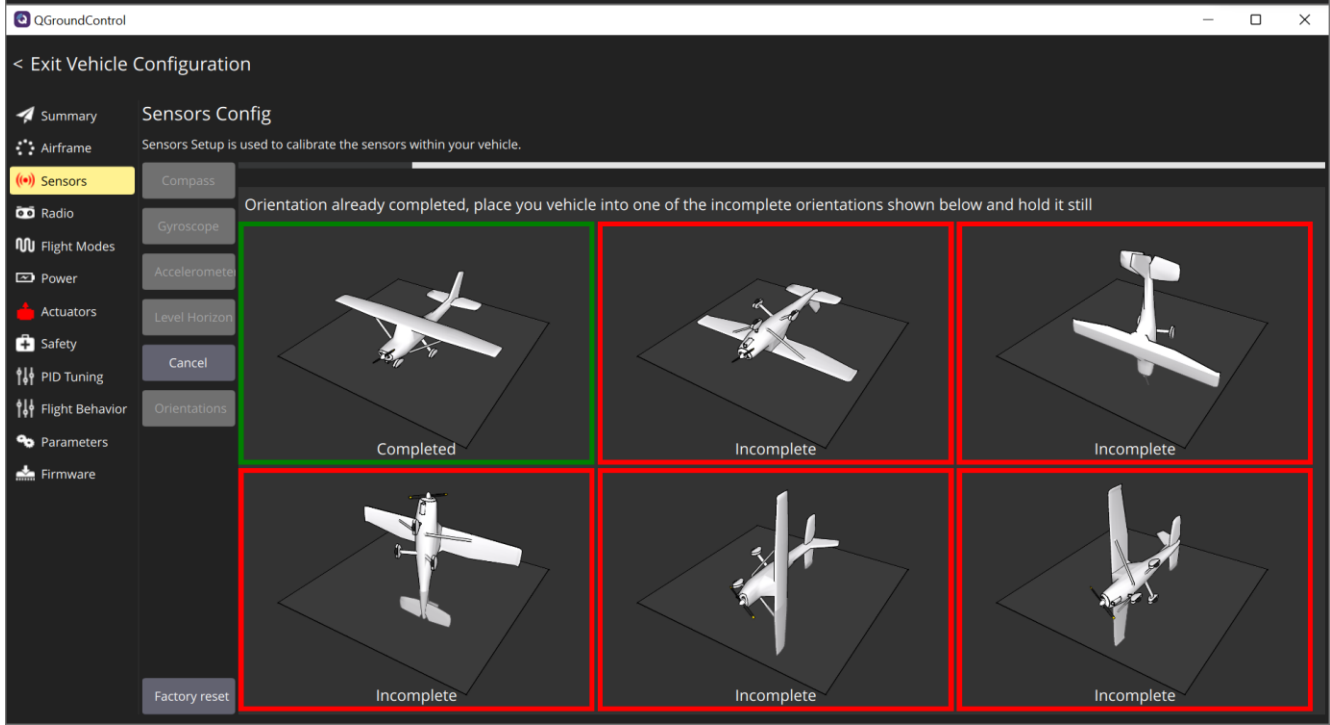




Compass Calibration

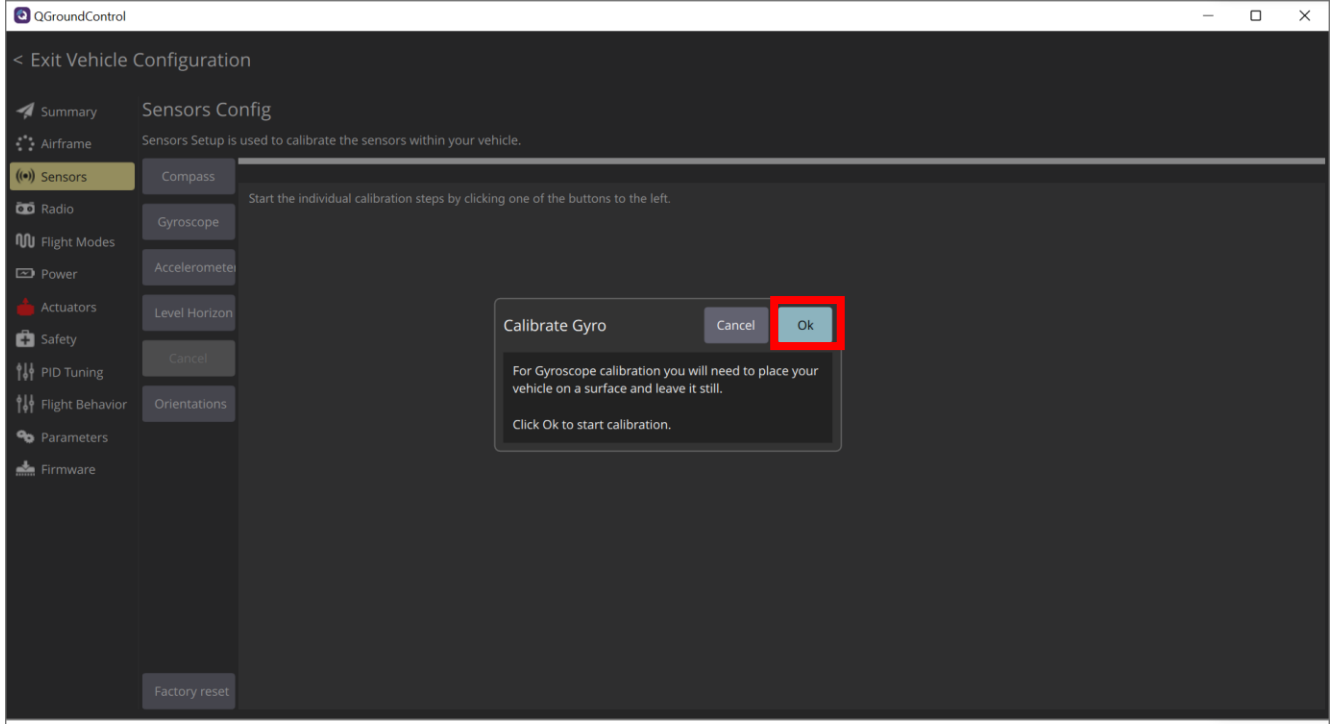
- Once all the orientation are completed click Reboot Vehicle to save the calibration
- You should not usually need to change Mag 1 Orientation
- This is set by the factory based on the magnetometer position and rotation relative to the whole flight controller

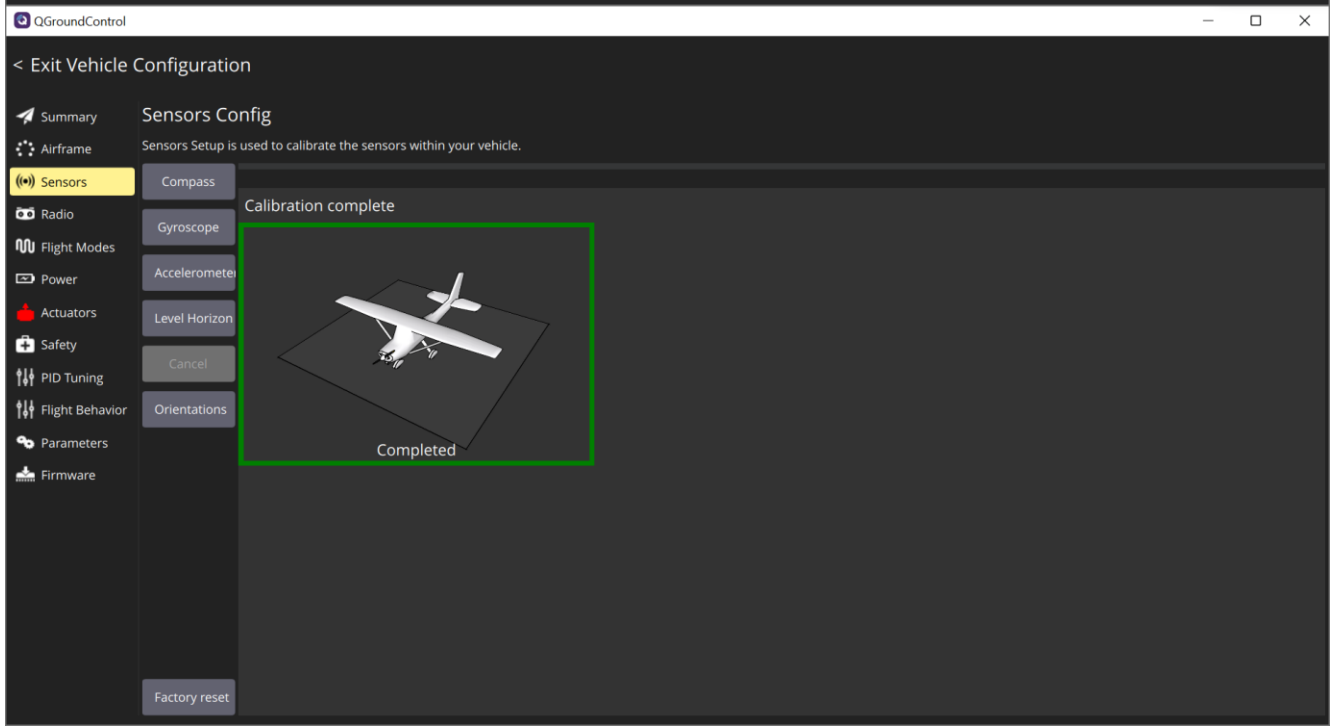




Gyro Calibration

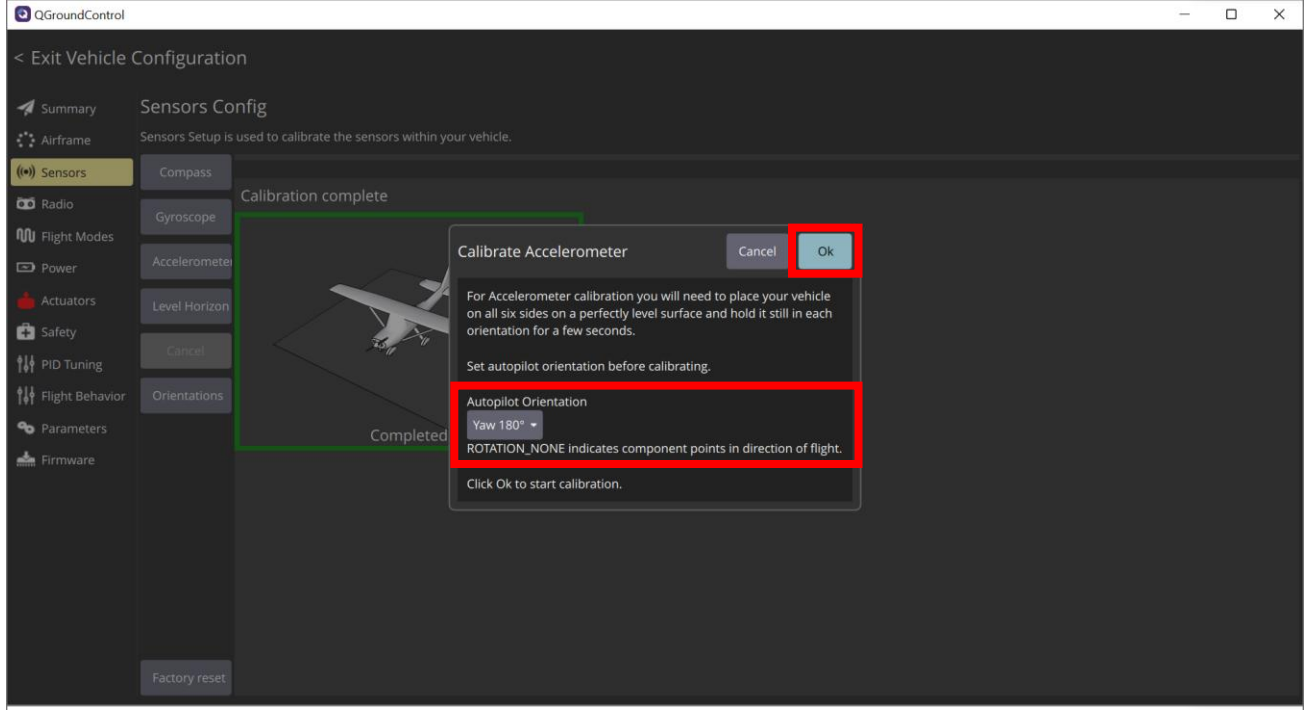
- This is very easy
- Put the drone on a level surface, leave it still
- Click OK
- The calibration should complete quickly
- If it does not it indicates you may have a bad gyro or too much vibration

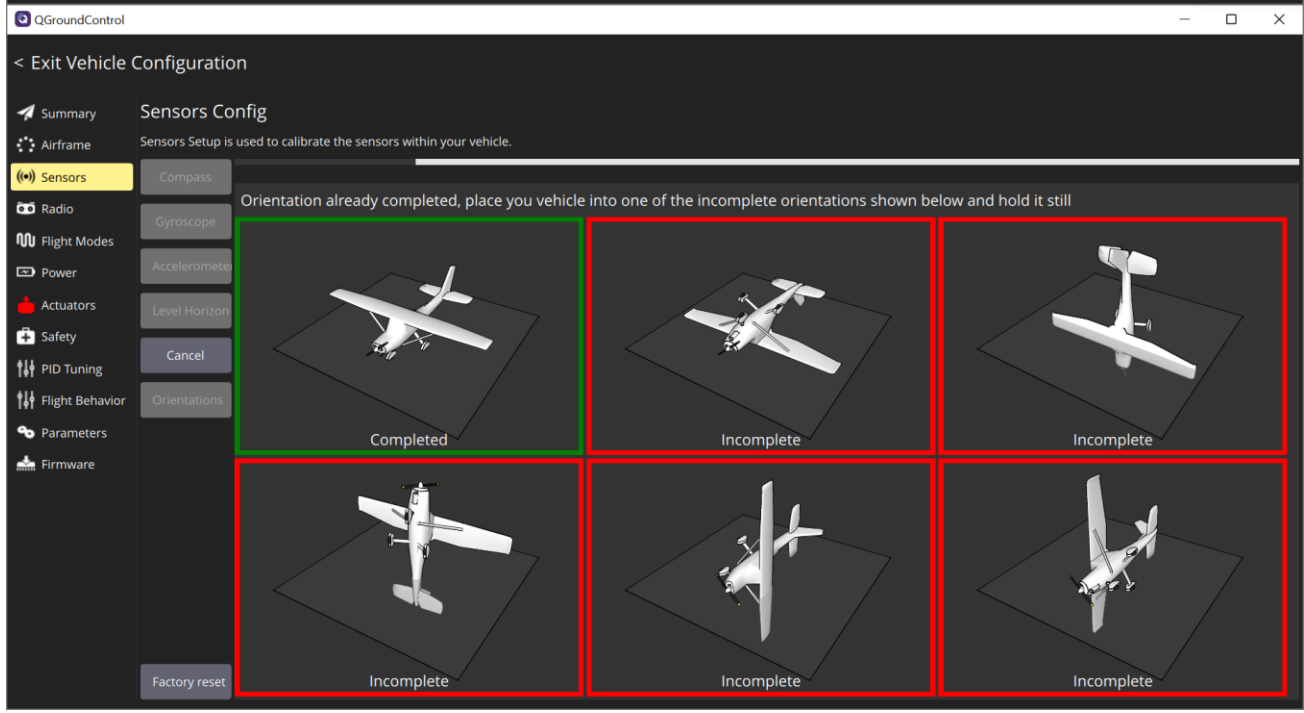




Accelerometer Cal.

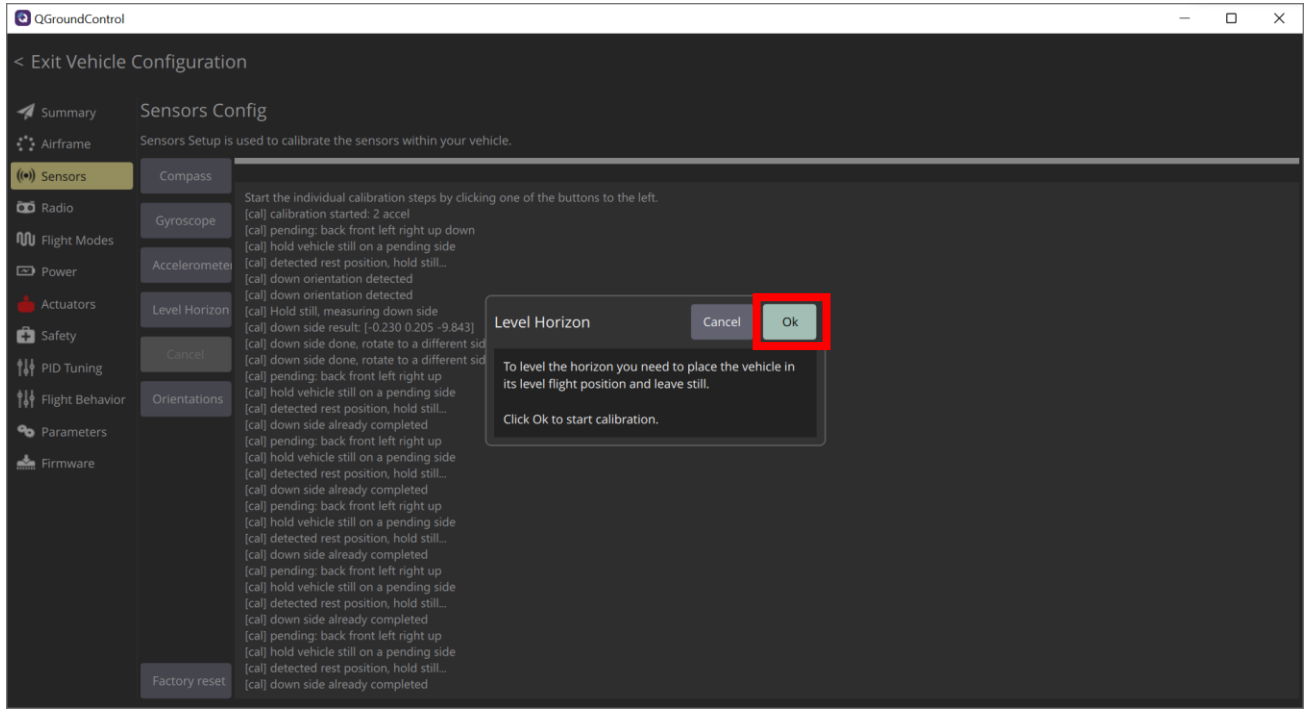
- This is **optional** as most accelerometers have a good factory calibration
- **It is also difficult to do if your drone won't sit flat on all six sides!**
- **I would suggest to skip this in most cases**
- Autopilot orientation should be set the same as for Compass Calibration (Yaw 180° for my drone, yours may be different)
- If you need to calibrate the accelerometer Click OK and follow the instructions





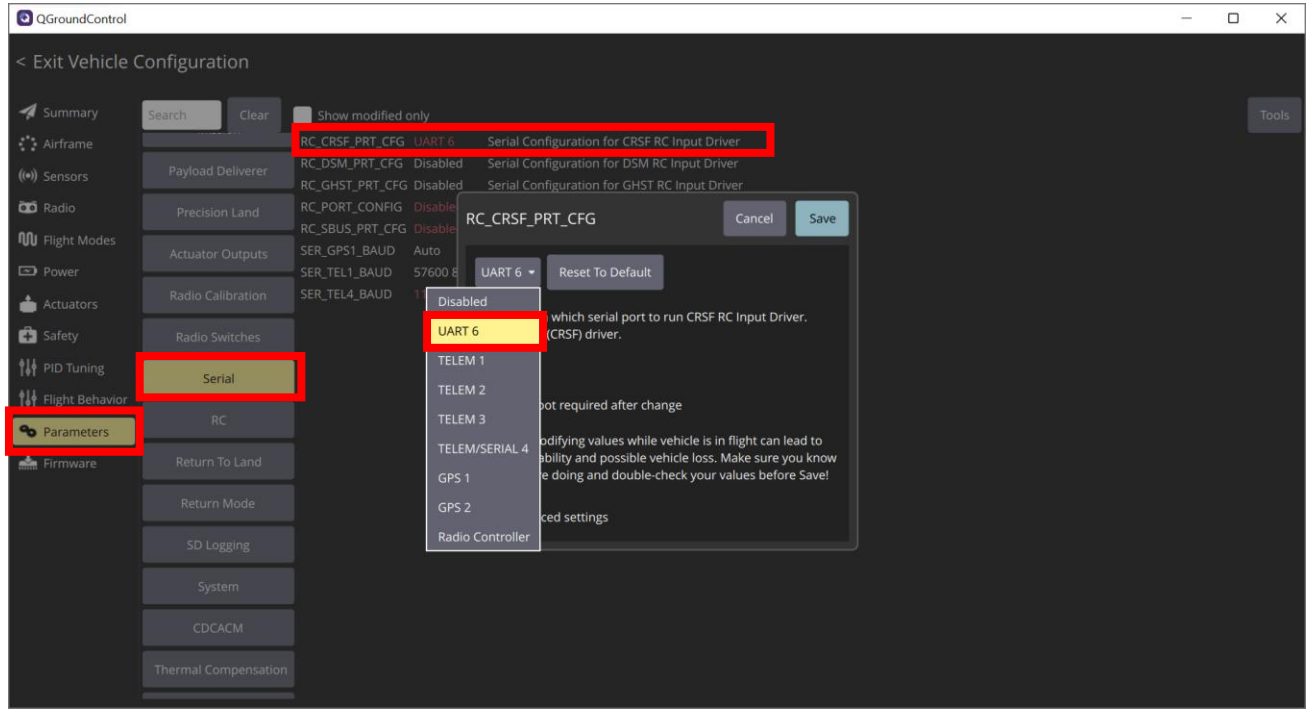
Level Cal.

- This quick, easy and helps reduce drift in stabilize mode
- Place the drone upright on a flat level surface
- Click OK
- The calibration should complete quickly
- If not this indicates an issue with your accelerometer or too much vibration



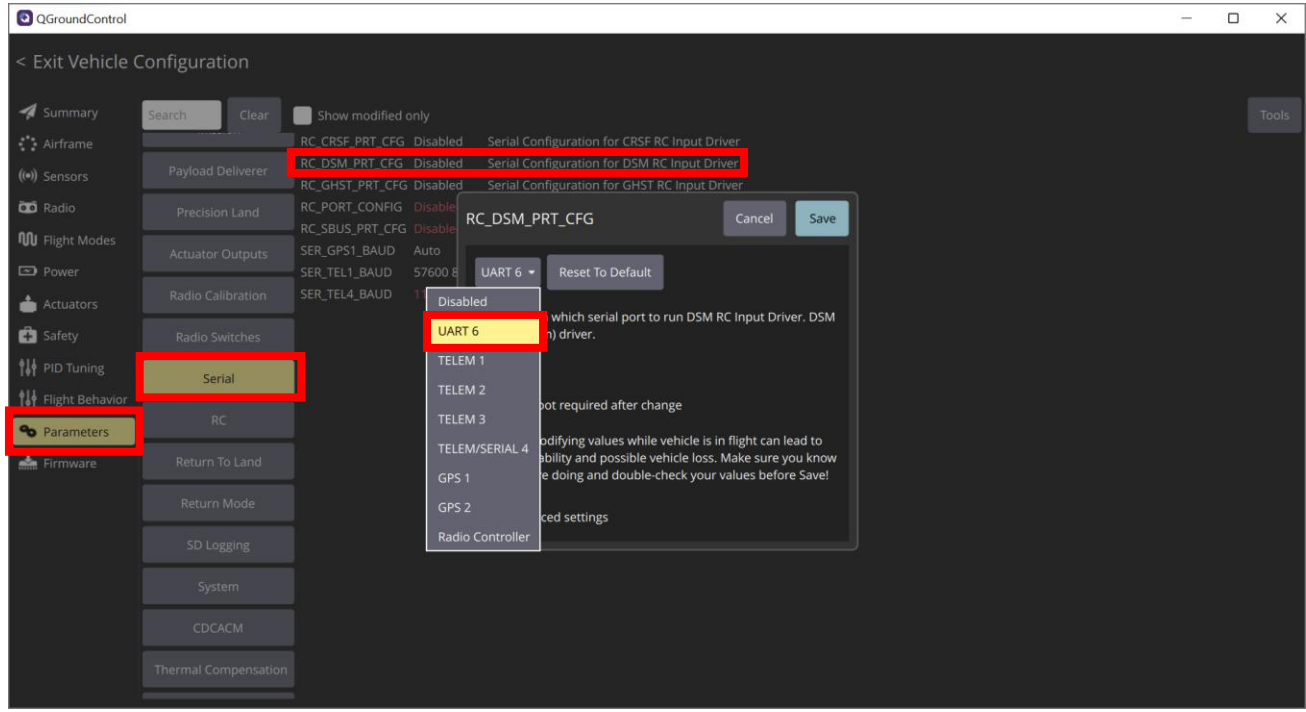
Radio Port - CRSF

- Before you can set up the Radio you will need to tell PX4 which port your receiver is connected to in **Parameters > Serial**
- If you are using a **CRSF** protocol receiver (TBS Crossfire, ExpressLRS) on **UART6**
- **RC_CRSF_PRT_CFG = UART6**
- **RC_DSM_PRT_CFG = Disabled**
- **RC_GHST_PRT_CFG = Disabled**
- **RC_PORT_CONFIG = Disabled**
- **RC_SBUS_PRT_CFG = Disabled**



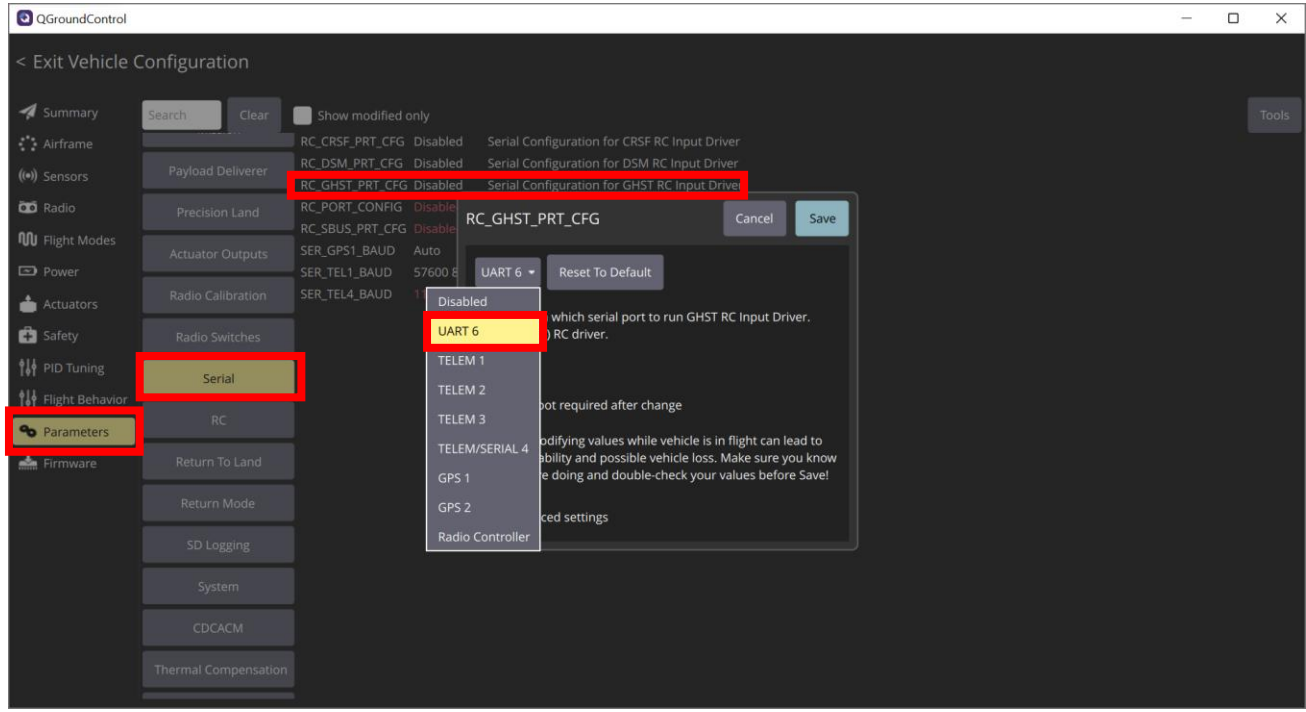
Radio Port – DSM (Spektrum)

- Before you can set up the Radio you will need to tell PX4 which port your receiver is connected to in **Parameters > Serial**
- If you are using a **DSM** protocol receiver (Spektrum Radios) on **UART6**
- **RC_CRSF_PRT_CFG = Disabled**
- **RC_DSM_PRT_CFG = UART6**
- **RC_GHST_PRT_CFG = Disabled**
- **RC_PORT_CONFIG = Disabled**
- **RC_SBUS_PRT_CFG = Disabled**



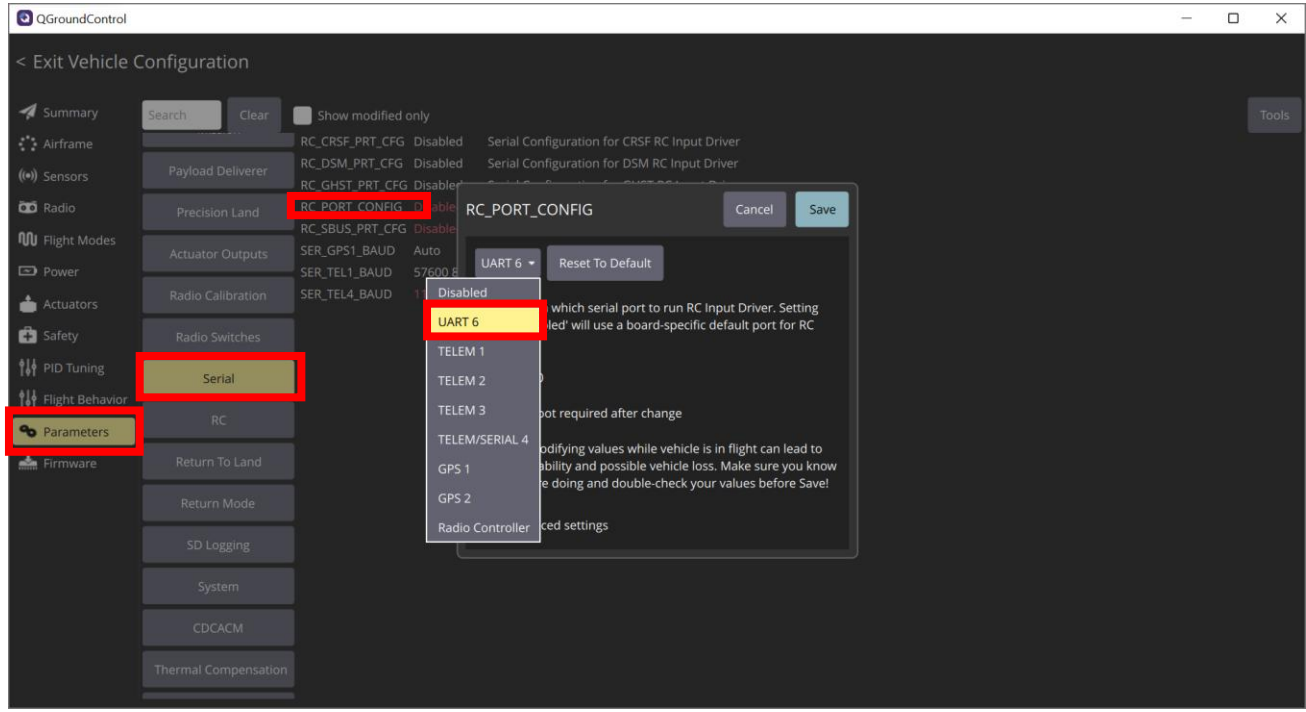
Radio Port – Ghost

- Before you can set up the Radio you will need to tell PX4 which port your receiver is connected to in **Parameters > Serial**
- If you are using a **Ghost** protocol receiver (ImpulseRC Ghost) on **UART6**
- **RC_CRSF_PRT_CFG = Disabled**
- **RC_DSM_PRT_CFG = Disabled**
- **RC_GHST_PRT_CFG = UART6**
- **RC_PORT_CONFIG = Disabled**
- **RC_SBUS_PRT_CFG = Disabled**



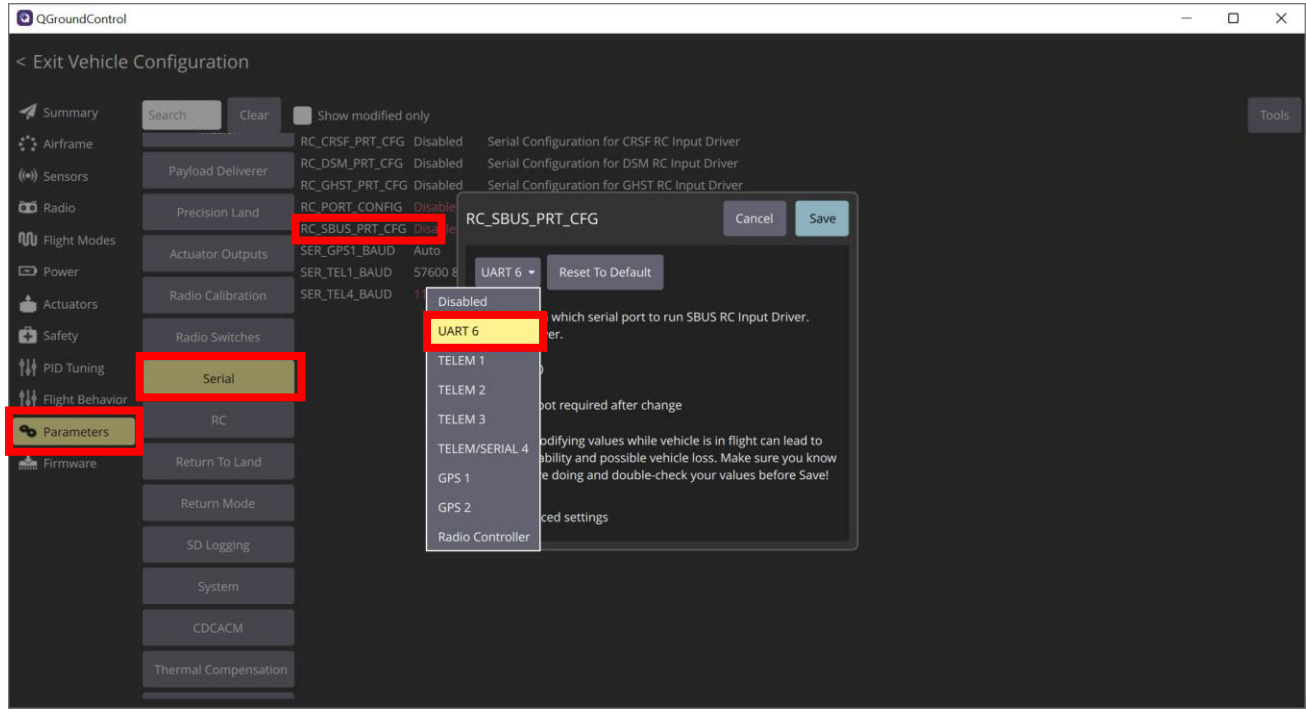
Radio Port – Autodetect

- Before you can set up the Radio you will need to tell PX4 which port your receiver is connected to in **Parameters > Serial**
- If you are using a **PPM, SUMD, or ST24** receiver on **UART6**
- **RC_CRSF_PRT_CFG = Disabled**
- **RC_DSM_PRT_CFG = Disabled**
- **RC_GHST_PRT_CFG = Disabled**
- **RC_PORT_CONFIG = UART6**
- **RC_SBUS_PRT_CFG = Disabled**



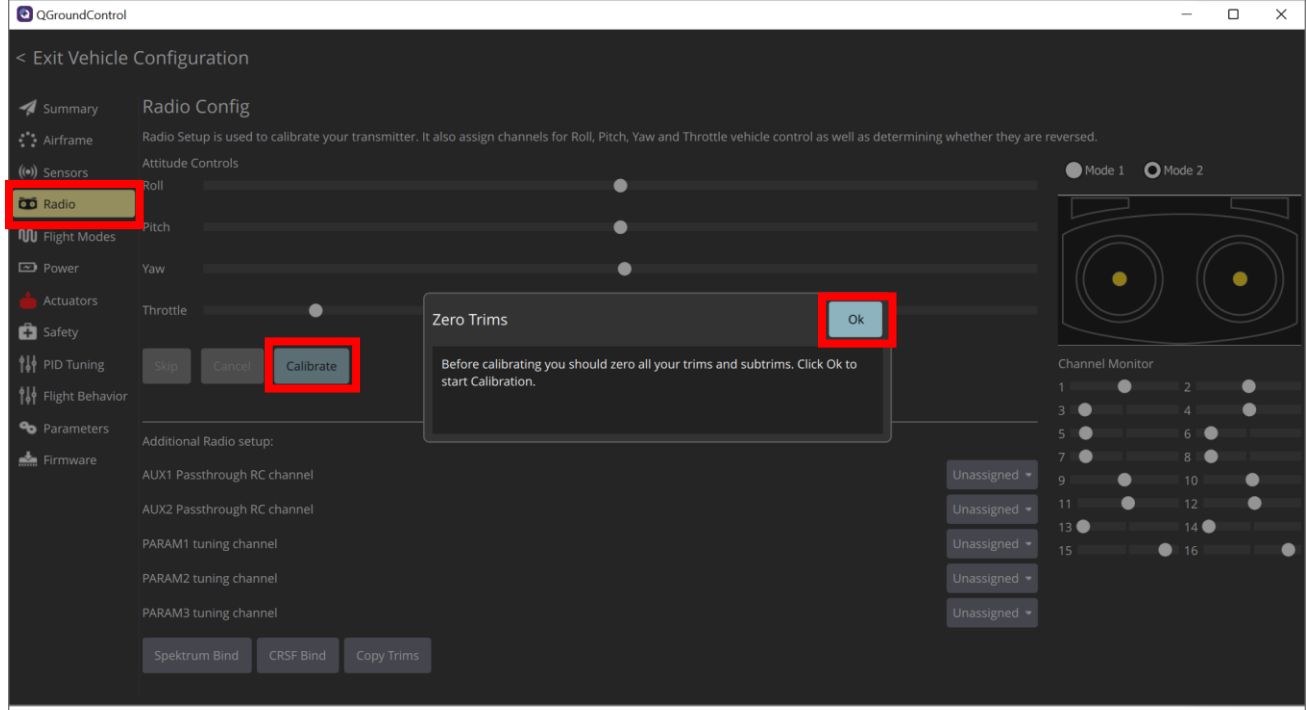
Radio Port - SBUS

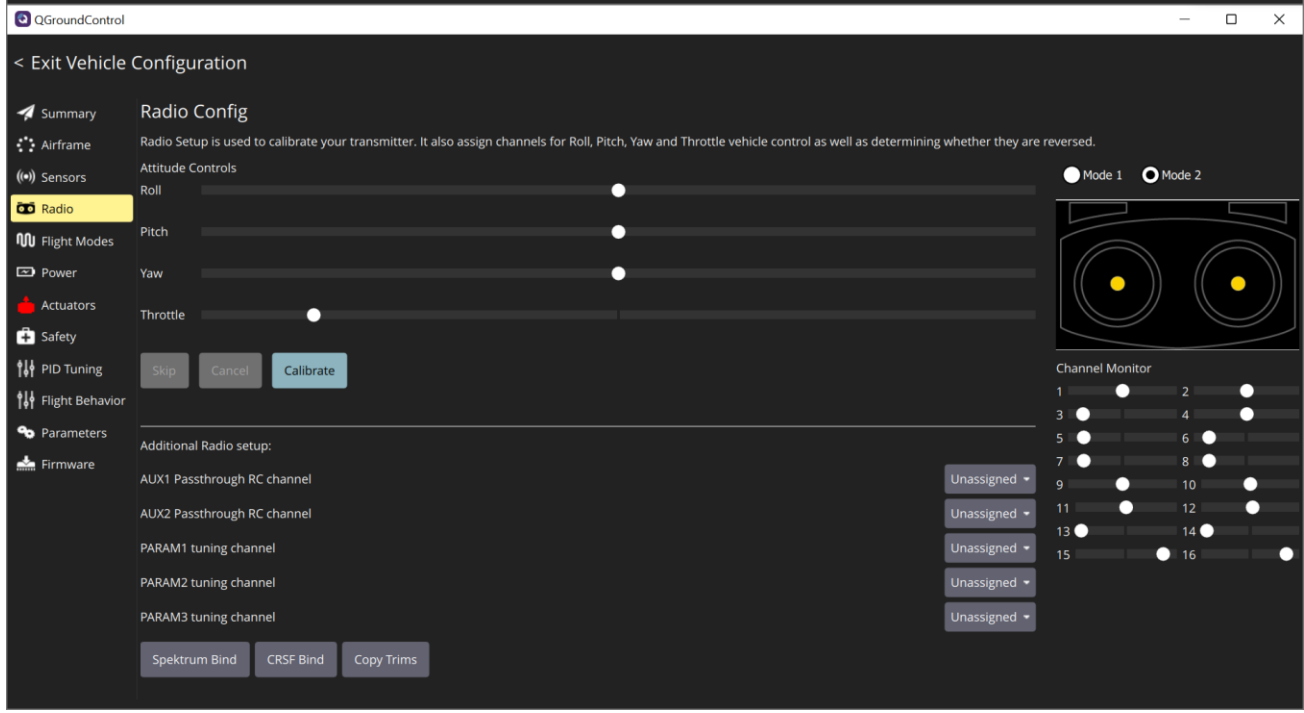
- Before you can set up the Radio you will need to tell PX4 which port your receiver is connected to in **Parameters > Serial**
- If you are using a **SBUS** protocol receiver (TBS Crossfire, ExpressLRS) on **UART6**
- **RC_CRSF_PRT_CFG = Disabled**
- **RC_DSM_PRT_CFG = Disabled**
- **RC_GHST_PRT_CFG = Disabled**
- **RC_PORT_CONFIG = Disabled**
- **RC_SBUS_PRT_CFG = UART6**



Radio Calibration

- You will need to go through the binding procedure for your receiver (for ELRS power cycle the receiver quickly 3+ times and look for the LED double flash)
- Centre your sticks and trims
- Click **Calibrate > OK**
- Follow the instructions to Map your stick channels
- Once calibration is complete ensure all channels/switches respond as expected and Roll, Pitch, and Yaw channels self-centre

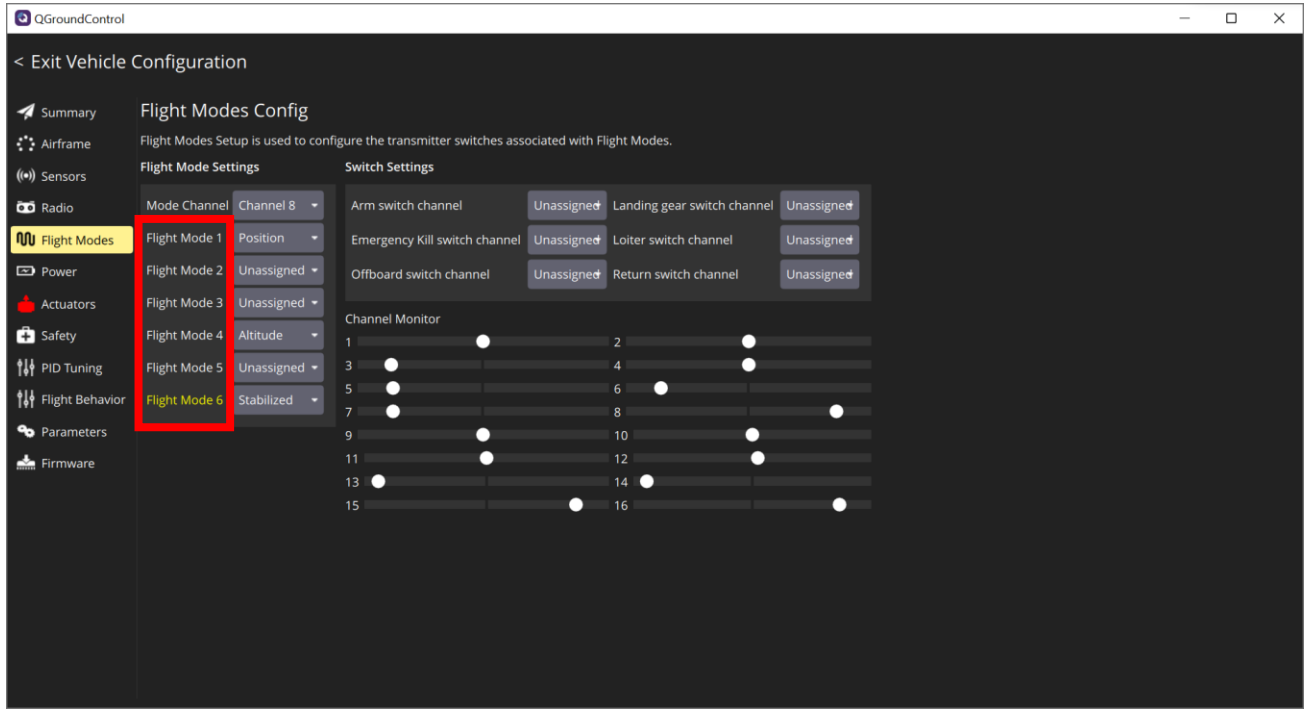




Flight modes

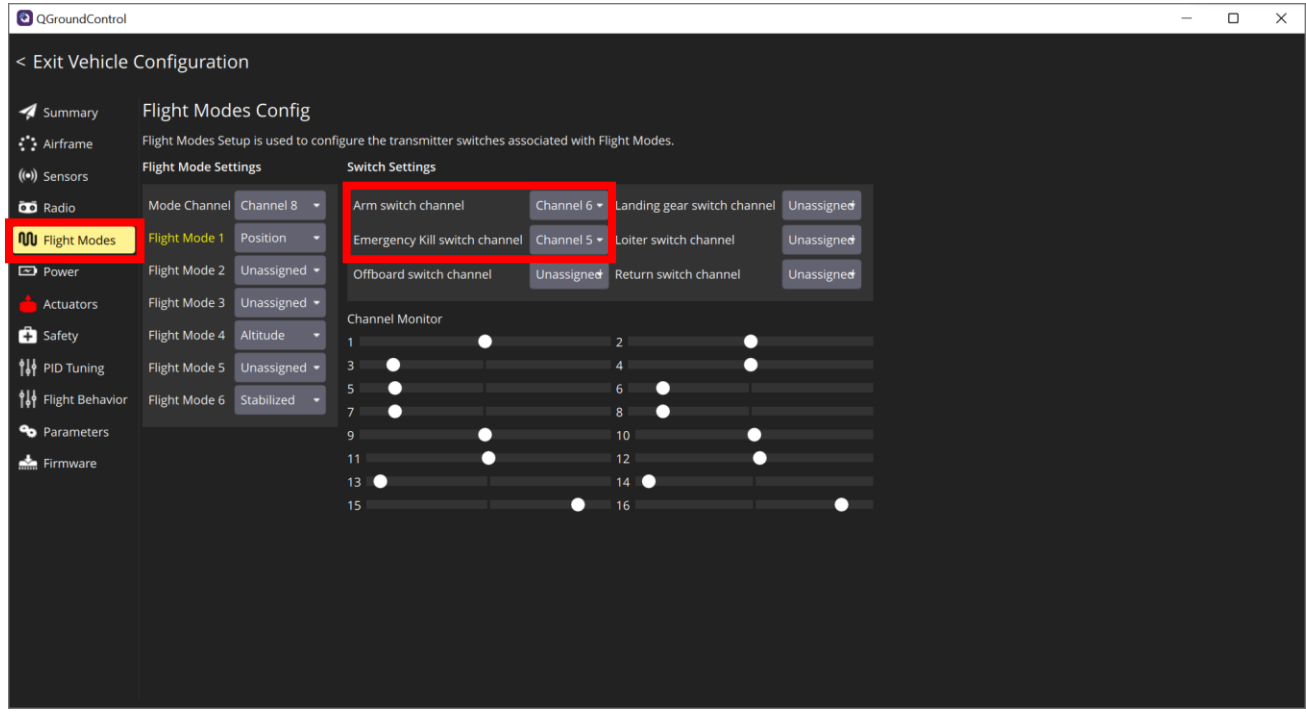
- This controls what mode the drone is in: Position, Stabilize, Altitude etc.
- Set the Mode Channel to the flight mode switch on your radio (or a multi-position switch)
- As you move that switch the flight modes should light up yellow
- Set up your flight modes as you prefer
- For tuning we will use Stabilize, Altitude and Position modes
- For safety the default position should be Position mode





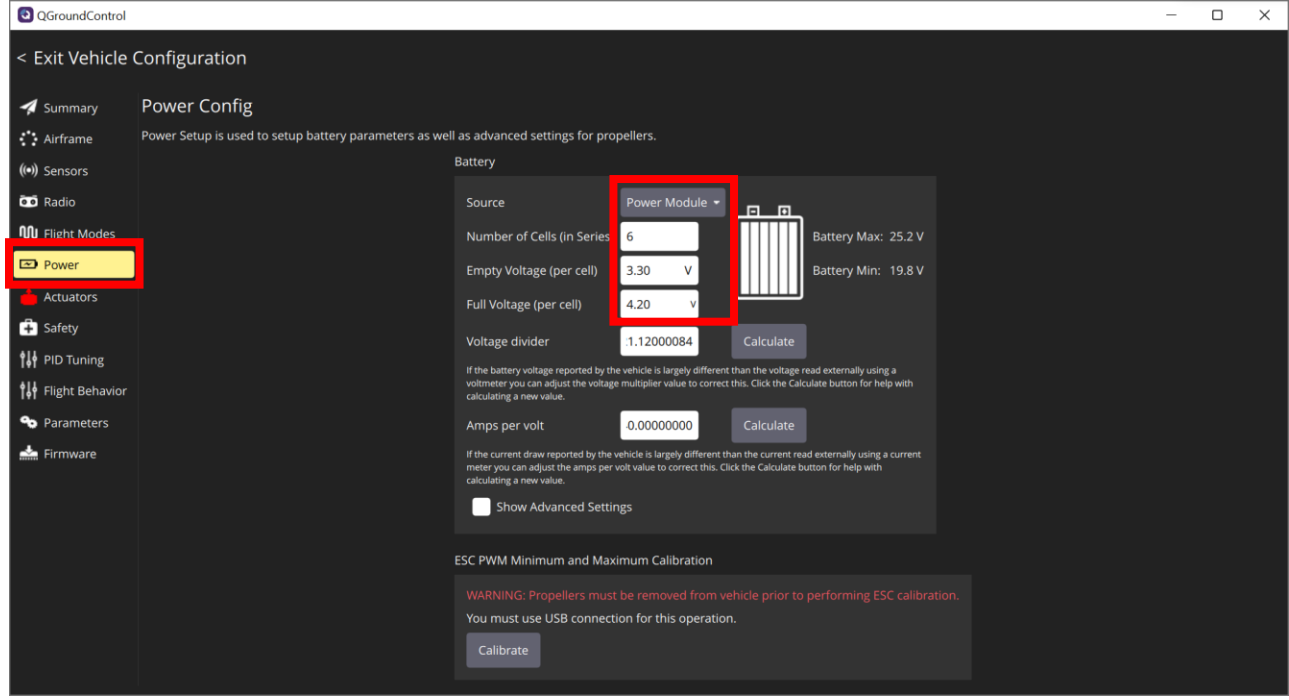
Kill Switch

- When tuning a drone it is good practice to have a kill switch
- If the drone starts to misbehave stick arming/disarming is too slow
- You can also set a separate arm switch if you prefer that to stick arming



Power

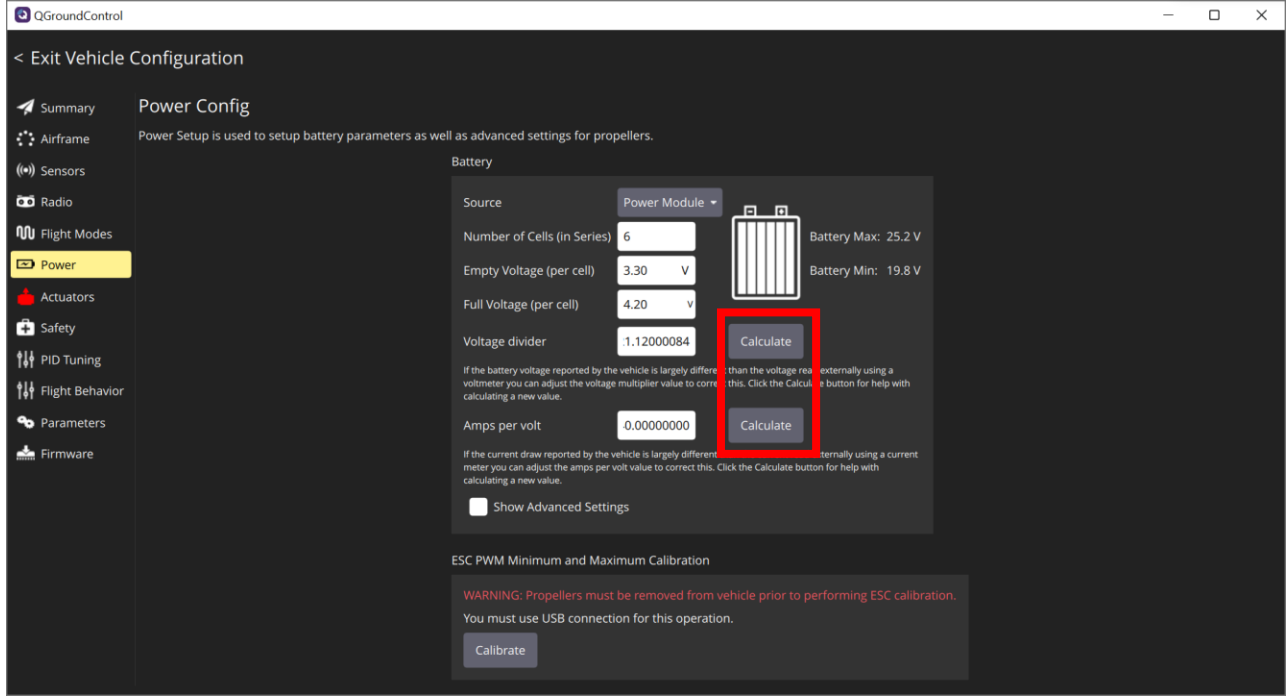
- Here you'll set your battery cell count, maximum, and minimum voltages
- If you're using the on board current meter on the FC set this to Power Module
- Set the number of cells in your battery and the max and min voltages



Chemistry	Max Voltage (V)	Min Voltage (V)
Li-Po	4.2	3.5 – 3.3
Li-Ion	4.2 – 4.3 (cell dependent)	2.5 – 3.3 (cell dependent)
Li-HV	4.35	3.5 – 3.3

Power

- Voltage divider and Amps per volt should be set correctly from the factory but if not you can calibrate them using the calculate buttons and an external Voltmeter or Ammeter
- For simple shunt resistors Amps per Volt is $1/R_{shunt}$

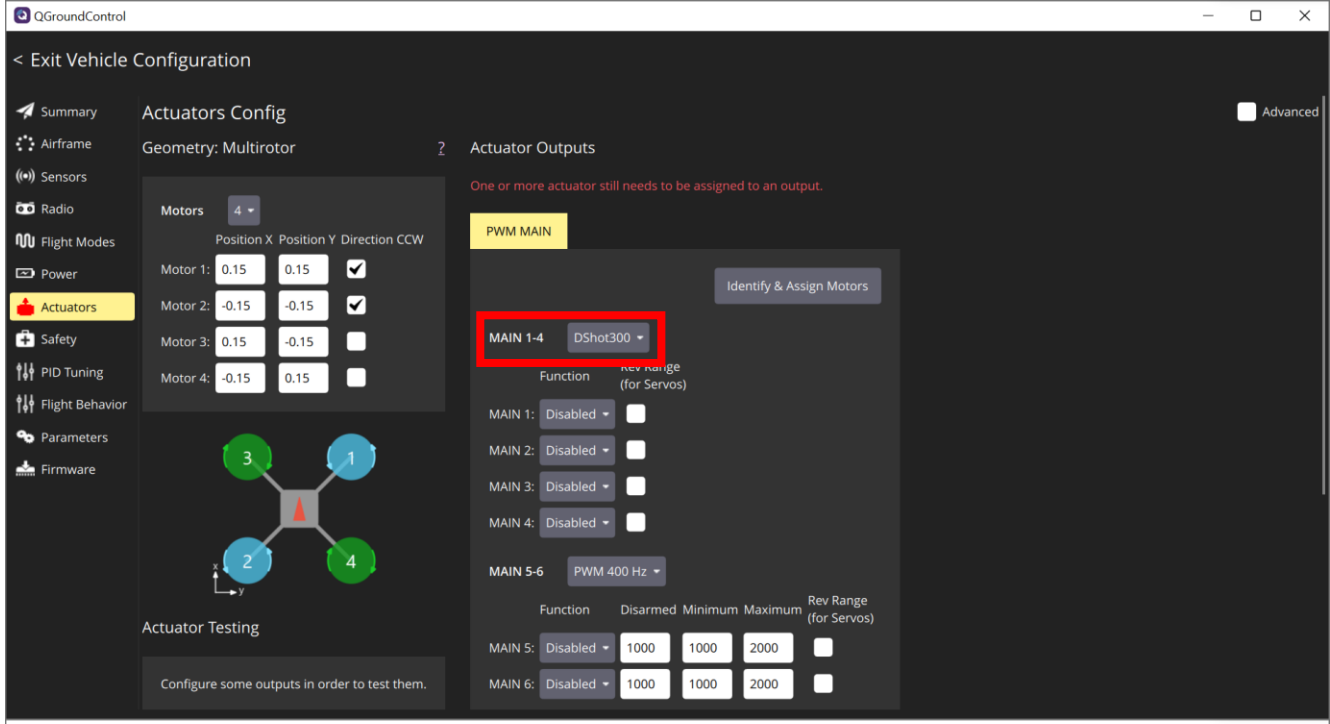


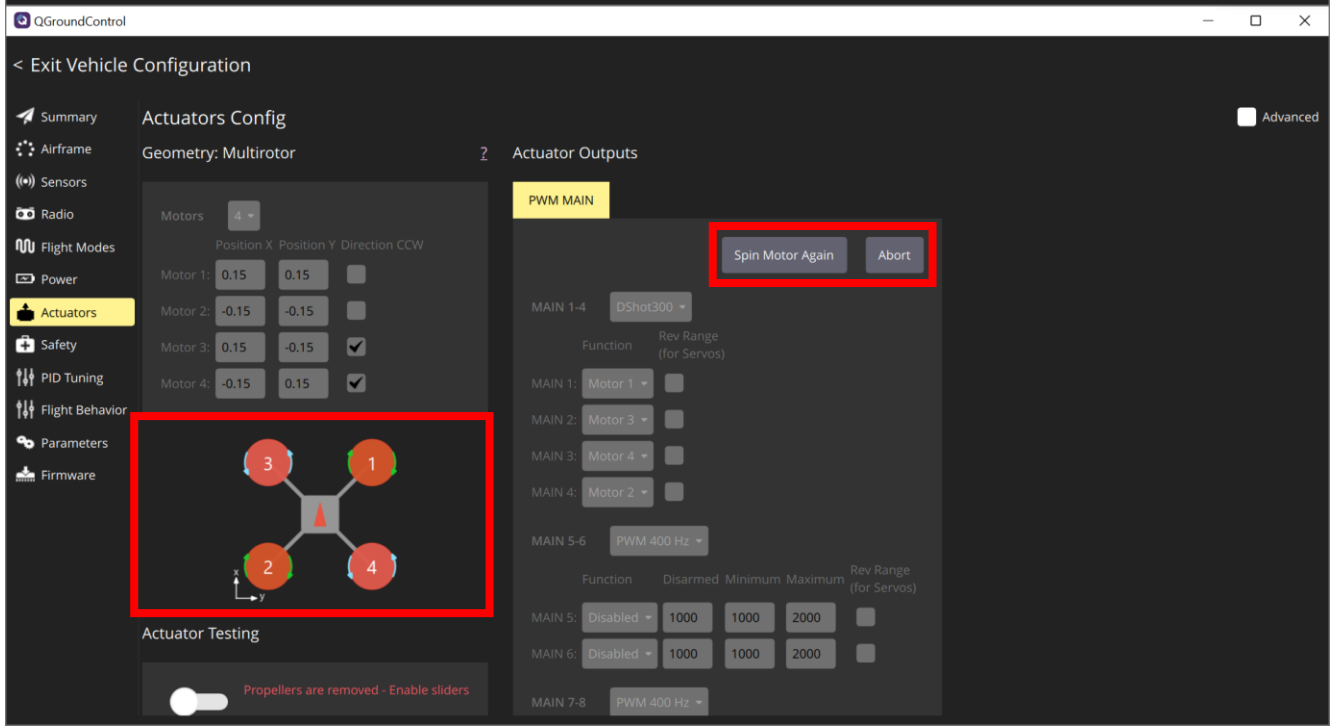
Resistor value (mΩ)	Amps per Volt (A/V)
10	100
20	50
25	40

Motor Protocol

- Before we can assign motor outputs we need to select the correct protocol
- DShot ESCs are strongly recommended!

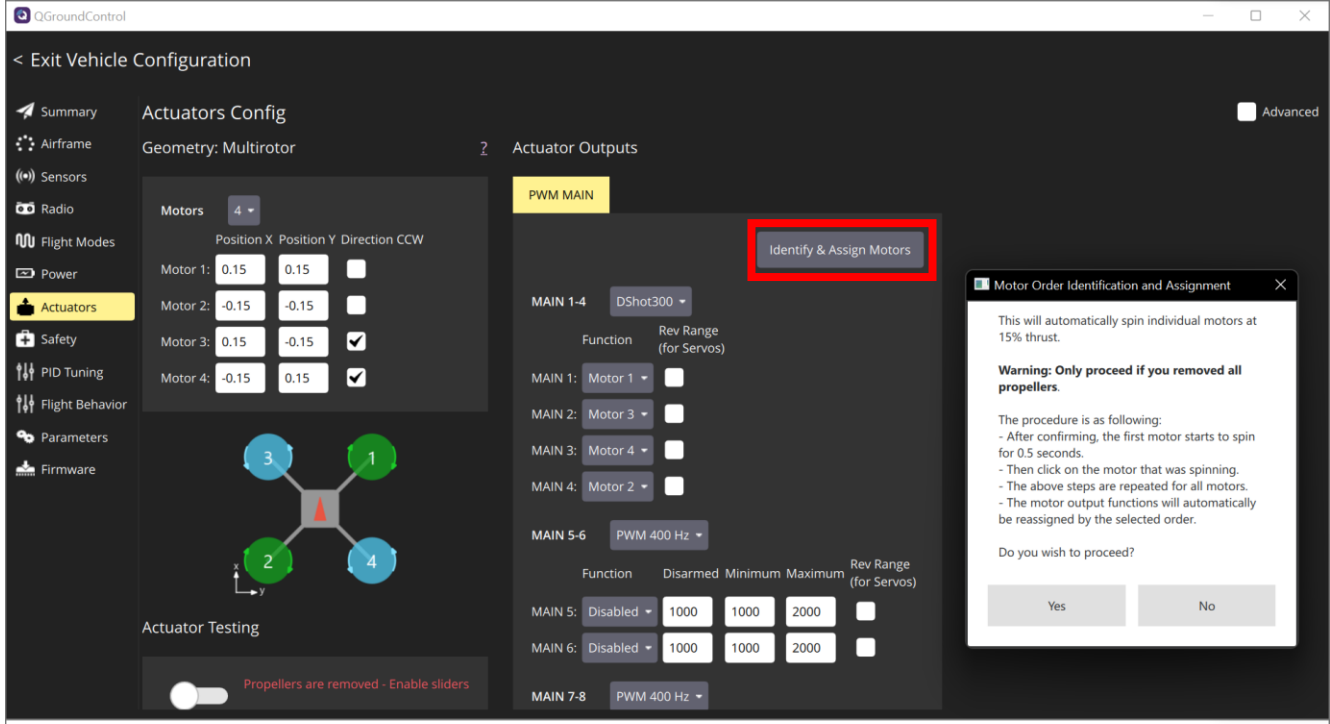
ESC type	Suggested setting
DShot ESCs	DShot300
PWM ESCs	PWM 400Hz (or the fastest rate supported)

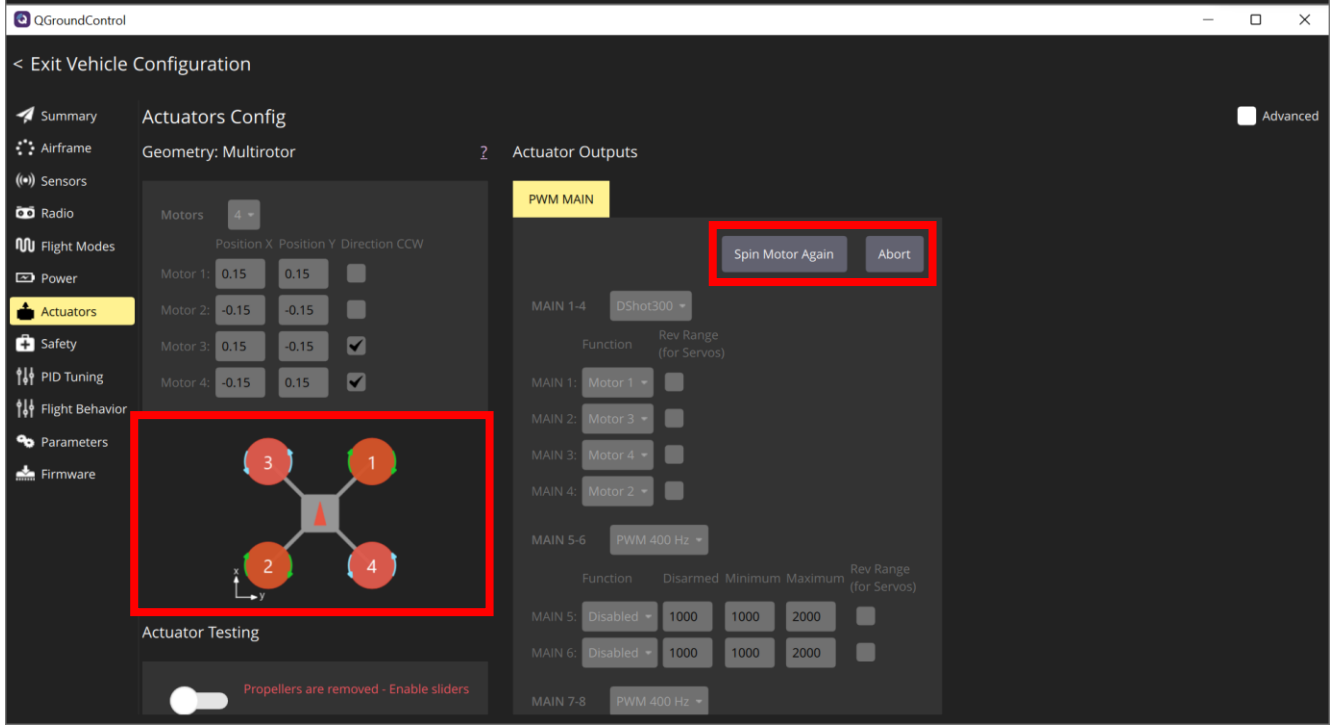




Motor Order

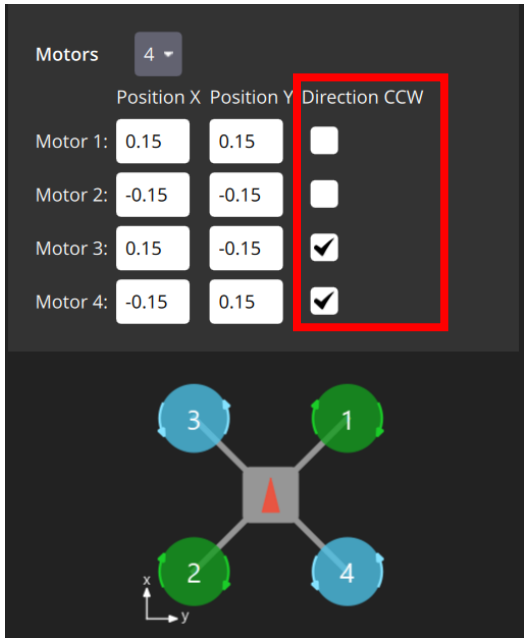
- Here we are going to configure the motor positions, directions and outputs
- **Remove props** and plug in a battery
- Click **Identify and Assign Motors**
- Check the warning then click **Yes** to proceed
- Click the currently spinning motor in the diagram
- Click **Spin motor again** as often as necessary
- **Abort** is self-explanatory!



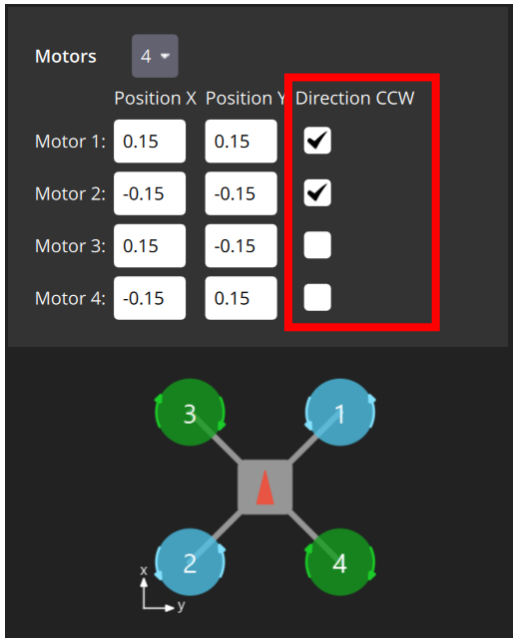


Motor Direction

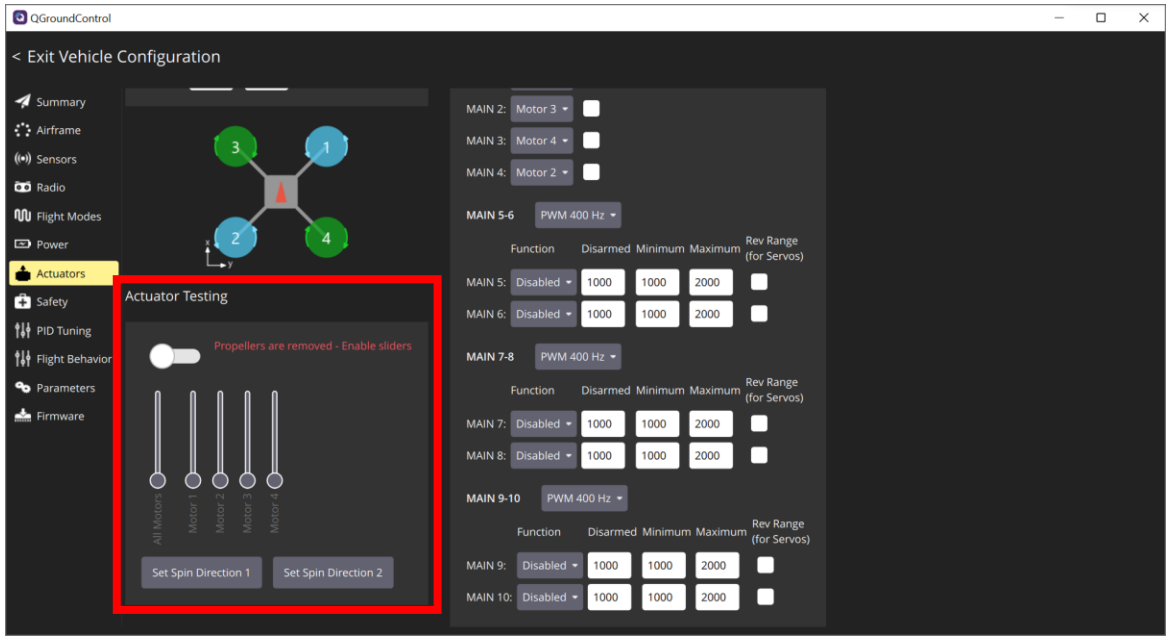
- Once motor order is set up correctly we need to configure motor direction
- There are two valid configurations for a quadcopter
- **Props IN** is usually the default
- **Props OUT** is preferred for maximum flight performance
- Use the **Actuator Testing** section to check motor direction matches the diagram exactly
- Tick the check-box and run each motor using the sliders



Props OUT

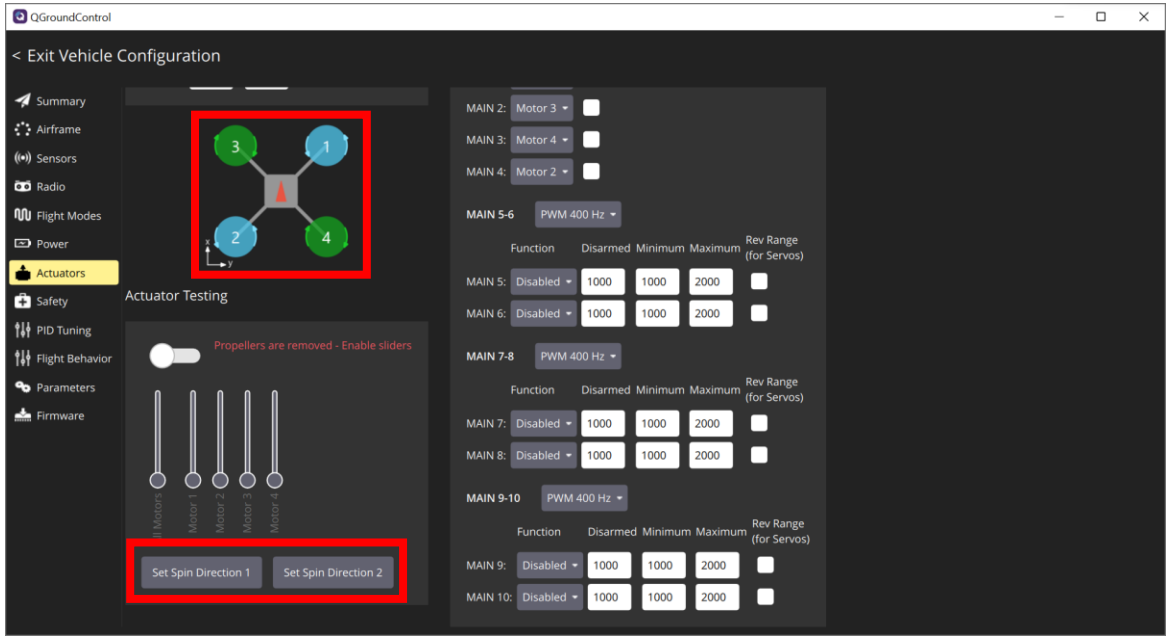


Props IN



Motor Direction

- For DShot ESCs motor direction can be changed by clicking **Set Spin Direction 1 (or 2)** and then choosing the correct motor based on the diagram
- **Spin Direction 1 is “default”**
- **Spin Direction 2 is “reversed”**
- Try both, one should be the correct direction
- If you have PWM ESCs then swap any two of the three motor wires to change the motor direction



Changing Motor Direction

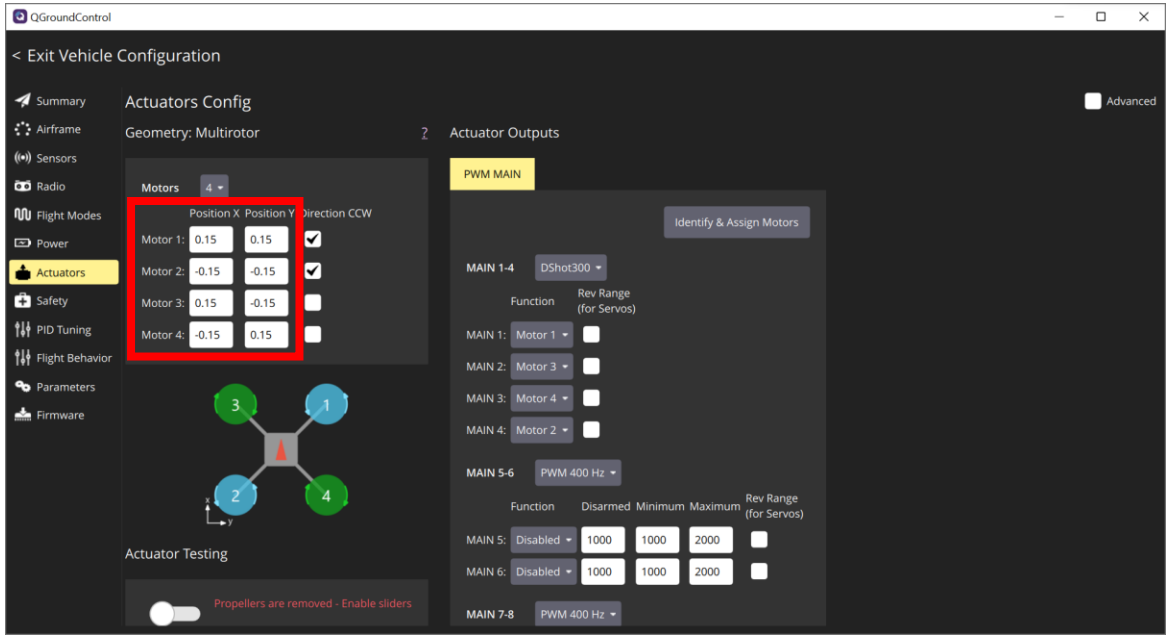
- If a motor is spinning in the wrong direction often the best solution is to swap two of the three motor wires soldered to the ESC
- This will reverse the motor direction
- It is also usually possible to do this by changing settings in the ESC firmware but a hardware fix can be more reliable in production





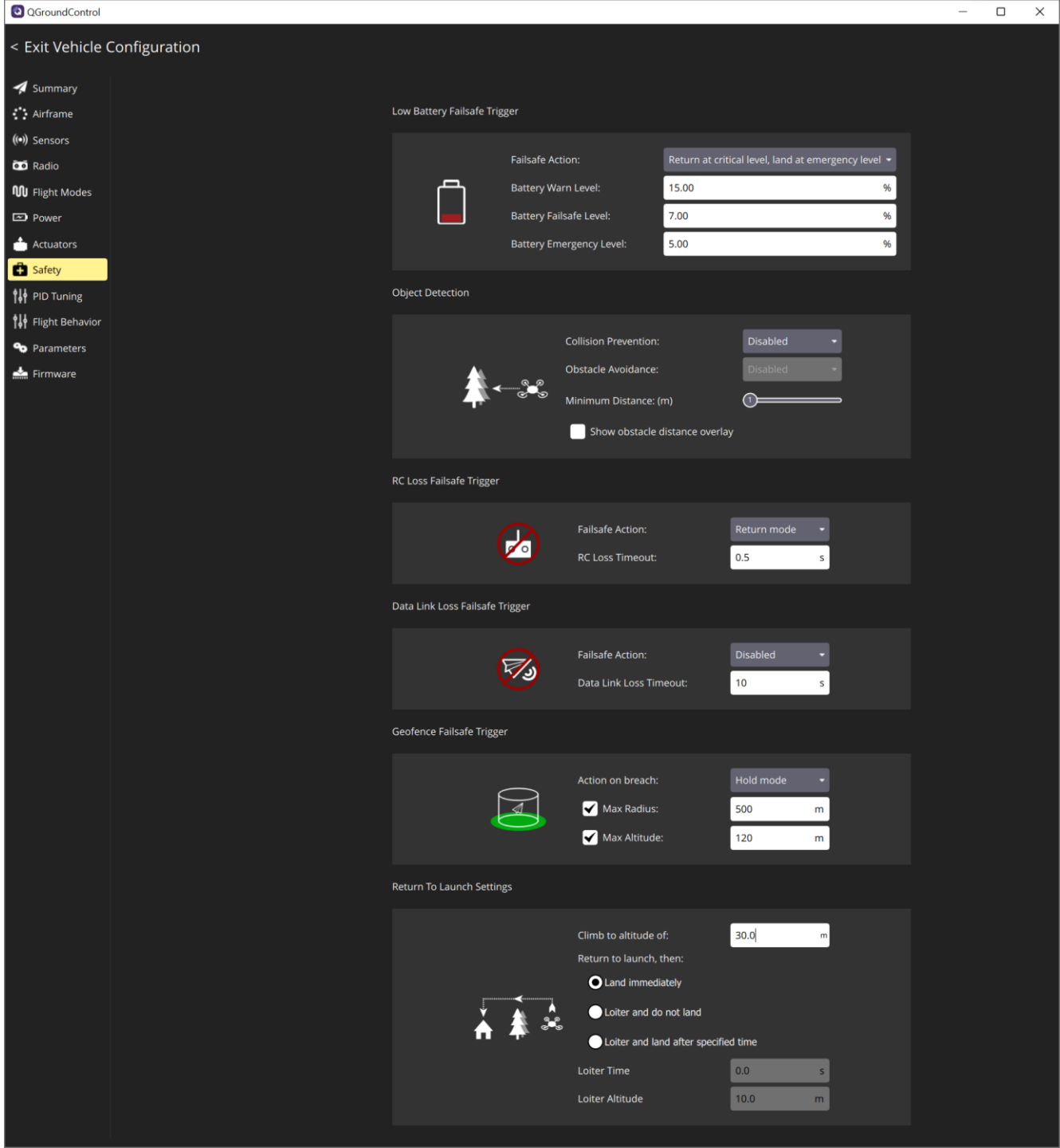
Motor Position

- To maximise flight performance it is useful to correctly set the motor positions
- Check the diagram for X and Y axes (X is Forward, Y is Right)
- Measure from the **Centre of Mass** of the drone



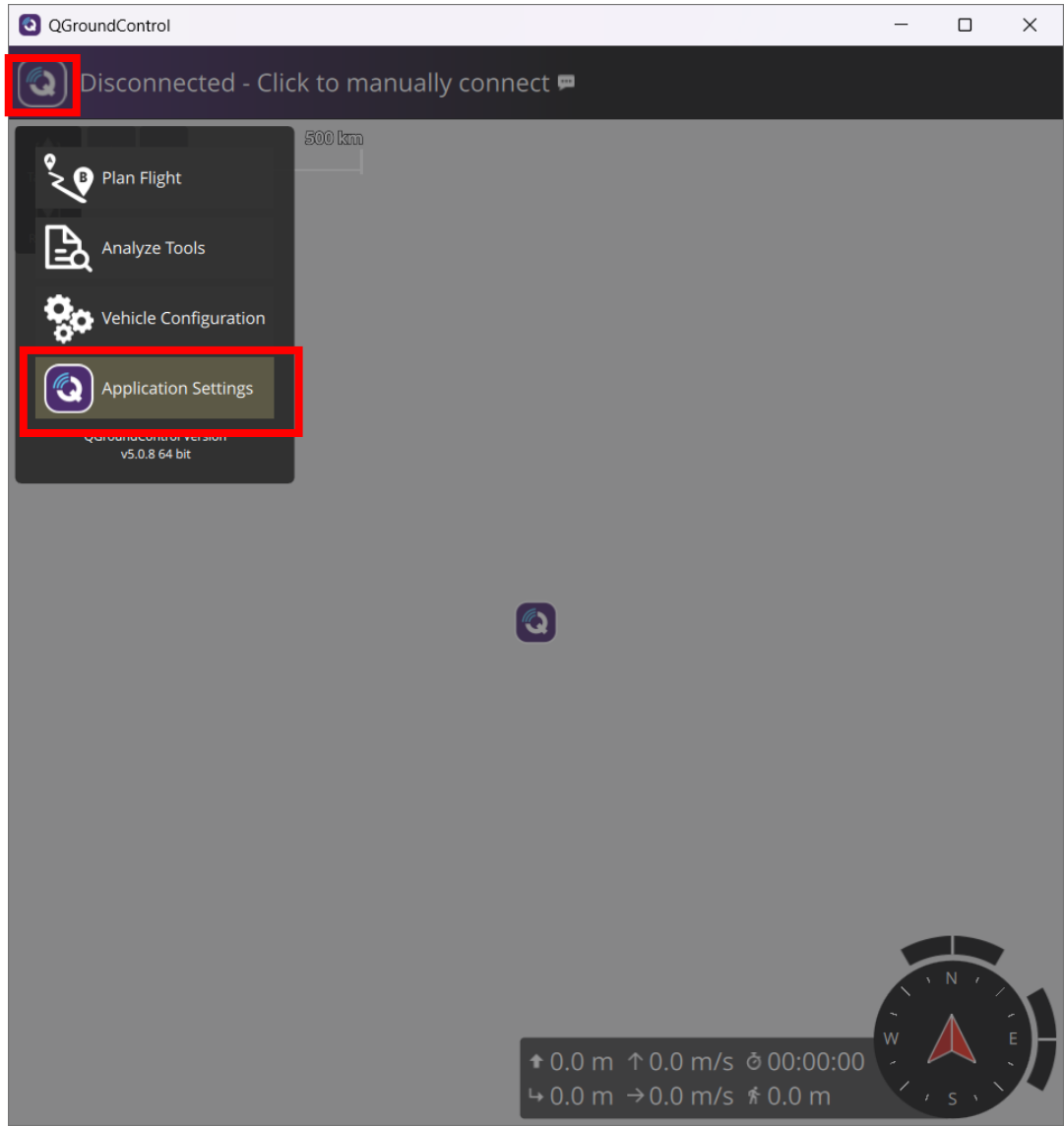
Safety

- Defaults are generally sensible
- Low Battery Failsafe can be set to **Return and Land**
- Geofence settings:
  - Radius = 500~700m (line of sight)
  - Altitude = 120m (400 ft)
- Return to Launch
  - Set altitude above the tallest obstacle



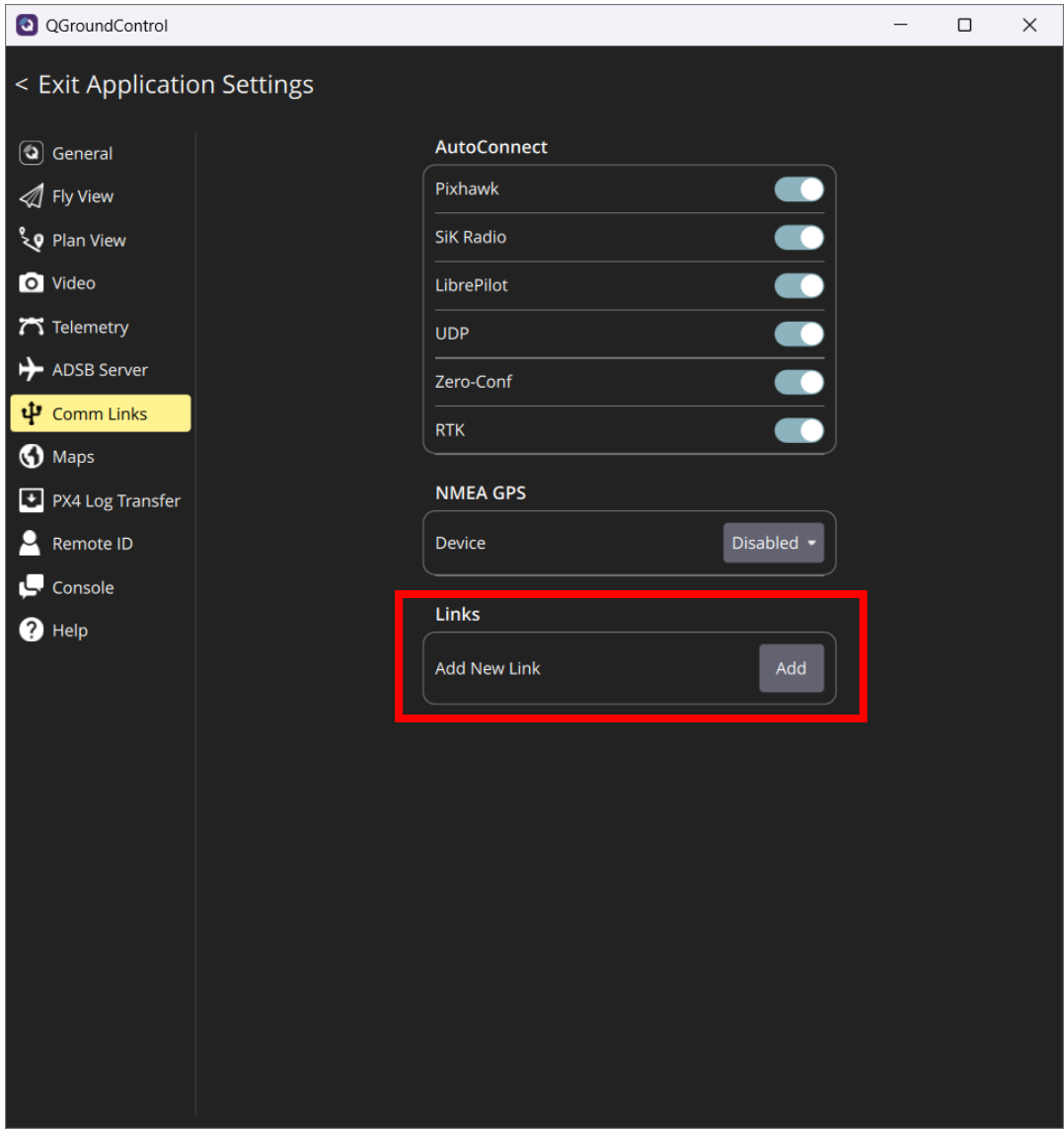
Connect to MicroAirV2 Over Bluetooth

- You'll need to add a new Comm Link in QGroundControl
- Click **Qlogo > Application Settings**
- Then Click the **Comm Links Tab**



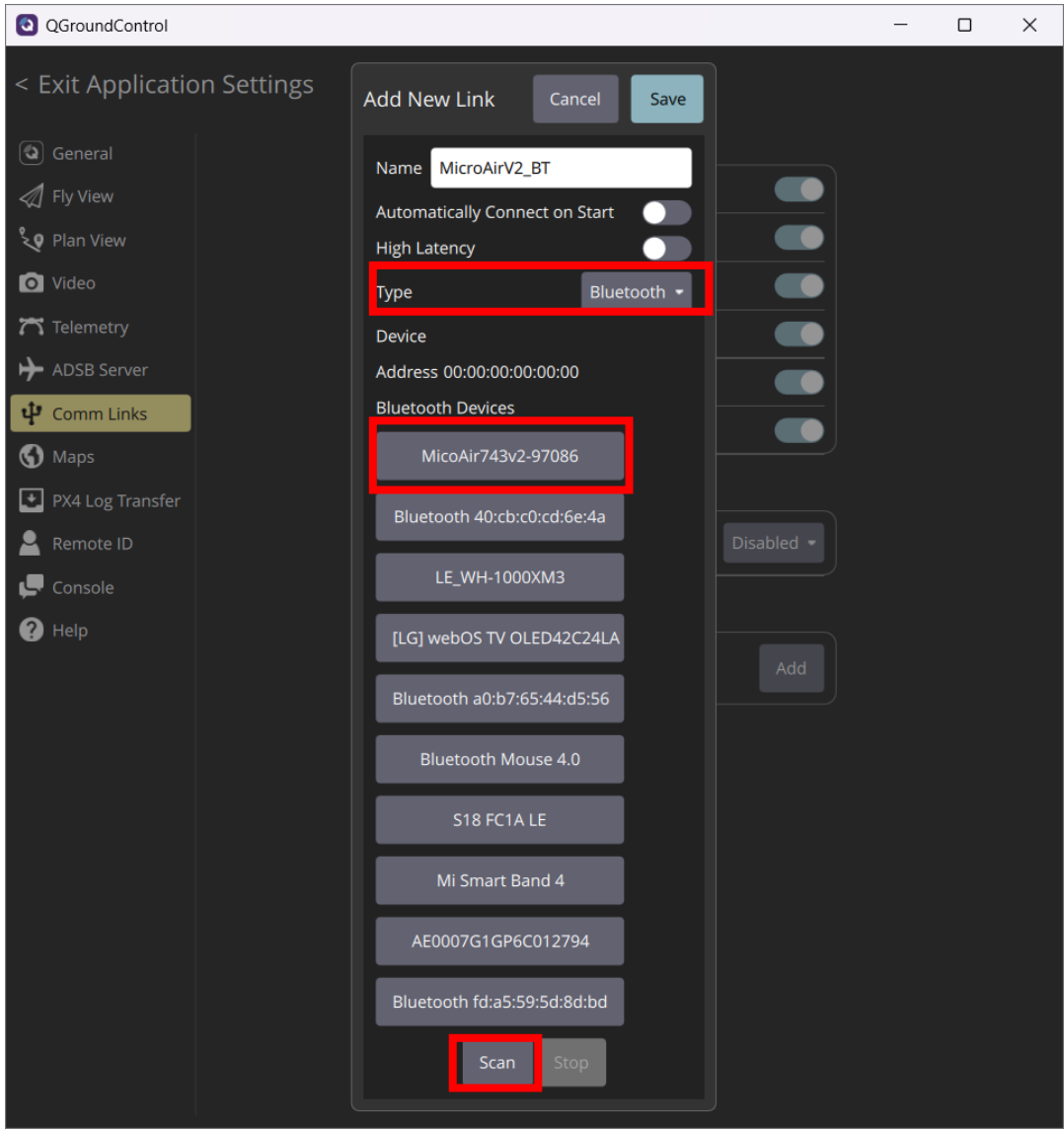
Connect to MicroAirV2 Over Bluetooth

- **Click to Add New Link**



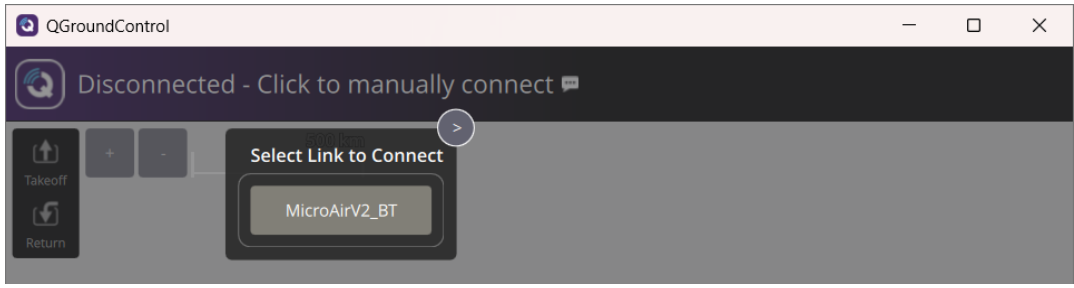
Connect to MicroAirV2 Over Bluetooth

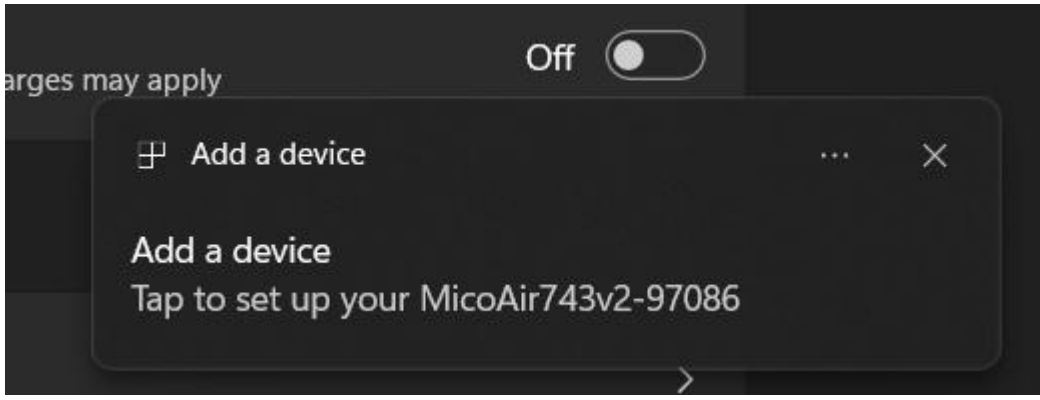
- **Click to Add New Link**
- Give it a memorable **Name**
- Set **Type** to **Bluetooth**
- Click **Scan**
- Choose **MicroAir743v2-XXXXX**
- Click **Save**

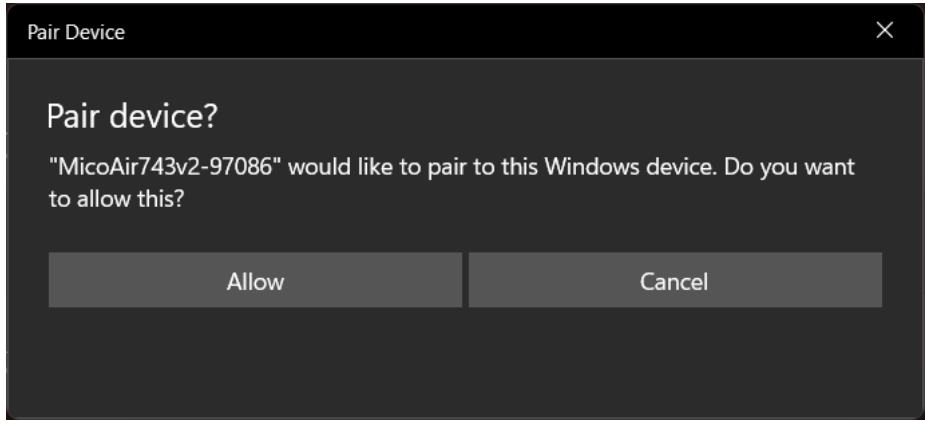


Connect to MicroAirV2 Over Bluetooth

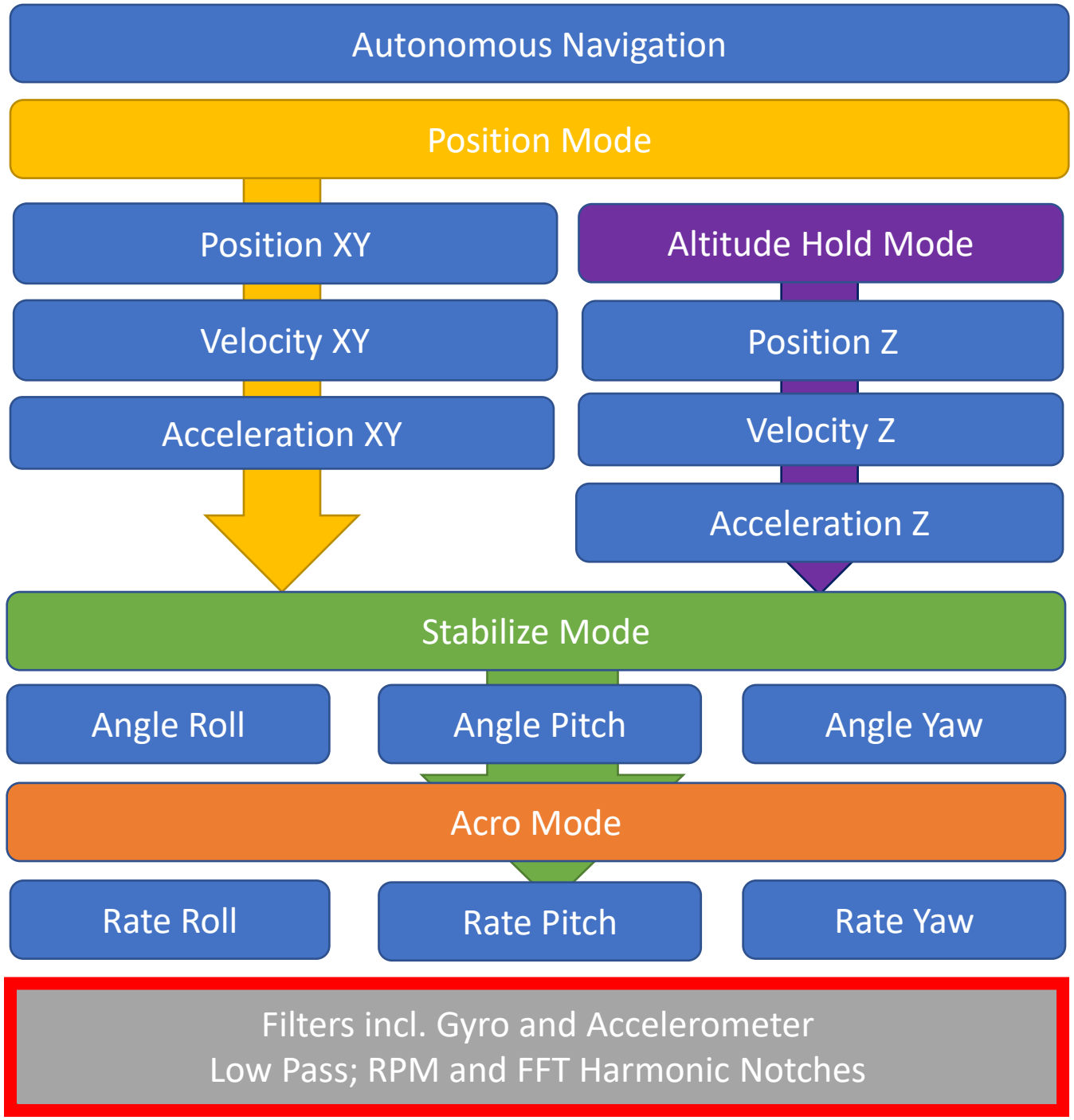
- Go back to the home page
- Click **Disconnected**
- Select the Link you just added MicroAirV2_BT in my case
- The Add a device notification will pop up > Click on it
- The Pair device notification will open > Click **Allow**
- This needs to be done reasonably quickly or the connection will fail







Filter Tuning in PX4


Autonomous Navigation
Position Mode
Position XY
Velocity XY
Acceleration XY
Altitude Hold Mode
Position Z
Velocity Z
Acceleration Z
Stabilize Mode
Angle Roll
Angle Pitch
Angle Yaw
Acro Mode
Rate Roll
Rate Pitch
Rate Yaw
Filters incl. Gyro and Accelerometer
Low Pass; RPM and FFT Harmonic Notches

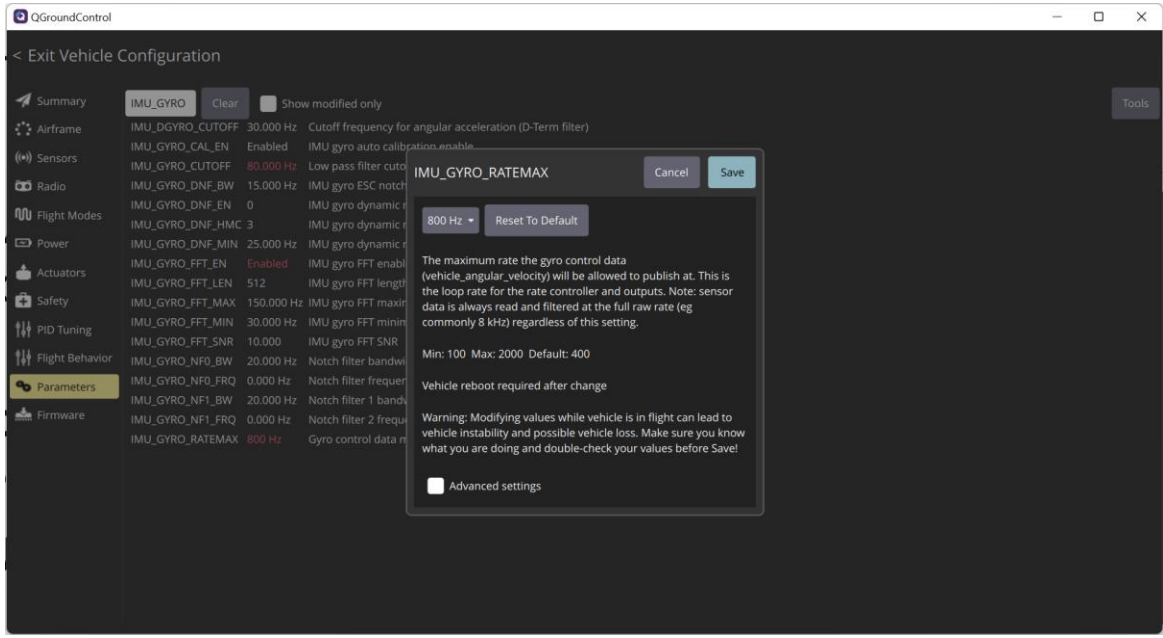
SD Card For Logging

- A fast microSD card is important to achieve high logging rates for filter tuning without drop-outs
- I recommend:
  - SanDisk Extreme 32 GB microSDHC Up to 100 MB/s, Class 10, UHS-I, U3, V30
- **Cards larger than 32GB are not supported**



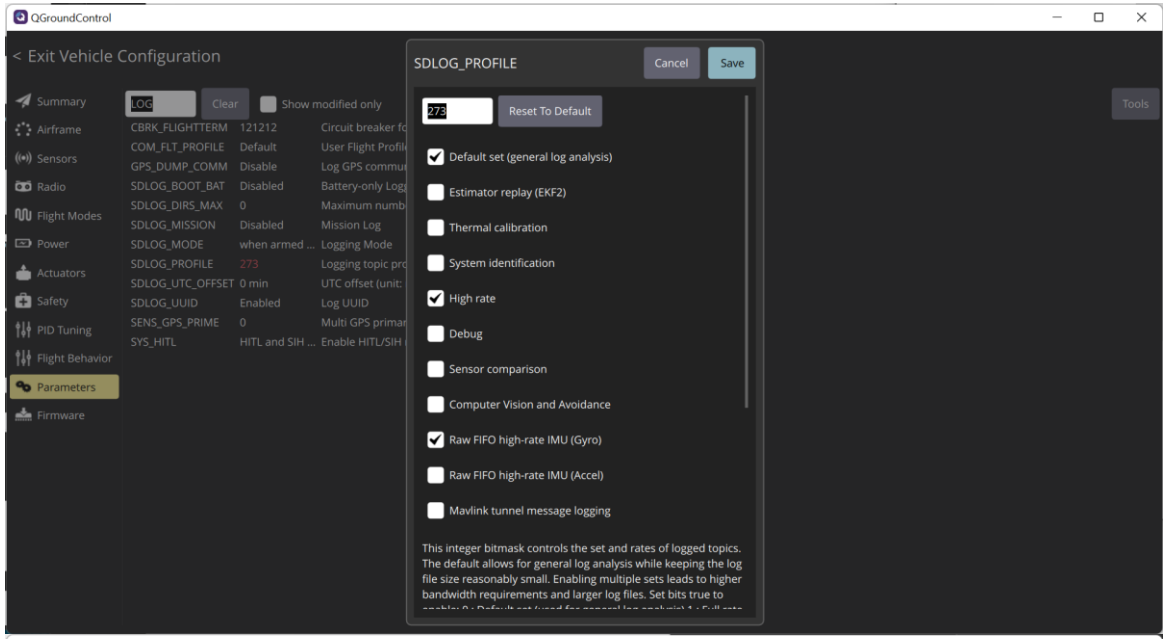
Logging Settings

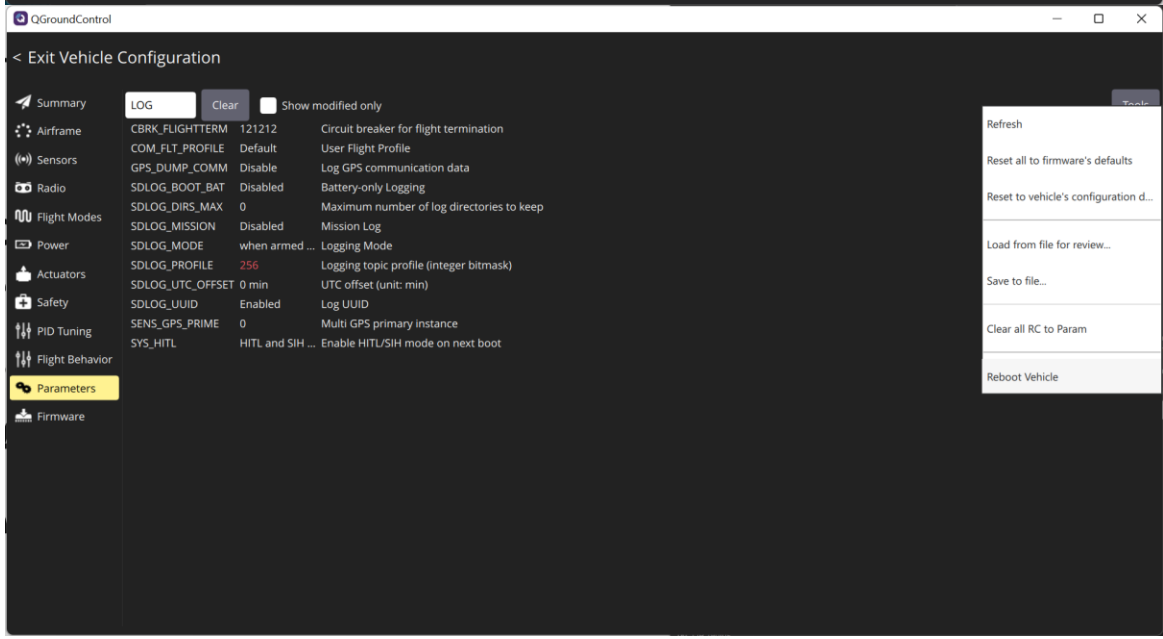
- Before we can start Filter Tuning we need to set up the correct logging settings
- In **Vehicle Configuration > Parameters** search **IMU_GYRO**
- Set **IMU_GYRO_RATEMAX = 800 Hz**
- **Save**



Logging Settings

- In **Vehicle Configuration > Parameters** search **LOG**
- Select **SDLOG_PROFILE**
- Tick:
  - **High Rate**
  - **Raw FIFO high-rate IMU (Gyro)**
  - **Default set**
- **Save**
- **Tools > Reboot Vehicle**



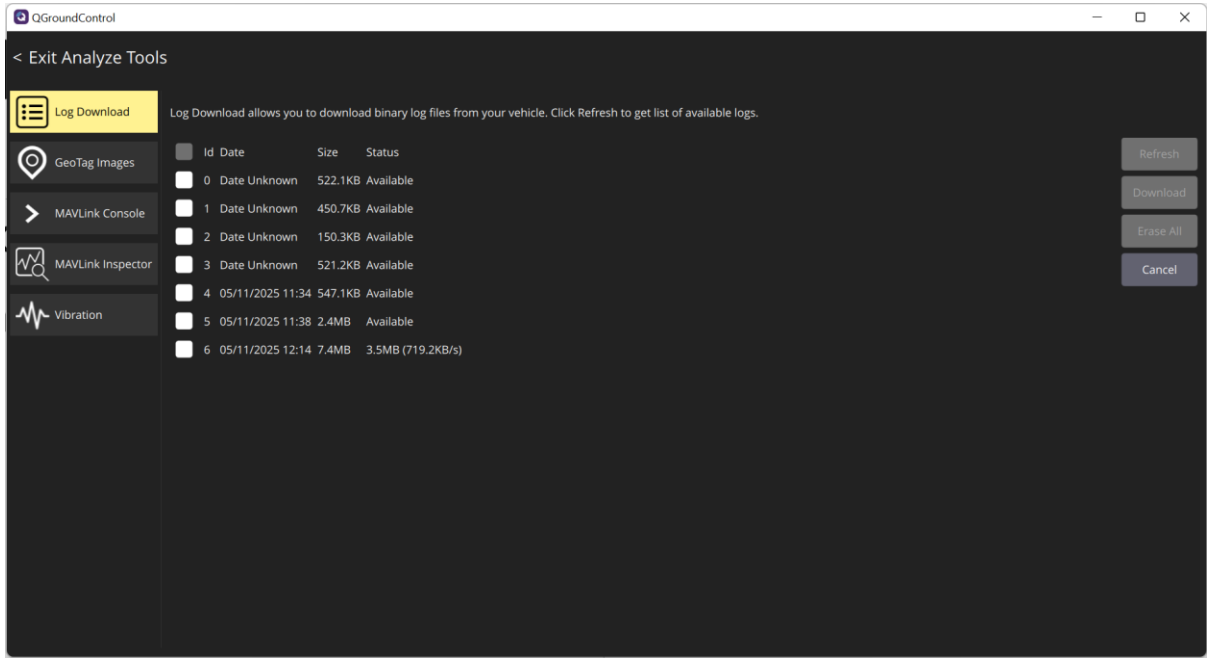


Hover Test

- Double check prop and motor direction, this is by far the biggest cause of crashes
- Verify your Kill Switch is working
- Do the hover test in the **Stabilized** Flight Mode. This will mean that you control the throttle directly and the drone returns to level when you release the pitch and roll sticks
- If possible secure the drone to a test stand that limits the range of motion but allows:
  - Take-off and ascend ~1m
  - Limited tilt on pitch and roll (±30 degrees)
  - Yaw rotation (at least ±90 degrees)
- If you don’t have a test stand conduct the first hover test in a wide open space and preferably over grass or a softer material
- Arm the drone and tilt it slightly on the ground first left and right (Roll) and then forward and backwards (Pitch) to check that the drone tilts in the direction you expect
- If all is well smoothly increase throttle until the drone lifts off
- Actively control the throttle to hold the drone in a hover for ~20s
- Reduce the throttle slightly so the drone descends and land gently, then disarm

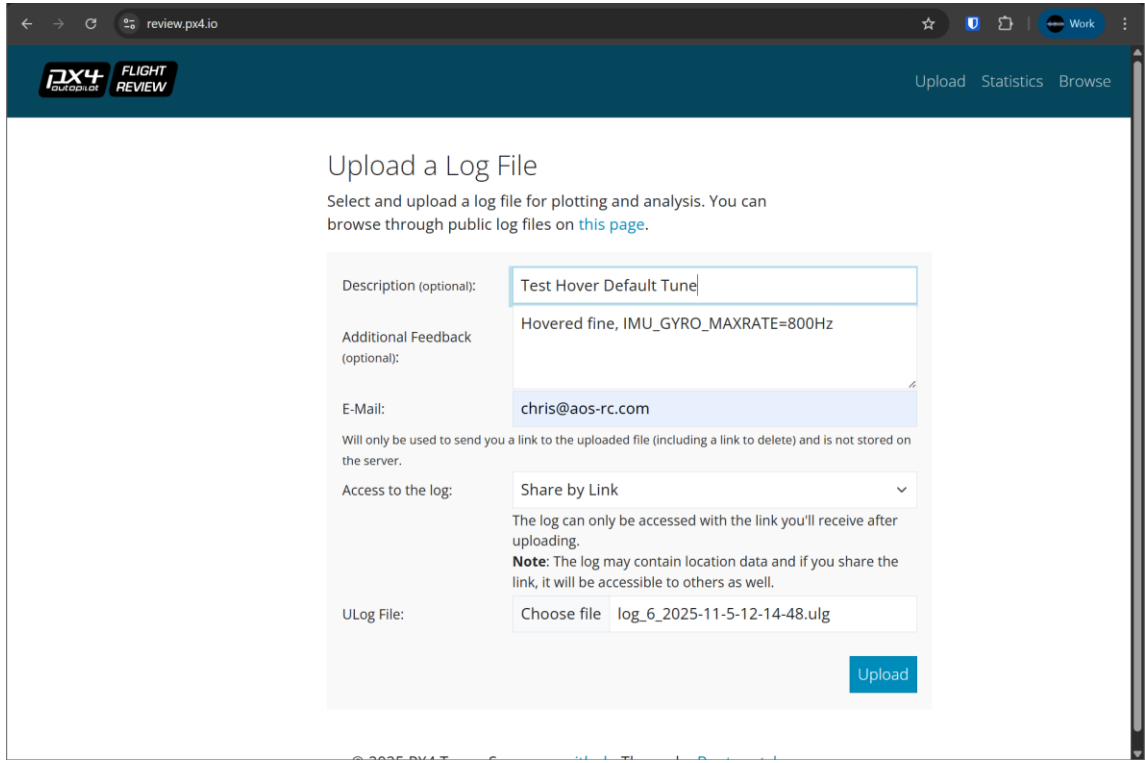
Hover + Log Download

- Hover for 30s or so
- Land and **Download the Log**
- **Q Logo > Analyze Tools > Log Download**
- **Refresh**
- Select the Log
- **Download**
- Select Folder



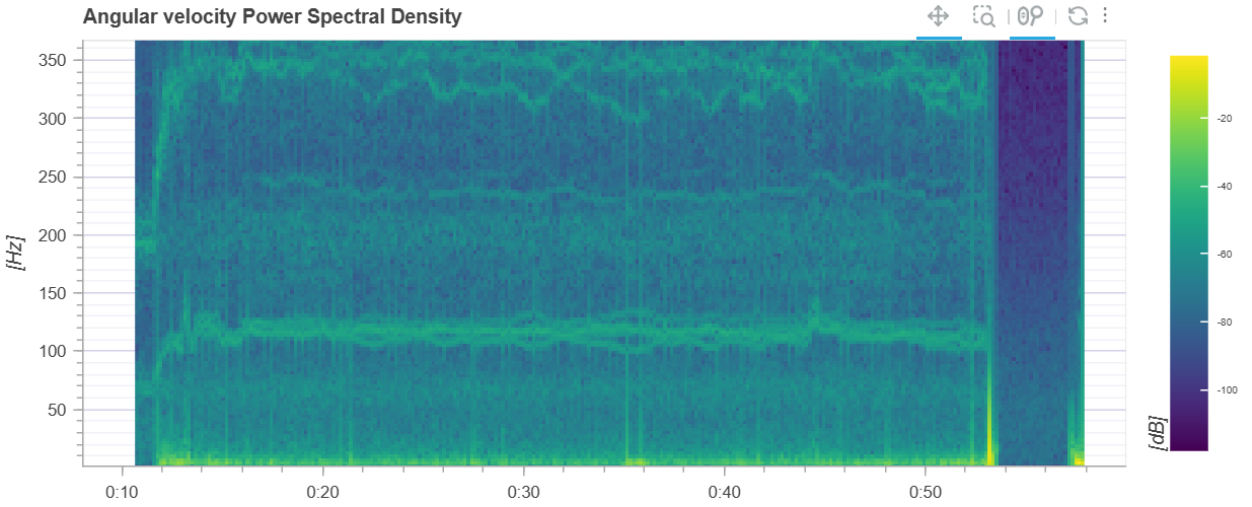
Upload Log to Flight Review

- Navigate to review.px4.io
- Fill out the form
- Set Access to **Share by Link**
- Choose your log file and upload

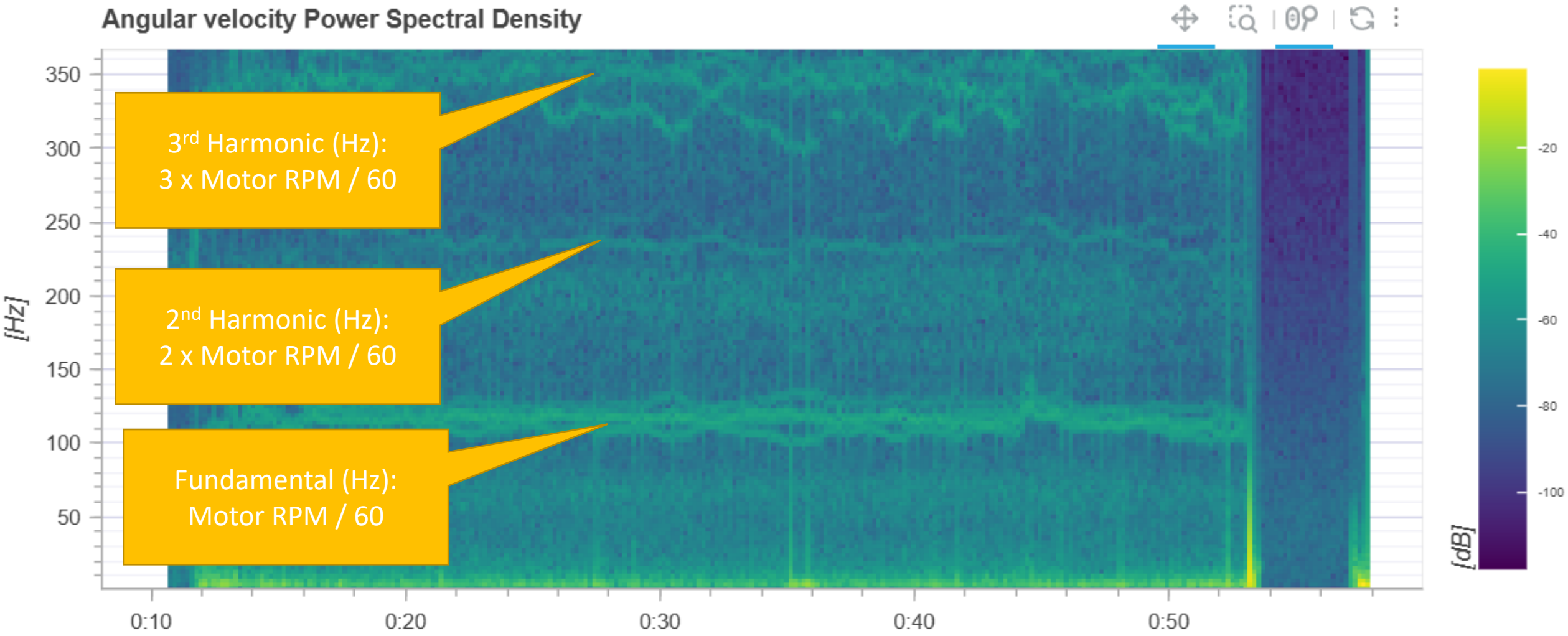


Check Gyro Noise Profile

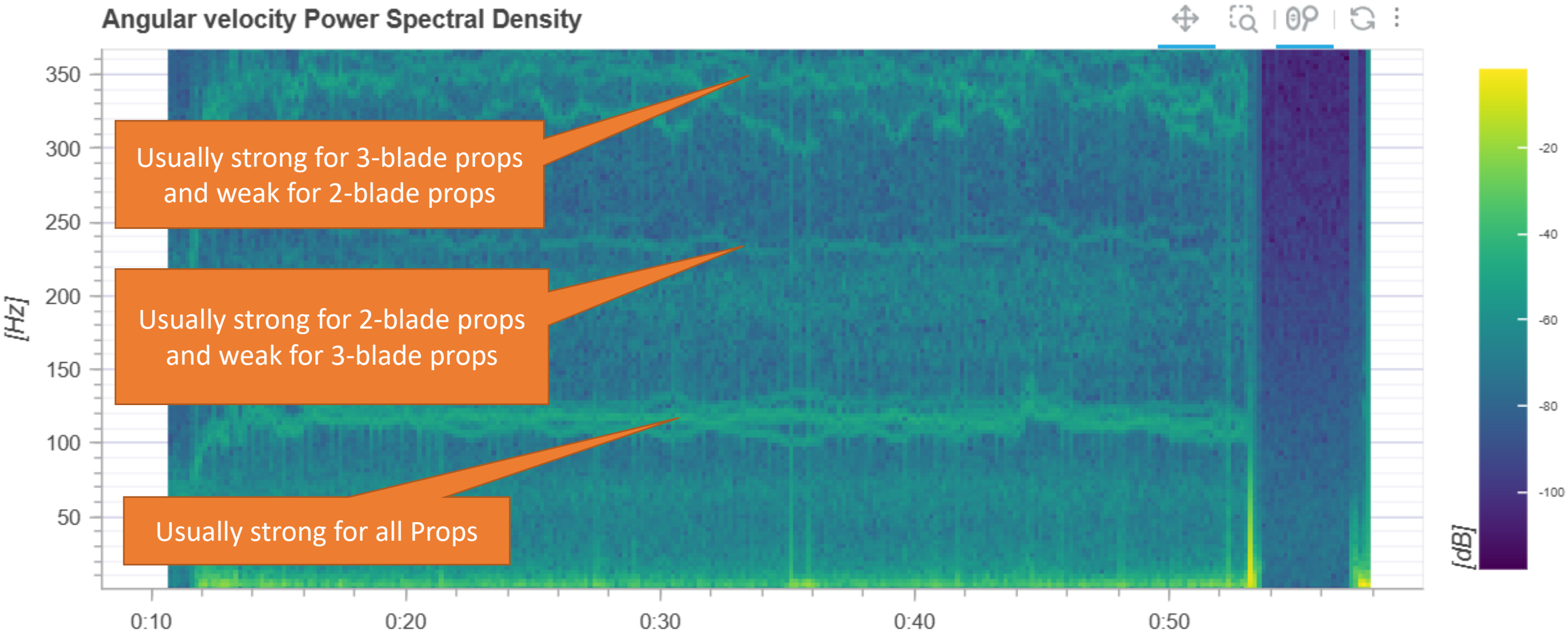
- If everything is working correctly with your logging you should see **Angular Velocity Power Spectral Density**
- The maximum frequency should be **~0.45 x IMU_GYRO_MAXRATE**



Typical Gyro Noise Profile

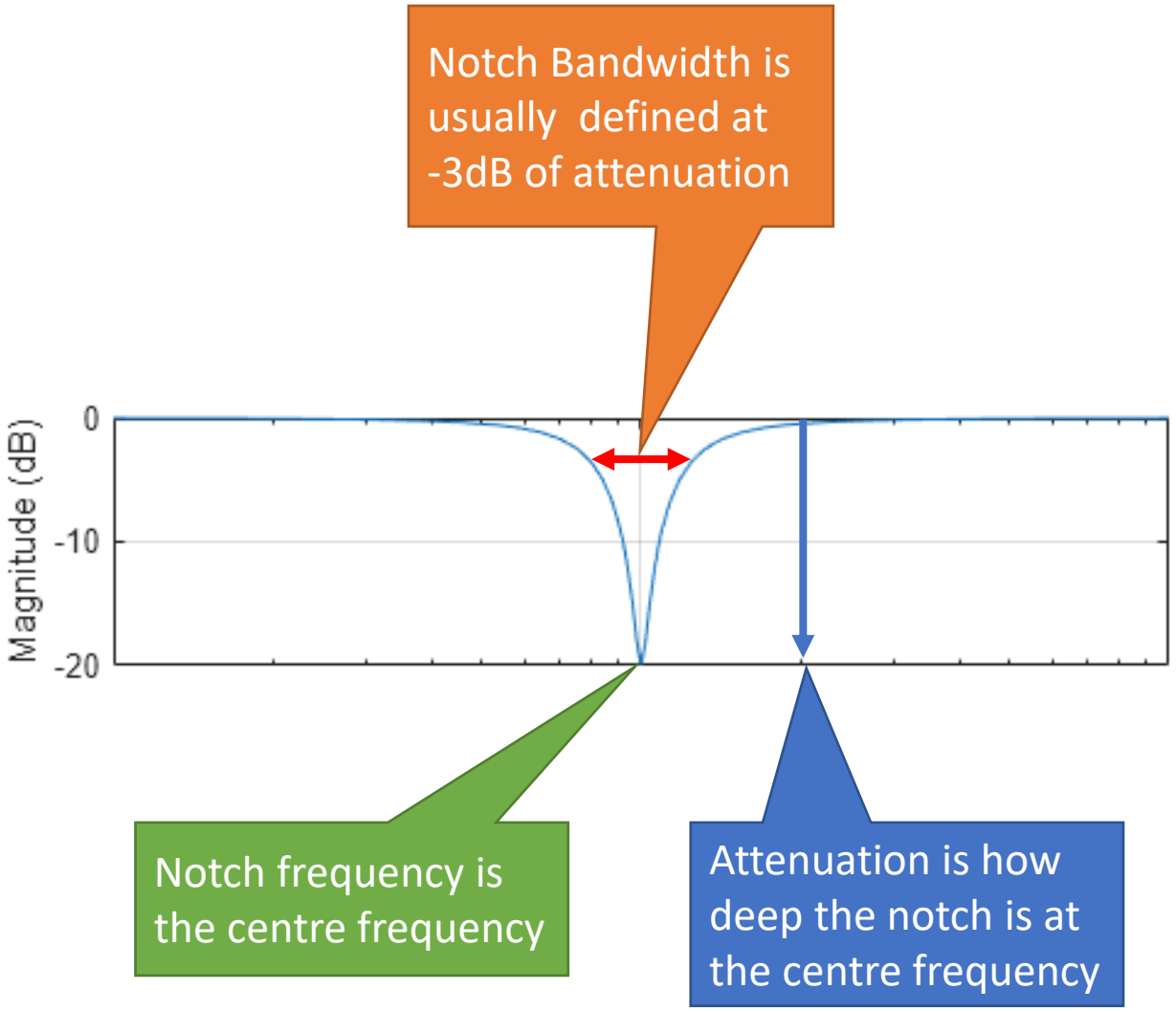


Typical Gyro Noise Profile



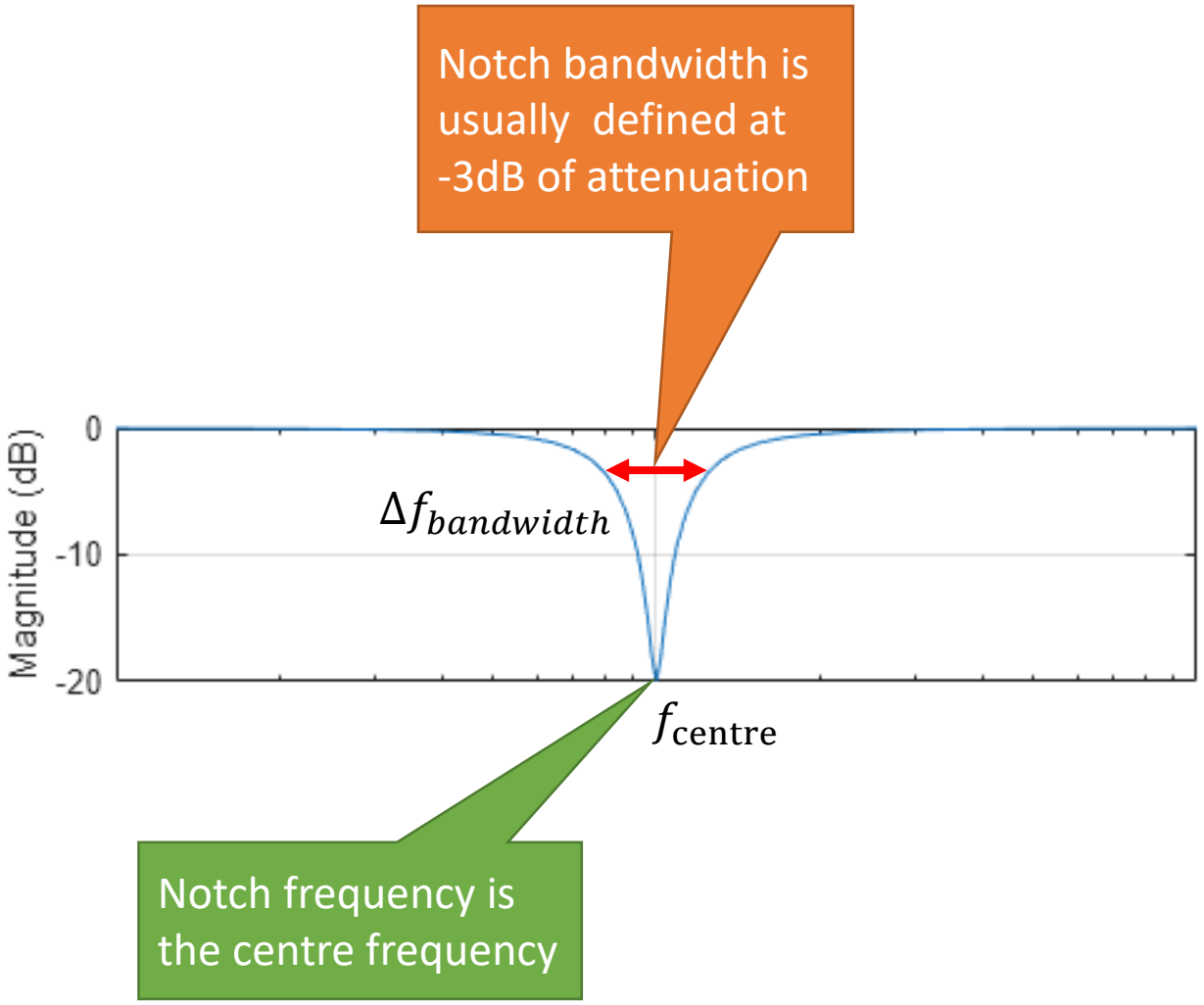
Filtering motor noise

- The best way to filter motor noise is using Notch Filters
- These produce strong filtering around a particular frequency
- The delay of the notch is dependent on:
  - Attenuation
  - Centre frequency
  - Bandwidth



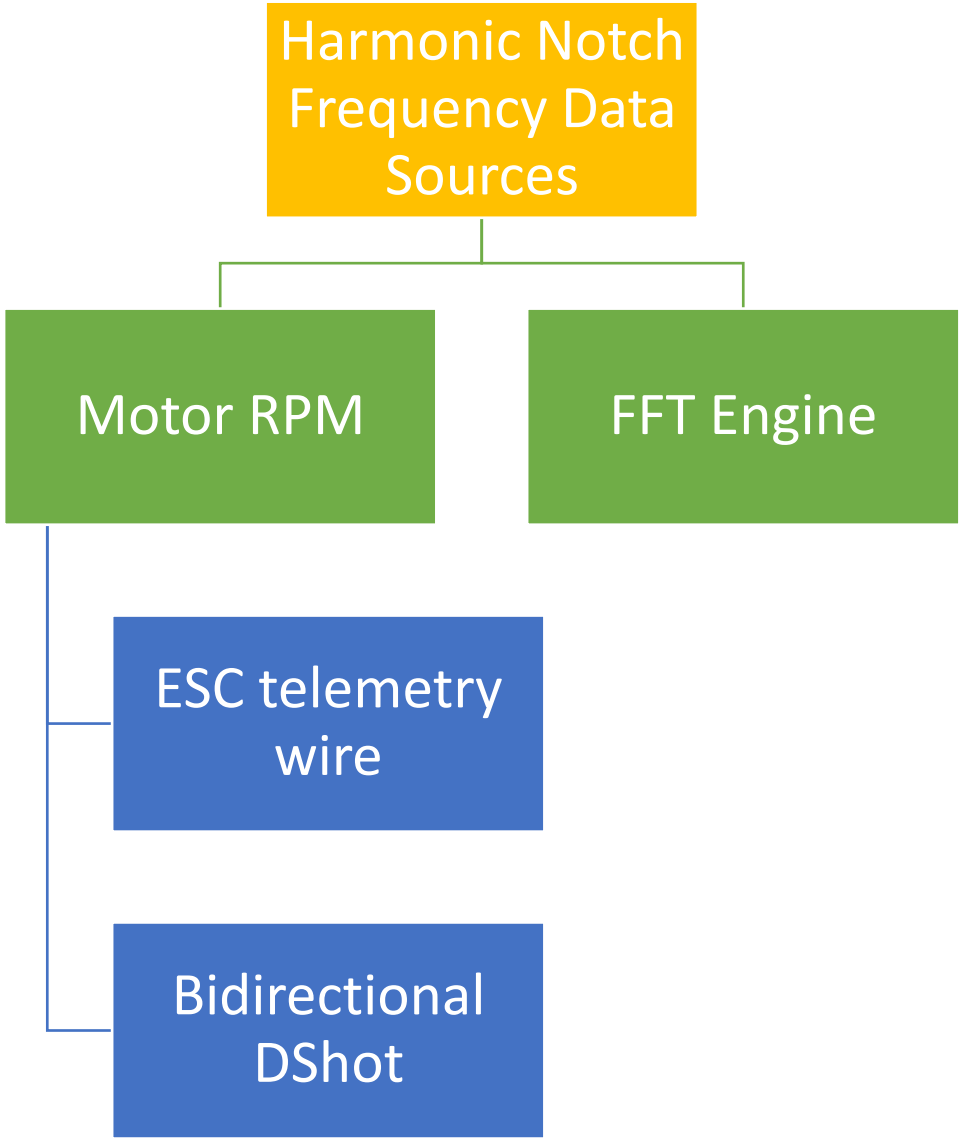
Notch filters

- There are 2-types of notches in PX4
  - Static:
    - Constant defined bandwidth
    - Constant centre frequency
  - Dynamic:
    - Varying bandwidth
    - Varying centre frequency
- For dynamic notches the Q-Factor is held constant
  - $Q_{Factor} = f_{centre}/\Delta f_{bandwidth}$



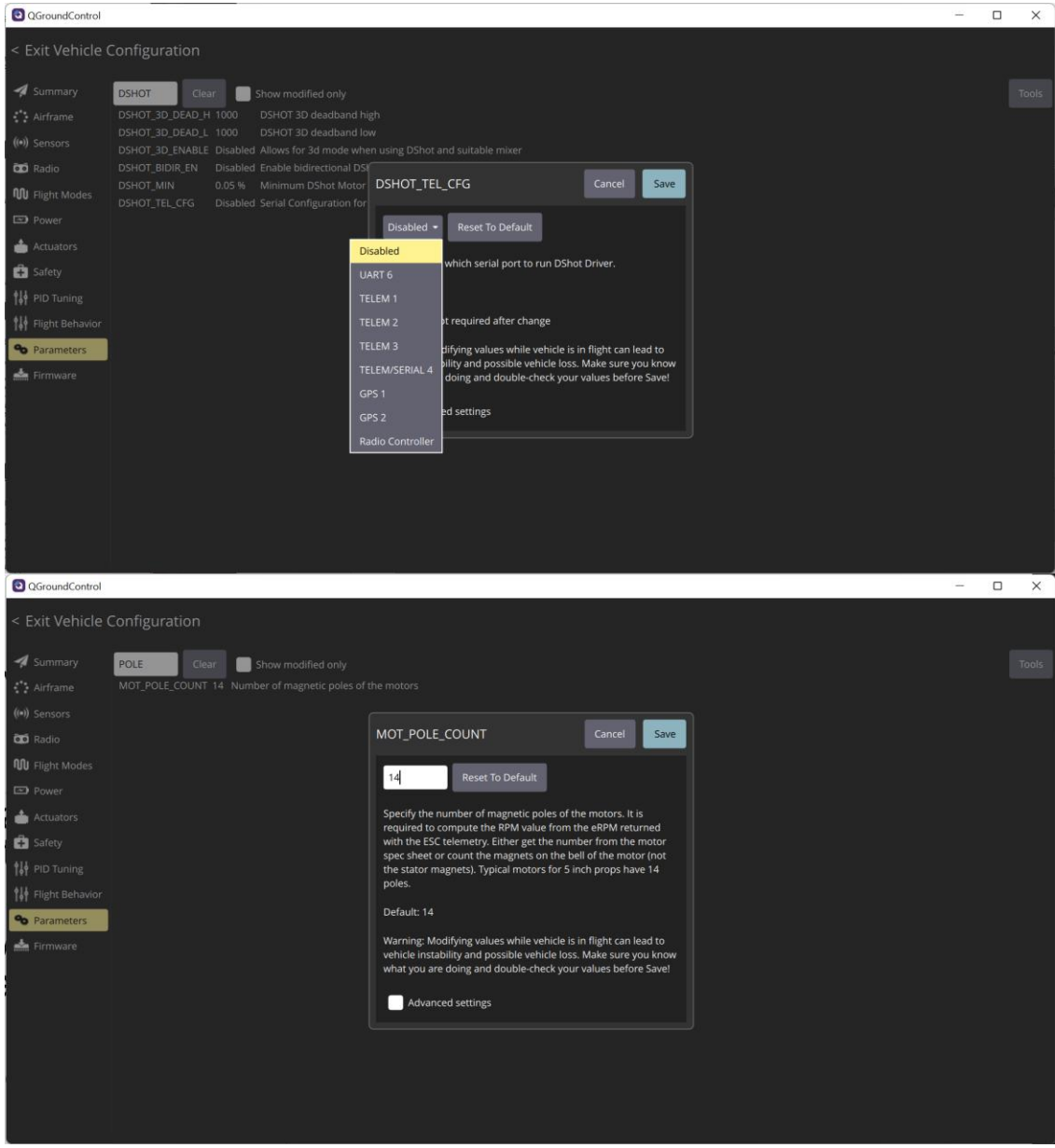
Dynamic Notch Filters

- There are two ways to drive dynamic Notch filters in PX4
  - RPM data from ESC telemetry
  - PX4s FFT (Fast Fourier Transform) Engine
- RPM data from ESC telemetry is preferred as the frequency is more accurate and the data has less delay
- RPM data can be acquired over the ESC telemetry wire or using Bi-directional DShot
- The FFT Engine is covered in detail in my Ultimate PX4 Tuning Course



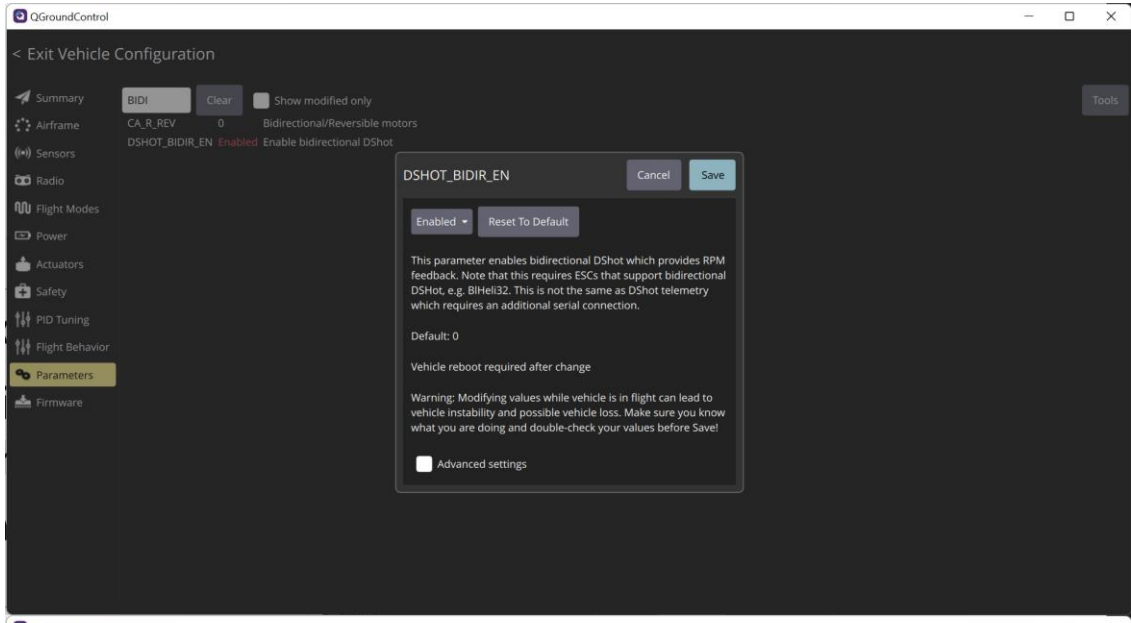
Set-up ESC telemetry

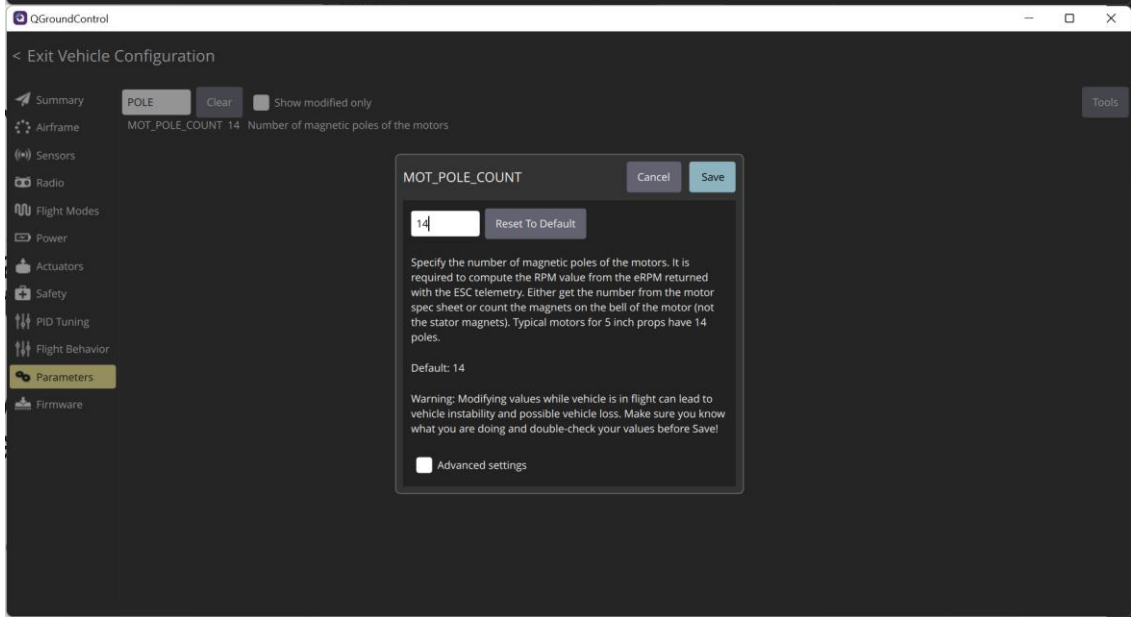
- The telemetry wires from all ESCs should be connected together to the RX pin of an unused UART
- To set up an ESC telemetry wire go to **Vehicle Configuration > Parameters** and search for **DSHOT**
- **Set DSHOT_TEL_CFG** to the correct UART
- Search for **POLE**
- **Set MOT_POLE_COUNT** equal to the number of magnets in your motor bell (14 is typical for smaller motors, larger motors often have 24, 28, or 42 poles)



Set-up Bidirectional DShot

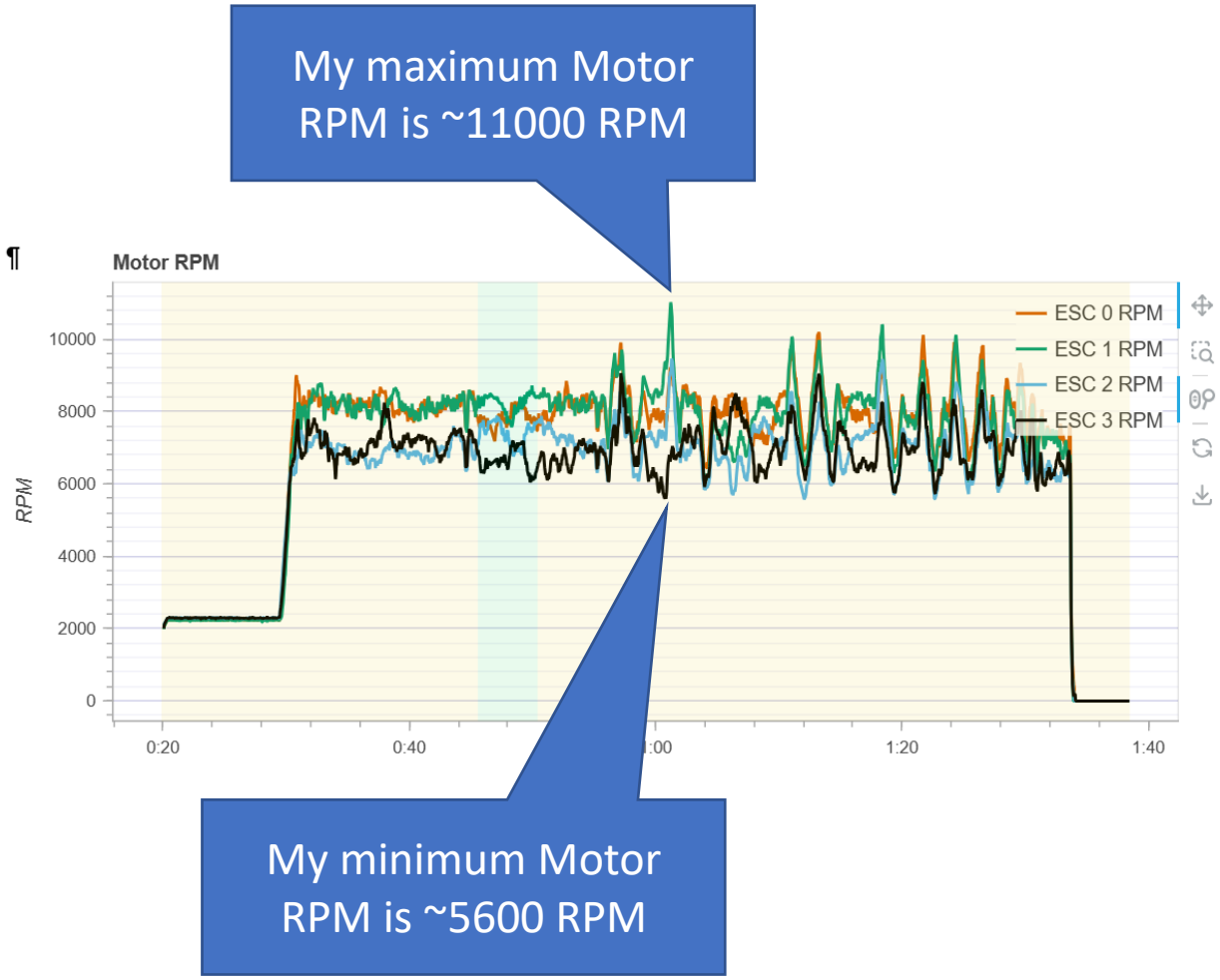
- If you have a modern DShot ESC it likely supports Bi-directional Dshot
- This provides RPM data in response to every DShot packet, this data is frequent and low latency
- To enable Bi-Directional Dshot
- **Vehicle Configuration > Parameters**
- Search **BIDI**
- Set **DSHOT_BIDIR_EN = Enabled** and **Save**
- **Tools > Reboot Vehicle**





Setup RPM Notch Filters

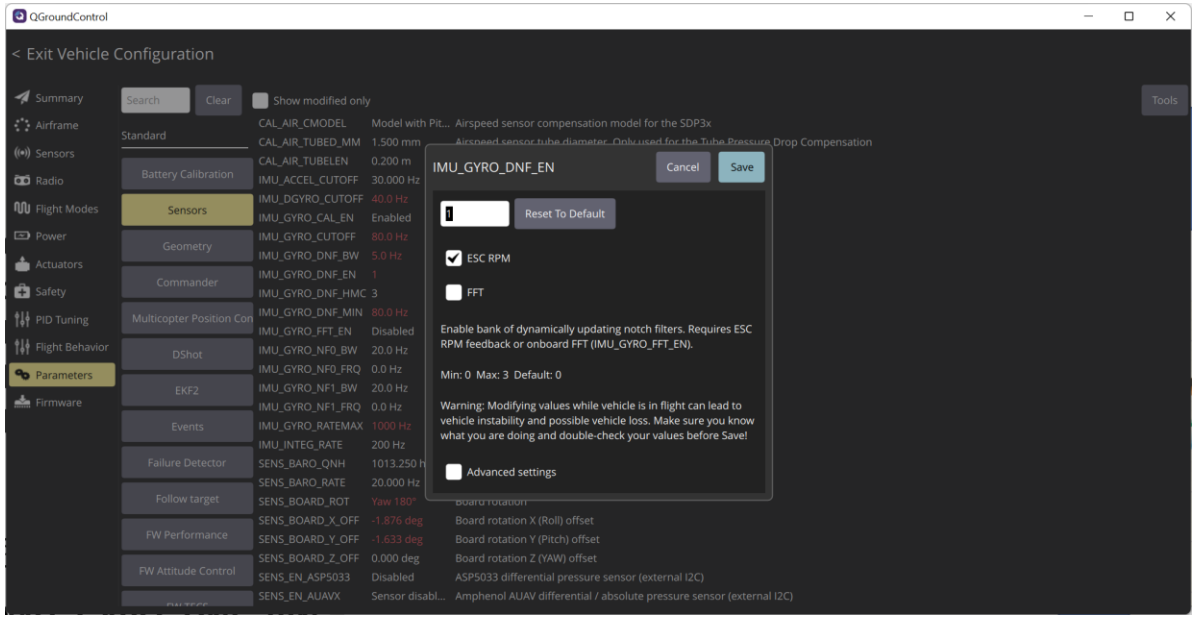
- With RPM data enabled do a test flight where you do several rapid climbs and descents in Position or Altitude Mode (or Stabilized if that is a use case for you)
- This will give an idea of the maximum and minimum frequency of motor noise during a typical flight

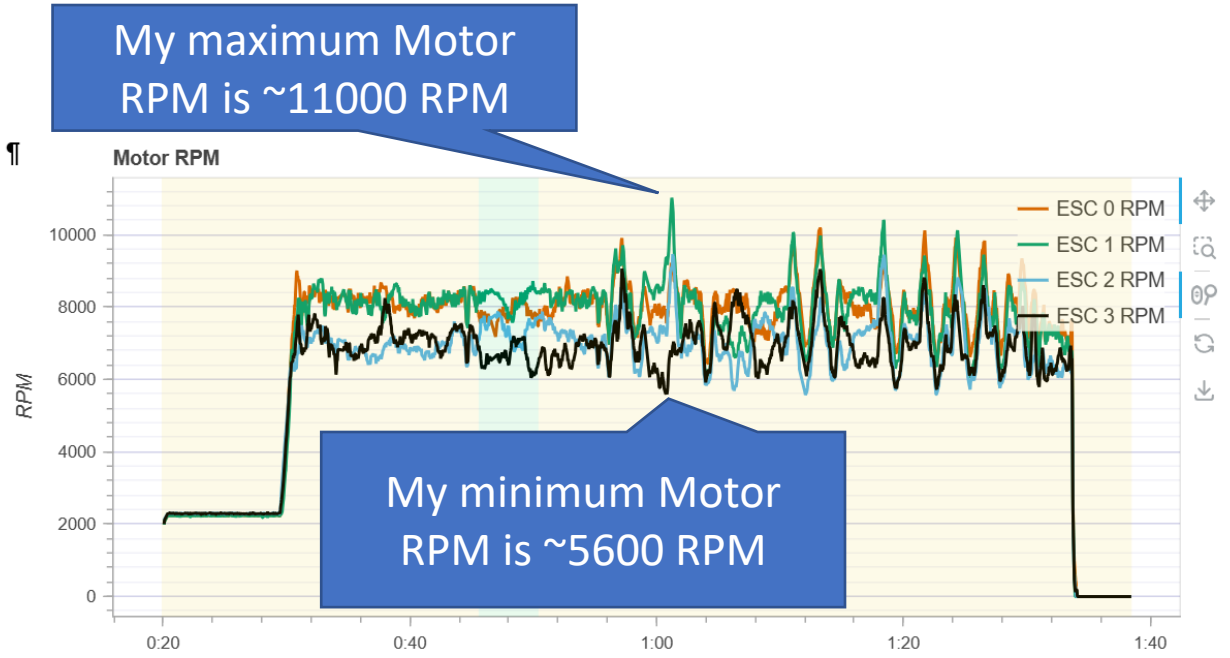


Setup RPM Notch Filters

- Now we'll set up the filters based on this data
- **Vehicle Configuration > Parameters**
- Search **IMU_GYRO**
- **Set IMU_GYRO_DNF_EN = 1 (ESC Telemetry)**
- **Set IMU_GYRO_FFT = Disabled**
- **Set IMU_GYRO_DNF_MIN** to about 90% of your minimum motor frequency in Hz

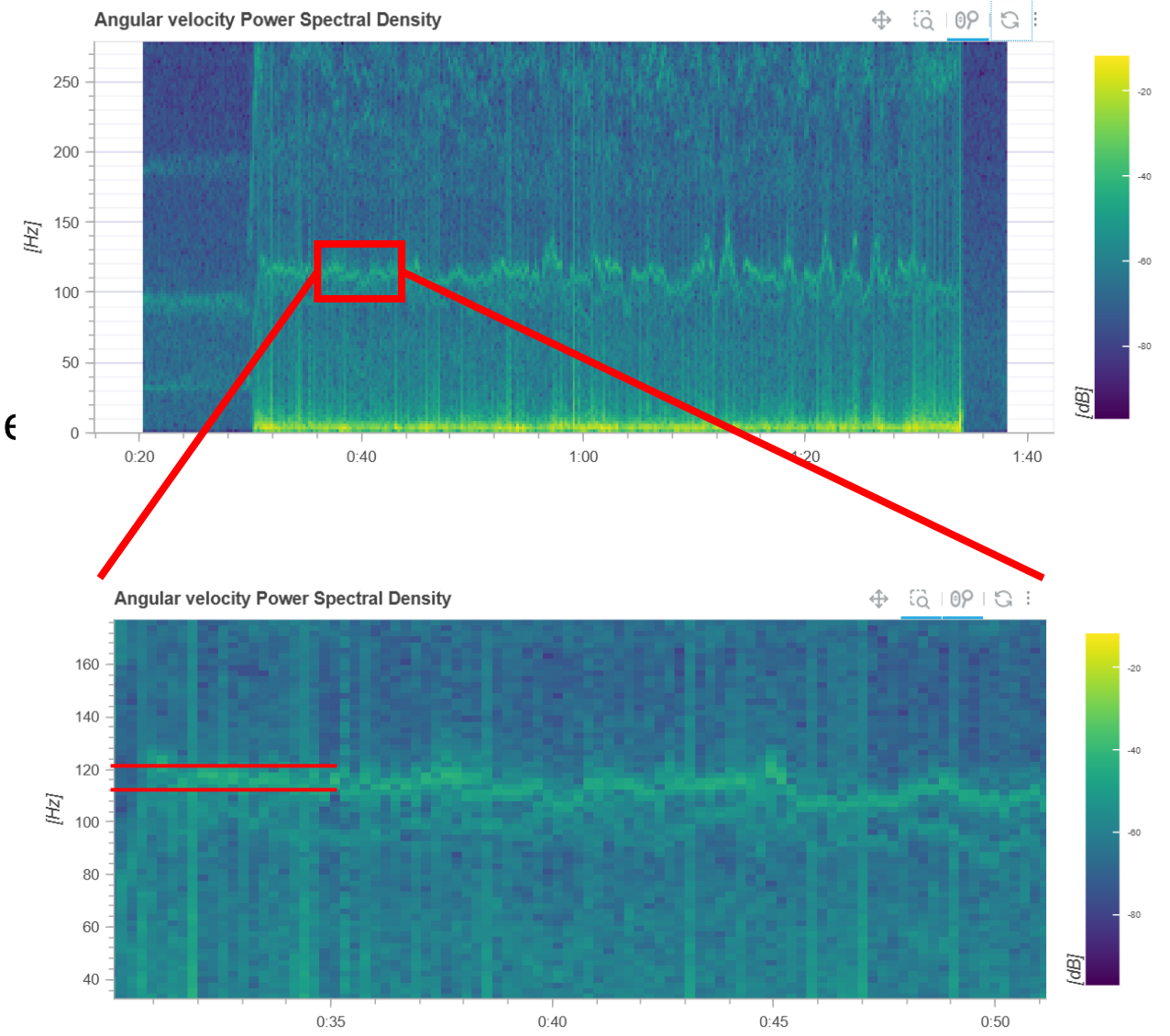
  E.g. $\frac{5600}{60} * 0.9 = 84$ Hz
- **Set IMU_GYRO_DNF_BW = IMU_GYRO_DNF_MIN * 0.20** (to start with)
  E.g. $84 * 0.2 = 17\ Hz$
- Set **IMU_GYRO_DNF_HMC** to **2** for 2-blade props or **3** for 3-blade props
- **Tools > Reboot Vehicle**





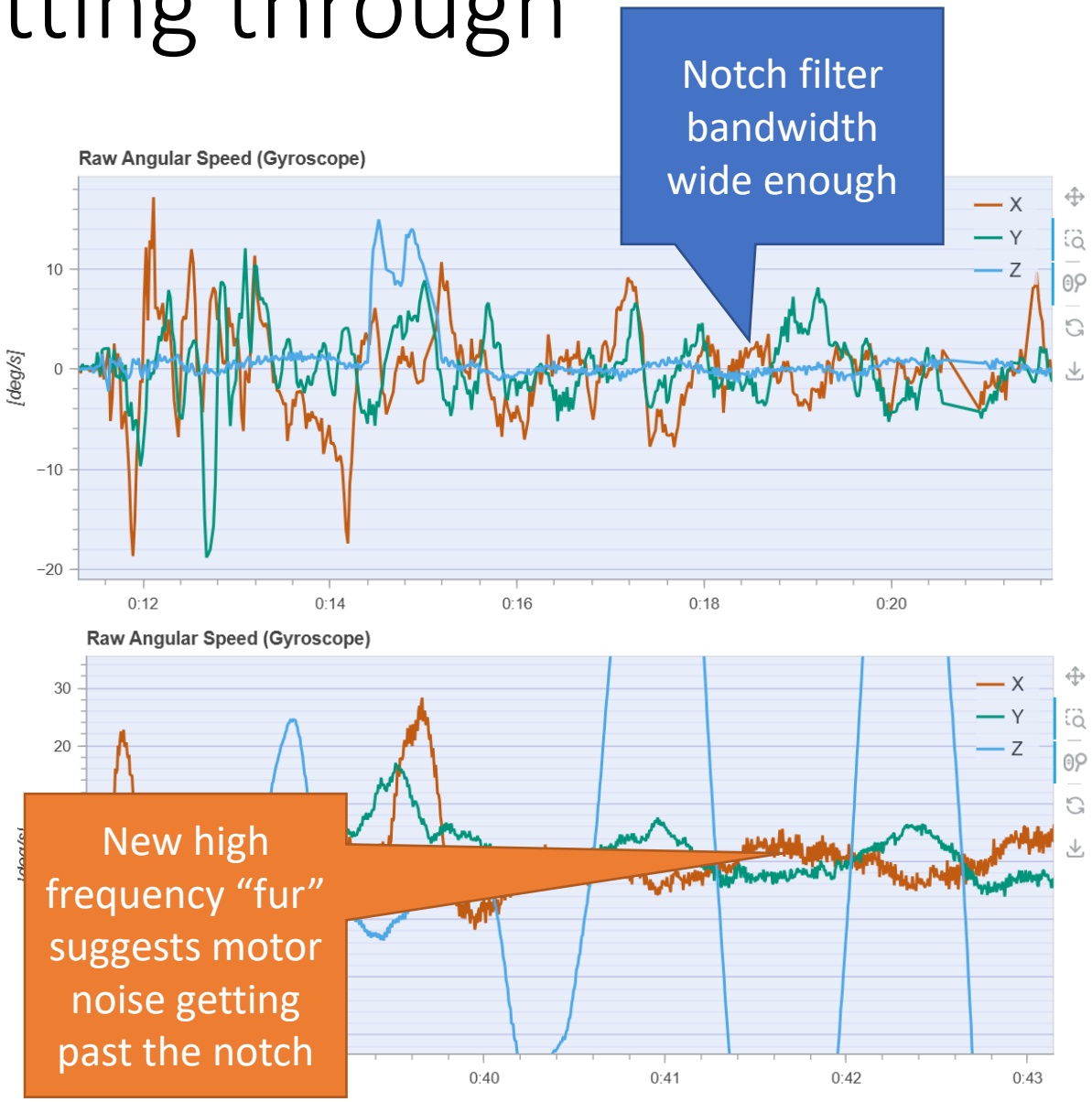
Tuning Notch Filters

- Tuning Notch Filters involves reducing the notch filter bandwidth to the minimum value that still captures all the noise
- Reducing the filter bandwidth reduces the filter delay which improves flight performance
- Zoom in on the Angular Velocity Power Spectral Density Chart and estimate the bandwidth of the noise
- In this log the bandwidth is ~5Hz so that would be a suitable value for an RPM notch
- Remember this is the bandwidth for the minimum centre frequency and the notch bandwidth at higher frequencies increases proportionally to the frequency



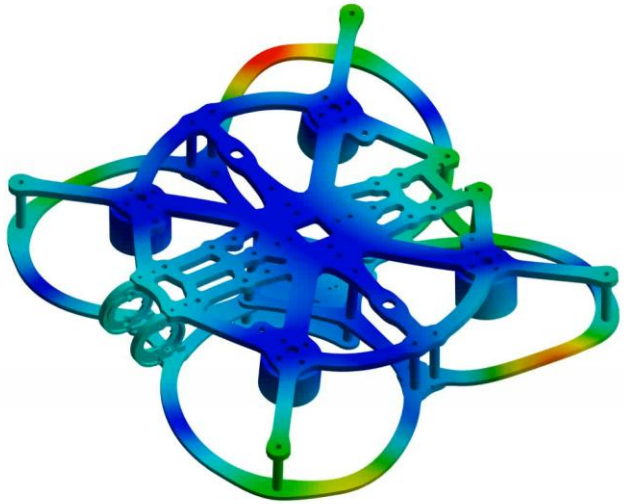
Check that noise isn't getting through

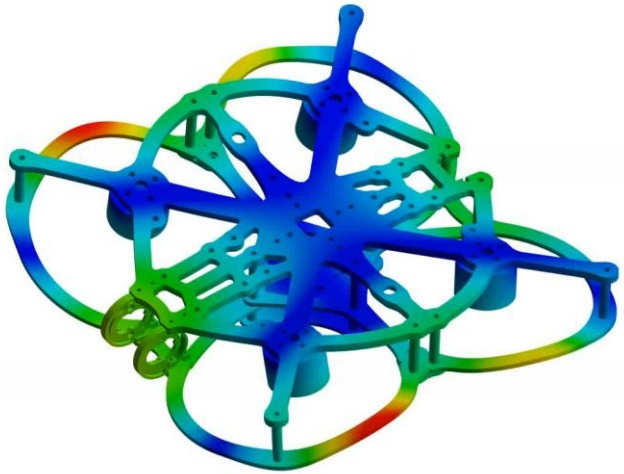
- As you reduce the Notch filter bandwidth you shouldn't see much change in the **Raw Angular Speed (Gyroscope)** plot
- If you start seeing a lot more high frequency "fur" on the traces that may indicate that your notch filter bandwidth is too small and noise is leaking past either side of the notch.



Gyro Low Pass Filter (IMU_GYRO_CUTOFF)

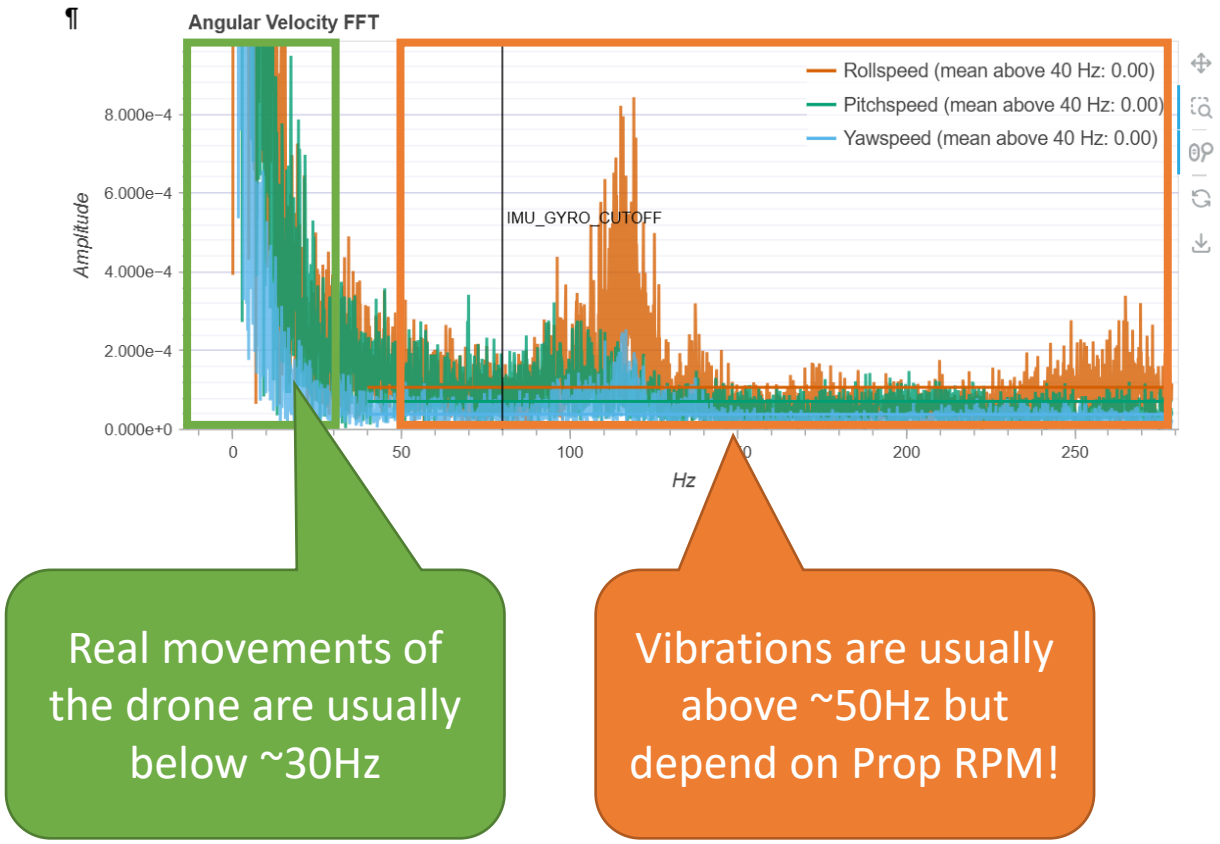
- The Gyro in a Drone is sensitive both to the rigid body movements of the drone (signal) but also to vibrations (noise)
- Vibration is generated by the spinning motors and props (and some camera gimbals)
- Resonance of the drone's airframe and payload can also amplify certain frequencies





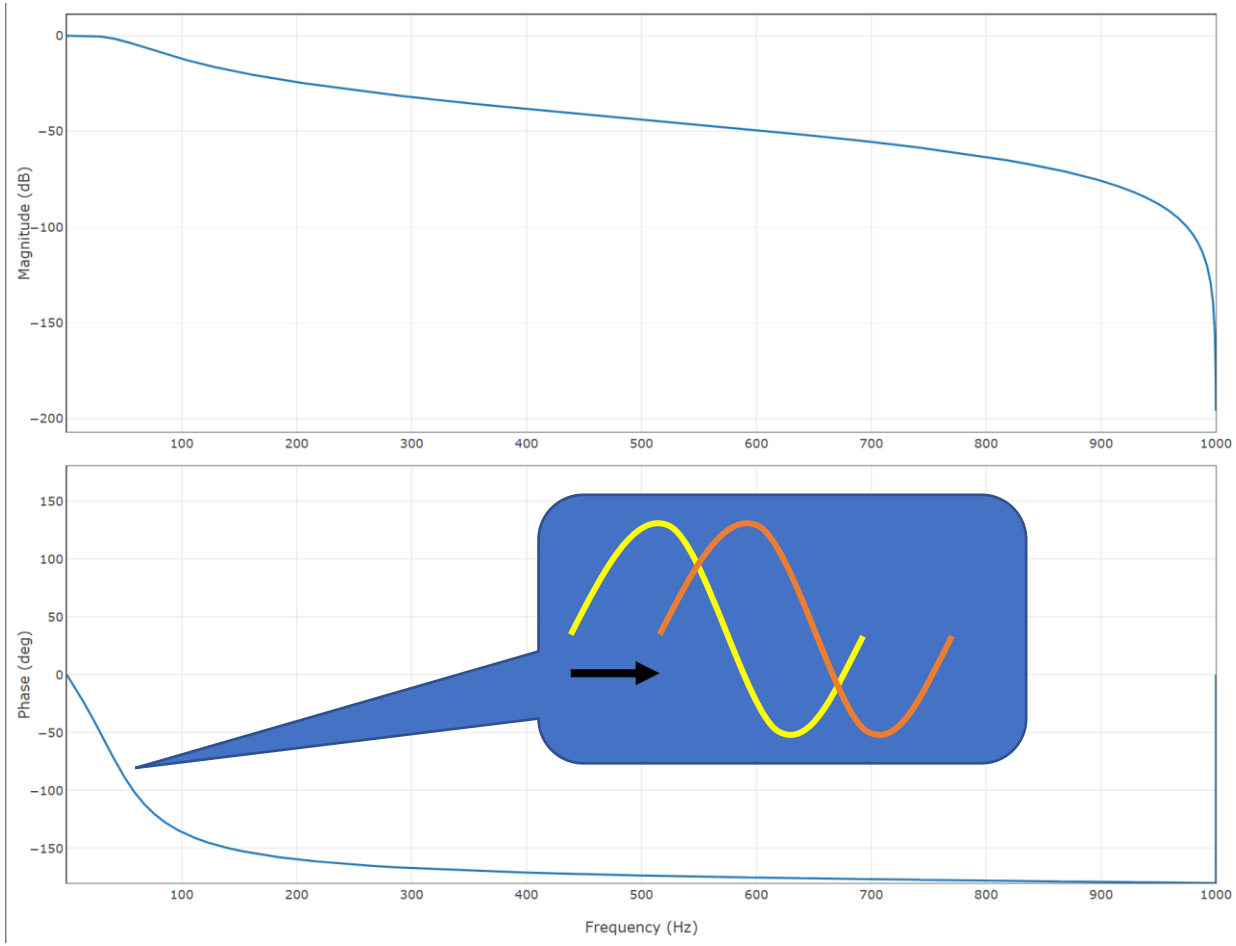
Gyro Low Pass Filter (IMU_GYRO_CUTOFF)

- The energy in vibrations (noise) is usually much higher than the energy in the rigid body movement of the drone (signal)!
- Fortunately the real movements are at a low frequency and the vibrations are at higher frequencies
- The Gyro Low Pass Filter takes advantage of this to reduce the noise without compromising the signal too much!



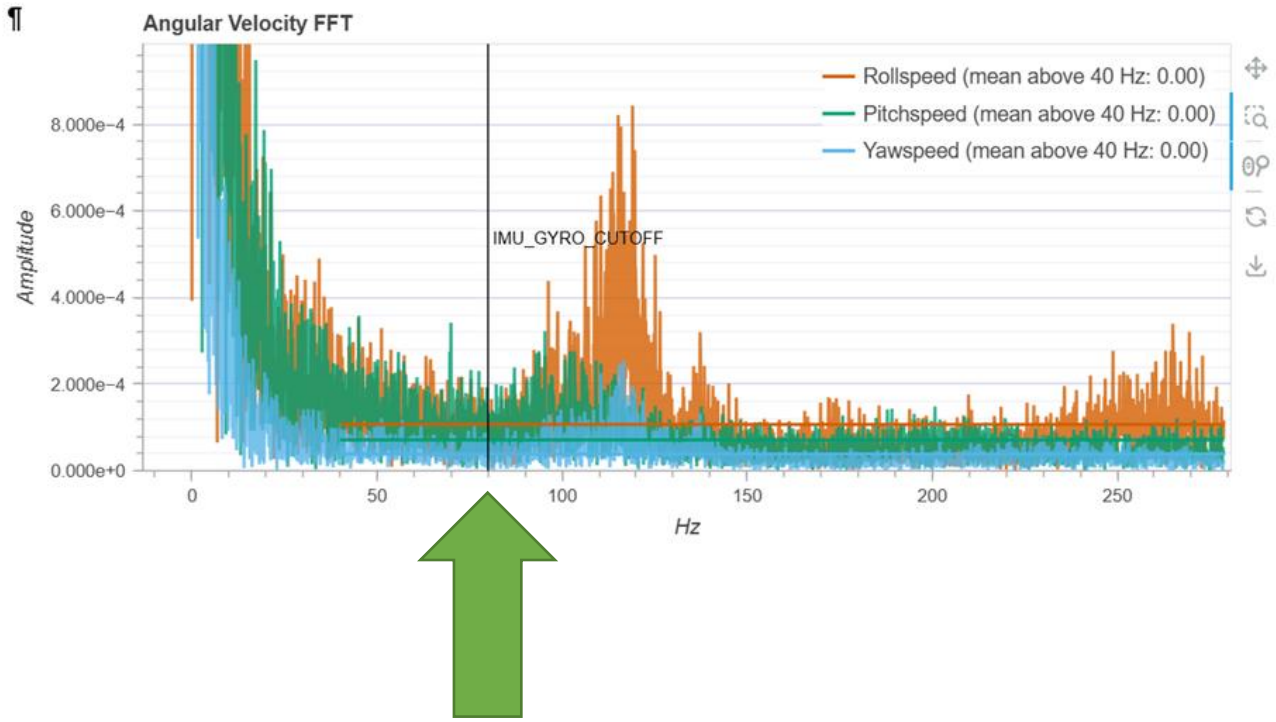
Gyro Low Pass Filter (IMU_GYRO_CUTOFF)

- Unfortunately the Gyro Low Pass Filter has some trade-offs
- Any low pass filter attenuates high frequencies but also adds delay
- The lower the 3dB cut-off frequency of the filter the more delay is added
- This delay slows the reaction of the drone to air turbulence and control inputs



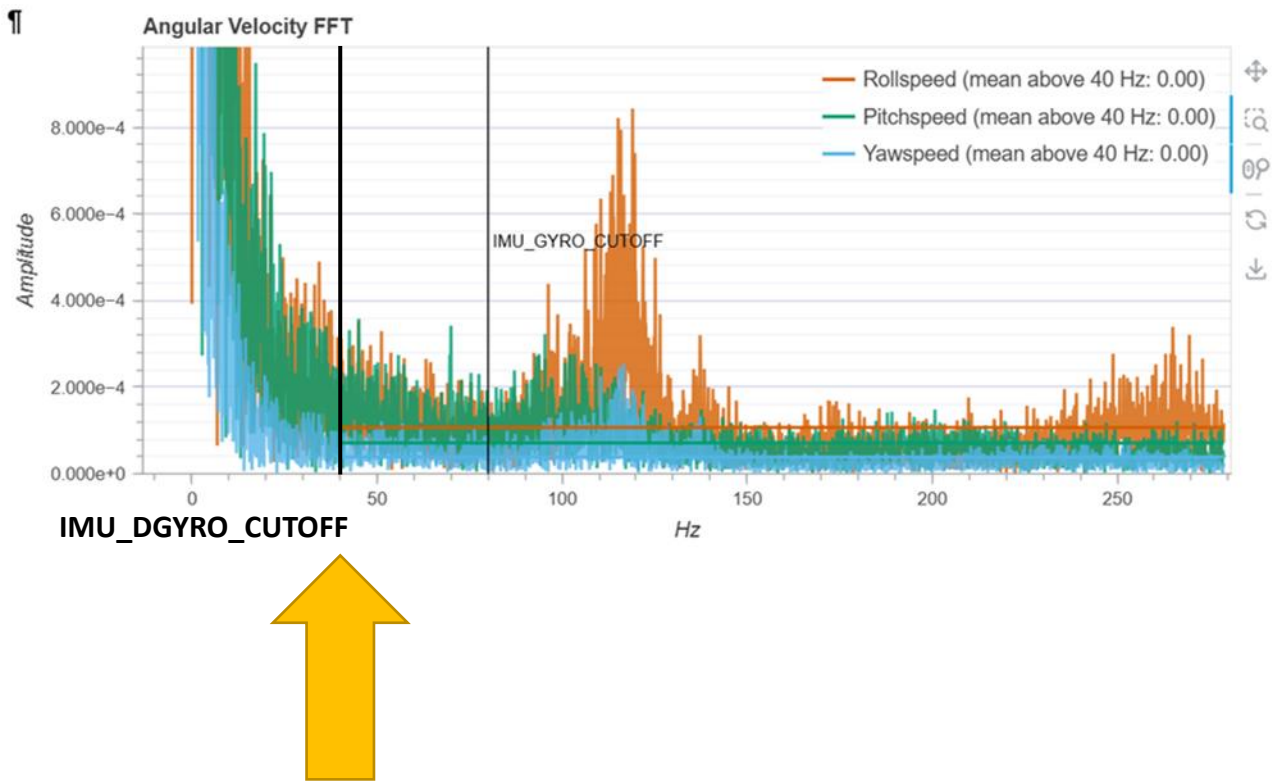
Tuning the Gyro Low Pass Filter

- We want to increase the cut-off frequency of the Gyro low pass filter as high as possible without letting noise through
- Our main adversary is the fundamental motor frequency / lowest frame resonance
- It is usually ideal to place the **IMU_GYRO_CUTOFF** just to the left of the motor noise peak (as shown) when the harmonic notch filters are working correctly



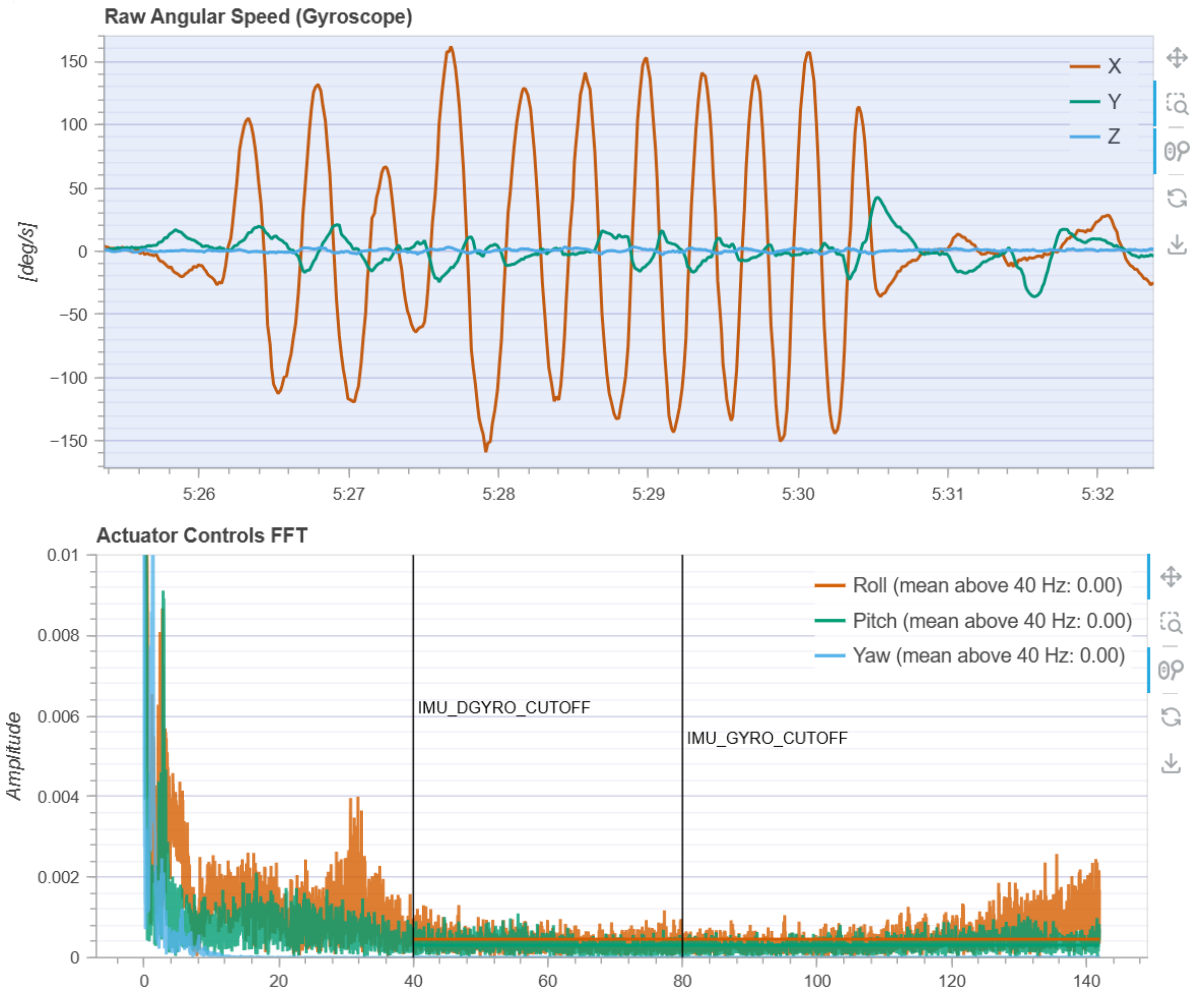
Tuning the Gyro D term Filter

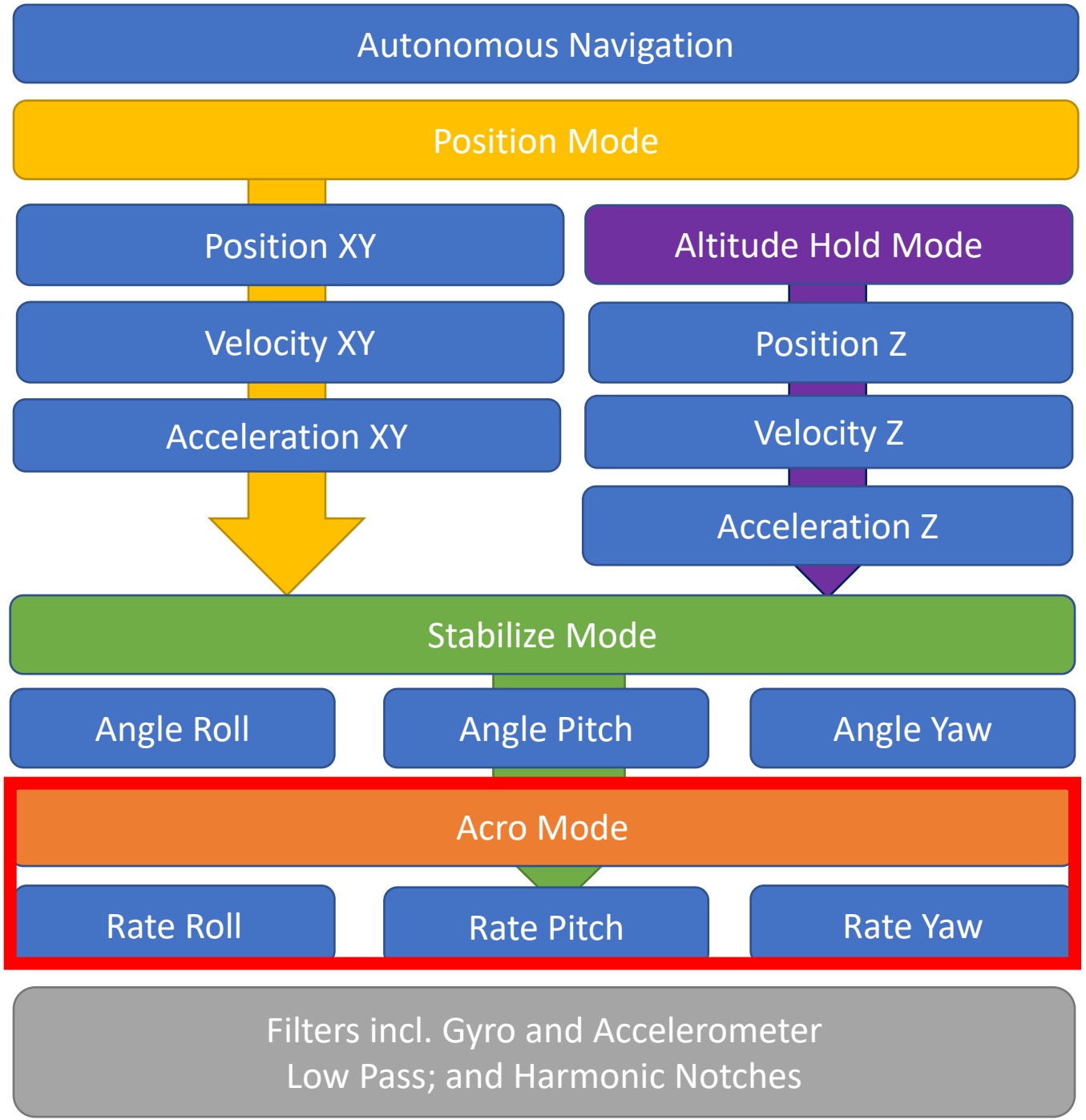
- When tuning the Gyro low pass filter it is also important to adjust the D-Term low pass filter
- When you change the gyro low pass filter cut-off set:
  IMU_DGYRO_CUTOFF = 0.5 * IMU_GYRO_CUTOFF
  E.g. $0.5 * 80 = 40Hz$ for my drone
- This rule is a good starting point for typical drones



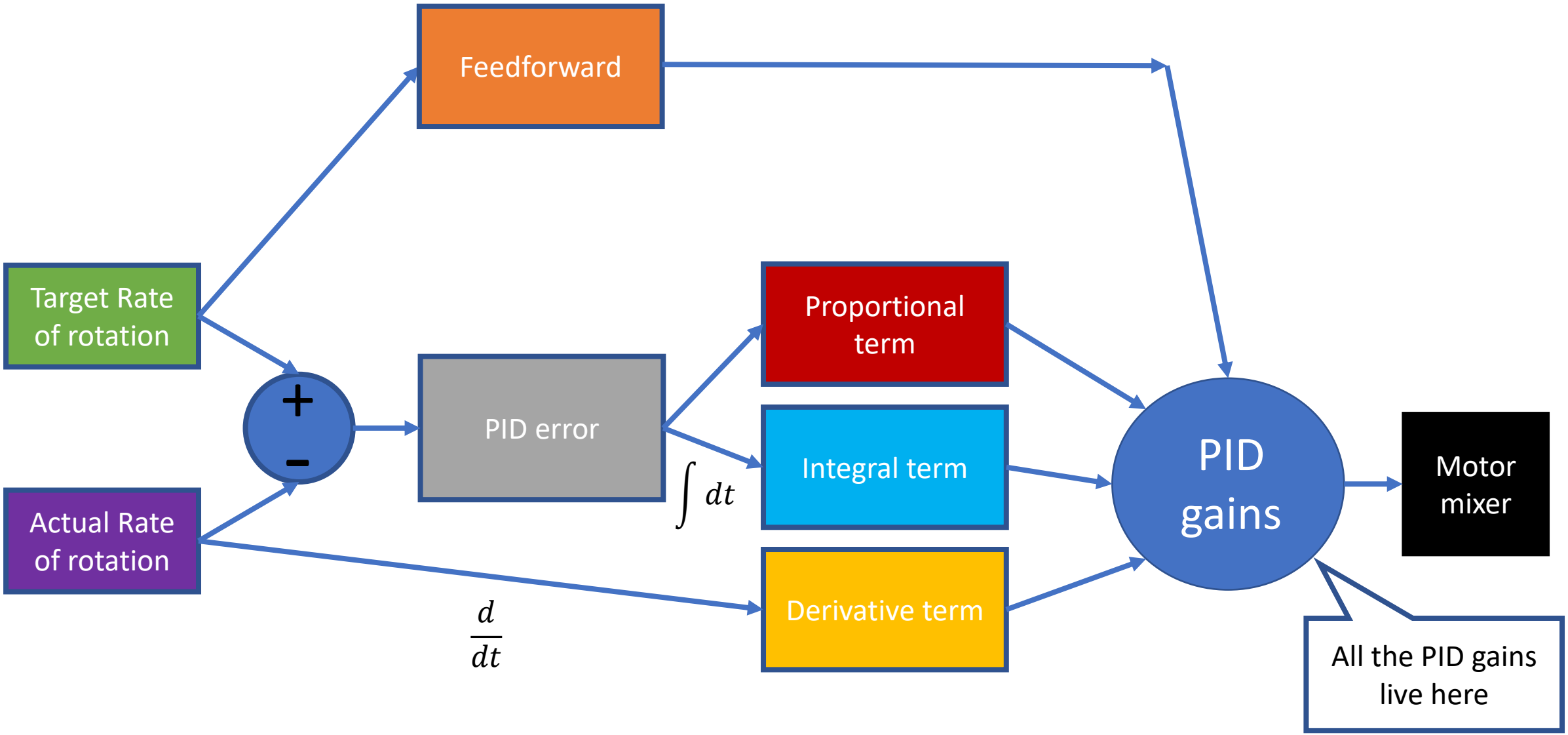
Increasing the IMU_GYRO_CUTOFF

- When agility and responsiveness is critical it can be advantageous to raise the IMU_GYRO_CUTOFF as high as possible
- This reduces PID loop delay
- As you raise the gyro cut-off (and dgyro cutoff) check the Raw Angular Speed and Actuator Controls FFT plots to ensure that noise is not passing into the PID loop




Autonomous Navigation
Position Mode
Position XY
Velocity XY
Acceleration XY
Altitude Hold Mode
Position Z
Velocity Z
Acceleration Z
Stabilize Mode
Angle Roll
Angle Pitch
Angle Yaw
Acro Mode
Rate Roll
Rate Pitch
Rate Yaw
Filters incl. Gyro and Accelerometer
Low Pass; and Harmonic Notches

The PX4 Rate PID-FF Controller



Proportional term



The P term acts like the spring in a suspension strut

PID error

PID error

PID error

PID error

PID error

PID error

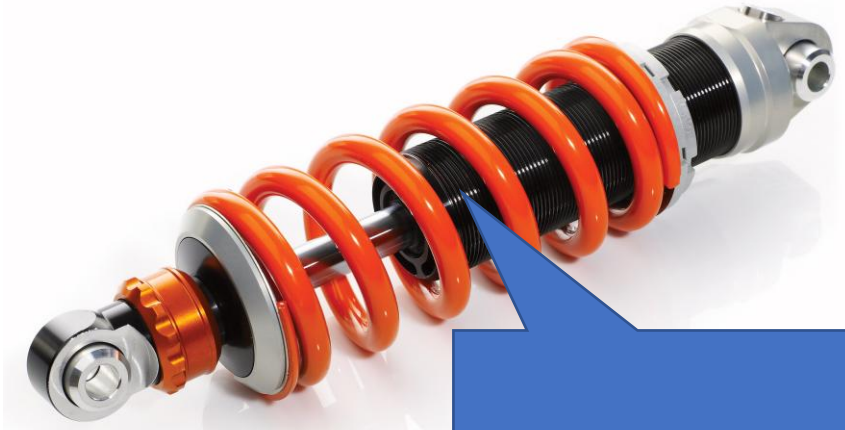
PID error

PID error

PID error

The bigger the error the harder the P term pushes back towards the Target

Derivative term



The D term acts like the shock absorber in a suspension strut

Actual

Actual

Actual

Actual

Actual

Actual

Actual

Actual

Actual

Derivative term wants the Actual value to stay the same and pushes against any change.

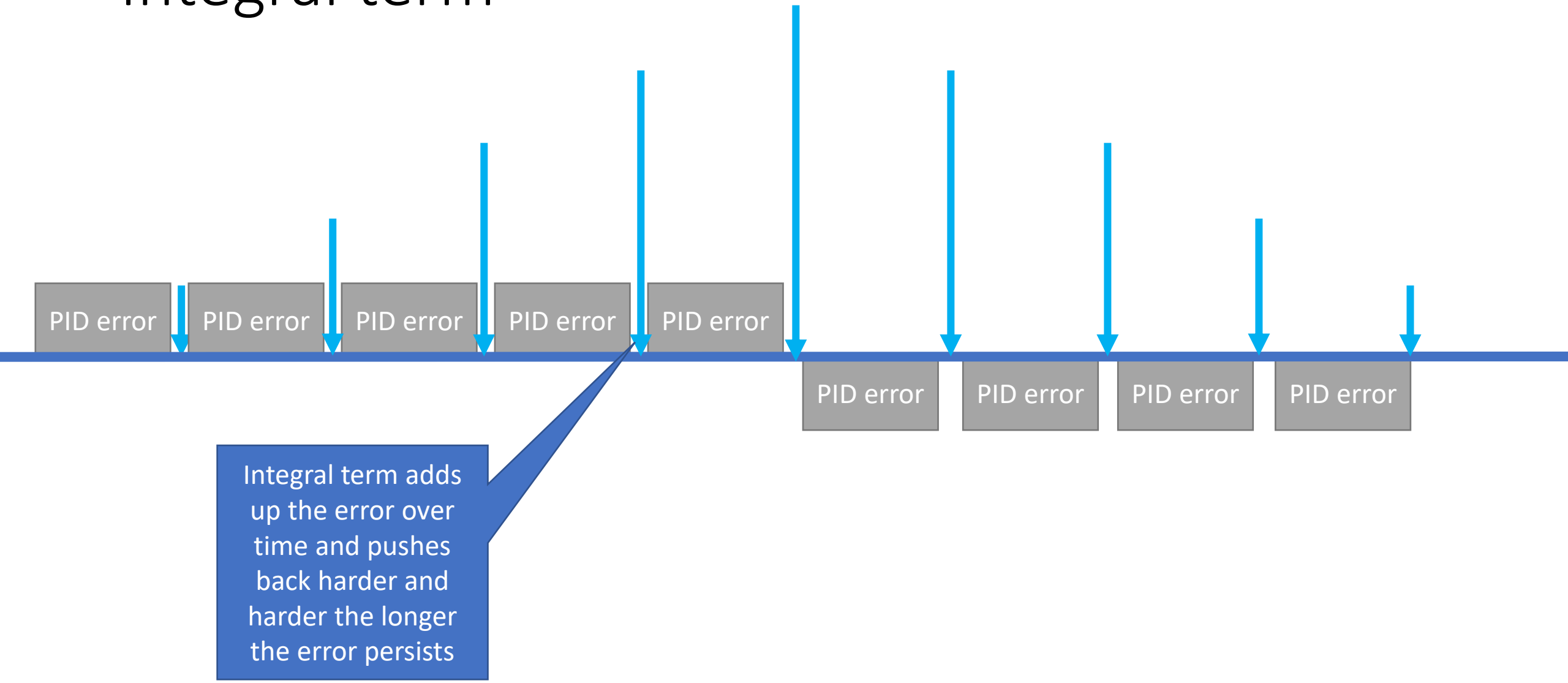
P:D balance

- Just like in a suspension strut the balance between the spring strength (P term) and the size of the shock absorber (D term) is very important
- The PX4 default P:D balance is usually a suitable starting point but not always!
- The right balance between P and D terms is the first step of a good PID tune



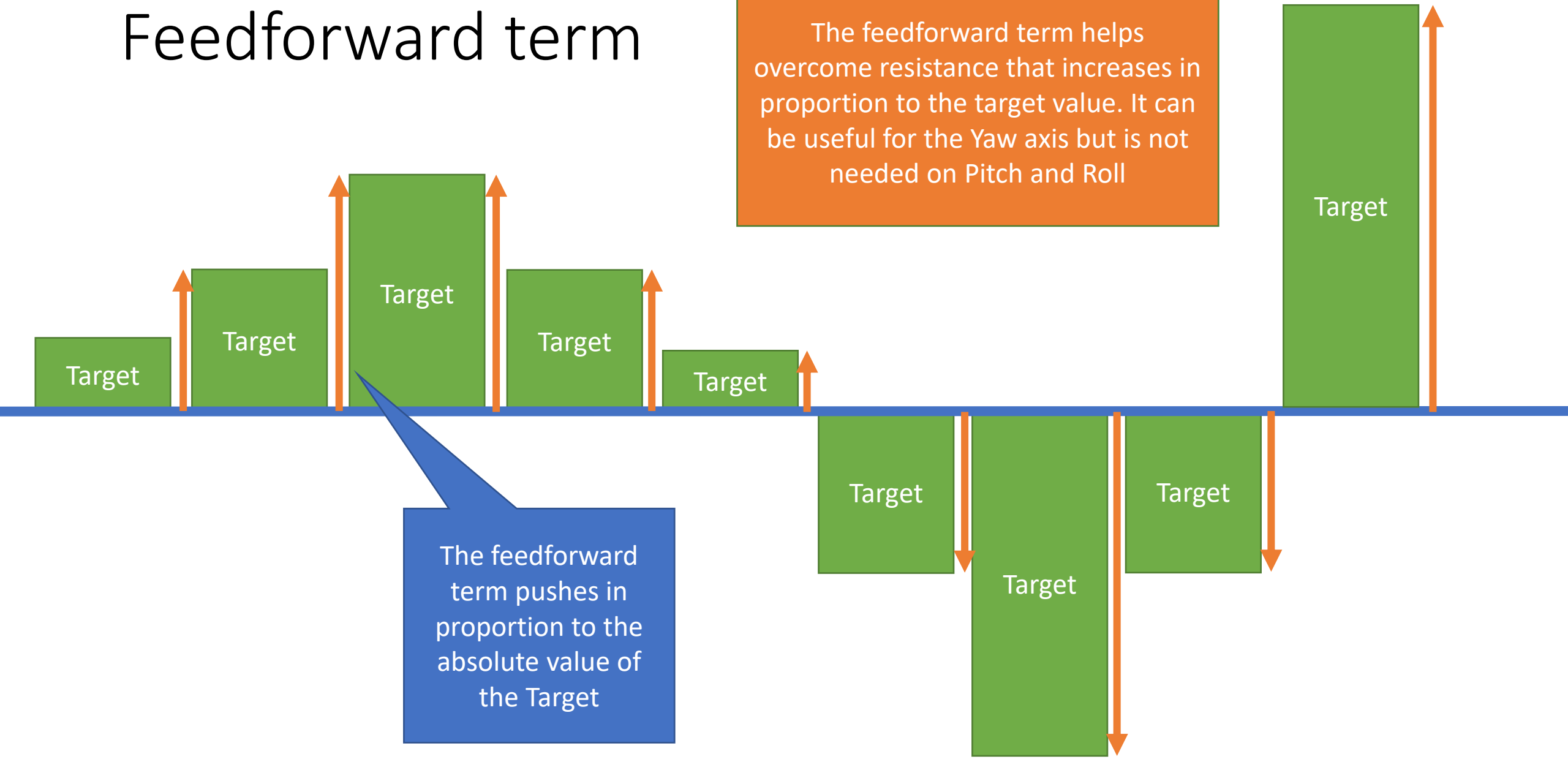


Integral term



Feedforward term

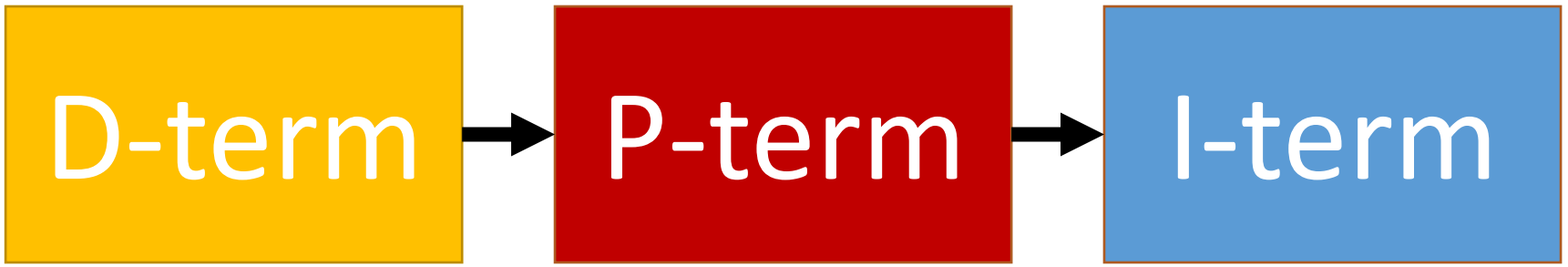
The feedforward term helps overcome resistance that increases in proportion to the target value. It can be useful for the Yaw axis but is not needed on Pitch and Roll



The feedforward term pushes in proportion to the absolute value of the Target

P:I balance and D->P->I->FF tuning approach

- In the same way as the D term prevents the P term oscillating the P term prevents the I term oscillating
- As a result there is an ideal P:I ratio for a controller that allows the maximum I term without oscillations
- This gives us the relationship D->P->I->FF which guides the tuning process: D term first, then P term, I term, and finally FF



There are 3 Rate Controllers in PX4

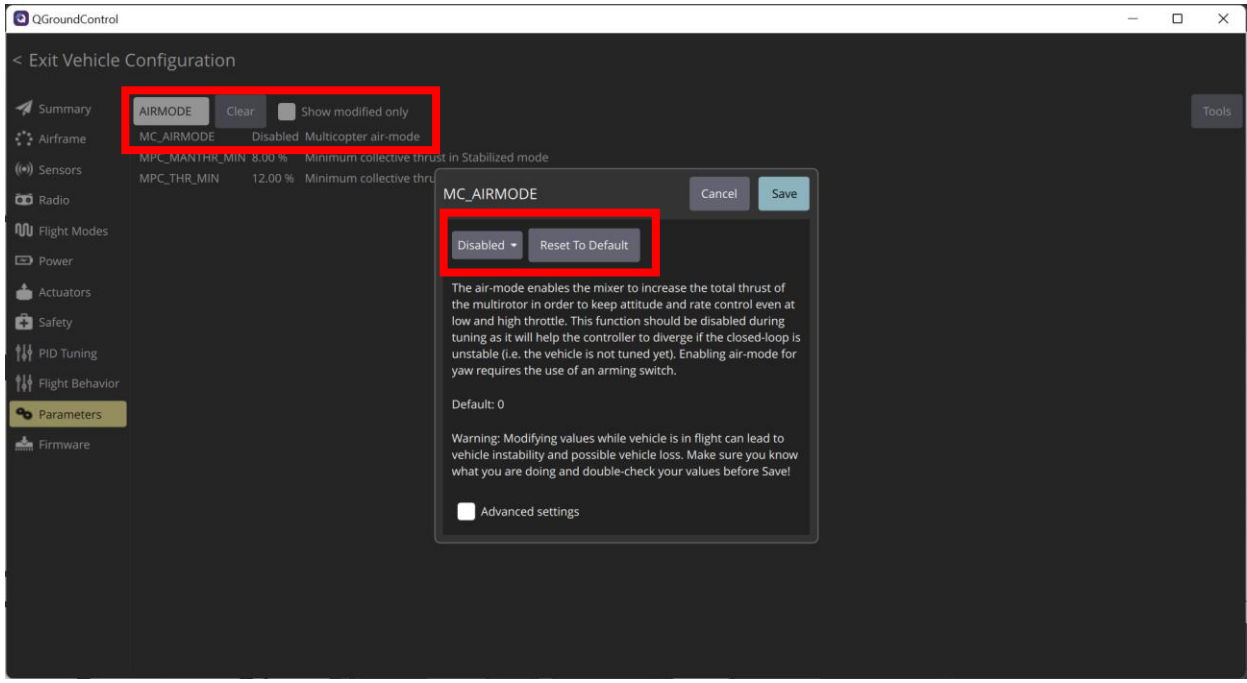
- One each for roll, pitch and yaw
- They all work the same way
- However, the derivative term on the yaw axis is usually set to 0 because the lack of authority on the yaw axis means it is usually over damped and does not require an additional derivative term
- The Yaw axis can often benefit from a small amount of Feedforward gain

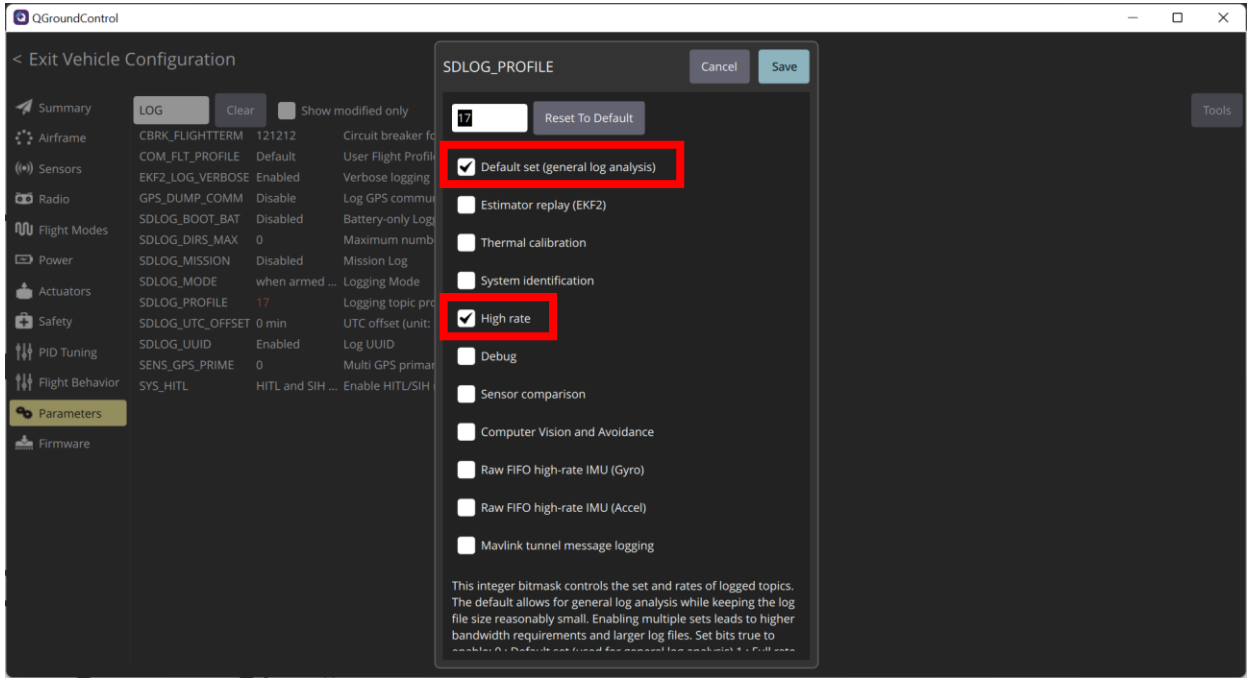
Tuning flight tips

- PID tuning flights consist of sharp wobbles on all 3 axis
- When you do sharp moves make sure that you keep hold of the stick as it returns to centre
- If you release the sticks, they can bounce around the centre as they spring back which will make the quad oscillate (and might make you think you have too much P gain when you don’t)
- You can do these sharp moves in Stabilize or Altitude mode (but **not** Position)

Setup for PID tuning

- **Vehicle Configuration > Parameters**
- Search **AIRMODE**
- Set **MC_AIRMODE = Disabled** (this prevents fly-aways during tuning)
- Search **LOG**
- Set SD_LOG_PROFILE = 17
  - Default + High Rate



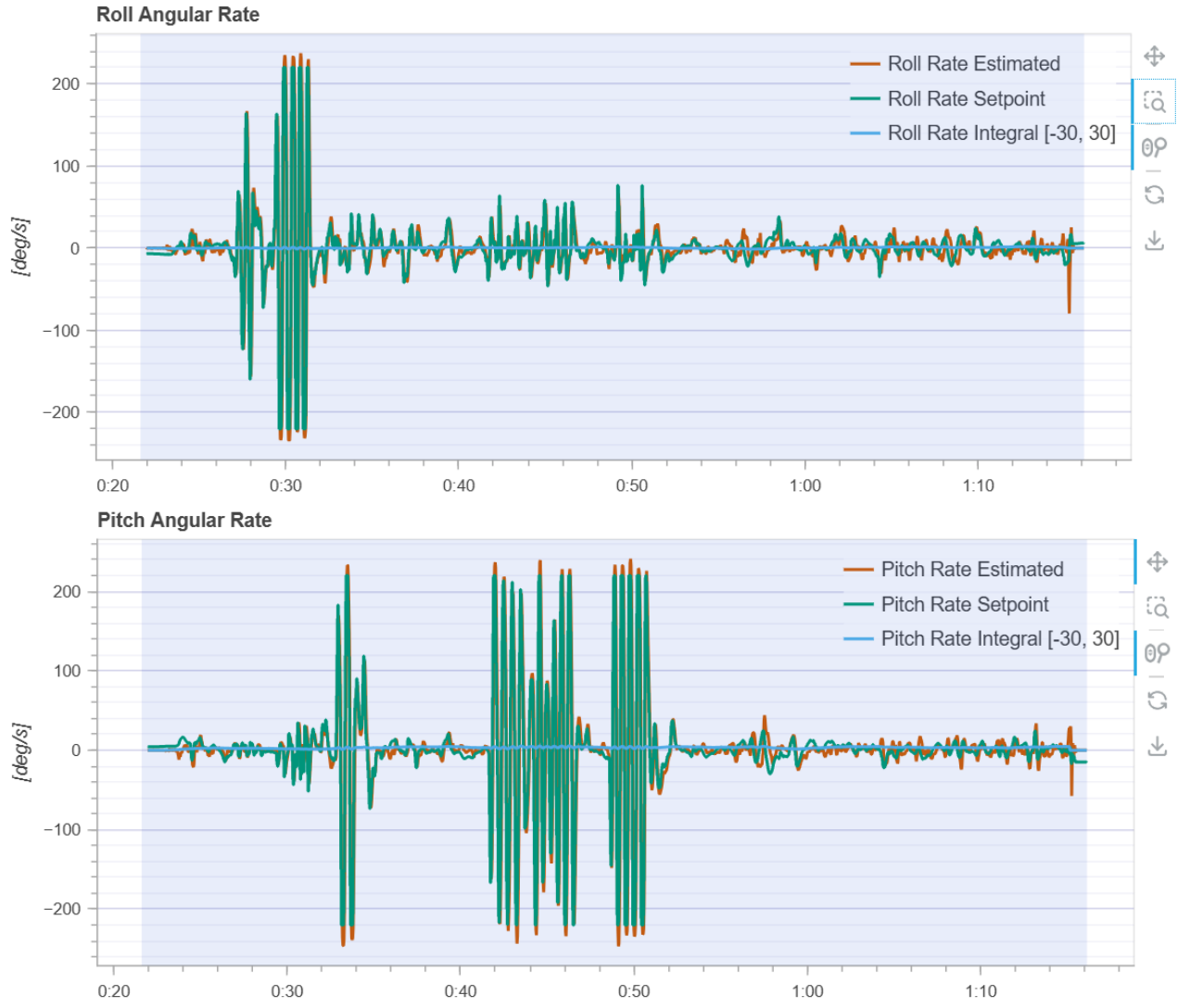


Test Flights

- Fly in Stabilized or Altitude Mode
- Wobble the drone quickly back and forth on Pitch, Roll and Yaw Axes
- 10s of wobbles on each axis is enough

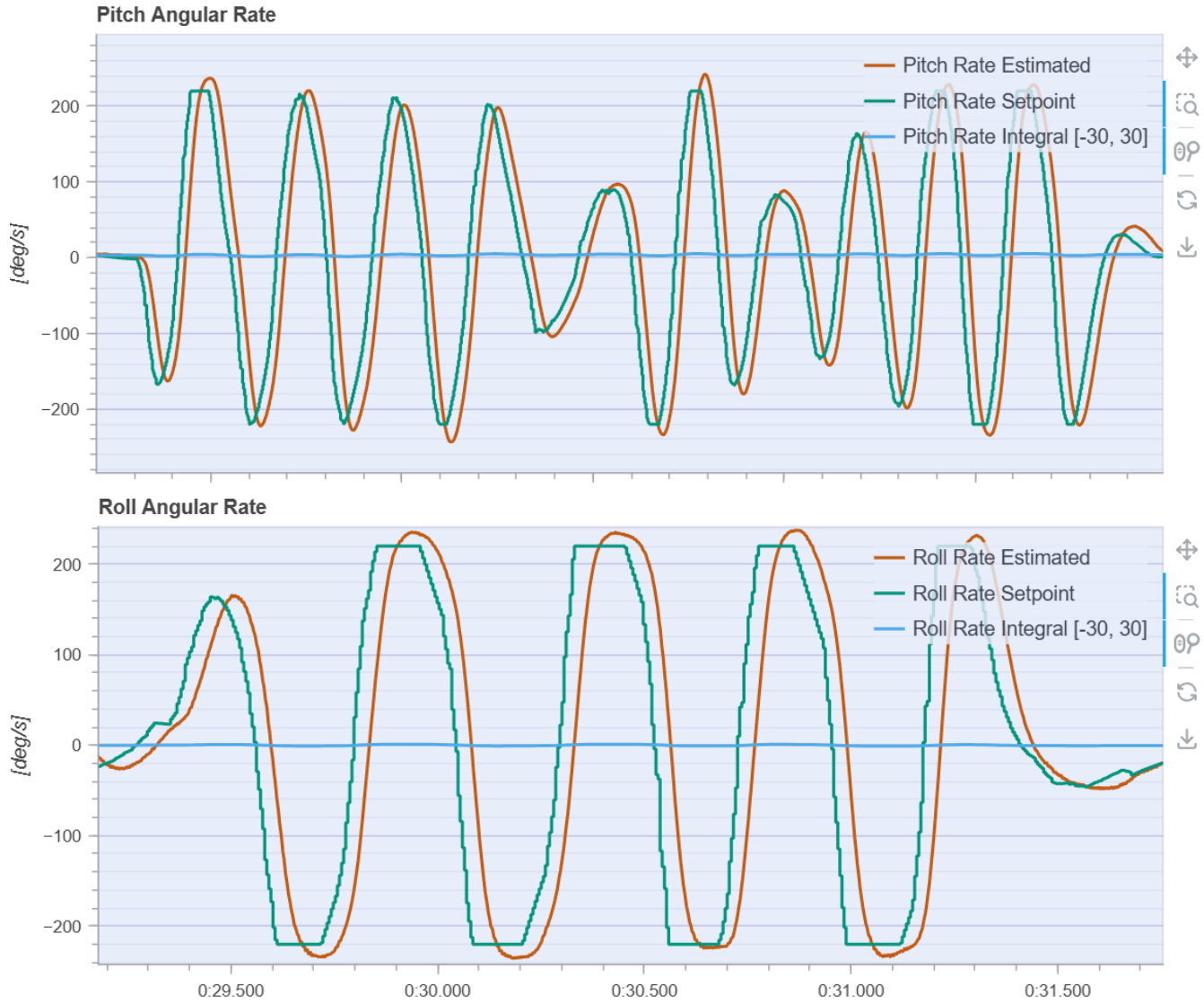
PID Review – Reading time series data

- Time domain data is unreadable at the default zoom level
- Zoom into the parts of the log where you wobbled the drone on the selected axis



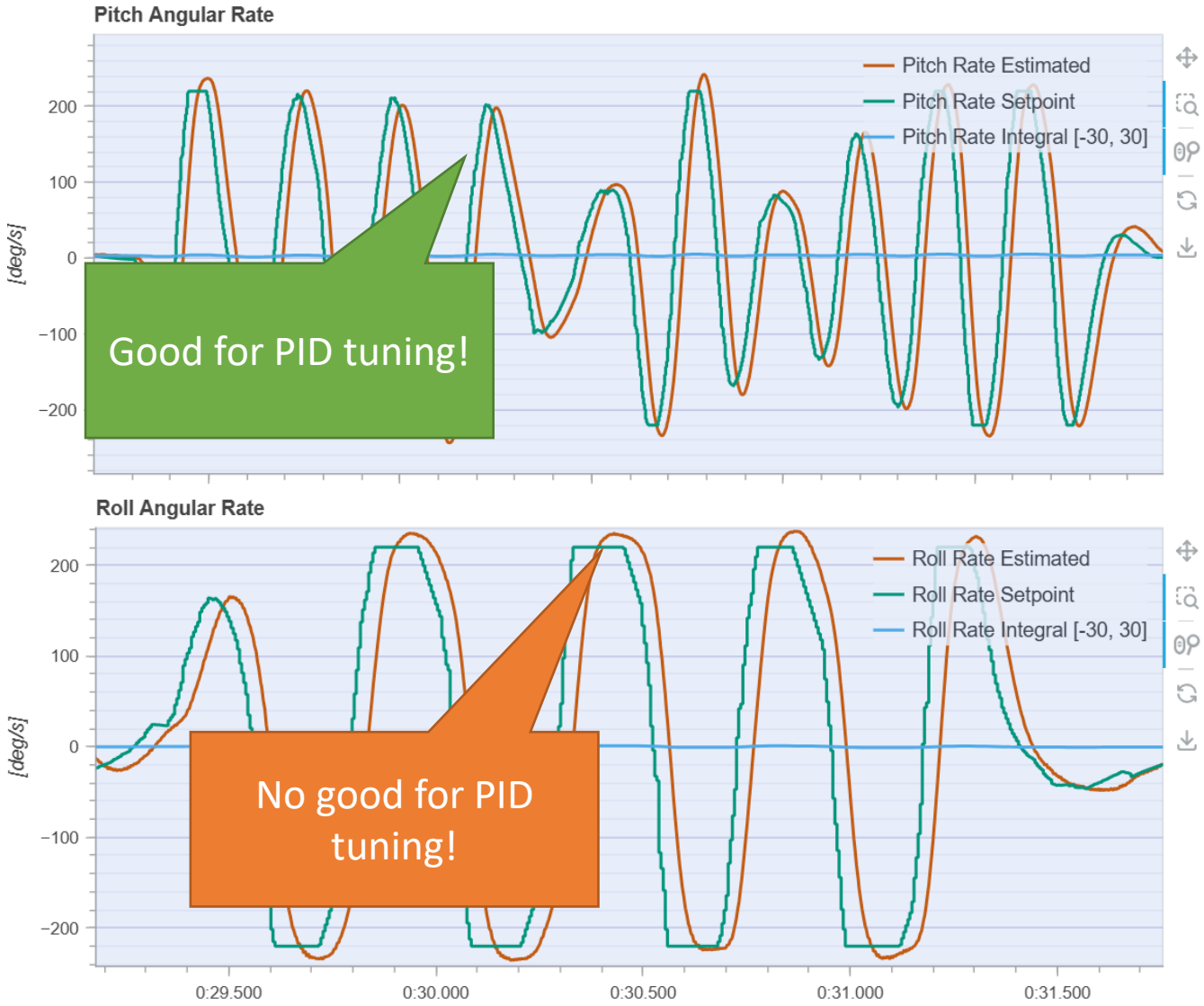
PID Review – Reading time series data

- At this zoom level we can see the individual movements
- The goal is to get the Actual and Target lines to be close to on top of each other
- It is possible to tune P:D balance only using time series data
- This is useful as for some controllers time series is all we have easy access to



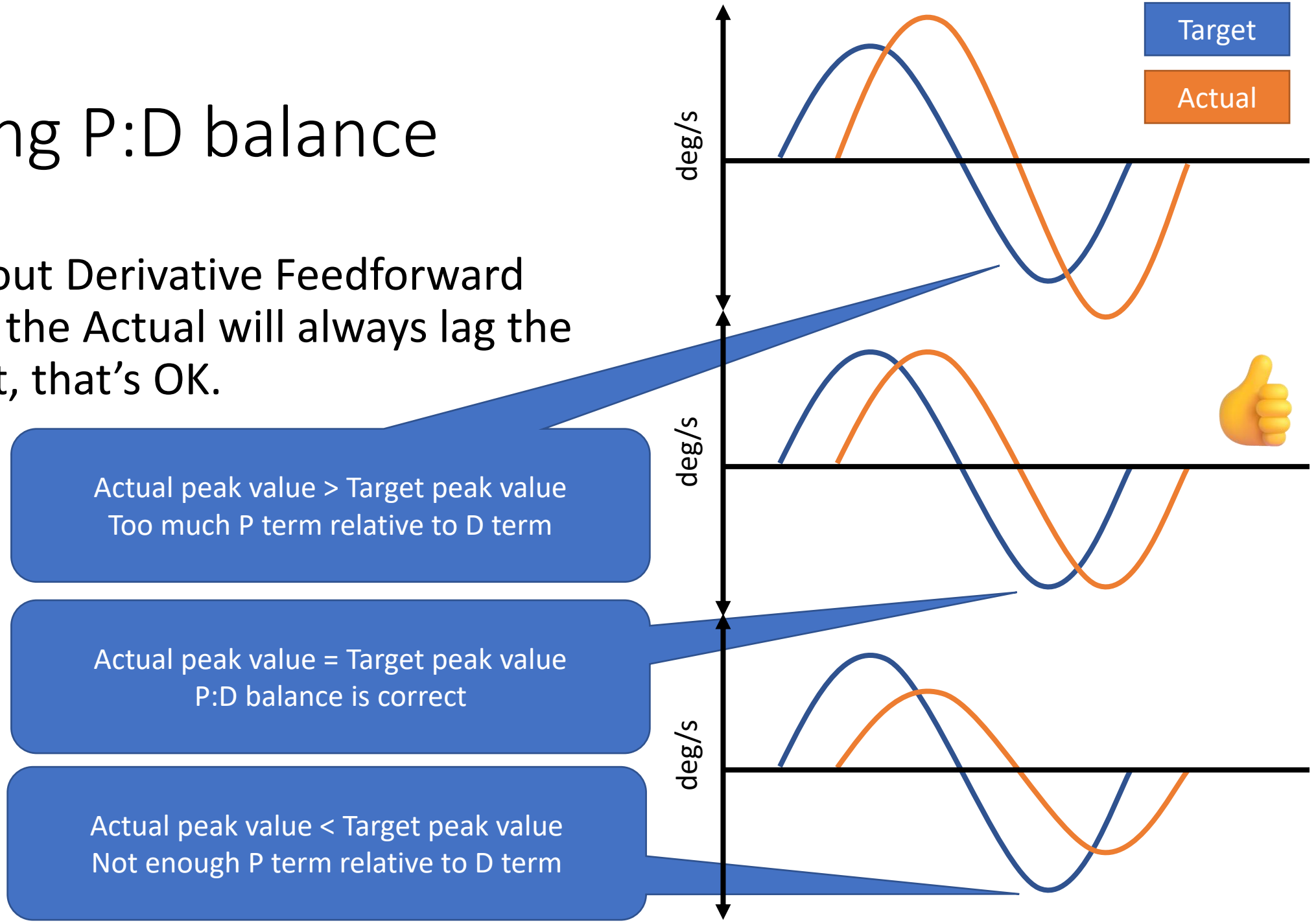
Avoid target saturation

- Although we want nice sharp moves for PID tuning we don’t want the moves to saturate the angular rate setpoint
- PX4 has a default 220 °/s limit for the rate setpoint
- If all your moves have flat tops in the setpoint graph redo the test flight with slightly gentler moves



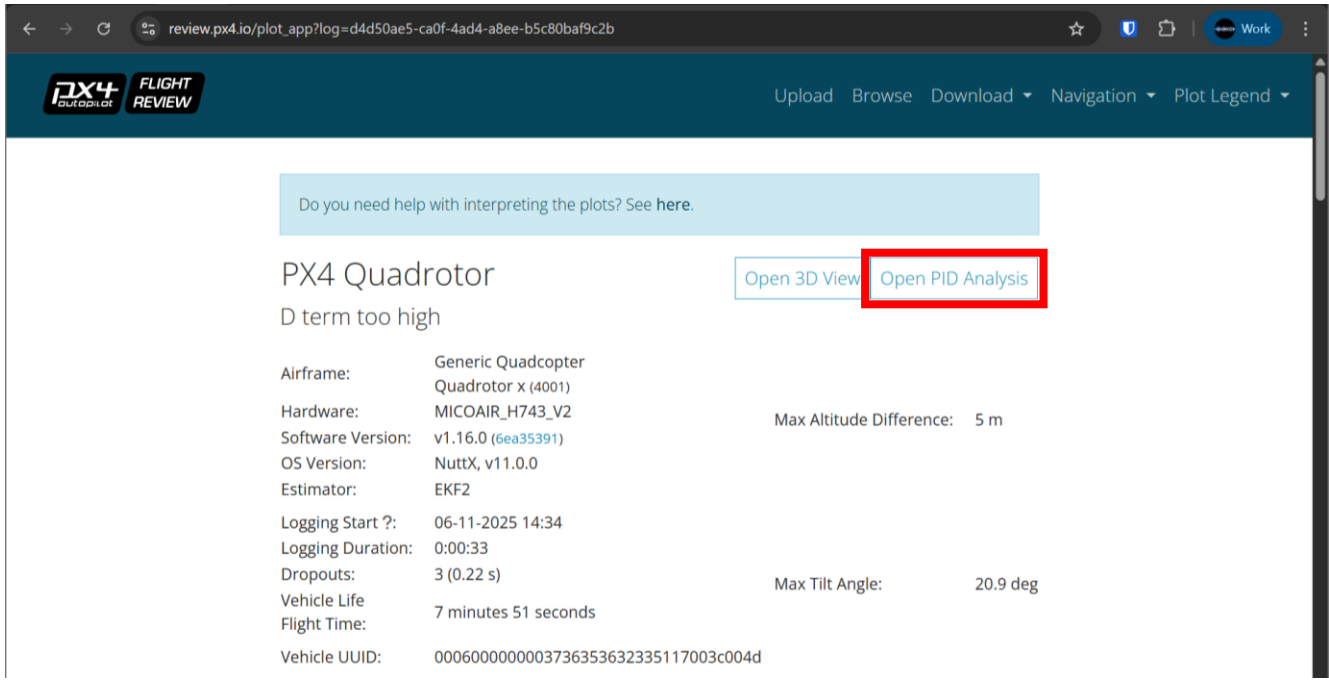
Tuning P:D balance

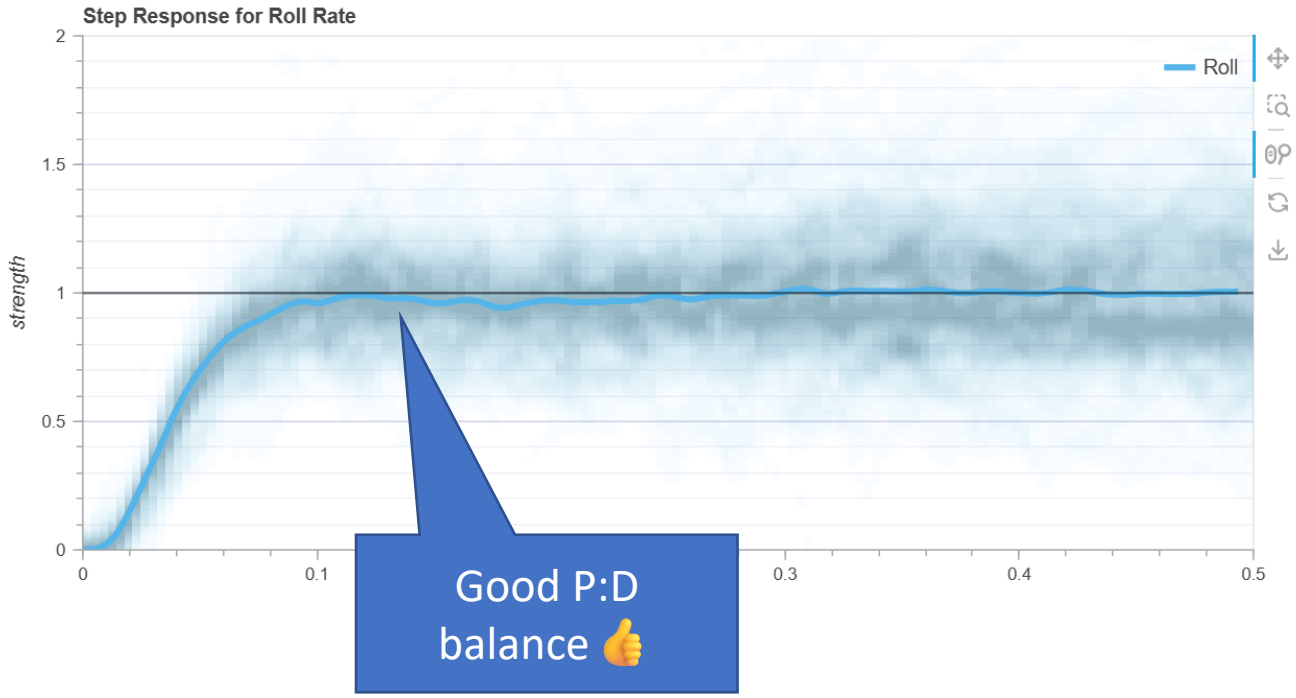
- Without Derivative Feedforward (DFF) the Actual will always lag the target, that's OK.



Using Step Responses

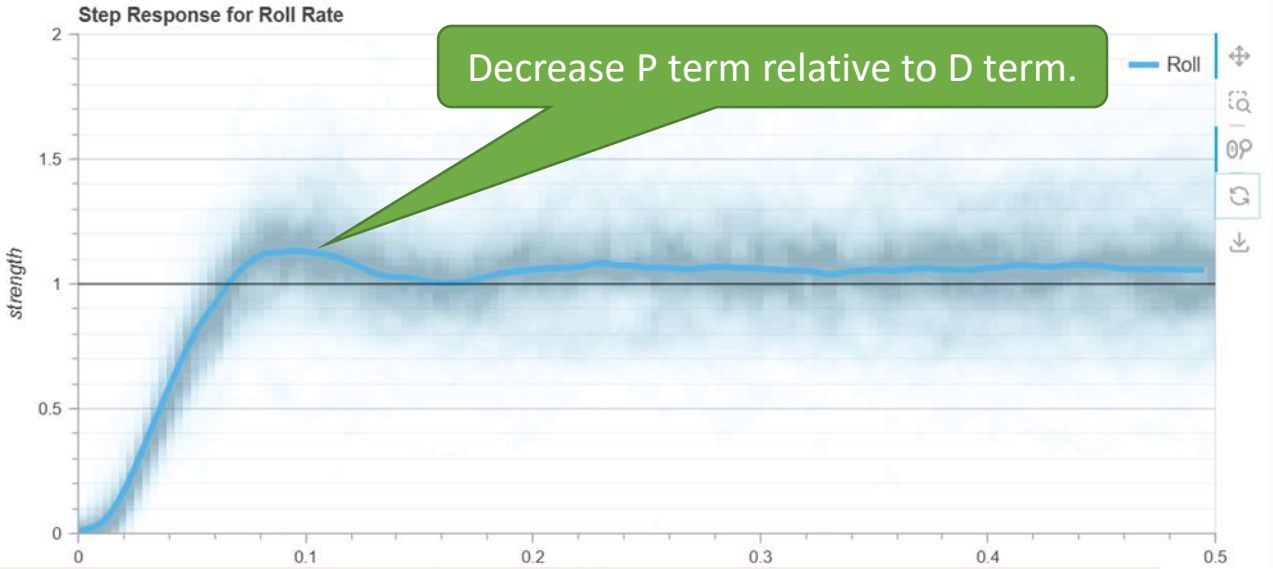
- You can access step response charts to assist with rate controller tuning
- Some pilots find these step response plots more useful than the time series data
- I generally prefer using time series data as a step response plot can sometimes hide tuning problems

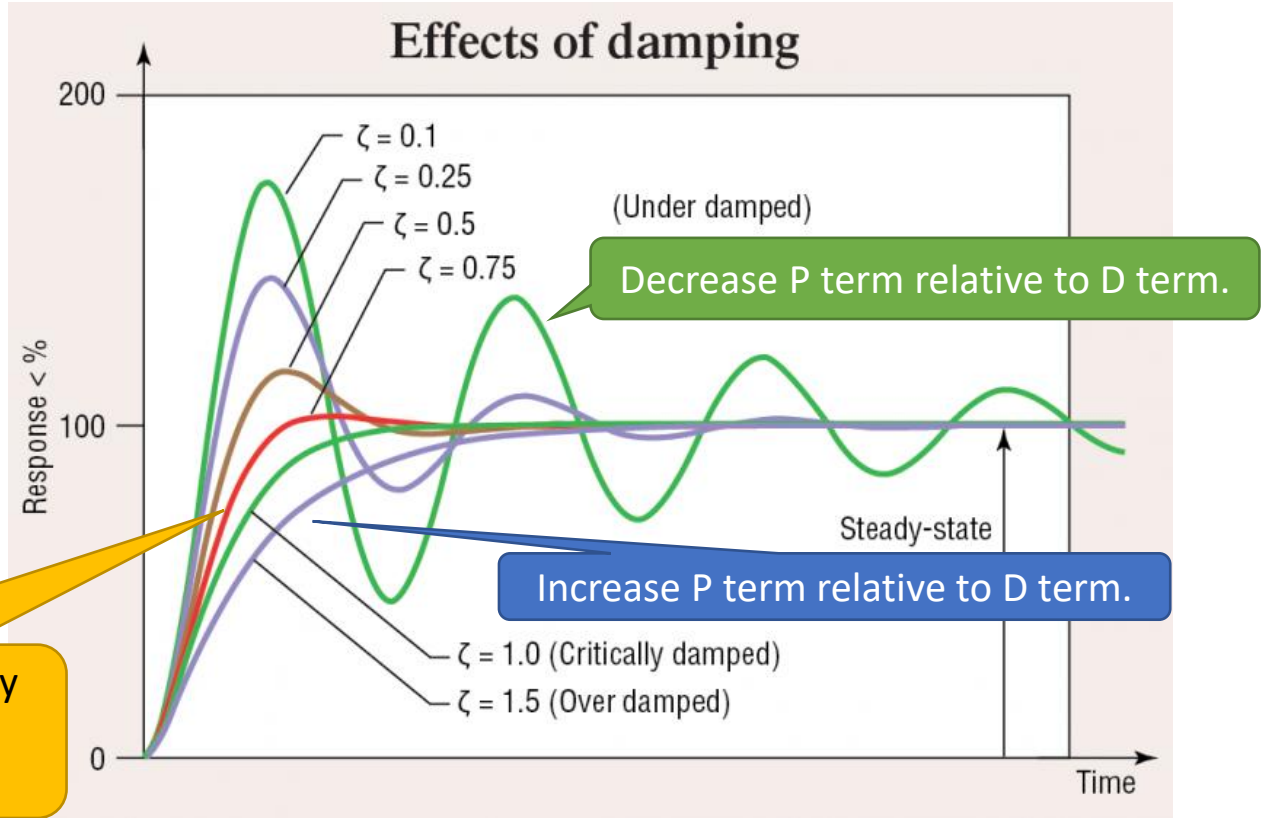




Tuning P:D balance

- In this guide we will use the term **Underdamped** which means not enough D term relative to P term
- We will use the term **Overdamped** which means not enough P term relative to D term
- We will use the term **Critically damped** for ideal P:D balance

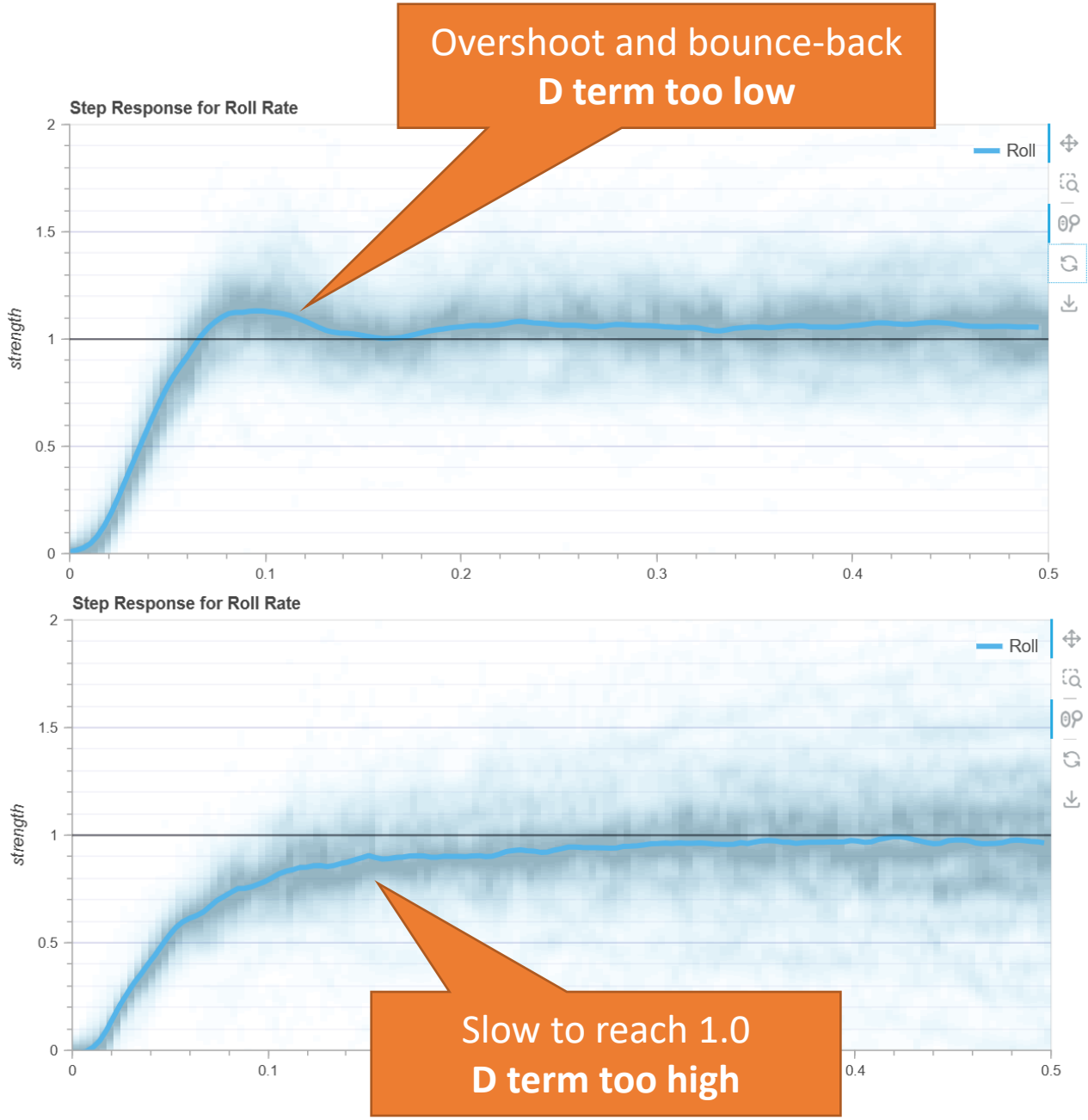




The red line is the goal! A tiny bit of overshoot is OK but absolutely no oscillation.

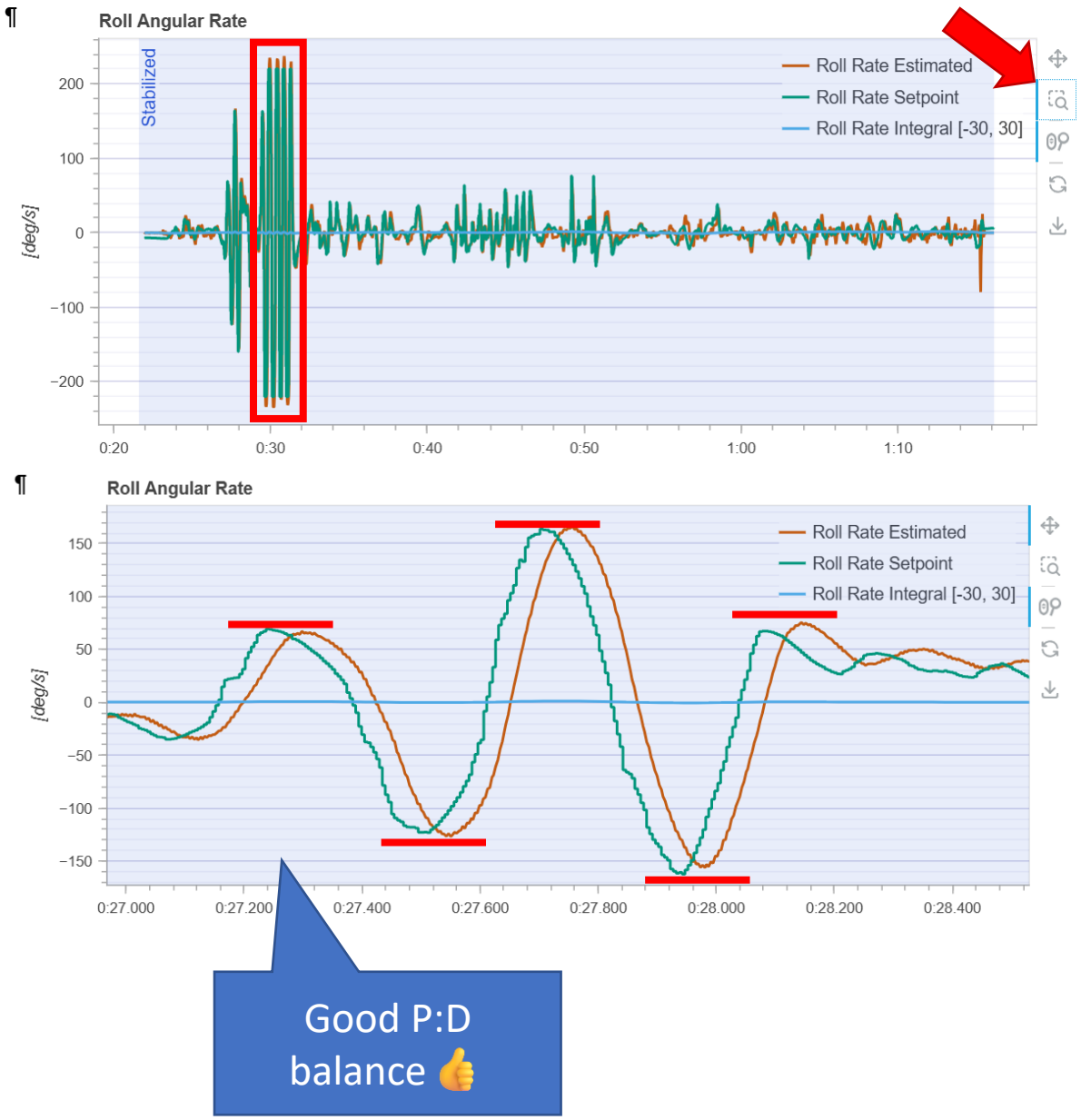
Using Step Responses

- Step response plots are not always available for every PID controller
- Try to become familiar with both methods but do not solely rely on step response plots for tuning!



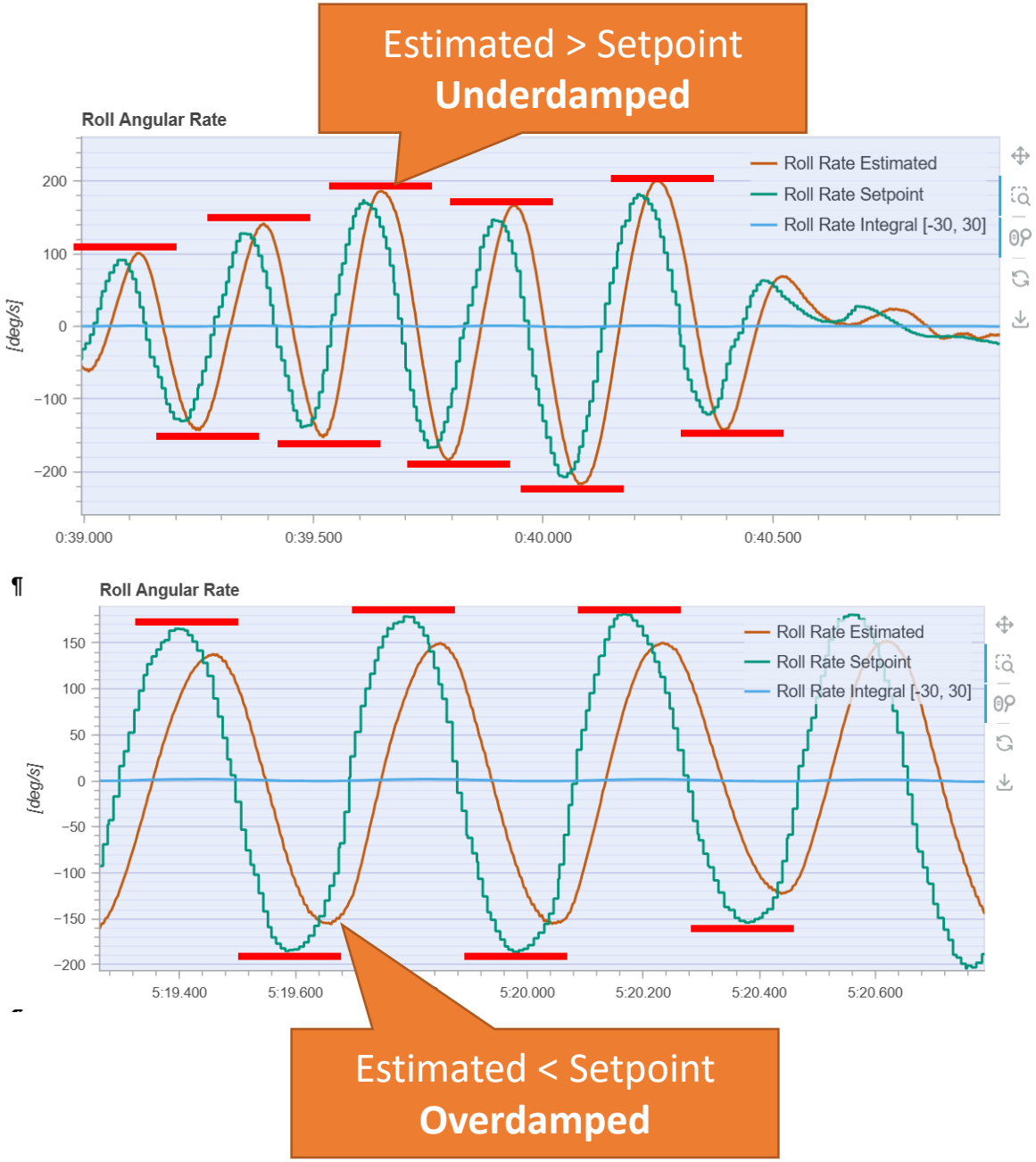
Tuning the P:D Balance

- Upload the log and look at the **Roll Angular Rate** Graph
- Use the zoom tools to zoom in on the wobbles
- The peaks on the **Roll Rate Estimated** graph should be the same height as the peaks on the **Roll Rate Setpoint** graph
- If they are then P:D balance is appropriate



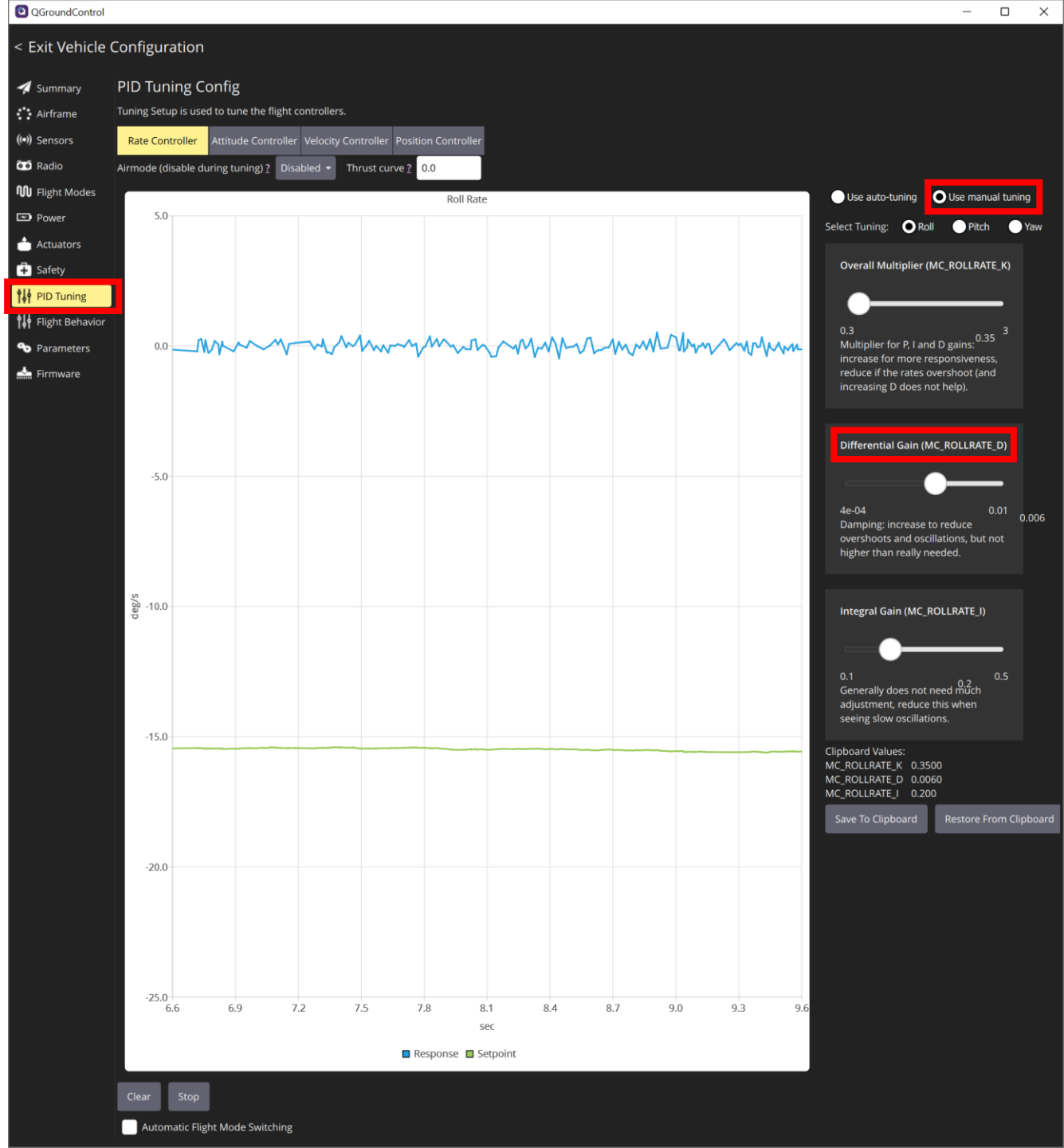
Tuning the P:D Balance

- If the peaks on the **Roll Rate Estimated** graph are bigger than the peaks on the **Roll Rate Setpoint** graph the **D term is too low (underdamped)**
- If the peaks on the **Roll Rate Estimated** graph are smaller than the peaks on the **Roll Rate Setpoint** graph the **D term is too high (overdamped)**



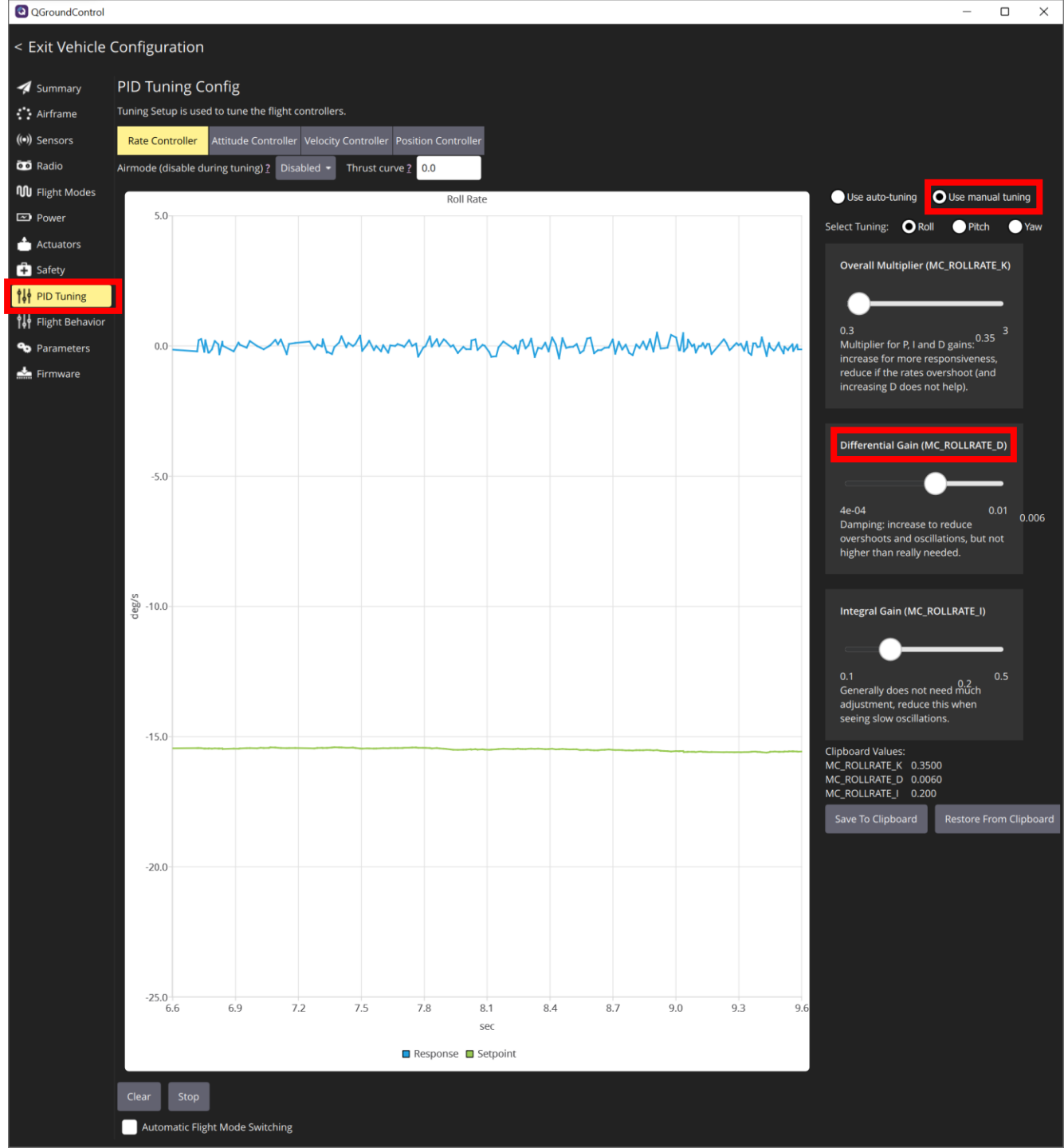
Adjusting P:D Balance

- **Vehicle Configuration > PID Tuning > Manual Tuning**
- Adjust **MC_ROLLRATE_D** in steps of **20%** up or down as necessary
- E.g. If the current value is 0.001 try 0.008 or 0.0012 **NOT** 0.002!
- As you get more experienced you may adjust in larger or smaller steps as necessary



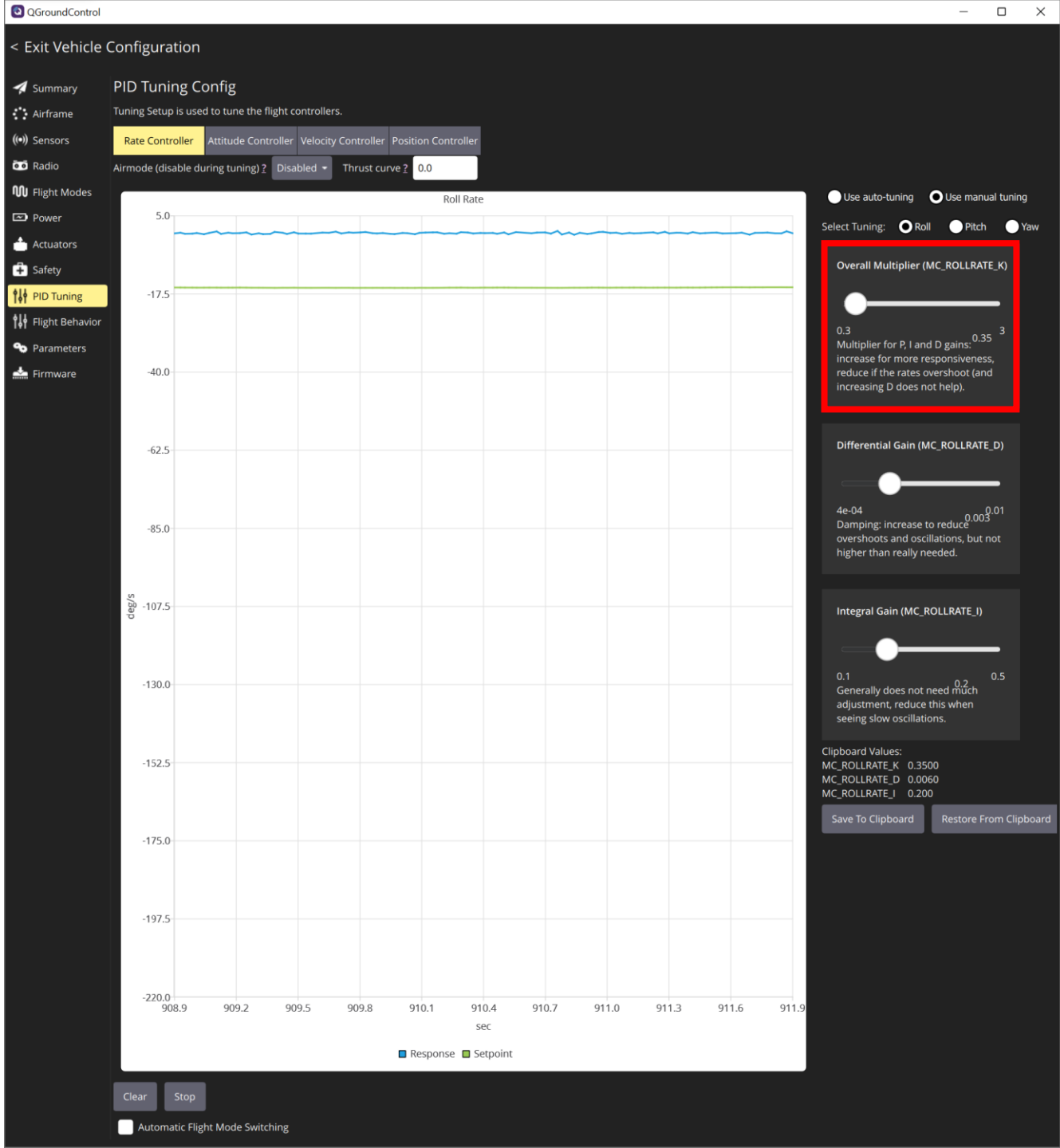
Roll and Pitch

- Roll and Pitch are independent
- Tune them one at a time
- Check the Roll and Pitch Angular Rate charts one at a time and adjust **MC_ROLLRATE_D** and **MC_PITCHRATE_D** up and down in steps of 20% as necessary
- As you get more experienced you will be able to tune them in parallel



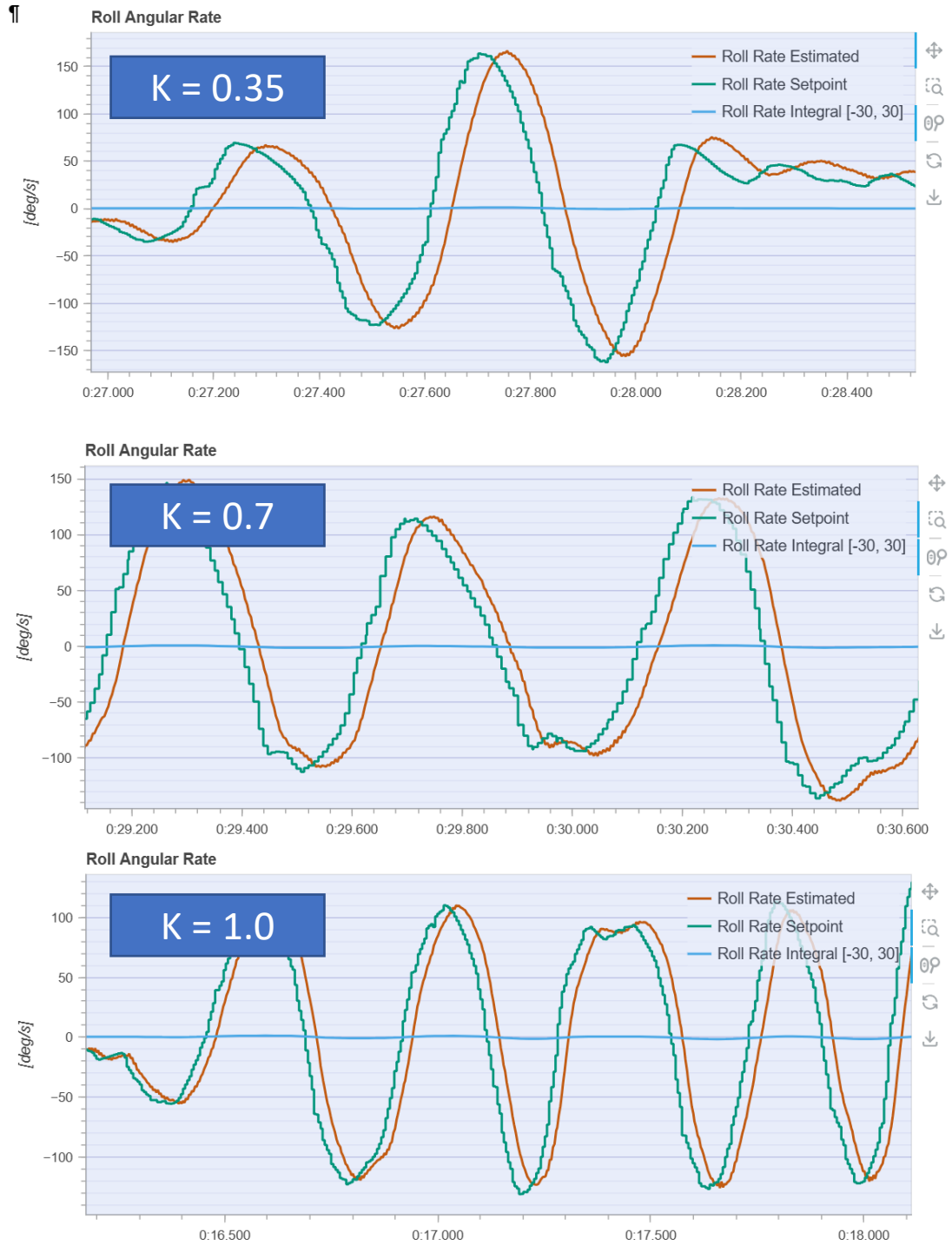
Overall Multiplier

- Once P:D balance is correct for Roll and Pitch we can move on to tuning the Overall Multiplier
- This increases the strength of the PID response and is equivalent to scaling P, I, and D together
- The goal is to increase this as far as possible without provoking oscillations
- Each axis is tuned independently



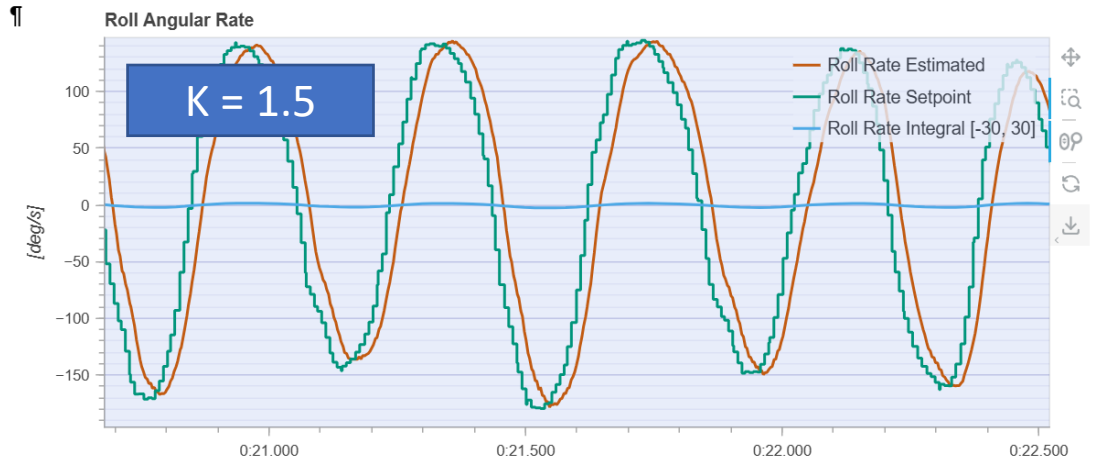
Overall Multiplier

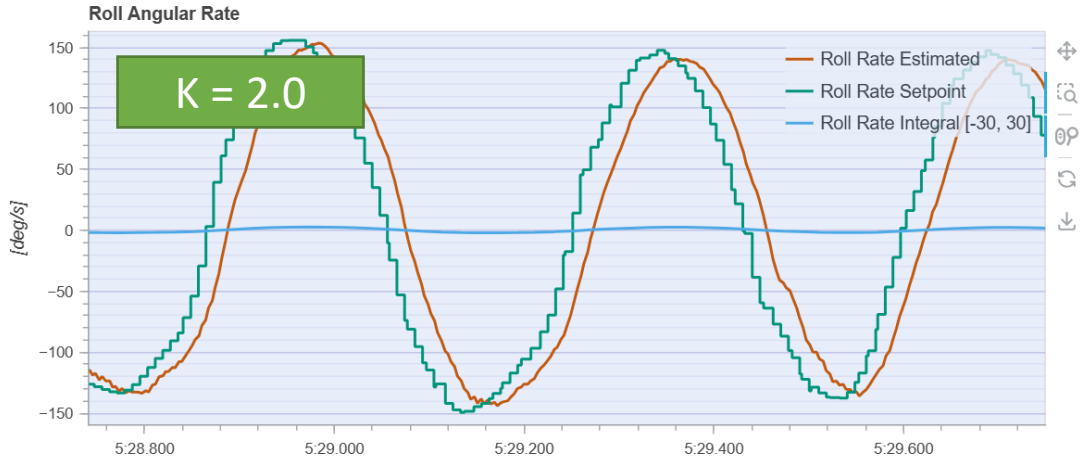
- As you increase MC_ROLLRATE_K and MC_PITCHRATE_K you should notice the flight performance of the drone improve (often substantially!)
- As PX4 does not have derivative feedforward (DFF) implemented in the PID loop the only way to reduce the delay between actual and setpoint is through increasing K

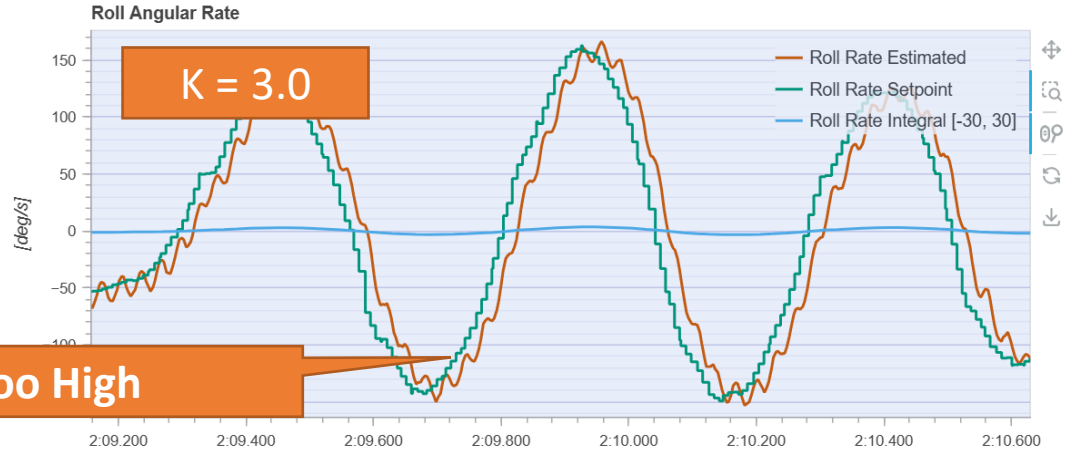


Overall Multiplier too high

- As you increase K it is likely you will start to notice diminishing returns (delay is not reducing much if at all with a larger K value)
- If you continue and Overall Multiplier gets too high you will see oscillations in the logs and hear them in flight



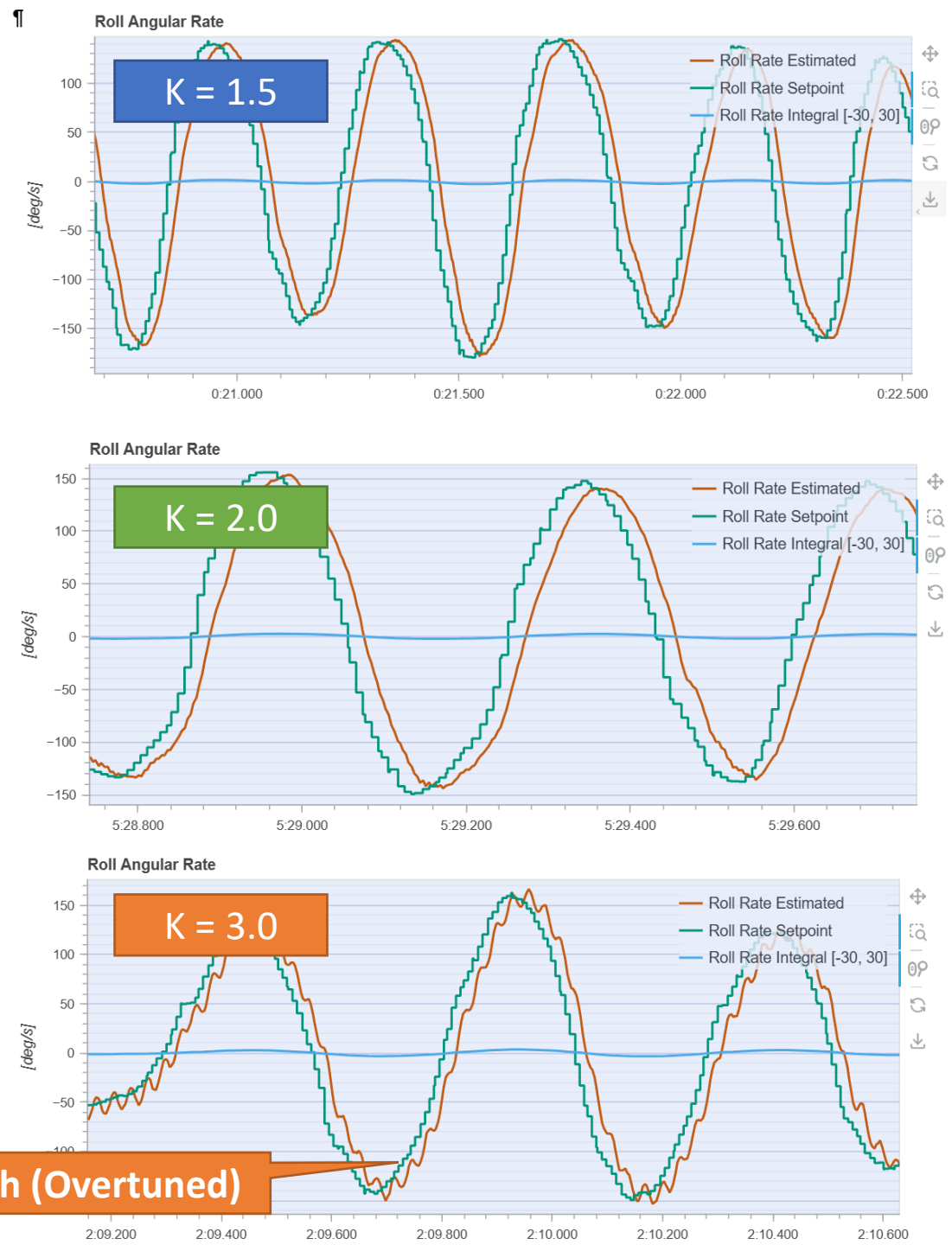




K too High

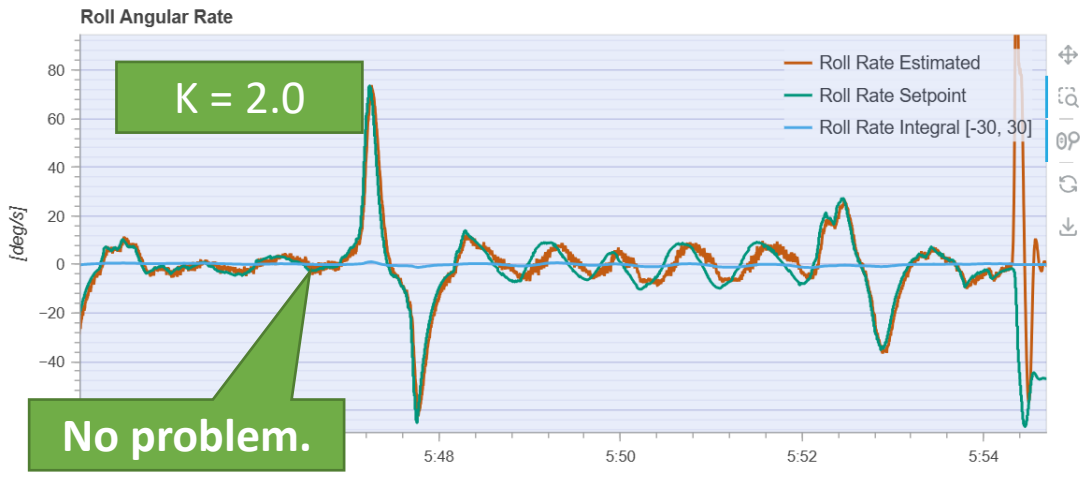
Under and Overtuned

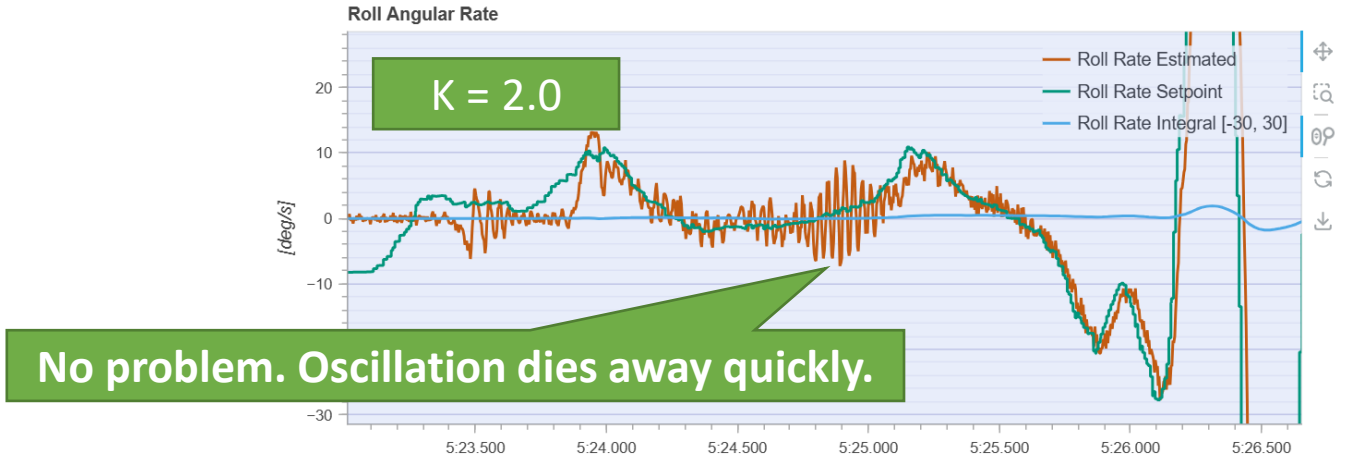
- In this guide we'll use the term **Undertuned** to indicate that PID gains are too low (too much delay between Estimated and Setpoint)
- In this guide we'll use the term **Overtuned** to indicate that PID gains are too high

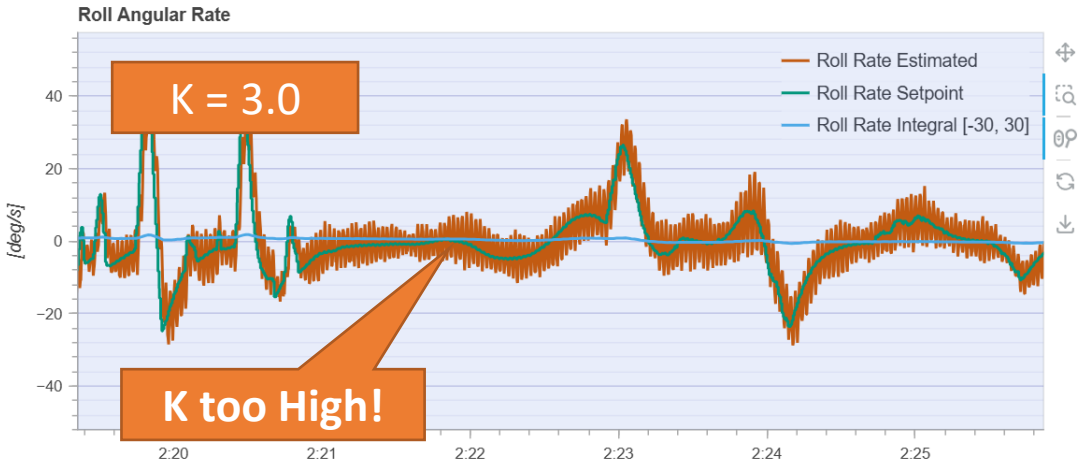


High K Oscillations

- These are potentially dangerous as they can cause ESCs and motors to fail!
- If you hear a **new** continuous oscillating/chittering sound from the motors upon take-off (especially if you've just raised K on one or more axes) **land immediately!**
- Check logs for micro-oscillations that do not die away
- This indicates that K is **too high** on one or more axes!

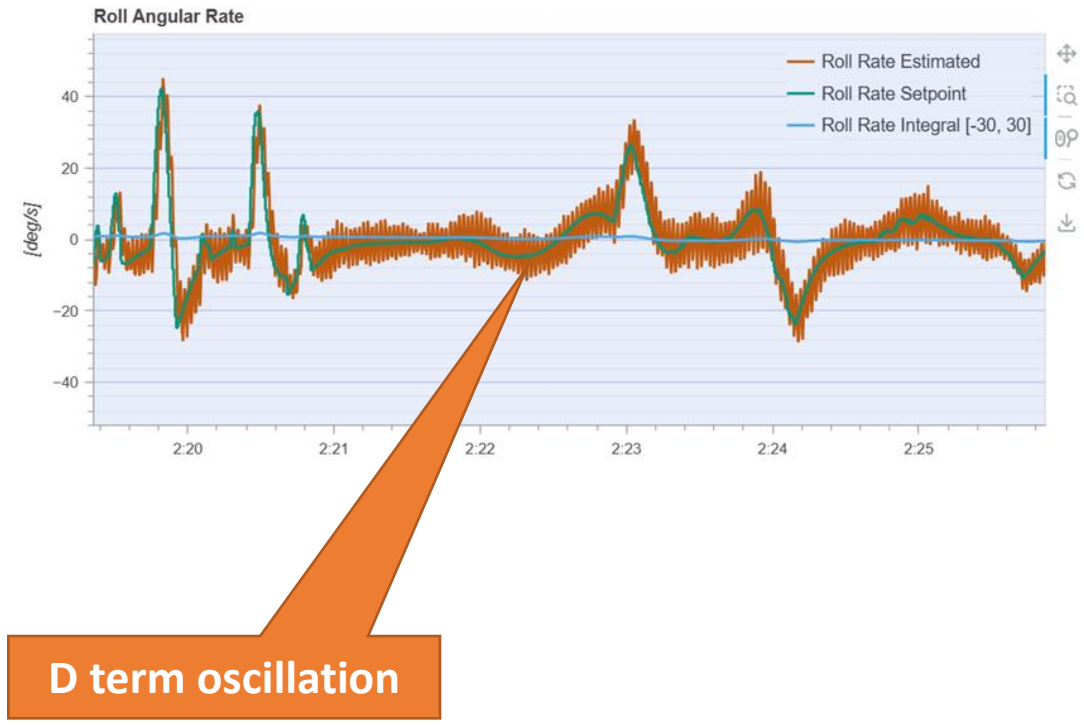






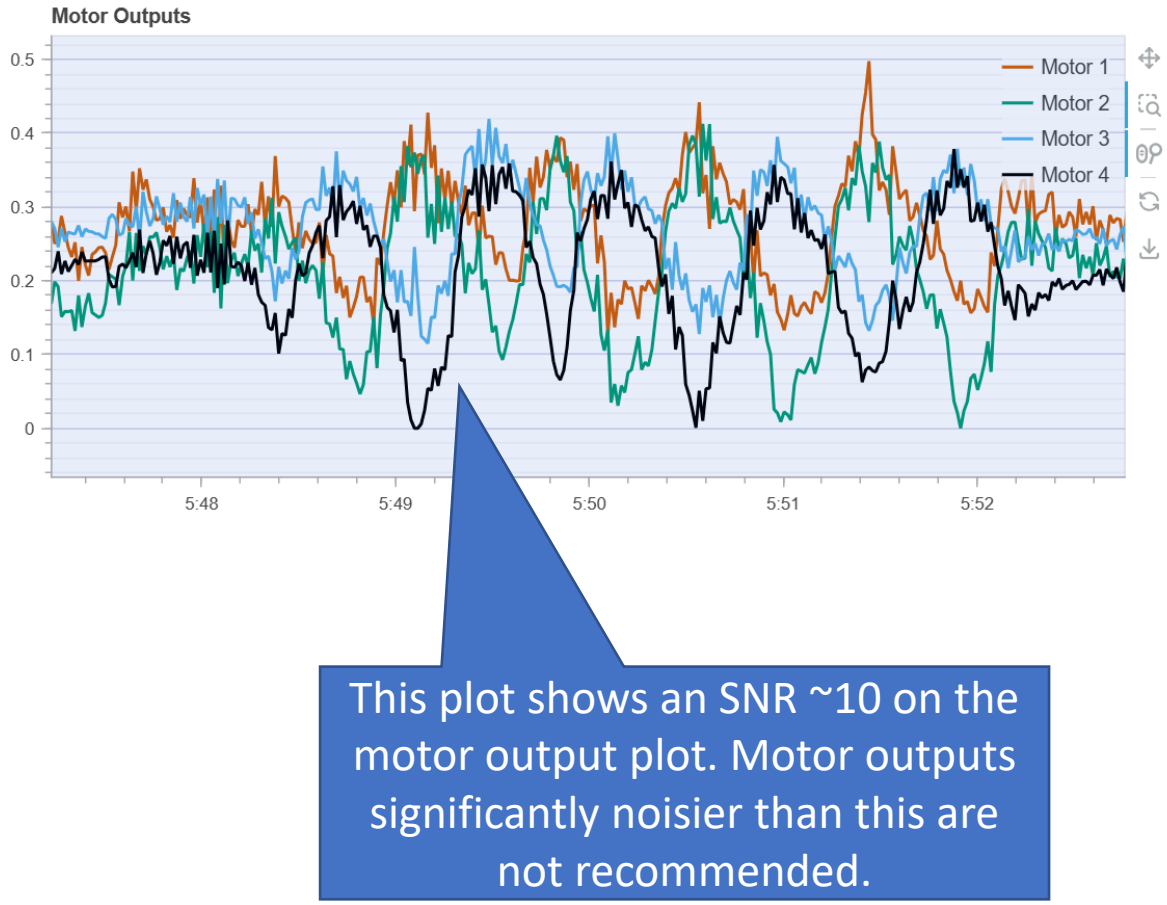
Troubleshooting Oscillation

- If you are suffering from oscillations that don’t improve when you reduce K :
- D term oscillation: High D gains can self-oscillate independent of the P term, to fix this reduce both the D and P terms in steps of 10%
- Excessive filter delay can also cause oscillations, double check that you cannot improve your gyro filter settings



PID Derivative Filtering

- Derivative filtering is essential on pitch and roll to prevent the D-Term amplifying any high frequency noise
- **IMU_DGYRO_CUTOFF = IMU_GYRO_CUTOFF/2** is a reasonable starting point
- PX4 does not provide plots that clearly show D term noise so it is best to look at the motor outputs
- Ideally these should have a signal to noise ratio (SNR) of ~10 or better
- Raise the **IMU_DGYRO_CUTOFF** in steps of 10-20%, continue until motor output SNR ~10 for smaller quads and ~15-20 for larger quads (larger motors and ESCs are more sensitive to noise)

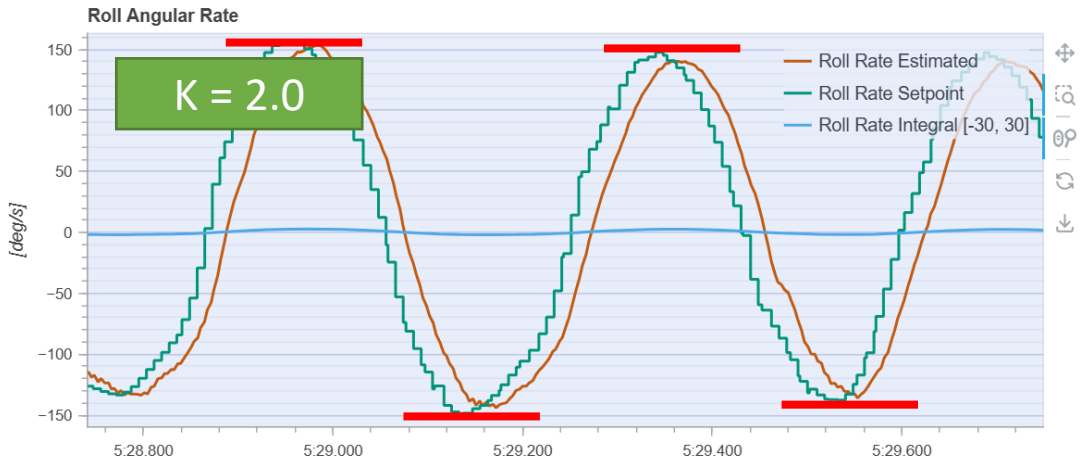


PID Tuning Tips

- Asymmetric drones may have different D and K values on Roll and Pitch
- It is best to tune P:D balance with the drone in it's heaviest configuration and K with the drone in it's lightest configuration
- Just because you've seen/heard micro oscillations on one axis doesn't mean that you can't continue to increase K on the other axes (they are independent)
- When you find a K value that starts to cause oscillations reduce K to provide some safety margin
- Usually 20%-30% is sufficient but it depends on your comfort level and the application (how much variation is there between drones of the same model and how much does the payload vary?)
- **Vehicle Setup > Parameters** allows you to increase K to 5.0, the PID tuning page maxes out at 3.0 😉 (this is especially useful for smaller quads)

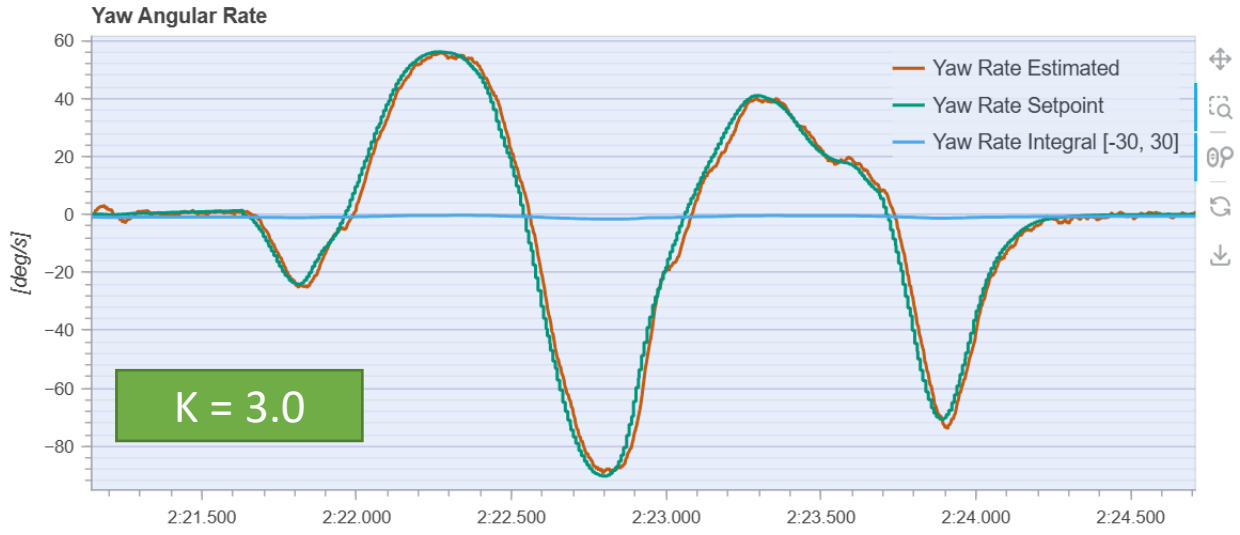
Double Check P:D Balance

- As you are tuning K keep checking that the P:D balance is still correct as sometimes **increasing K can make a P:D imbalance more visible**
- You can slightly adjust D if necessary during the K tune but *you shouldn't usually need to make big changes*
- Be aware that increasing D may precipitate D-oscillations at a lower K value as they multiply inside the controller ($K * D$)



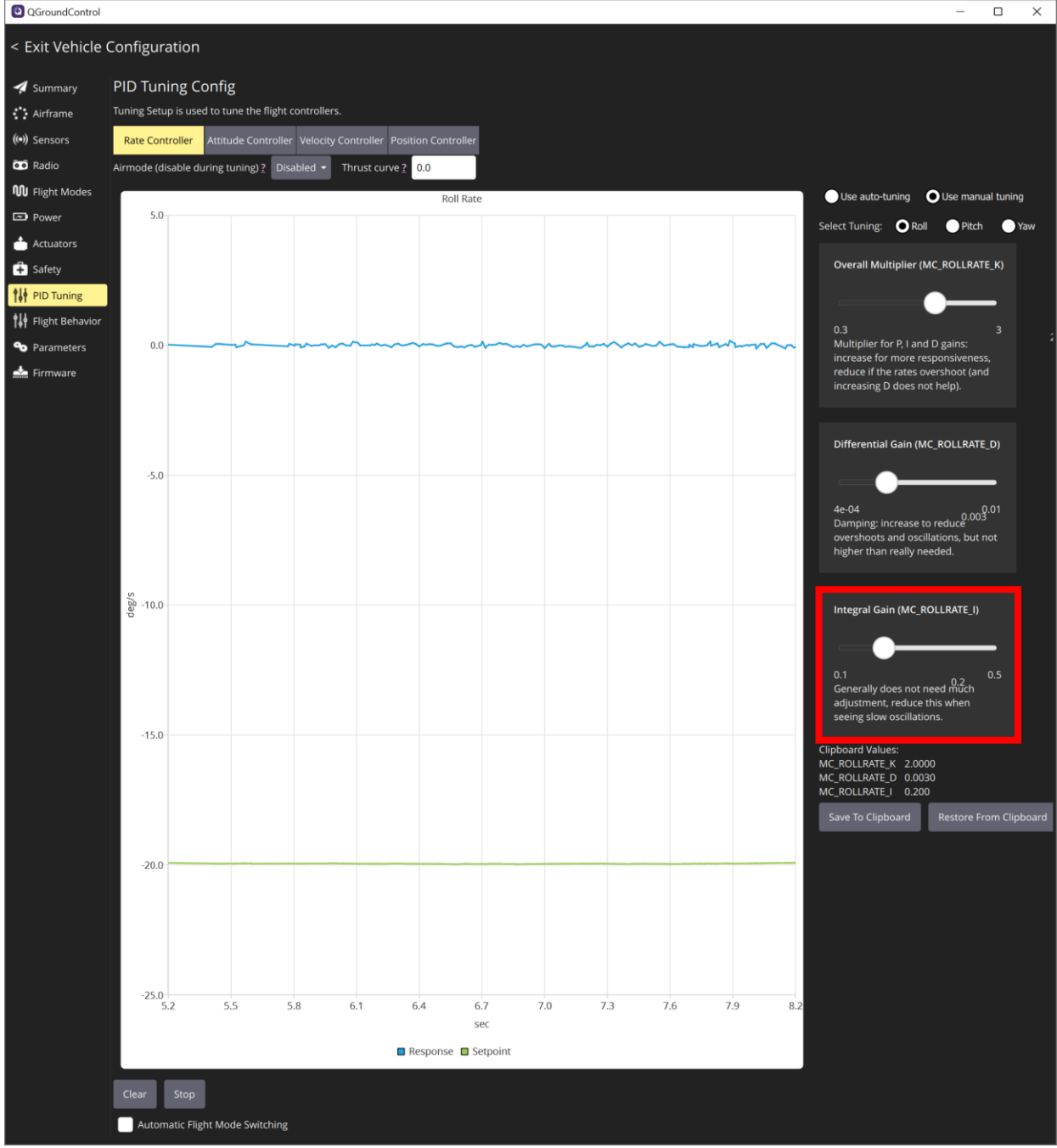
Yaw

- The Yaw axis doesn’t use a D term so you won’t need to tune P:D balance
- Yaw is K tuned just the same as Pitch and Roll
- Increase K watching for micro oscillations then when they become visible reduce K by 20-30% (or more) to provide a safety margin



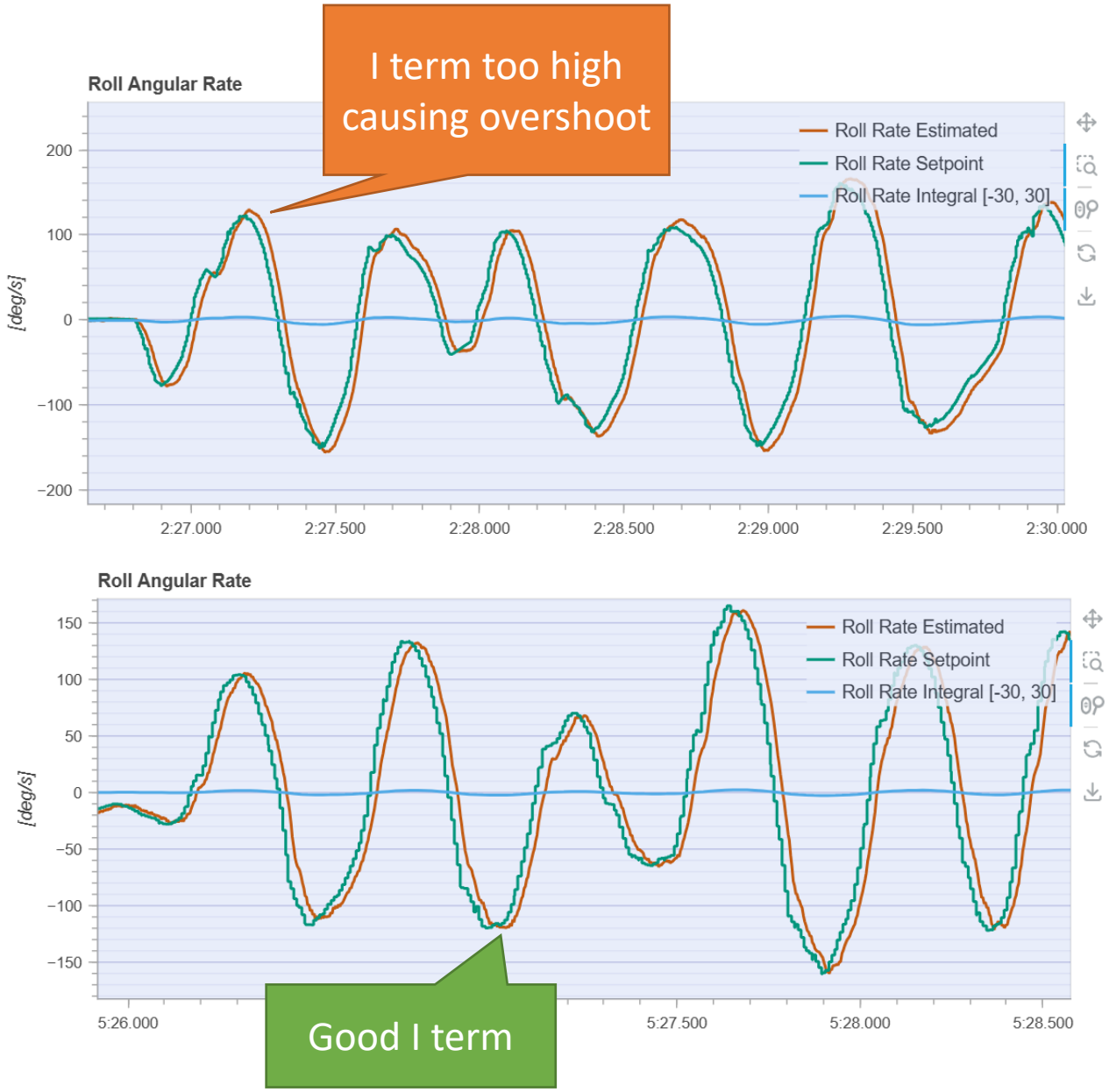
I term tuning

- The I term typically has a wide tuning window
- K Term tuning will naturally increase I gain in proportion to the P and D gains
- Unlike P and D gain more I gain does not make the I term stronger, instead it makes it respond *more quickly*
- For high performance tunes it may be important to have the I term respond more quickly to correct for wind etc.
- In this case we can increase **MC_ROLLRATE_I, MC_PITCHRATE_I,** and **MC_YAWRATE_I** to speed up the I term



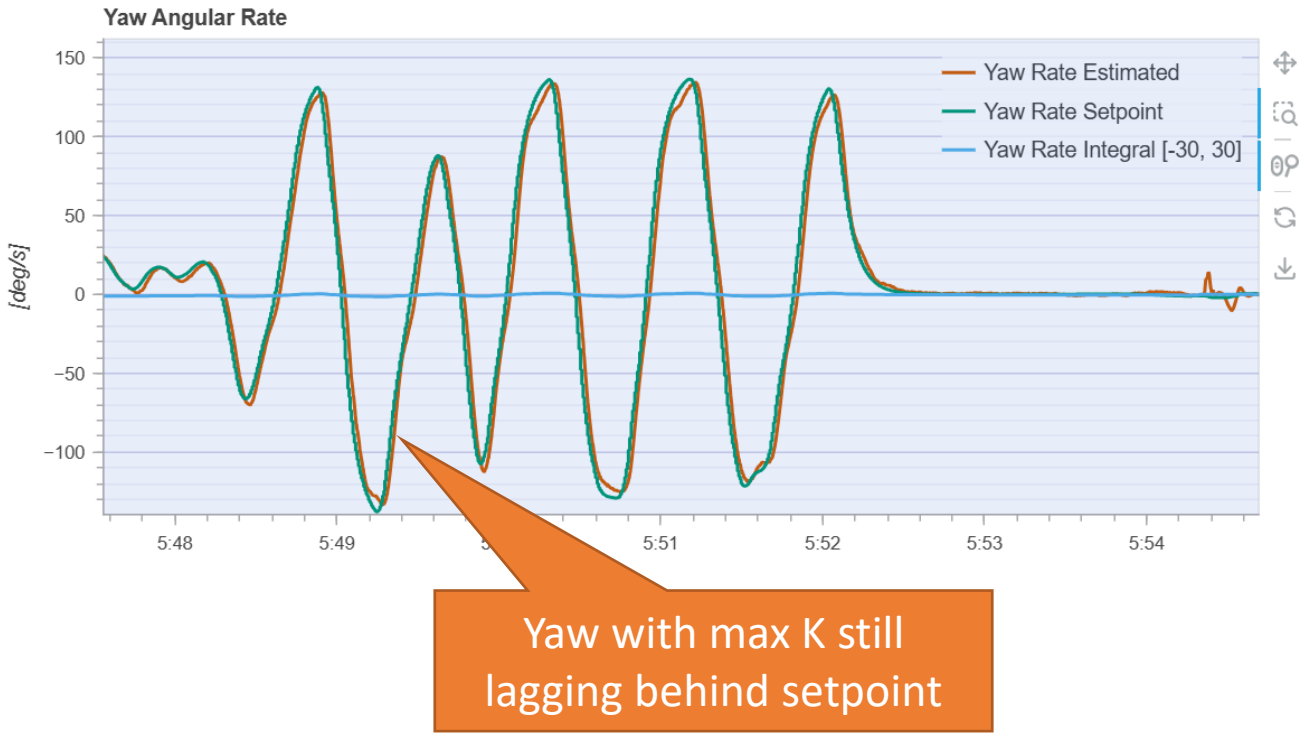
I Term too high

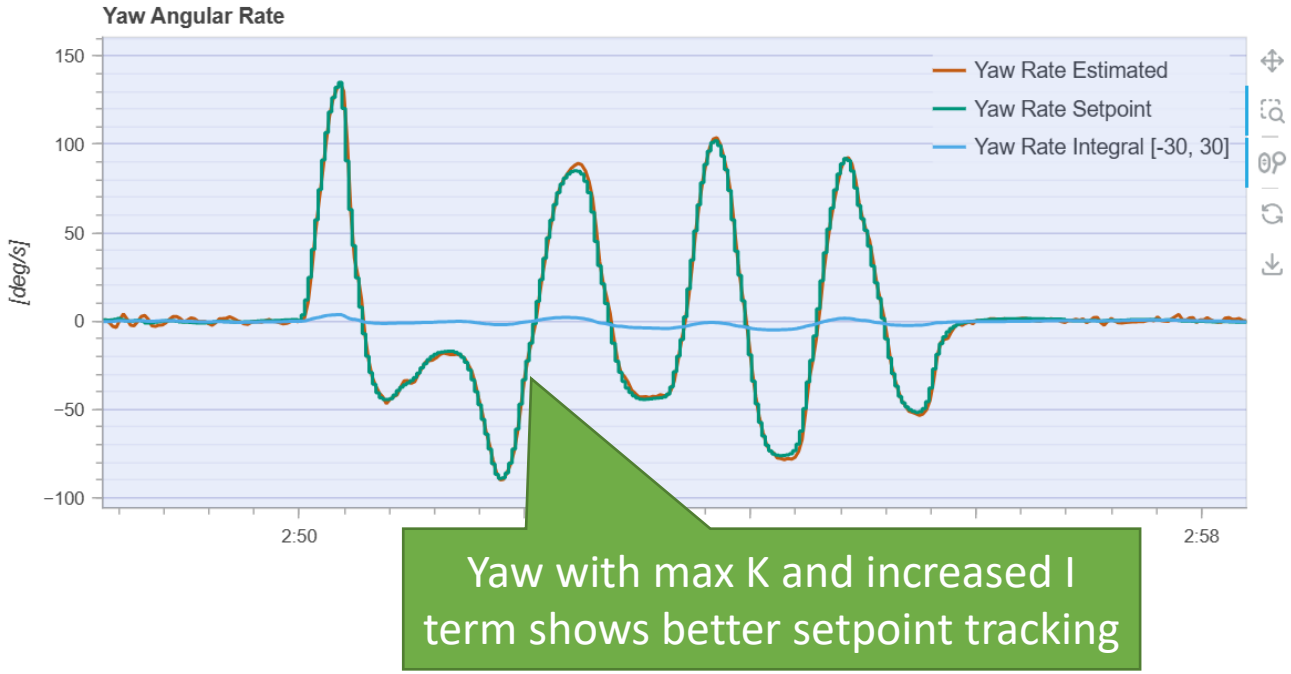
- I term that is too large will cause overshoot as I term is building up too much **during** a fast move
- This is undesirable and will reduce the accuracy of the angle controller
- I term should not usually need much if any adjustment from default as it will be increased appropriately during K tuning



Yaw I term tuning

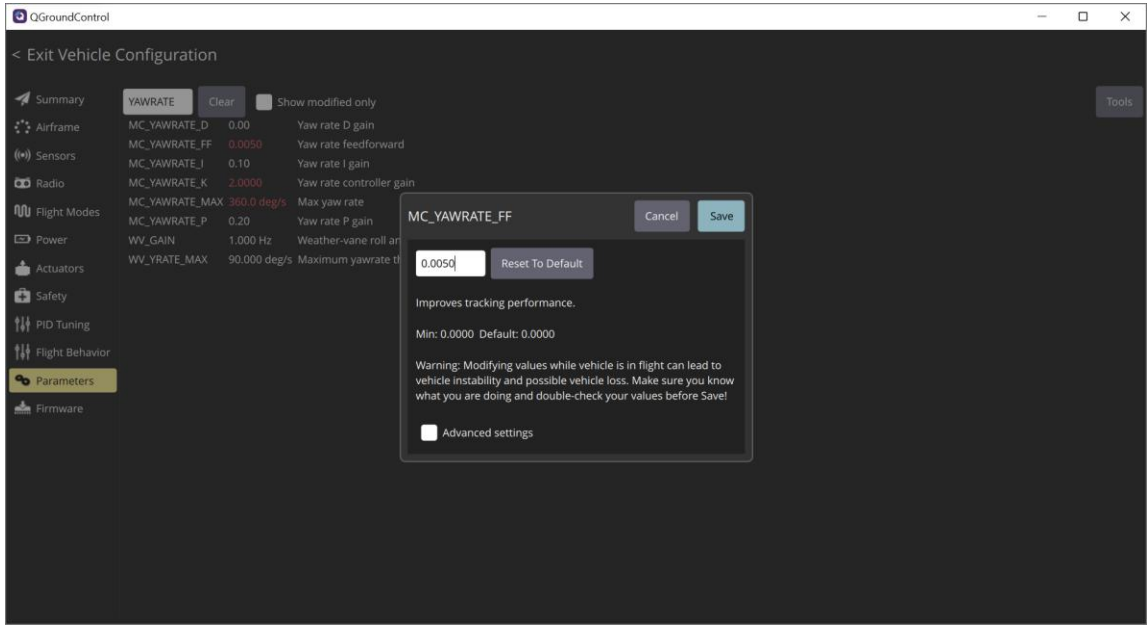
- Tuning I term on Yaw is slightly different from Pitch and Roll
- Increasing I term on Yaw helps ensure the estimated Yaw rate reaches the setpoint
- It may be necessary to increase I term on Yaw to improve setpoint tracking
- Unlike on roll and pitch we sometimes need I to build up during sharp moves on Yaw to help track setpoint





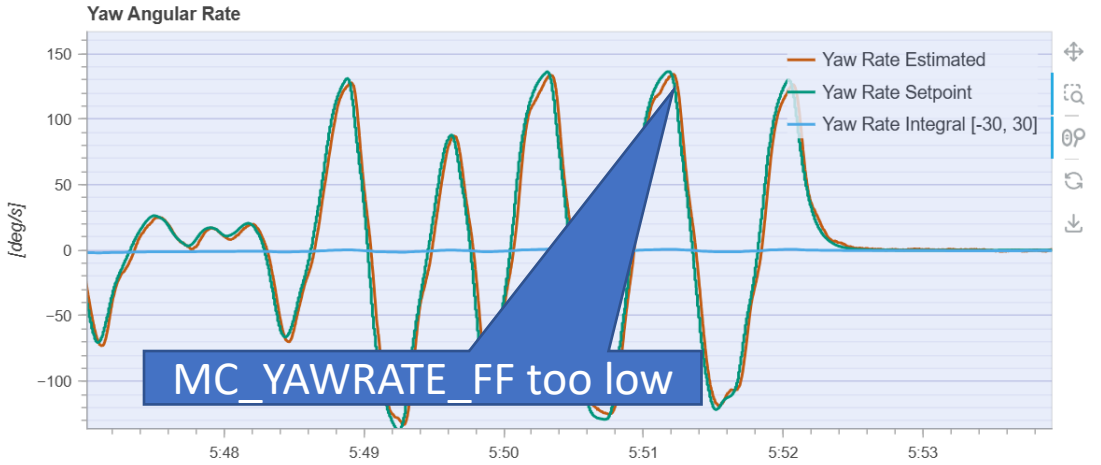
Yaw Feedforward tuning

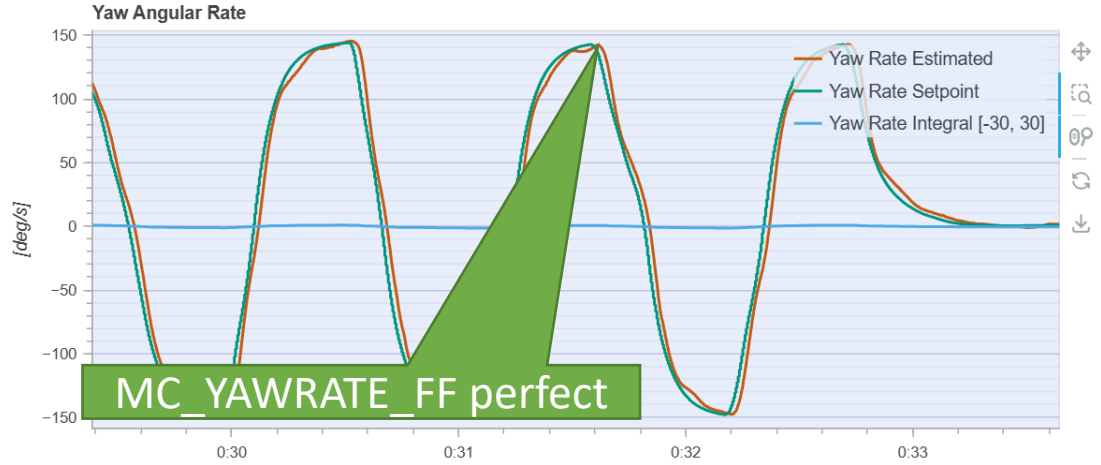
- Feedforward is a useful tool for tuning **the Yaw axis only**
- The Yaw axis is usually controlled through differential motor torque rather than thrust it can benefit from Feedforward
- I don’t recommend using feedforward on Pitch or Roll for Multicopter drones (it is mainly used for control surfaces on fixed wing)

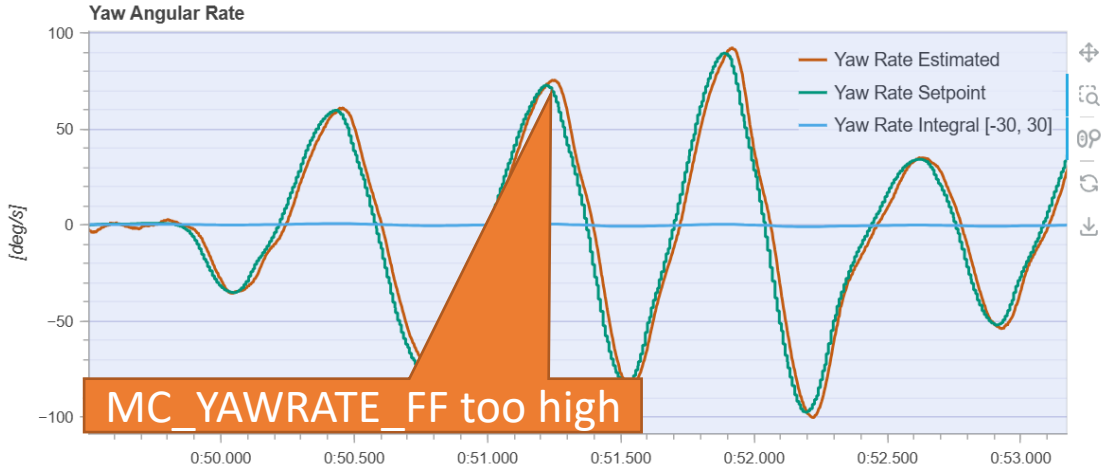


Yaw Feedforward tuning

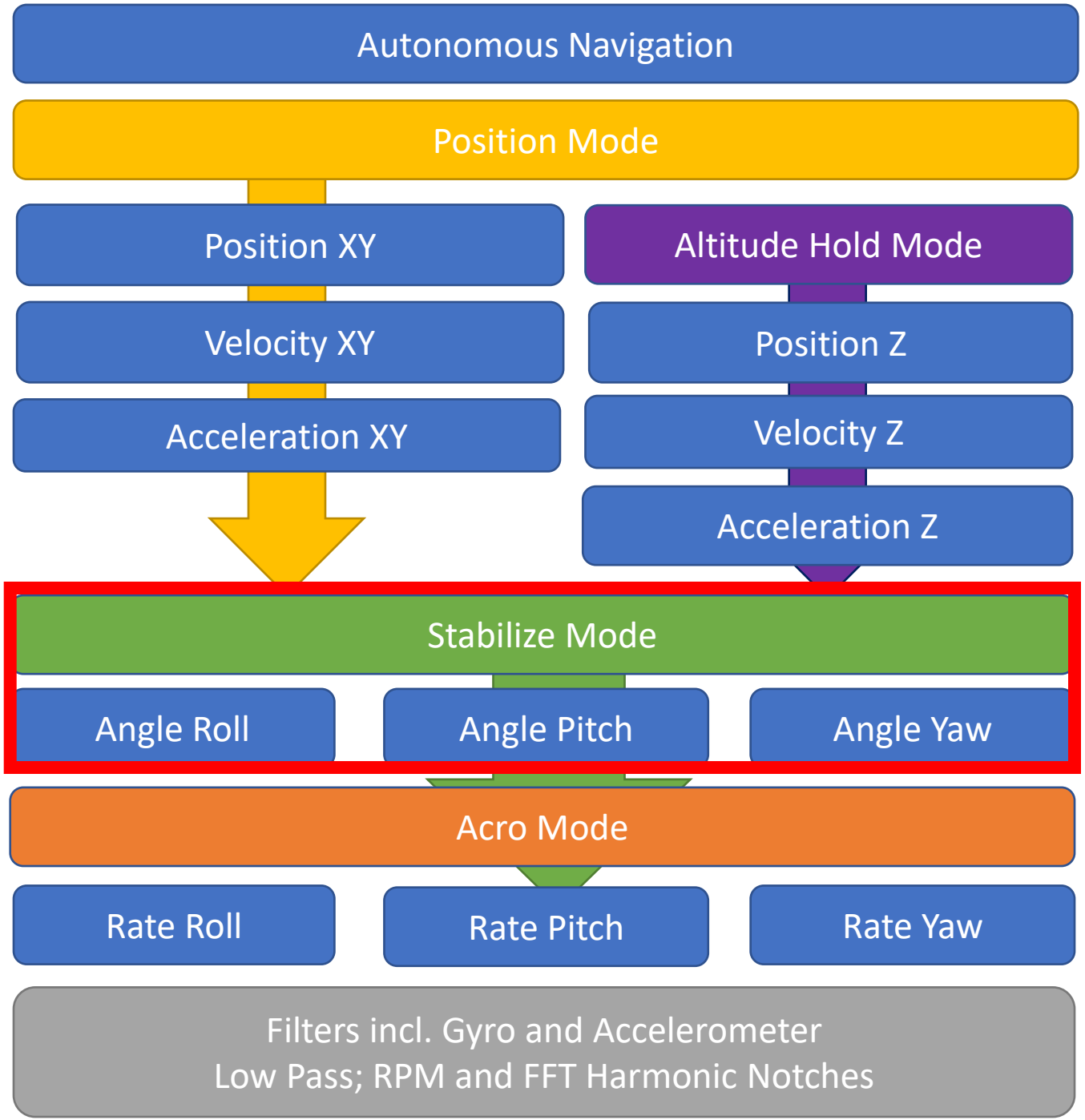
- Start with **MC_YAWRATE_FF = 0.005** which is usually close to the right value
- Increase **MC_YAWRATE_FF** until the estimated yaw rate peak is equal to the Yaw rate setpoint
- If you see overshoot decrease **MC_YAWRATE_FF**
- I usually suggest adjusting in steps of 20% of the current value (multiply by x1.2 or x0.8)







Angle Controller Tuning

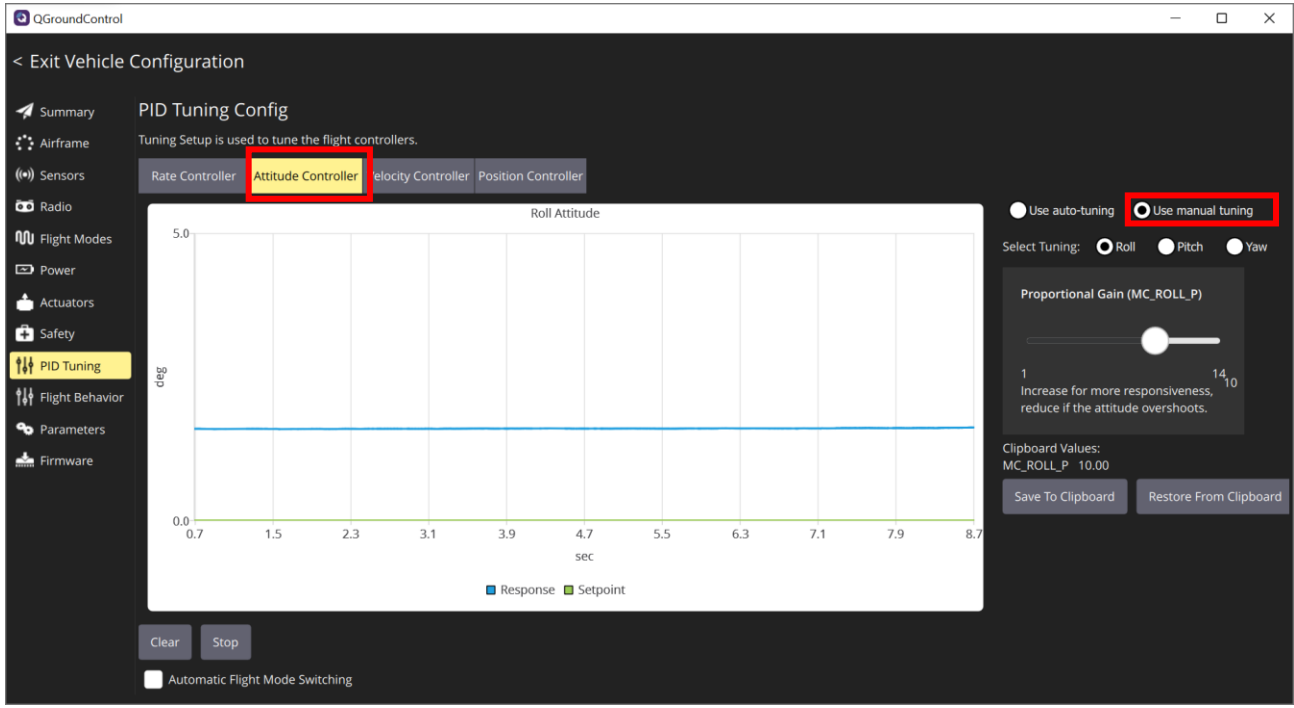

Autonomous Navigation
Position Mode
Position XY
Velocity XY
Acceleration XY
Altitude Hold Mode
Position Z
Velocity Z
Acceleration Z
Stabilize Mode
Angle Roll
Angle Pitch
Angle Yaw
Acro Mode
Rate Roll
Rate Pitch
Rate Yaw
Filters incl. Gyro and Accelerometer
Low Pass; RPM and FFT Harmonic Notches

Intro to angle controller tuning

- Once the Rate controller is dialled in it is time to tune the angle controller
- This controller converts a target angle into a target rate of rotation and is a simple proportional controller
- The goal with tuning is to find the proportional gain value that ensures the Estimated Roll peak height matches the Target roll peak height (usually with a small delay)

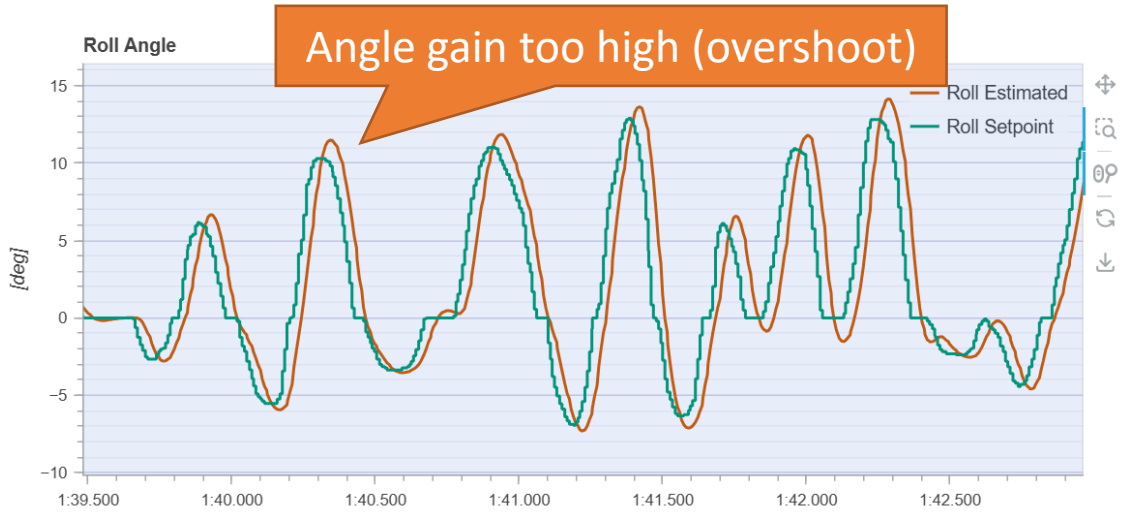
Adjusting Angle Controller Gain

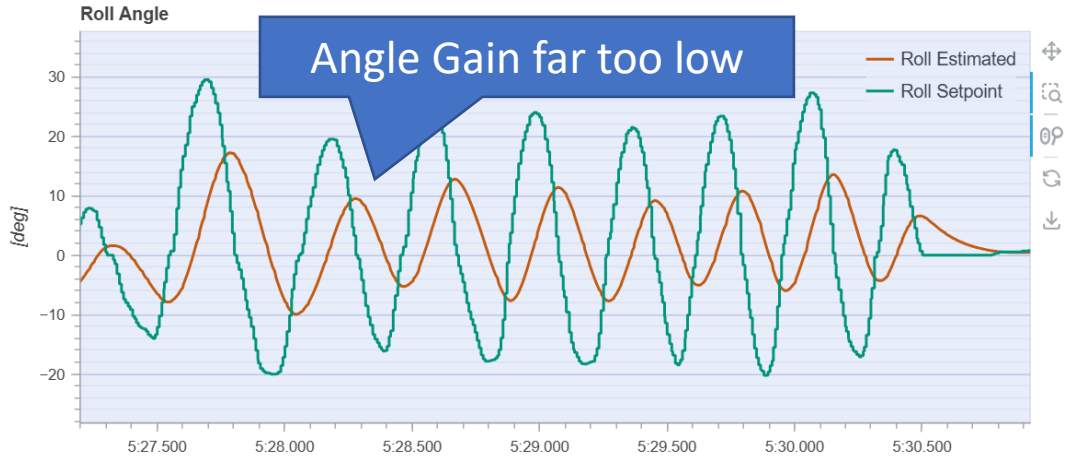
- Angle controller gain can be adjusted in:
  Vehicle Configuration > Parameters > PID Tuning > Attitude Controller > Use Manual Tuning > Roll/Pitch/Yaw

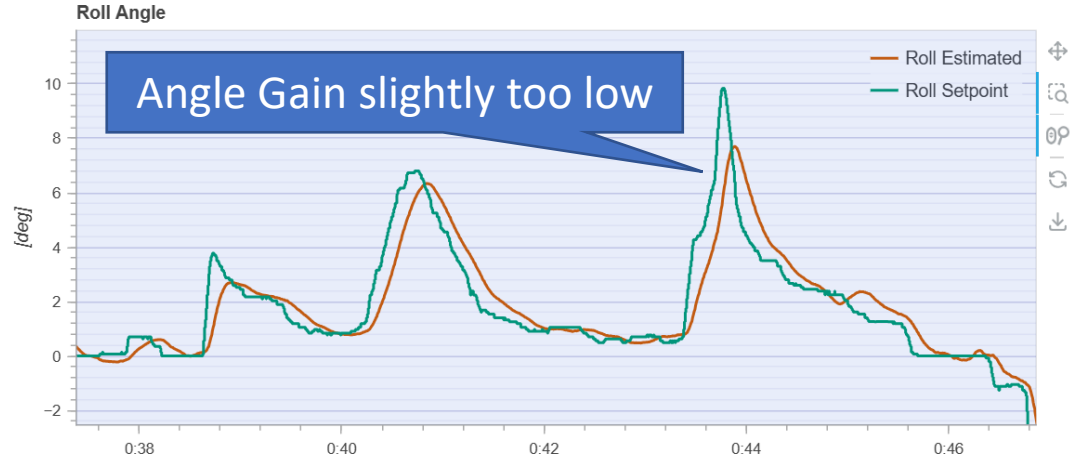


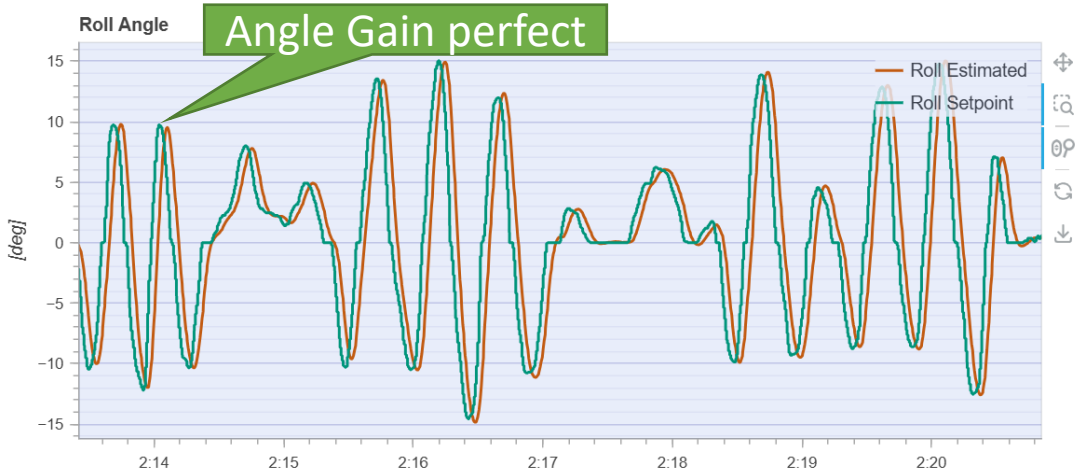
Adjusting Angle Gain

- The data for tuning the Angle Controller is visible in PX4 PID Review:
  - Roll Angle
  - Pitch Angle
  - Yaw Angle

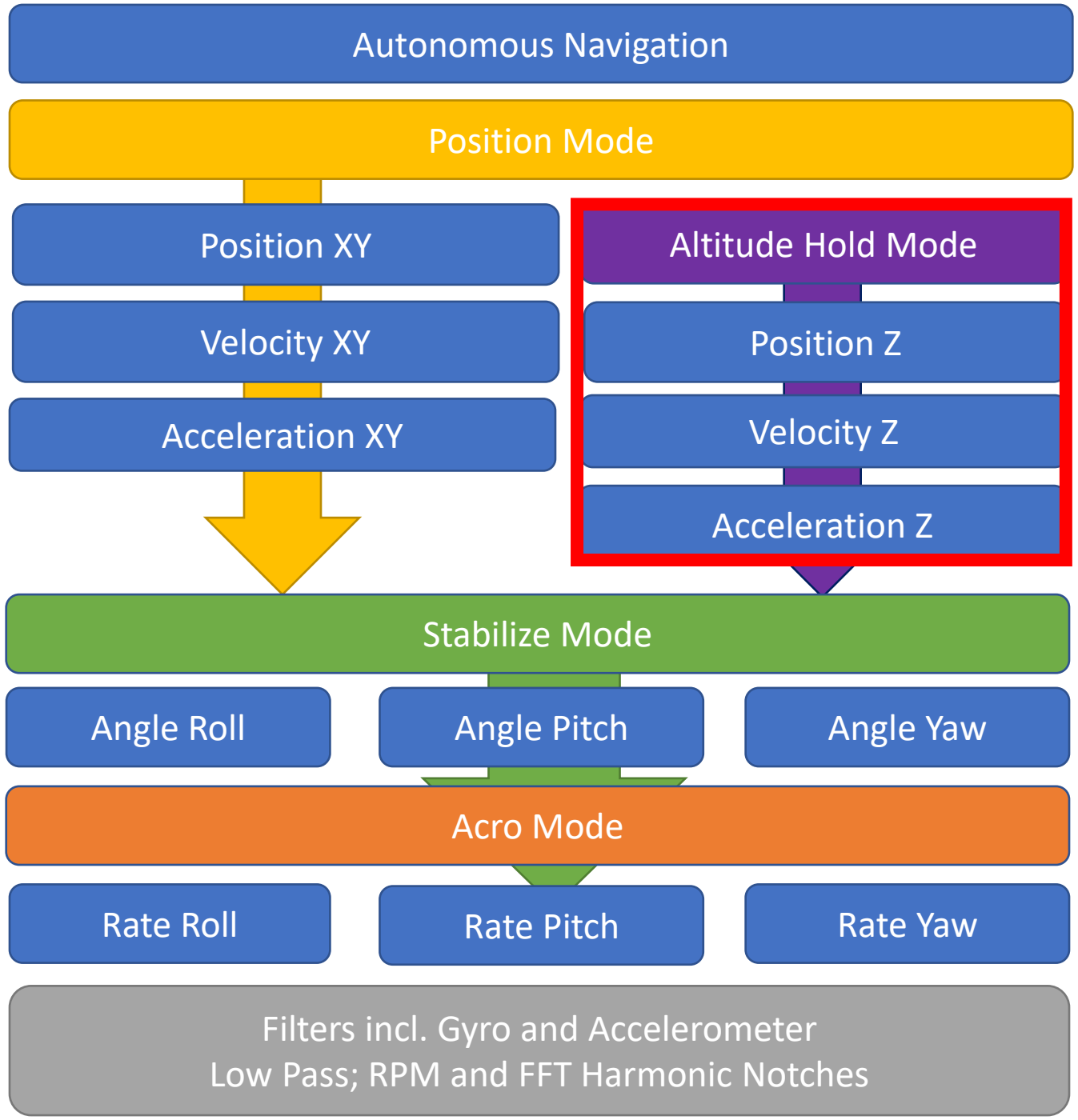








Tuning Vertical Velocity

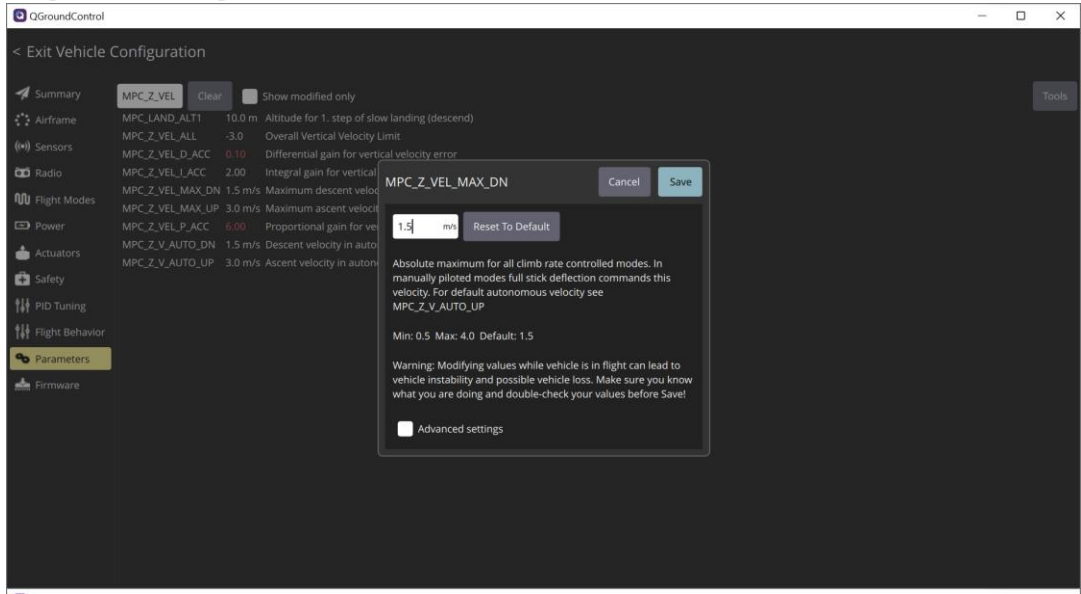

Autonomous Navigation
Position Mode
Position XY
Velocity XY
Acceleration XY
Altitude Hold Mode
Position Z
Velocity Z
Acceleration Z
Stabilize Mode
Angle Roll
Angle Pitch
Angle Yaw
Acro Mode
Rate Roll
Rate Pitch
Rate Yaw
Filters incl. Gyro and Accelerometer
Low Pass; RPM and FFT Harmonic Notches

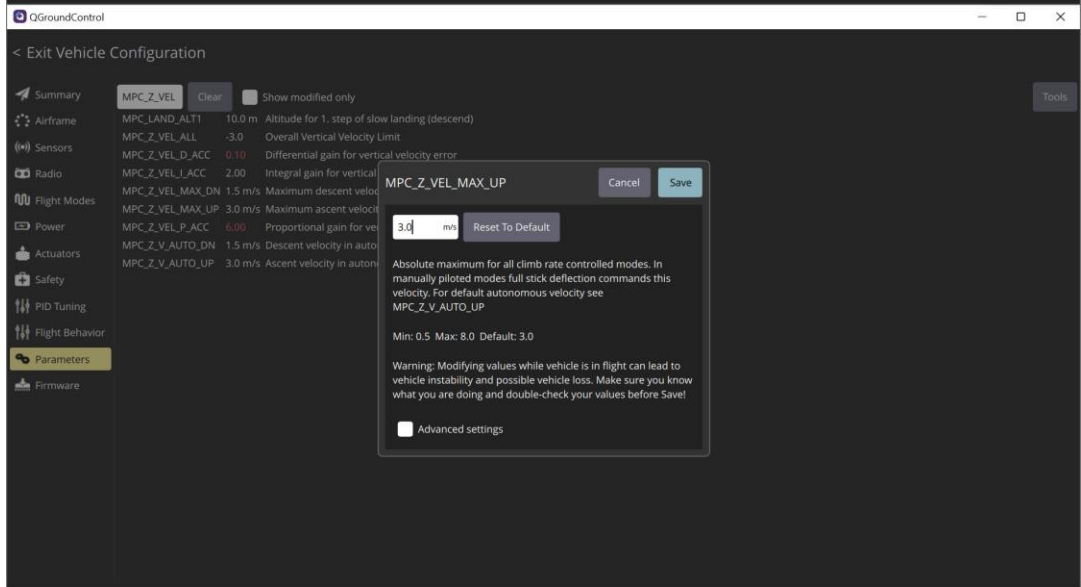
Intro to vertical velocity controller tuning

- Once the Angle controller is dialled in it is time to tune the vertical velocity controller
- This controller converts a target vertical velocity into a target throttle setting
- The goal with tuning is to find the correct PID values to minimise the error between target and actual vertical velocity

Setting up for Velocity tuning flights

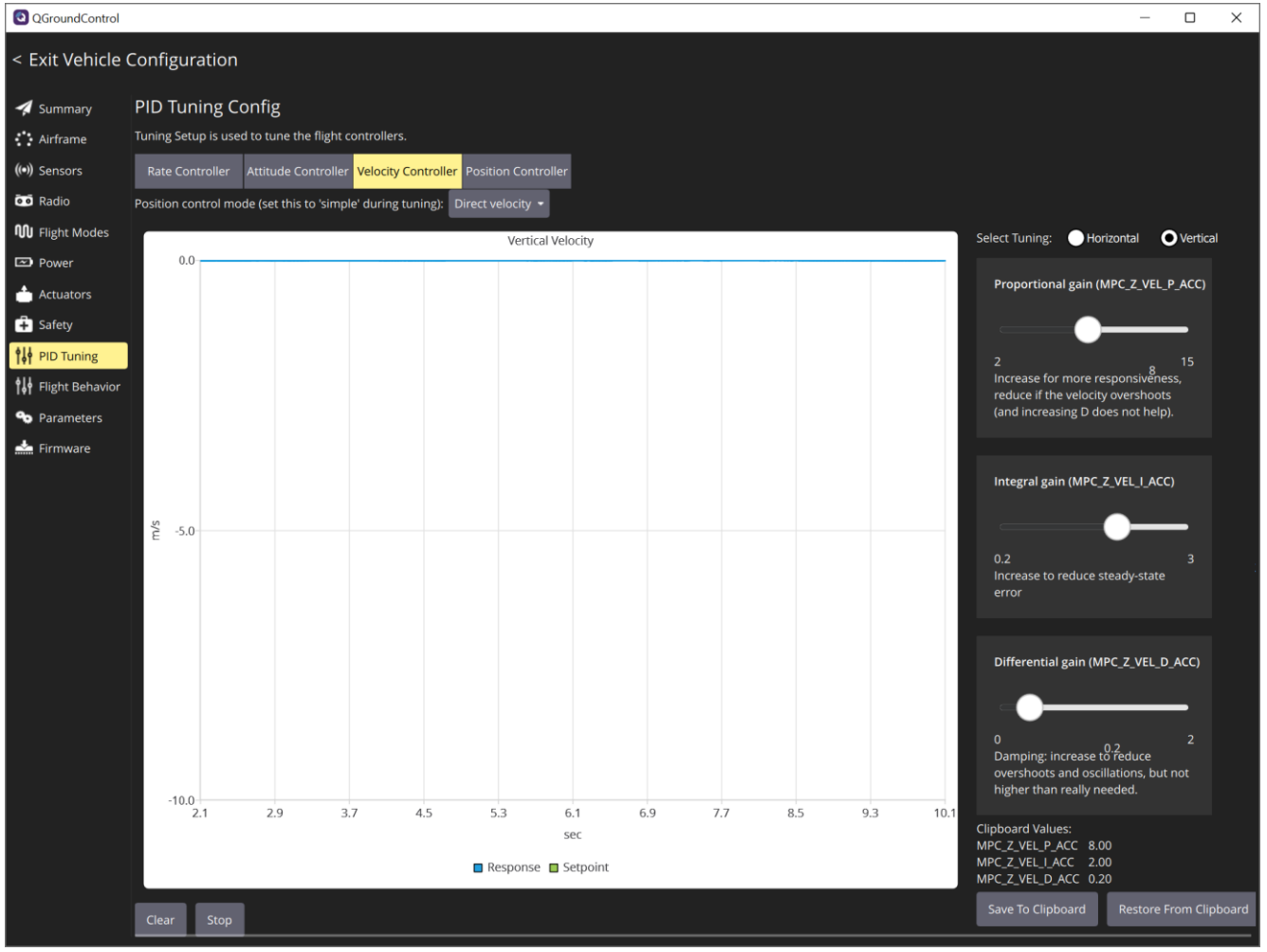
- **Vehicle Configuration > Parameters**
- **Search MPC_Z_VEL**
- Set **MPC_Z_VEL_MAX_DN = 1.5 m/s**
- Set **MPC_ Z_VEL_MAX_UP = 3 m/s**
- **Save**





Adjusting Vert. Vel. Controller Parameters

- You can find the velocity controller PID parameters in **Vehicle Configuration > PID Tuning > Velocity Controller > Vertical**
- We will start by adjusting **MPC_Z_VEL_P_ACC** to find the right P:D balance
- Set position control mode to **Direct Velocity** for tuning

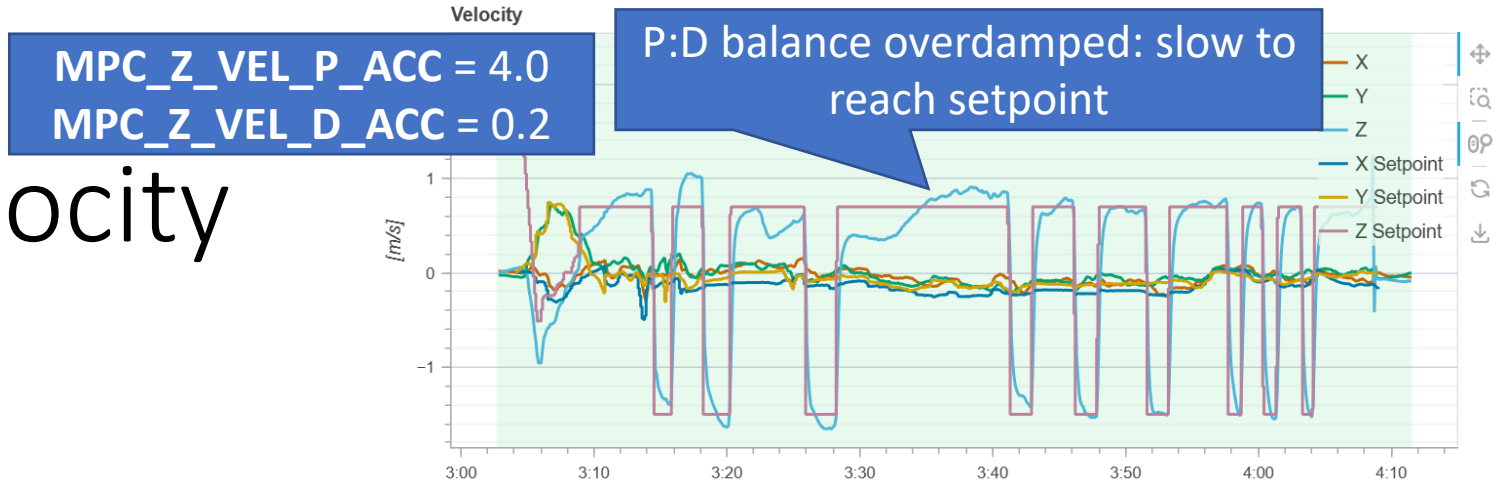


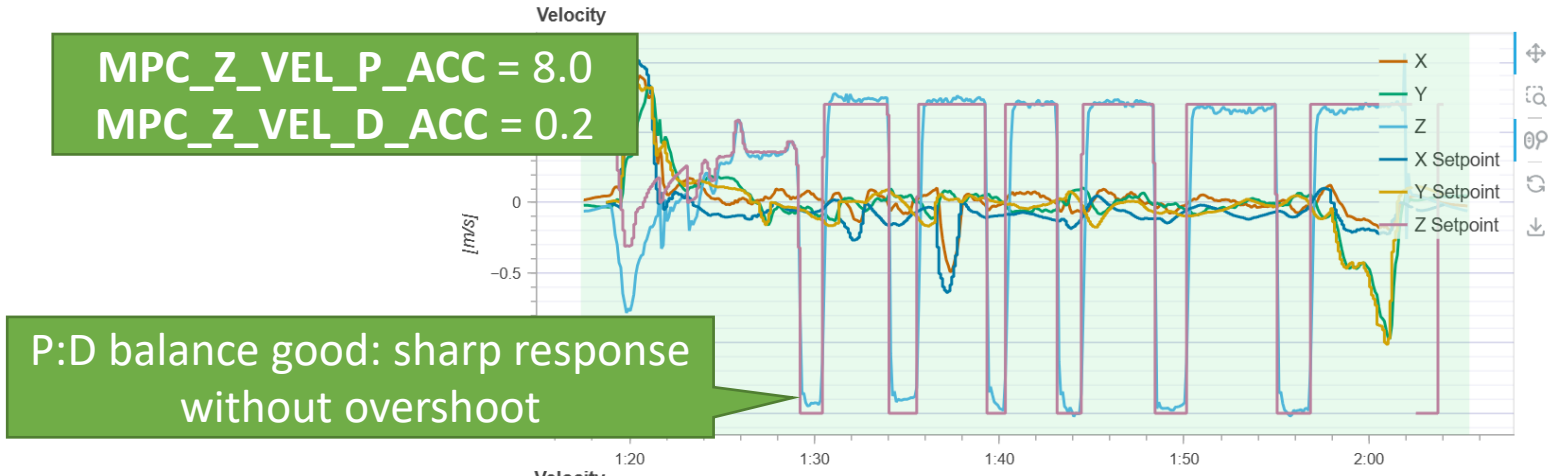
Tuning flights

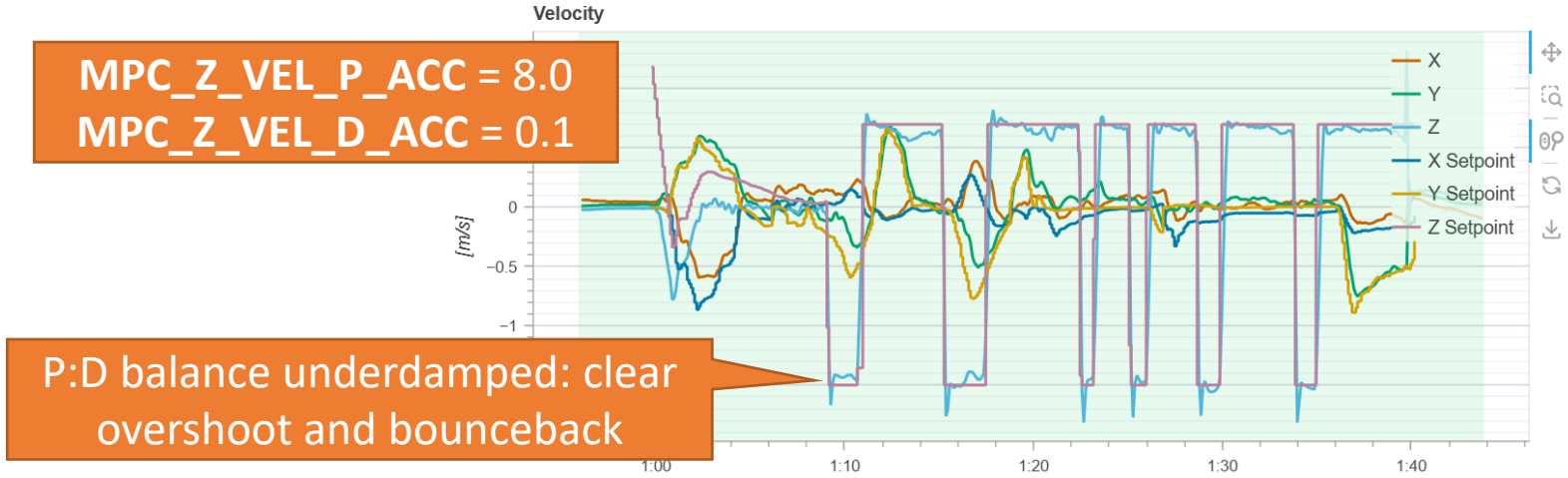
- Tuning flights for the vertical velocity controller are done in Altitude mode
- They consist of **sharp** full throttle ascents and descents
- In velocity controller tuning the quad needs to accelerate to full speed in one direction then decelerate and accelerate to full speed in the opposite direction
- Make sure to allow the quad time and space to reach its maximum climb and descent rates for at least 1 second
- In **Direct Velocity** mode your stick position corresponds to the velocity setpoint so you can give close to step inputs to the controller

Tuning Vertical Velocity

- We follow the same process as tuning the Horizontal Velocity controller
- Increase P in steps of 20% until there is visible overshoot then back off to largest value that did not show overshoot

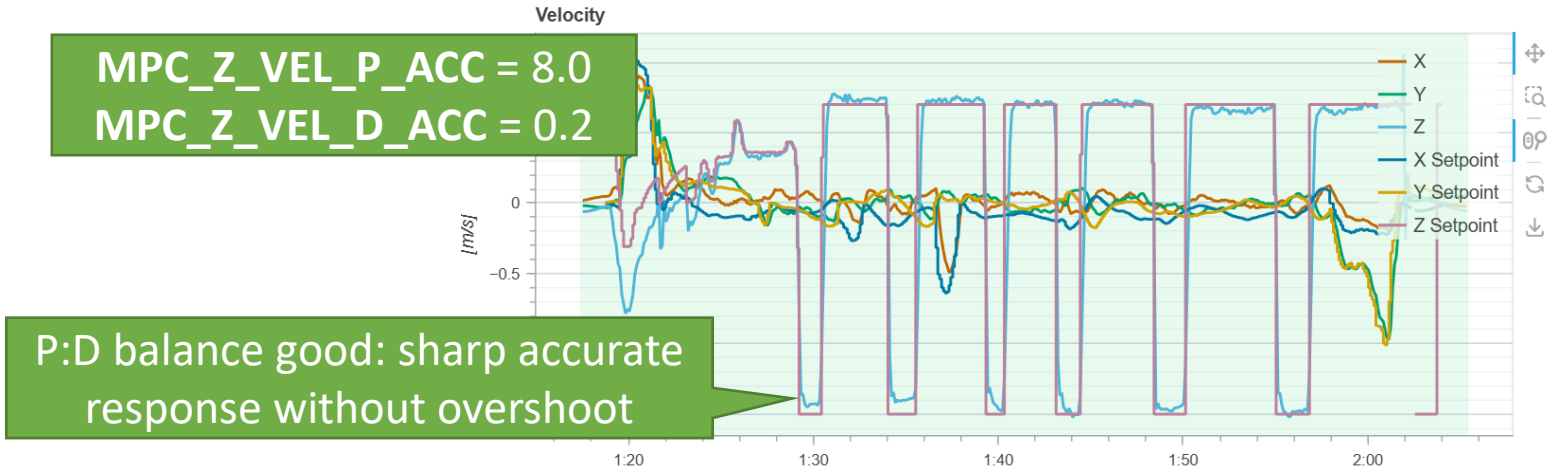


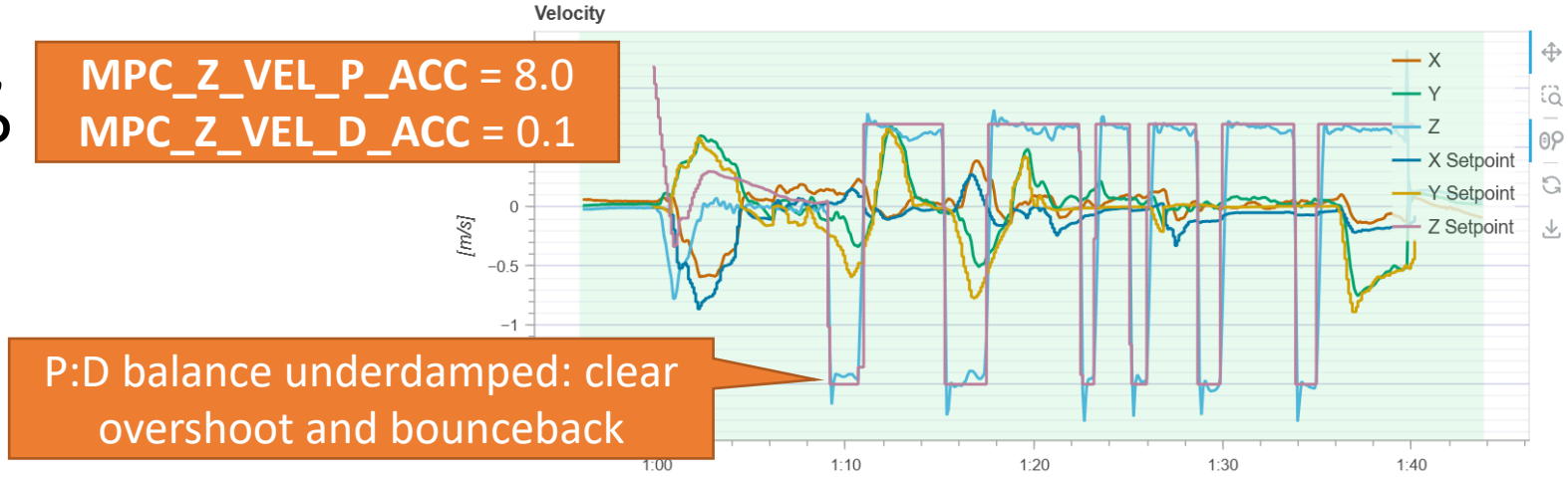




Tuning Vertical Velocity

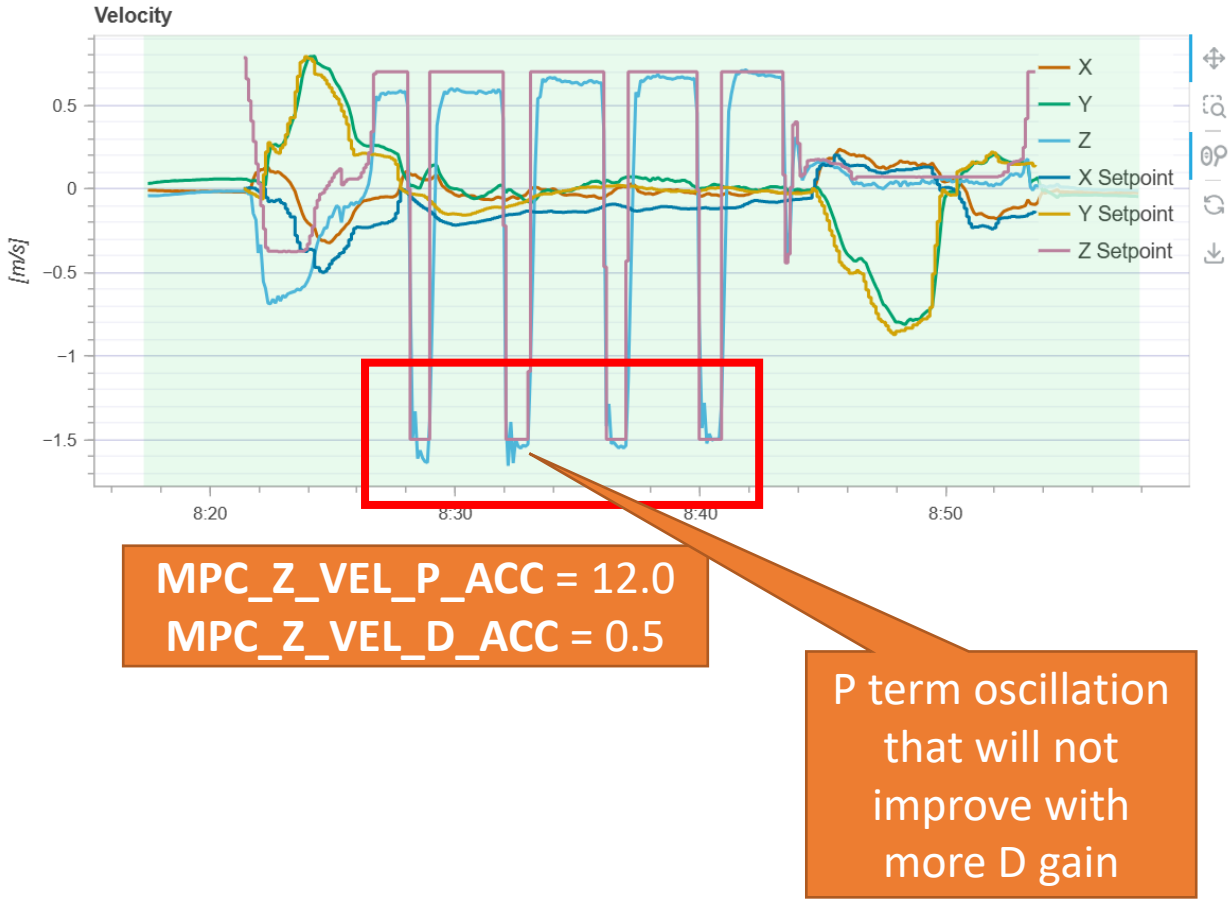
- Once correct PD balance is found continue to increase P, I and D together until the response is sharp and accurate
- You may find as you do this that the response becomes underdamped, in this case just reduce P gain slightly then continue



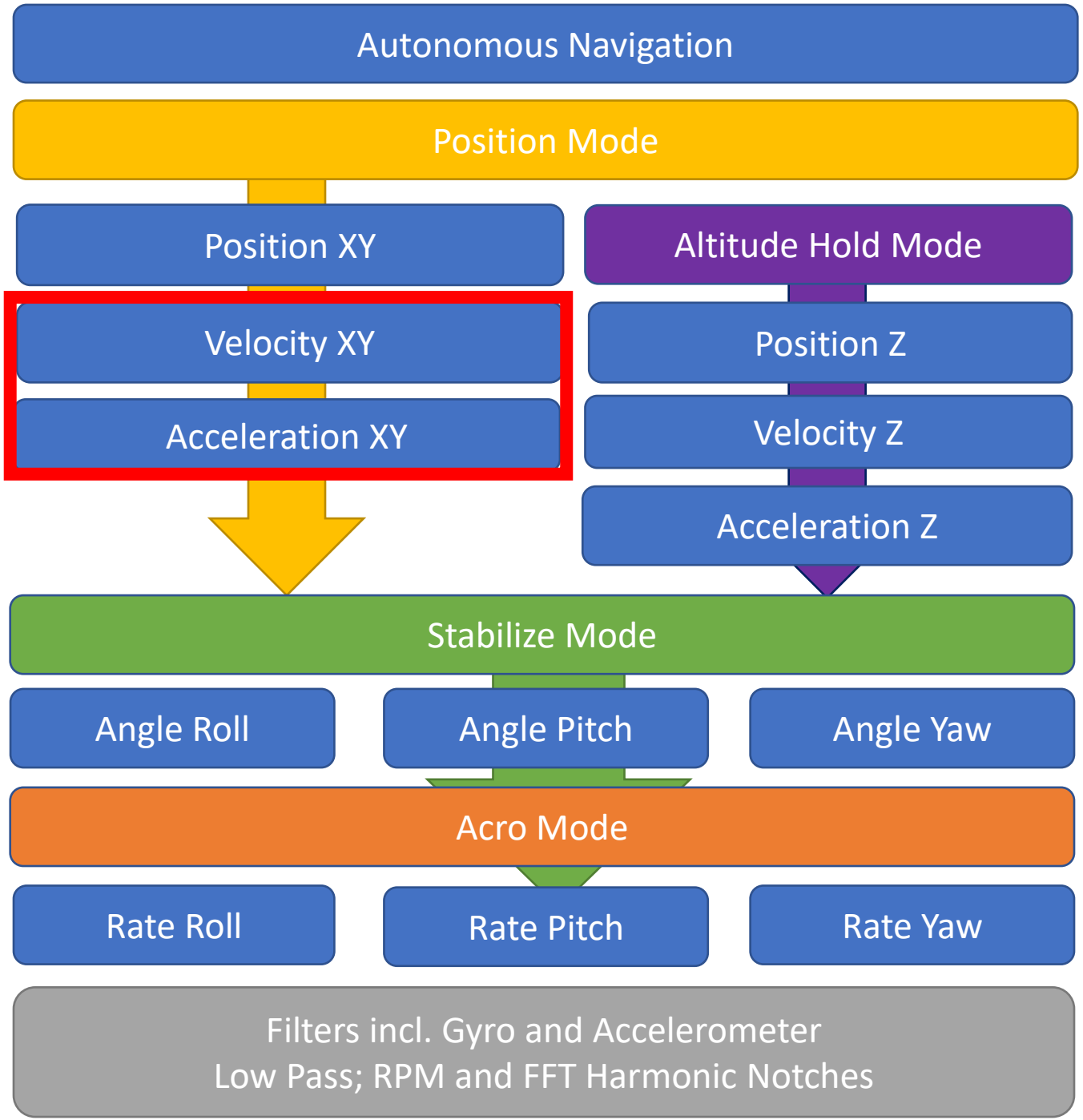


Vert. Velocity controller P oscillation

- The vertical velocity controller exhibits the same rapid oscillations as the horizontal velocity controller above a critical P gain
- Due to the slow loop rate of the velocity controller (50Hz) rigid body oscillations can occur at or above the nyquist limit (25Hz) and therefore cannot be addressed with D gain
- This is not a problem for tuning but you may find that you hit an oscillation limit with P gain regardless of D gain
- If this happens reduce P gain by at least 30%
- You can then reduce D term (at max safe P gain) to get a less damped response
- In my experience this gives good flight performance
- **Increasing the loop rate of the velocity controller in PX4 might help to address this**



Horizontal velocity controller tuning

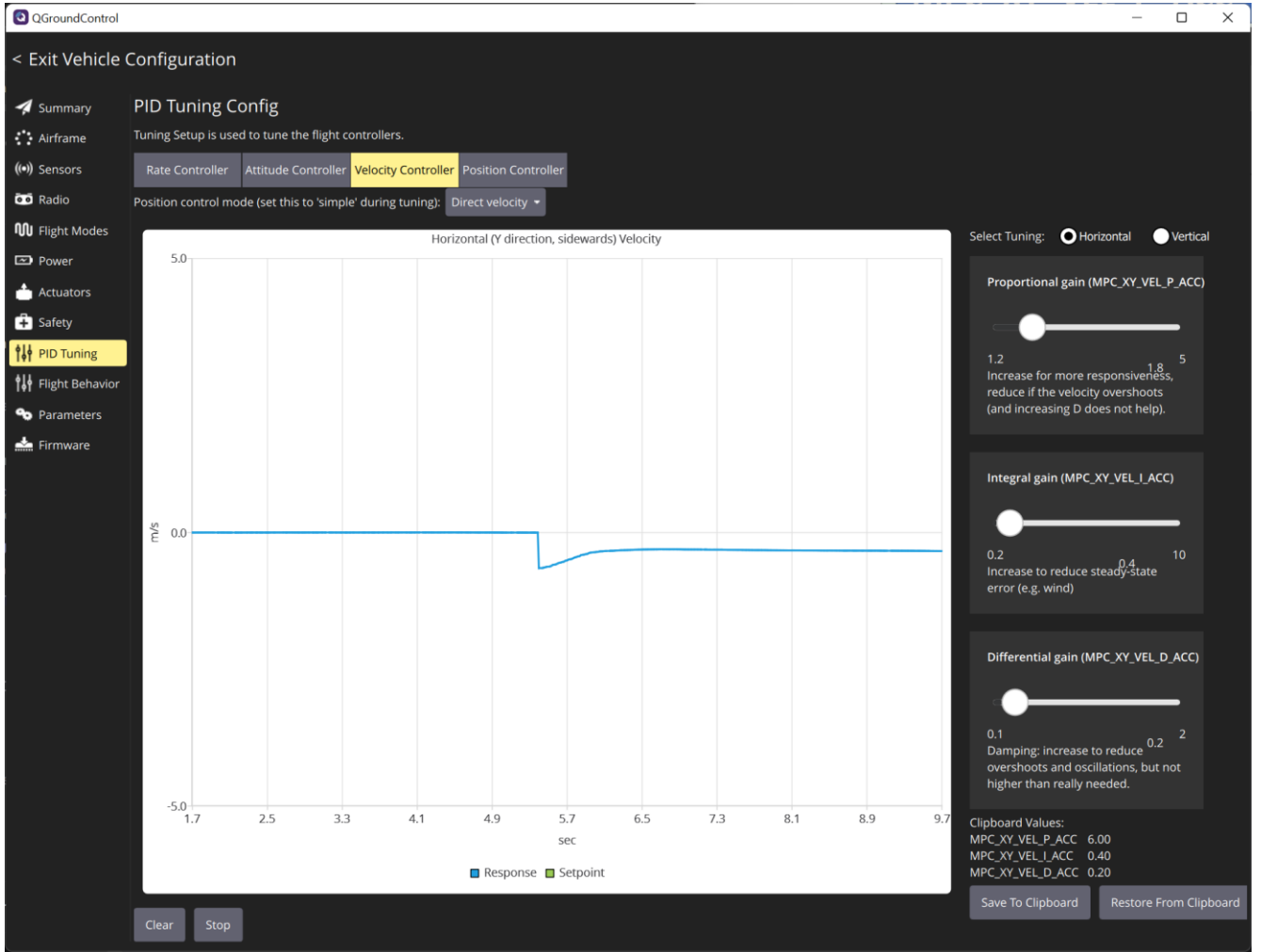

Autonomous Navigation
Position Mode
Position XY
Velocity XY
Acceleration XY
Altitude Hold Mode
Position Z
Velocity Z
Acceleration Z
Stabilize Mode
Angle Roll
Angle Pitch
Angle Yaw
Acro Mode
Rate Roll
Rate Pitch
Rate Yaw
Filters incl. Gyro and Accelerometer
Low Pass; RPM and FFT Harmonic Notches

Intro to horizontal velocity controller tuning

- Once the Altitude controller is dialled in it is time to tune the horizontal velocity controller
- This controller converts a target horizontal velocity into a target pitch and roll angle
- The goal with tuning is to find the correct PID values to minimise the error between target and actual horizontal velocity

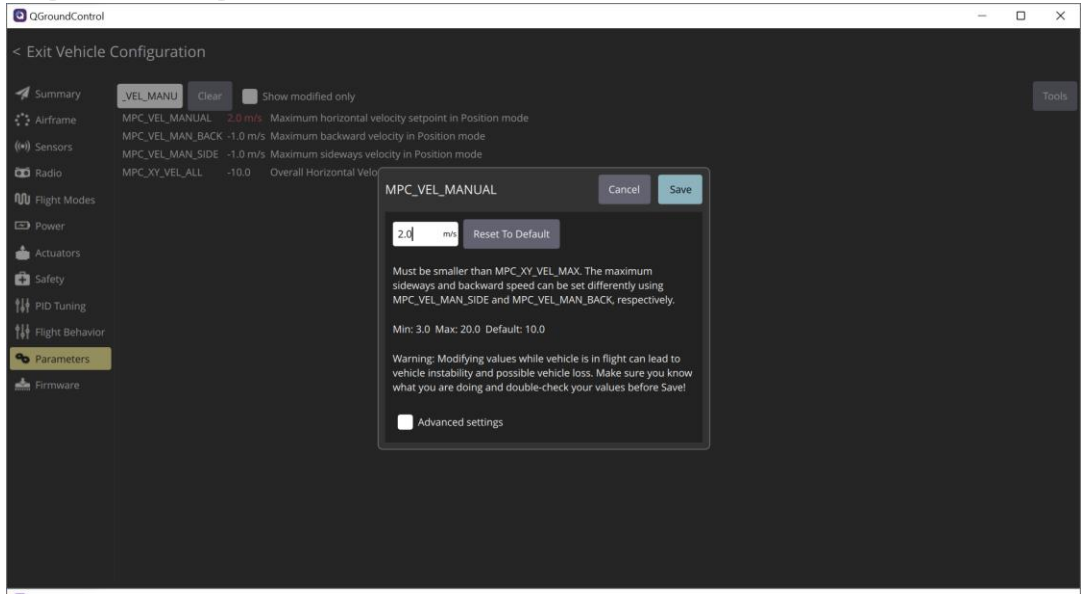
Adjusting Horiz. Vel. Controller Parameters

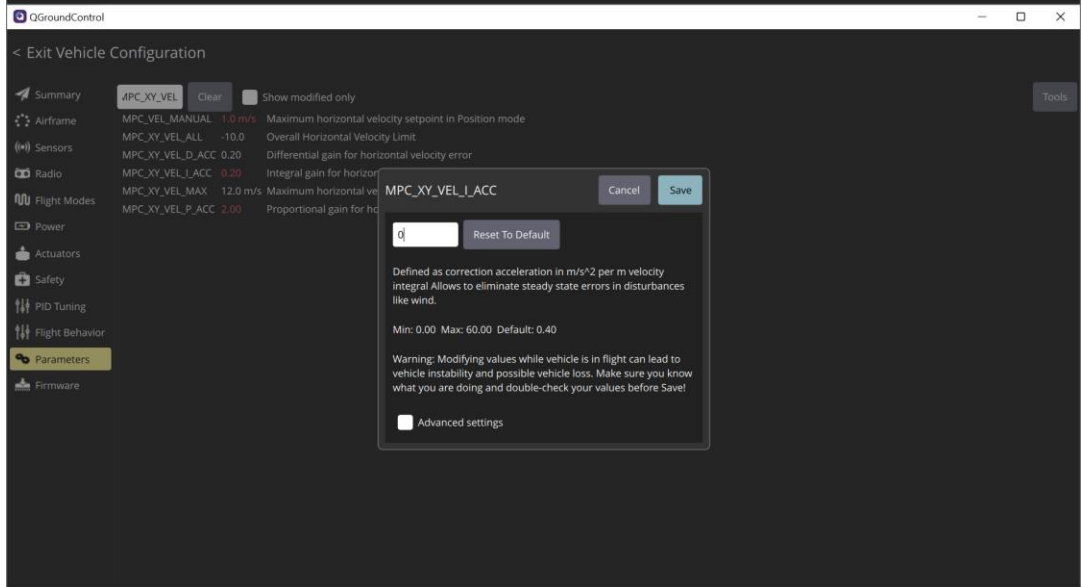
- You can find the velocity controller PID parameters in **Vehicle Configuration > PID Tuning > Velocity Controller > Horizontal**
- We will start by adjusting **MPC_XY_VEL_P_ACC** to find the right P:D balance
- Set position control mode to **Direct Velocity** for tuning



Setting up for Velocity tuning flights

- **Vehicle Configuration > Parameters**
- **Search MPC_VEL**
- Set **MPC_VEL_MANUAL** to 2.0 m/s
- **Force Save**
- **Search MPC_XY_VEL**
- Set **MPC_XY_VEL_I_ACC = 0.2**
- **Save**



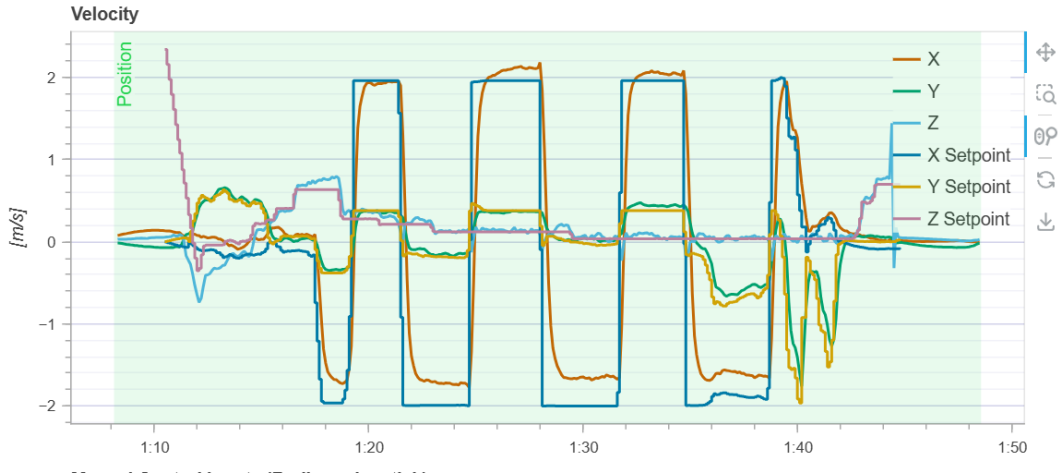


Tuning flights

- Tuning flights for the horizontal velocity controllers are done in Position mode
- They consist of **sharp, full stick** movements horizontally left/right and optionally forward/back
- These moves are slightly different to angle mode tuning
- In angle mode tuning the quad stays relatively still over a point on the ground
- In velocity controller tuning the quad needs to accelerate to full speed in one direction then decelerates and accelerates to full speed in the opposite direction
- In **Direct Velocity** mode your stick position corresponds to the velocity setpoint so you can give close to step inputs to the controller

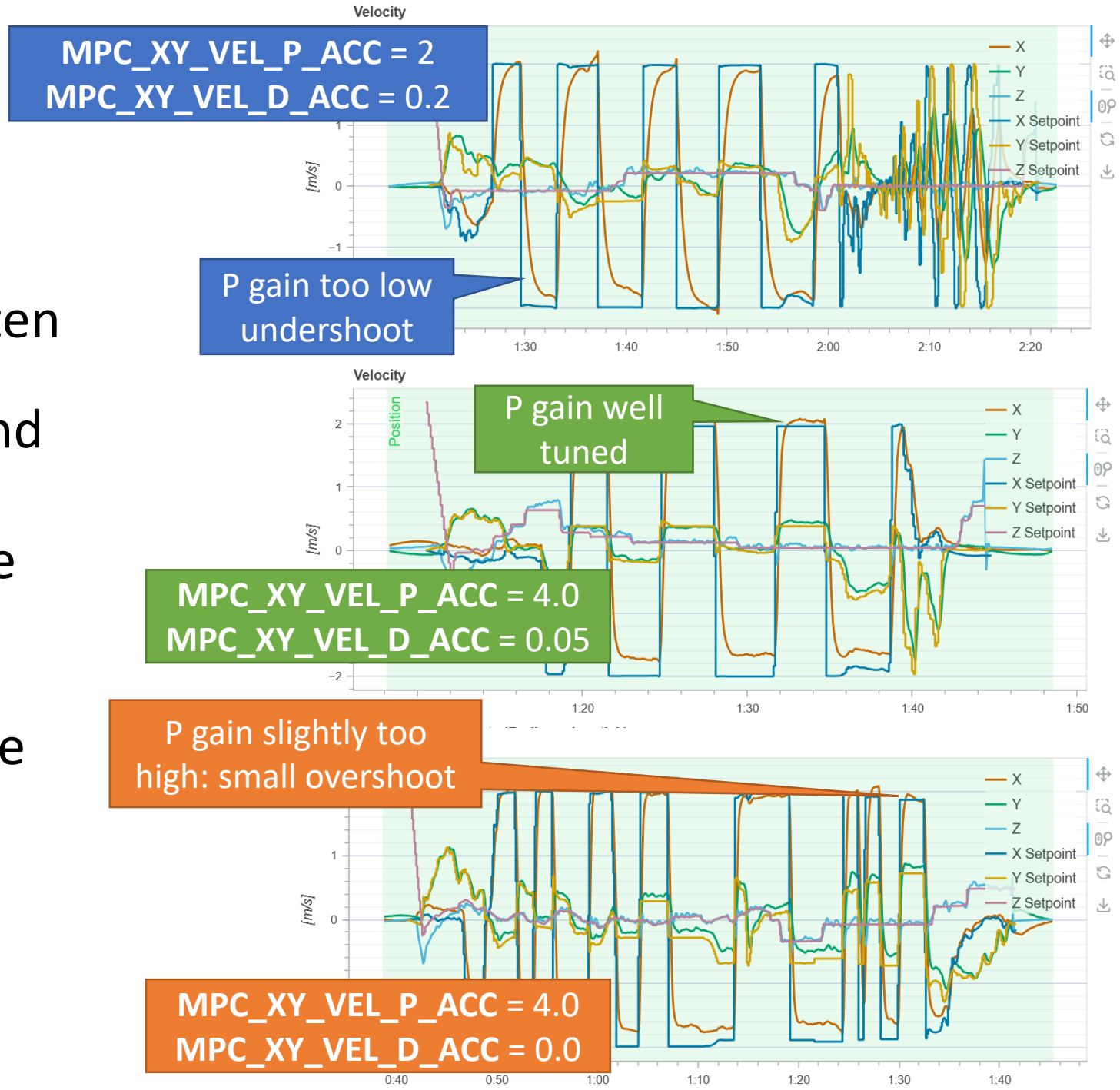
Tuning XY Velocity – PID Review

- To tune the velocity controller you will want to look at the **Velocity** chart in PID review
- If you have flown the tuning flight correctly you should see near step inputs to the velocity controller e.g. X Setpoint



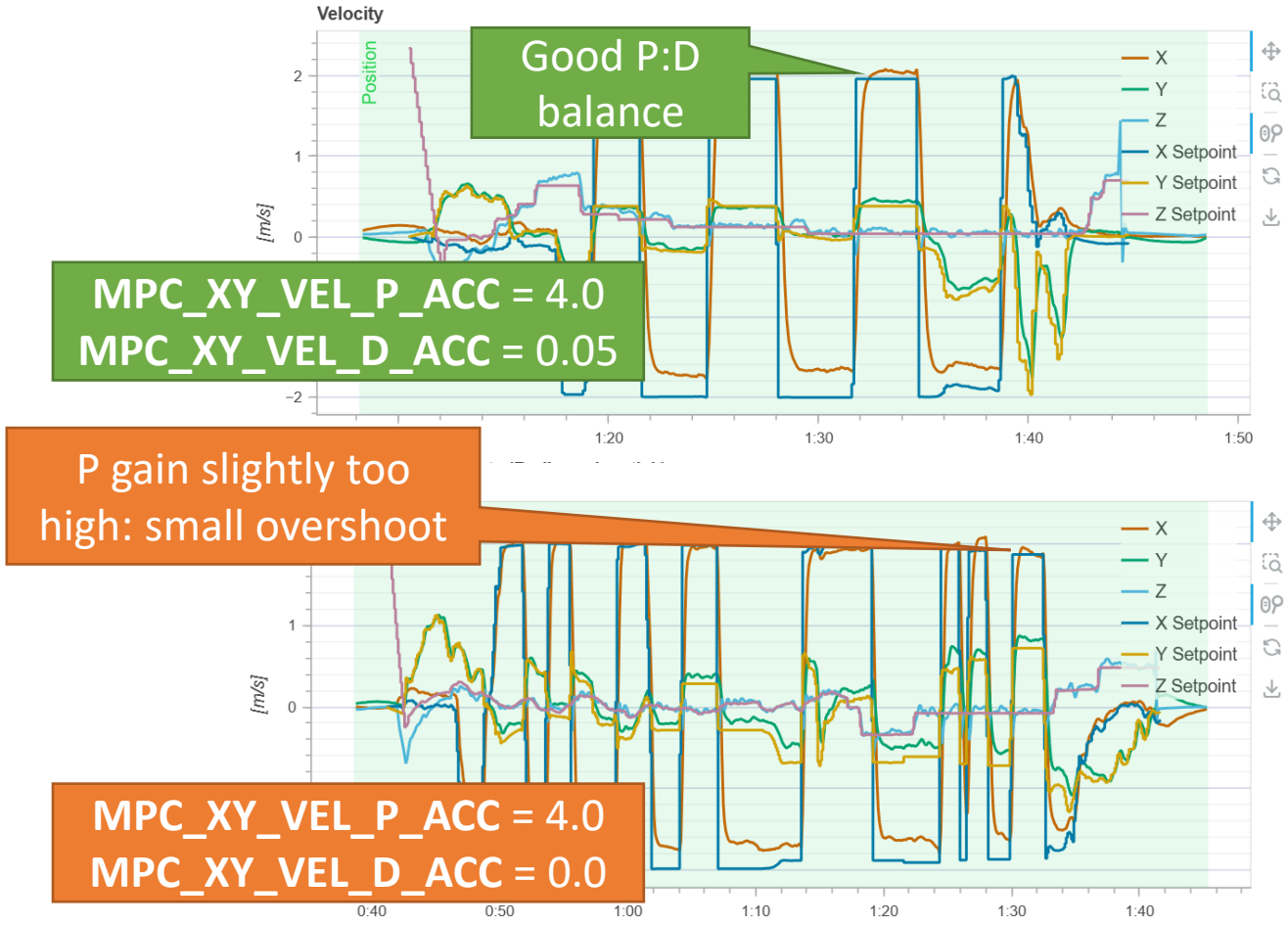
Tuning XY Velocity

- The Velocity controller is often overdamped on default settings (too much D gain and not enough P gain)
- If your log looks like the blue example start by reducing D gain
- On some copters you may be able to reduce D to zero because there is enough natural damping



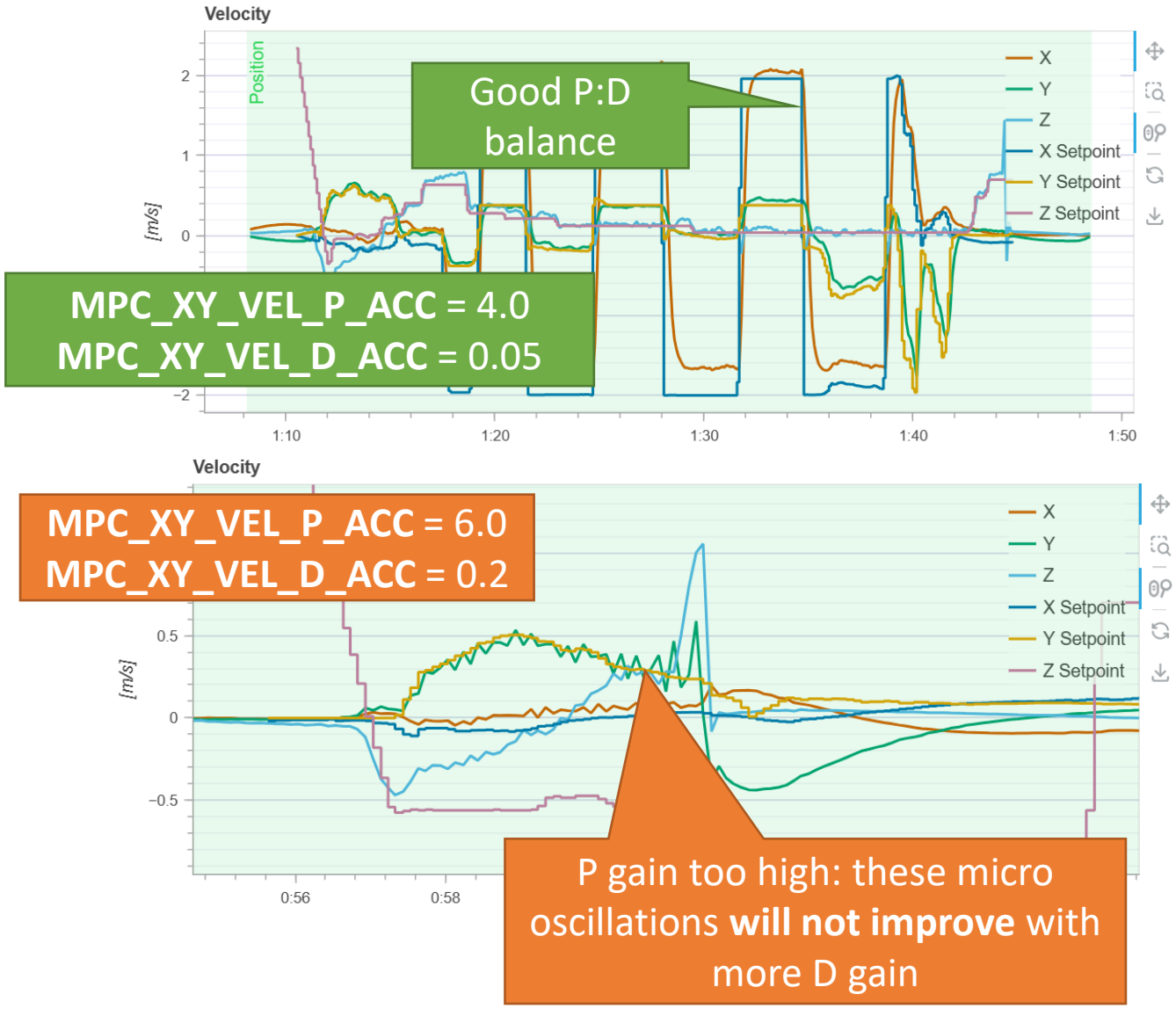
Tuning XY Velocity

- Once you have found the correct P:D balance (or reduced D to zero) you can start to increase P, I and D together in proportion
- Do this by multiplying all the gains by 1.2 (to maintain the ratio)



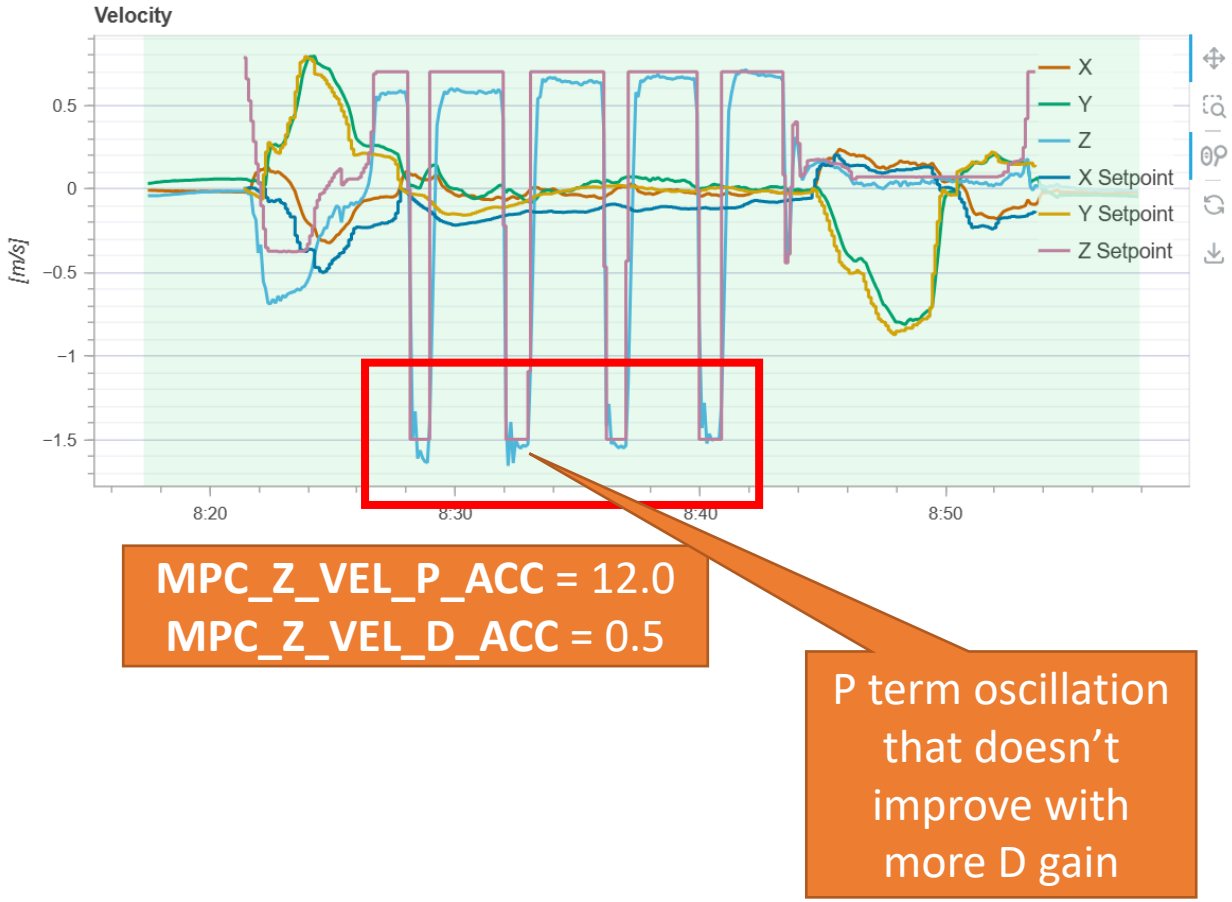
XY Velocity P term oscillation

- Unlike for the rate controllers the velocity controllers in PX4 can exhibit rapid oscillation above a critical P gain that does not improve with additional D gain
- Due to the slow loop rate of the velocity controller (50Hz) rigid body oscillations can occur at the nyquist limit (25Hz) and therefore cannot be addressed with D gain
- If this happens reduce P gain by at least 30% for safety
- You can then reduce D term (at max safe P gain) to get a less damped response
- In my experience this gives good flight performance
- **PX4 could consider increasing the loop rate of the velocity controller to address this if possible**

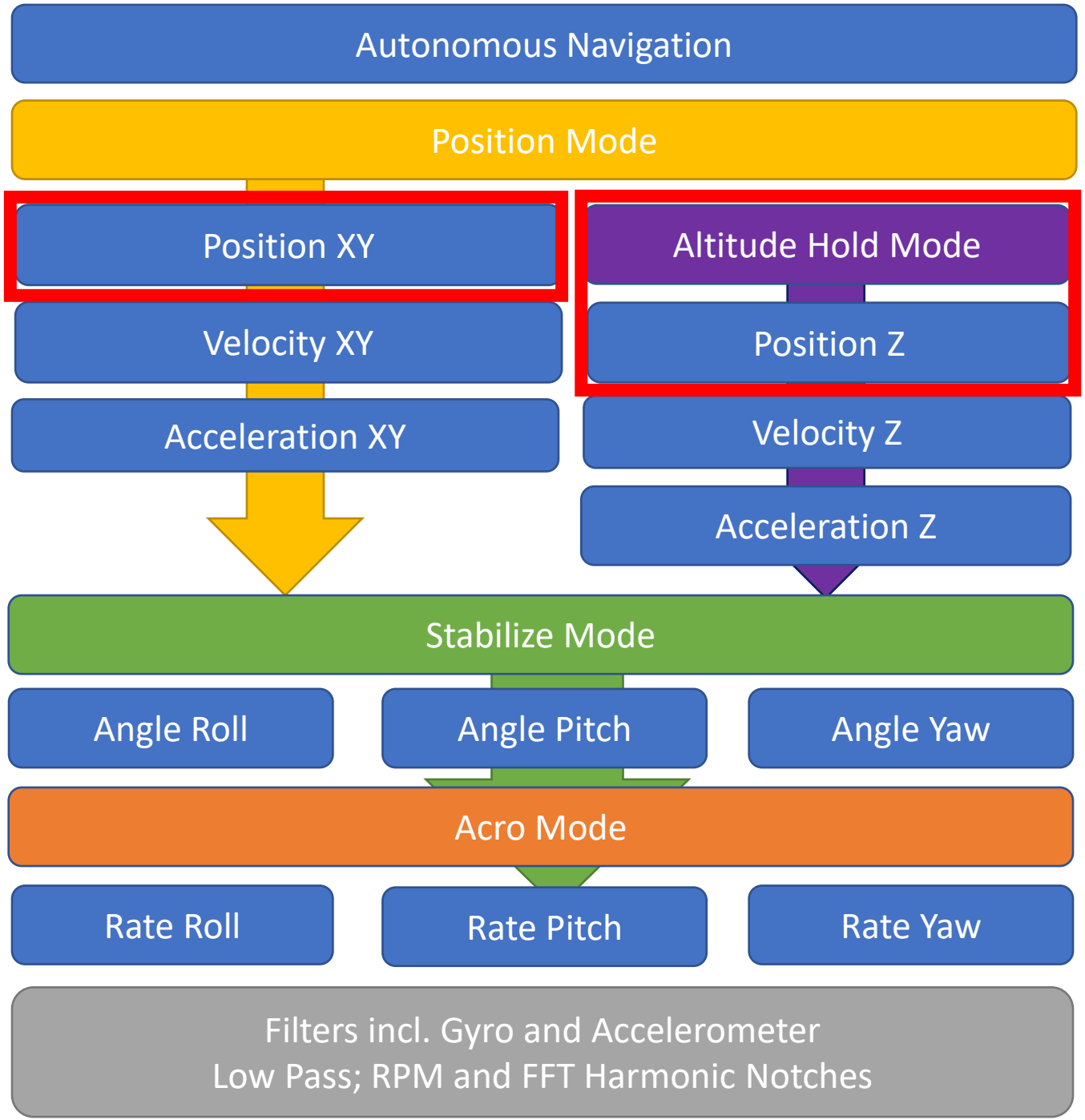


Vert. Velocity controller P oscillation

- Unlike for the rate controllers the velocity controllers both exhibit a rapid P term oscillation above a critical P gain that does not improve with additional D gain
- I think this is due to the relatively slow loop rate of the velocity controller (50Hz) and default 5Hz low pass filter for MPC_Z_VEL_D_ACC
- This is not a problem for tuning but you may find that you hit an oscillation limit with P gain regardless of D gain
- If this happens you can reduce D term (at max safe P gain) to get a less damped response
- In my experience this gives good flight performance

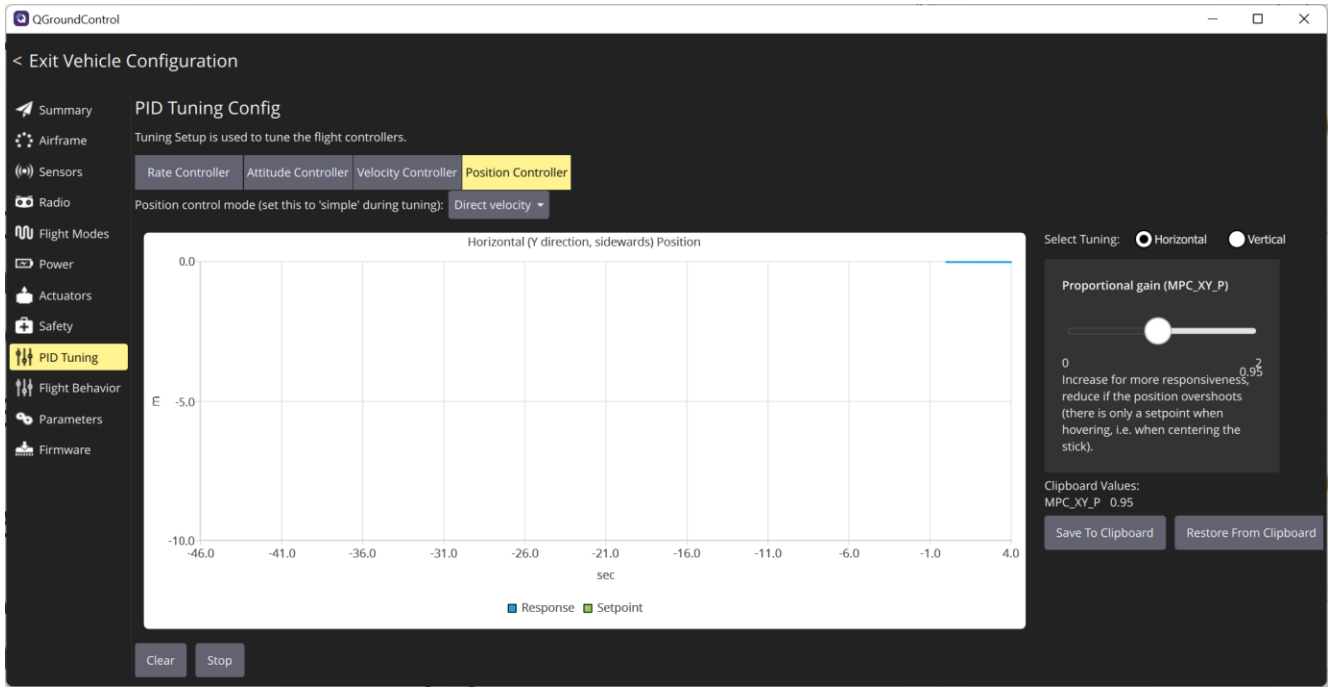


Position Controller


Autonomous Navigation
Position Mode
Position XY
Velocity XY
Acceleration XY
Altitude Hold Mode
Position Z
Velocity Z
Acceleration Z
Stabilize Mode
Angle Roll
Angle Pitch
Angle Yaw
Acro Mode
Rate Roll
Rate Pitch
Rate Yaw
Filters incl. Gyro and Accelerometer
Low Pass; RPM and FFT Harmonic Notches

Adjusting Horiz. Pos. Controller Parameters

- You can find the horizontal position controller PID parameters in **Vehicle Configuration > PID Tuning > Position Controller > Horizontal**
- We will start by adjusting **MPC_XY_P**
- Set position control mode to **Direct Velocity** for tuning



Adjusting Vert. Pos. Controller Parameters

- You can find the vertical position controller PID parameters in **Vehicle Configuration > PID Tuning > Position Controller > Vertical**
- We will start by adjusting **MPC_Z_P**
- Set position control mode to **Direct Velocity** for tuning

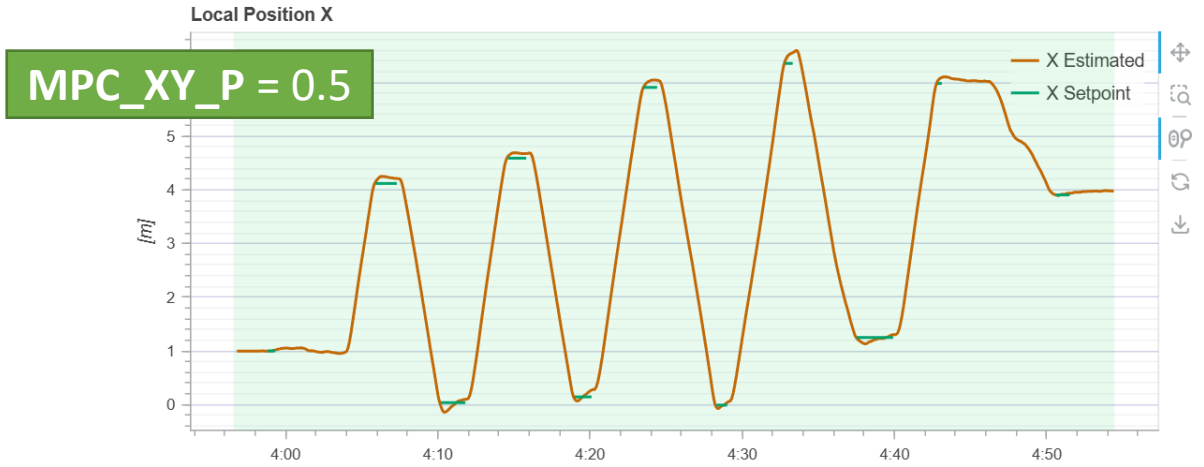


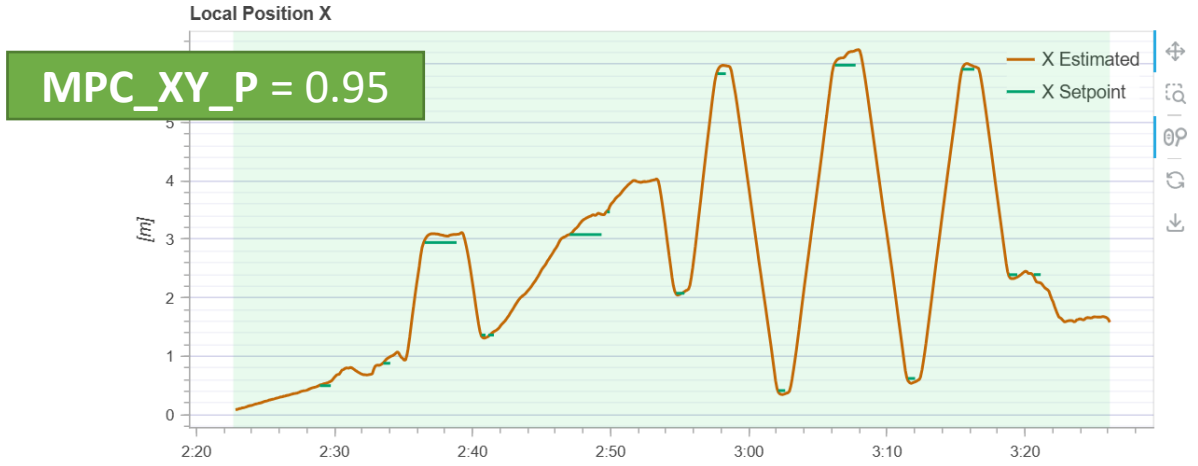
Position Controller Tuning Flights

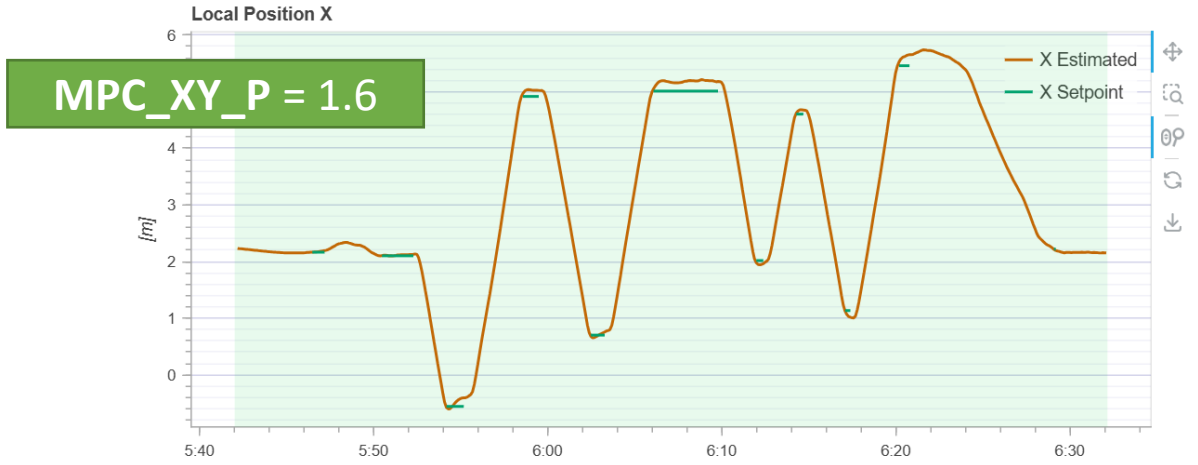
- Tuning flights for the horizontal position controller involve doing sharp moves left, right, forward or backwards and then centring the pitch/roll stick and waiting for the drone to settle between each move
- Tuning flights for the vertical position controller involve doing sharp moves upwards or downwards and then centring the throttle stick and waiting for the drone to settle between each move

Tuning Horiz. Position P

- With a tightly tuned velocity controller there should be very little for position P gain to do
- As a result it can have a wide tuning window and you may not need to adjust it much if at all
- Increase P gain until vertical position hold is accurate but no further
- Further increasing P gain doesn't give any benefit and increases the risk of oscillations

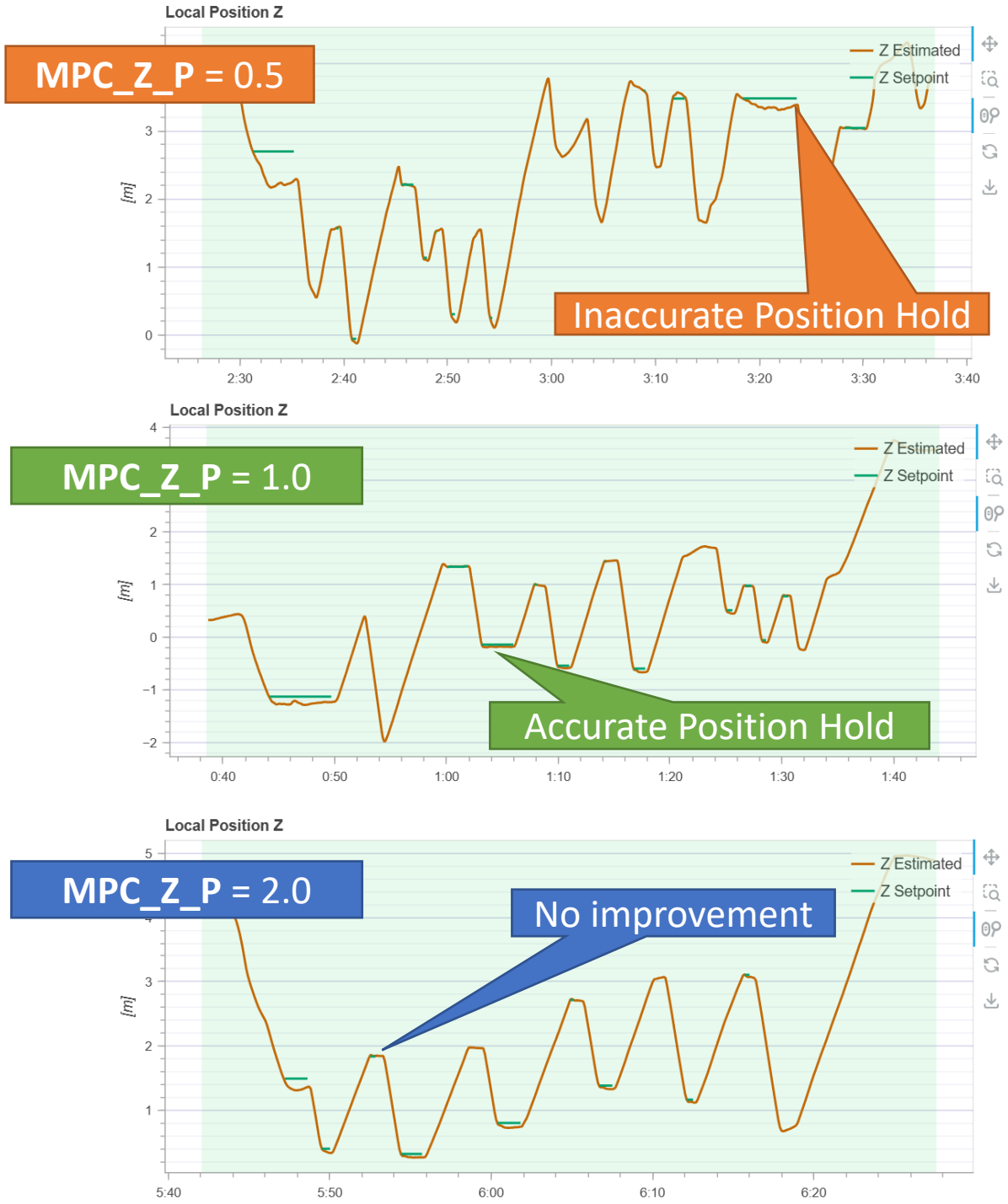






Tuning Vert. Position P

- With a tightly tuned velocity controller there should be very little for position P gain to do
- Increase P gain until vertical position hold is accurate but no further
- Further increasing P gain doesn't give any benefit and increases the risk of oscillations



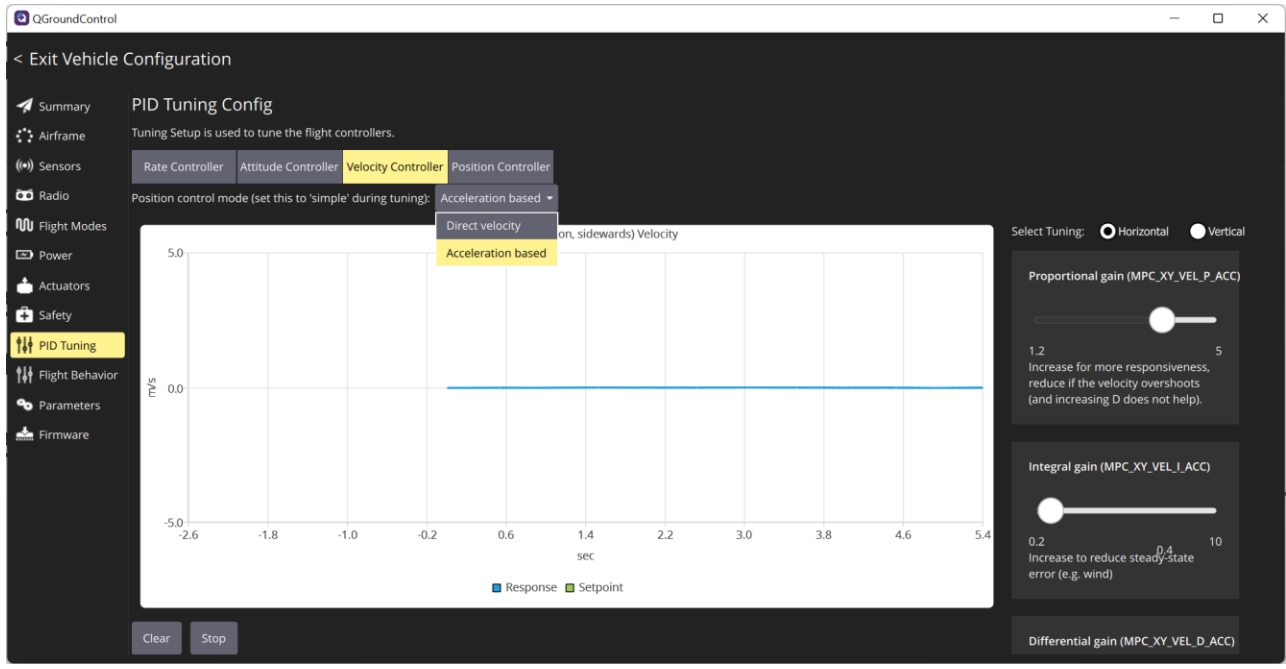
Tuning Flight Behaviour

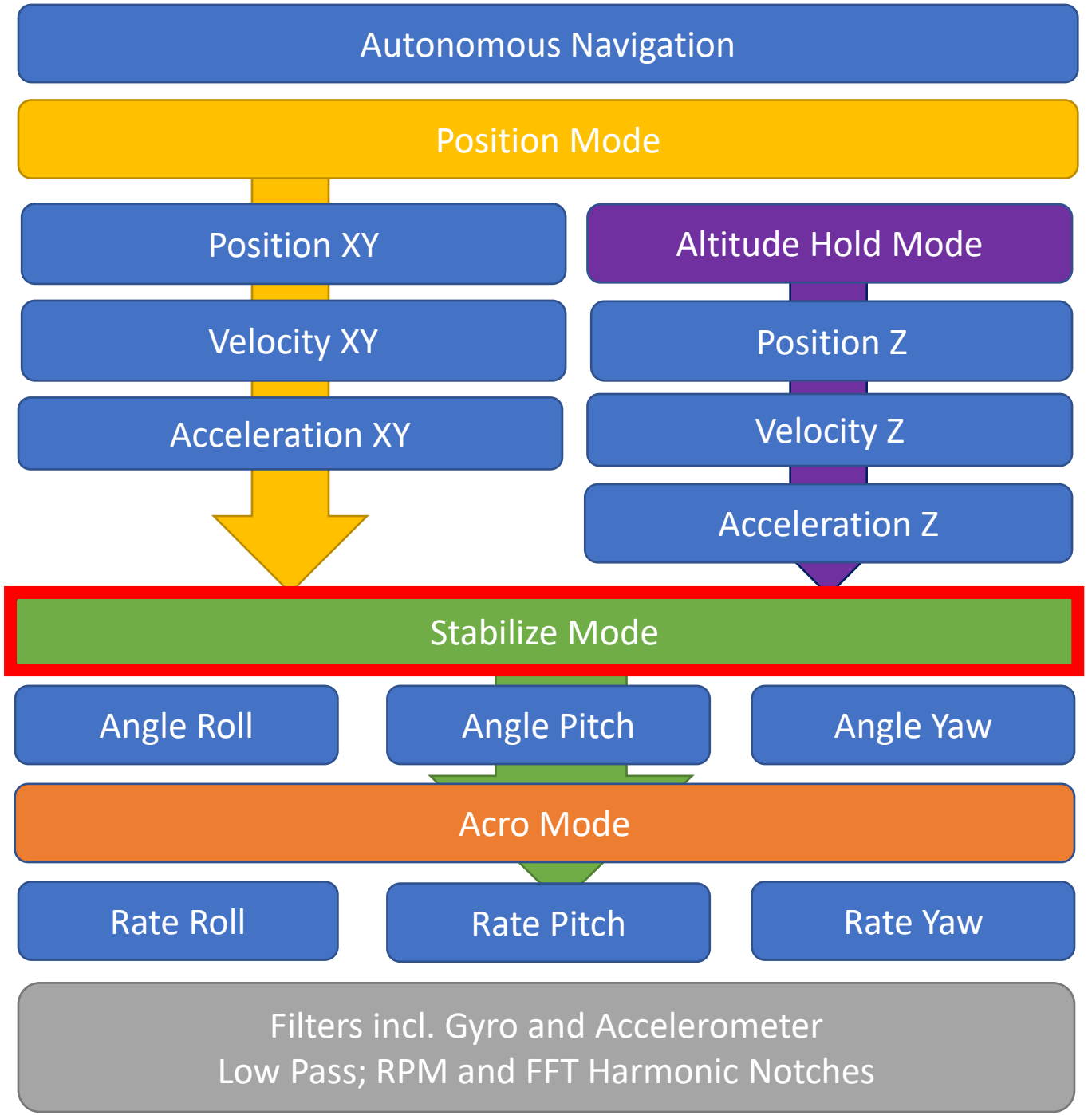
Scientific Tuning Complete

- At this stage PX4’s control stack is well tuned
- The drone should react sharply to stick inputs (especially in **Direct Velocity** mode) but without overshoot or oscillation
- The drone may react too sharply for your application, don’t worry we will calm everything down by tuning the kinematic path generation parameters

Reset Velocity Control to Acceleration Based

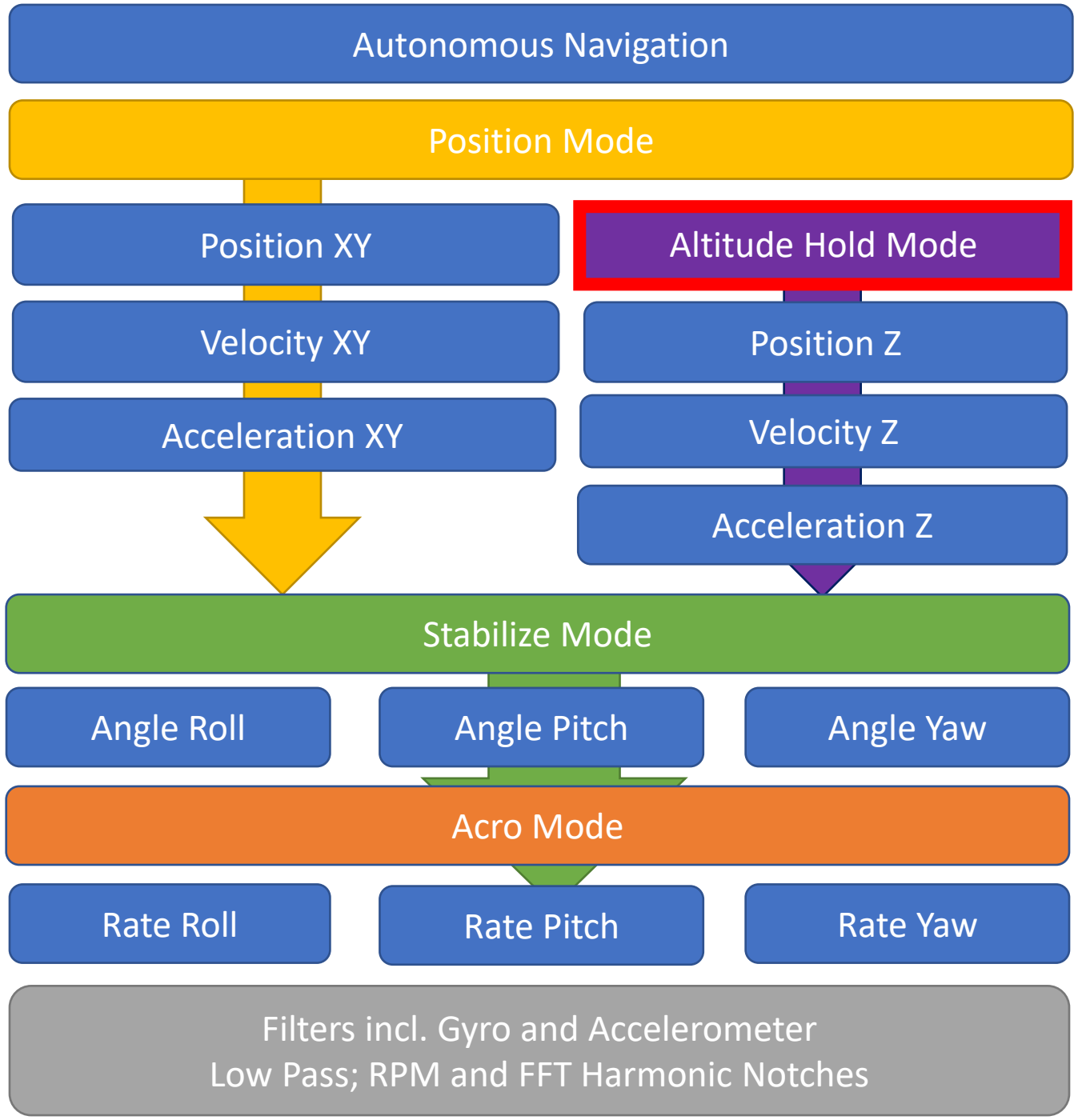
- **Vehicle Configuration > PID Tuning**
- Set **Position Control Mode** to **Acceleration Based**
- This is the usual mode that should be used when flying in position and altitude mode




Autonomous Navigation
Position Mode
Position XY
Velocity XY
Acceleration XY
Altitude Hold Mode
Position Z
Velocity Z
Acceleration Z
Stabilize Mode
Angle Roll
Angle Pitch
Angle Yaw
Acro Mode
Rate Roll
Rate Pitch
Rate Yaw
Filters incl. Gyro and Accelerometer
Low Pass; RPM and FFT Harmonic Notches

Flight Behaviour Parameters – Stabilize

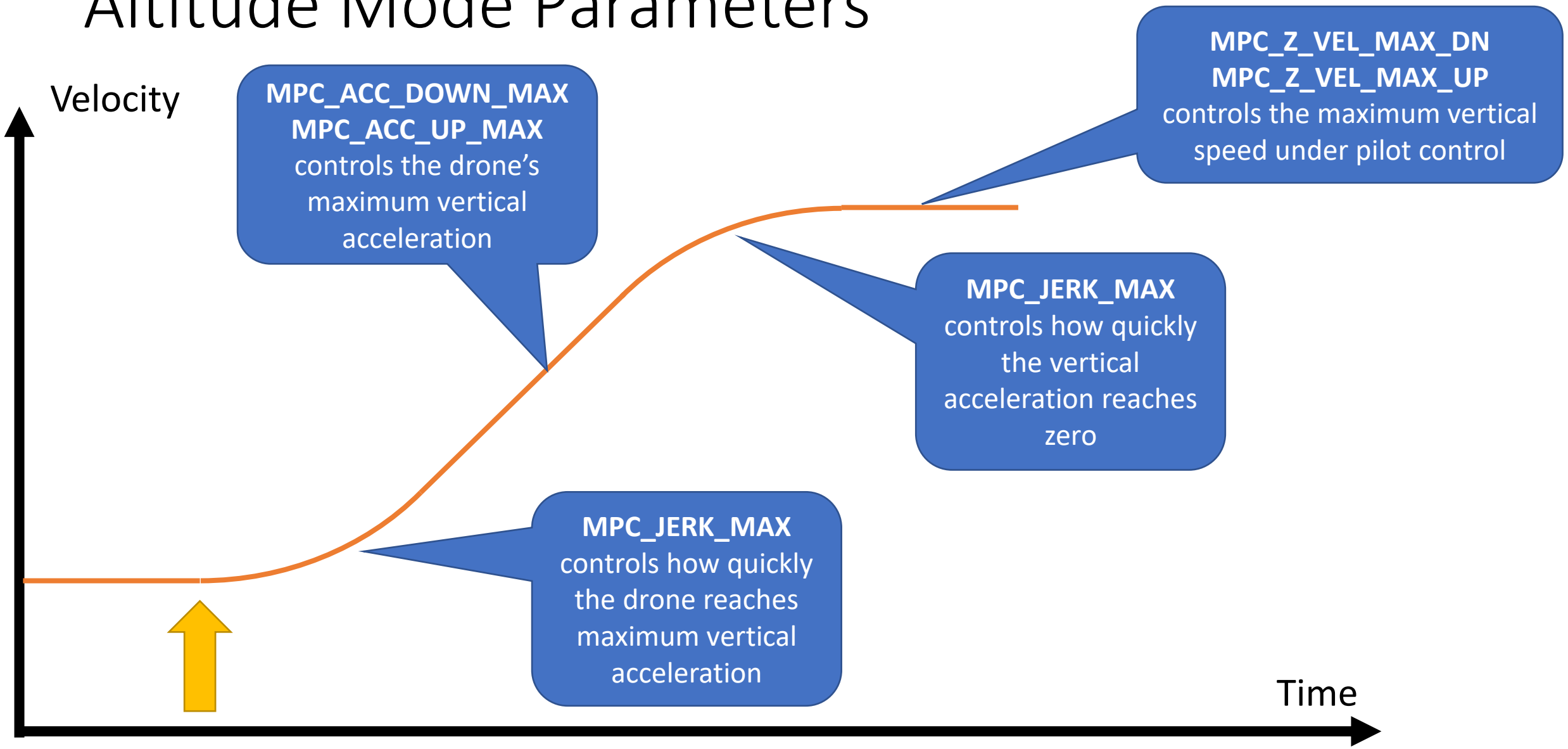
Parameter	
MPC_MAN_TIL_TAU	Provides a low pass filter on pitch/roll inputs in stabilized mode to smooth out jittery stick movements. The larger the time constant the smoother the input signal. If the drone is too sharp in Stabilized this parameter will smooth it out.
MPC_MAN_Y_TAU	Provides a low pass filter on yaw inputs in stabilized mode to smooth out jittery stick movements. The larger the time constant the smoother the input signal.
MPC_MANTHR_MIN	The minimum throttle setting sent to the ESC in stabilized mode. Set lower to allow quad to drop more quickly (but not below 7% as that risks motor stall)
MPC_TILTMAX_AIR	The maximum tilt angle in the air
MPC_MAN_TILT_MAX	The maximum tilt angle in all manual flight modes.
MPC_MAN_Y_MAX	Max yaw rate in manual mode

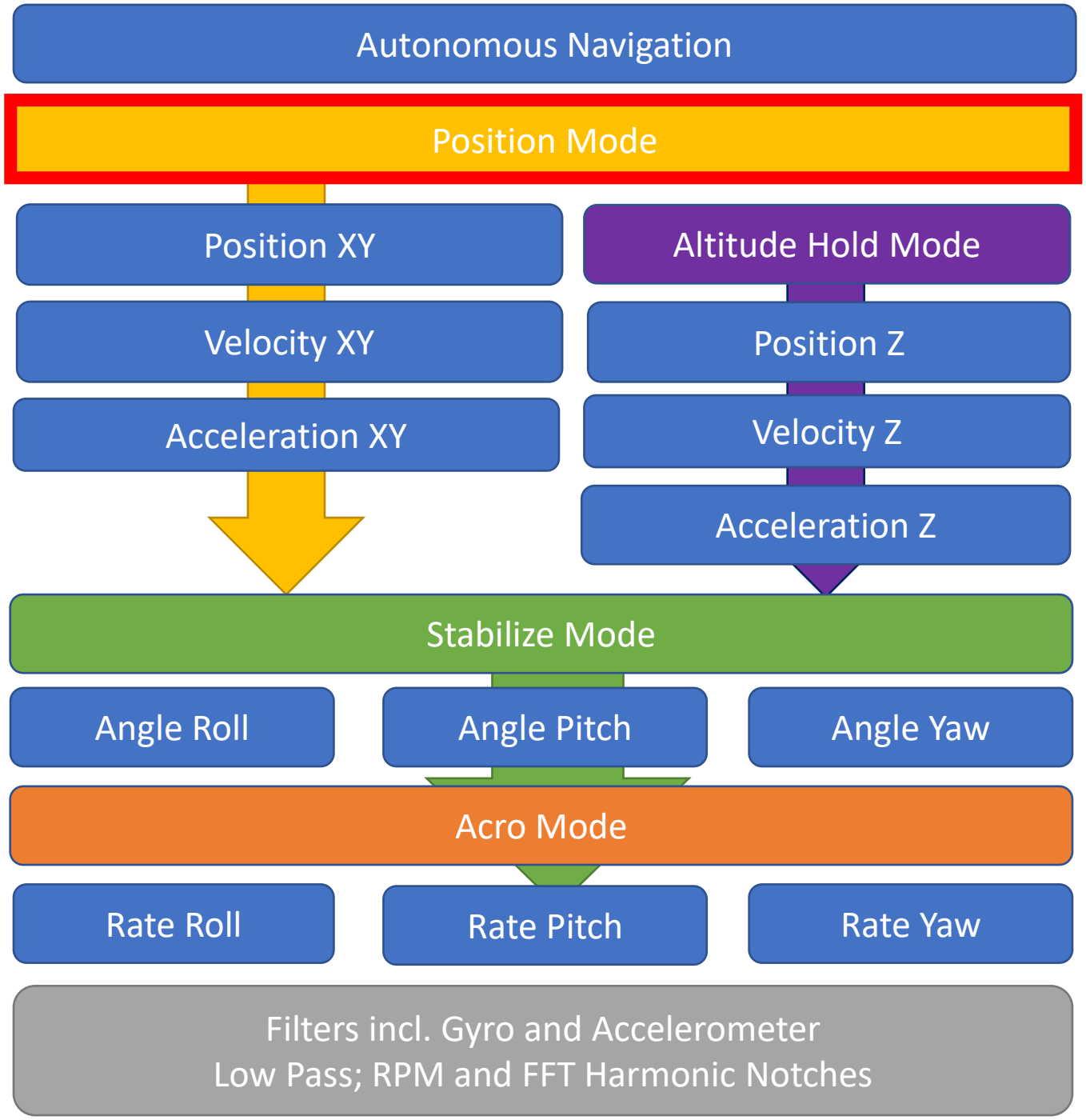

Autonomous Navigation
Position Mode
Position XY
Velocity XY
Acceleration XY
Altitude Hold Mode
Position Z
Velocity Z
Acceleration Z
Stabilize Mode
Angle Roll
Angle Pitch
Angle Yaw
Acro Mode
Rate Roll
Rate Pitch
Rate Yaw
Filters incl. Gyro and Accelerometer
Low Pass; RPM and FFT Harmonic Notches

Flight Behaviour Parameters – Altitude

Parameter	
MPC_ACC_DOWN_MAX	Maximum Downwards acceleration in altitude and position modes
MPC_ACC_UP_MAX	Maximum Upwards acceleration in altitude and position modes
MPC_ALT_MODE	**Altitude Following:** Relative to Sea Level **Terrain Following:** Relative to height above terrain (distance sensor required) **Terrain Hold:** Horizontal speed > MPC_HOLD_XY >>> Altitude following Horizontal speed < MPC_HOLD_XY >>> Terrain following
MPC_JERK_MAX	Maximum horizontal and vertical jerk in Position and Altitude Mode
MPC_THR_MAX	Maximum throttle in Position and Altitude Mode
MPC_THR_MIN	Minimum throttle in Position and Altitude Mode
MPC_Z_VEL_MAX_DN	Maximum descent velocity
MPC_Z_VEL_MAX_UP	Maximum ascent velocity
MPC_HOLD_MAX_Z	Maximum vertical velocity below which position control is enabled (rather than velocity control)

Altitude Mode Parameters

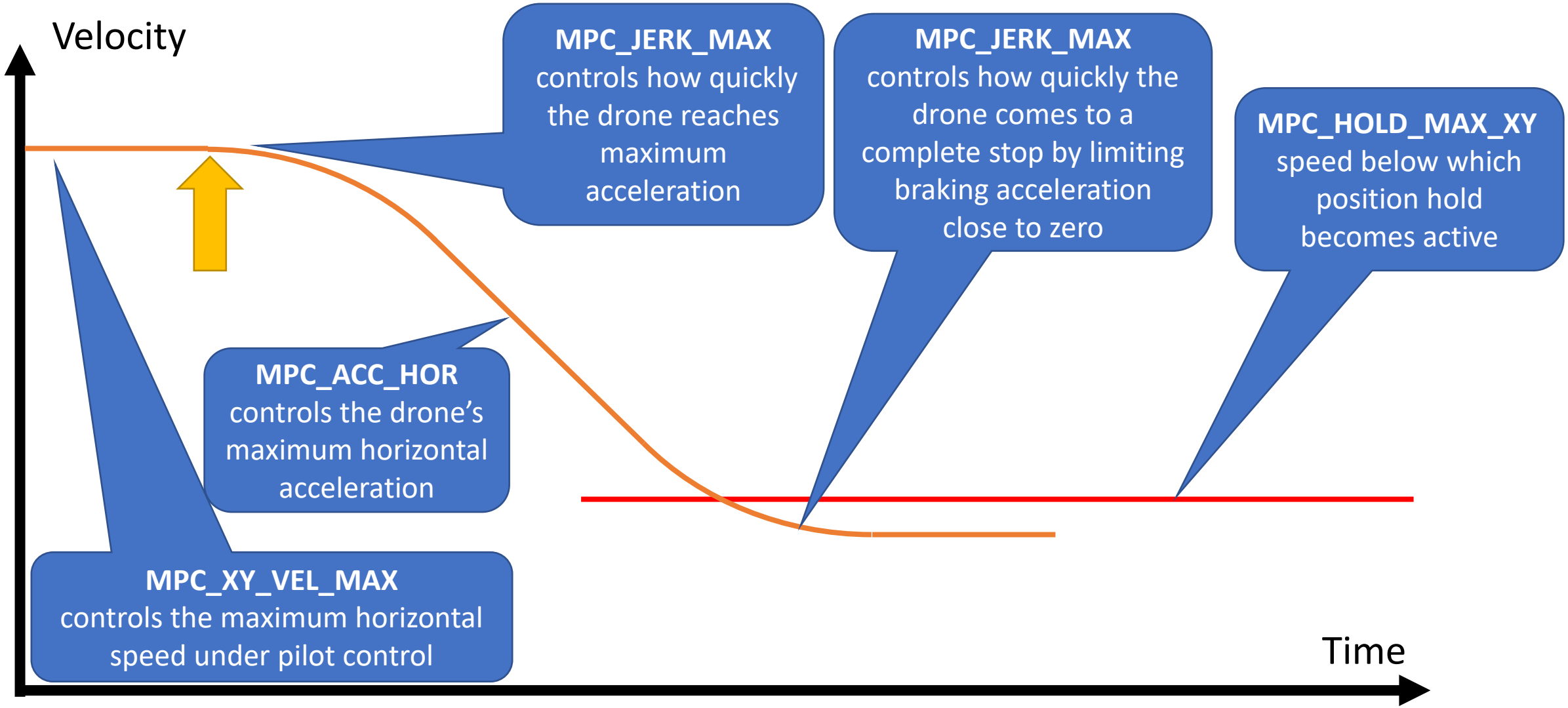


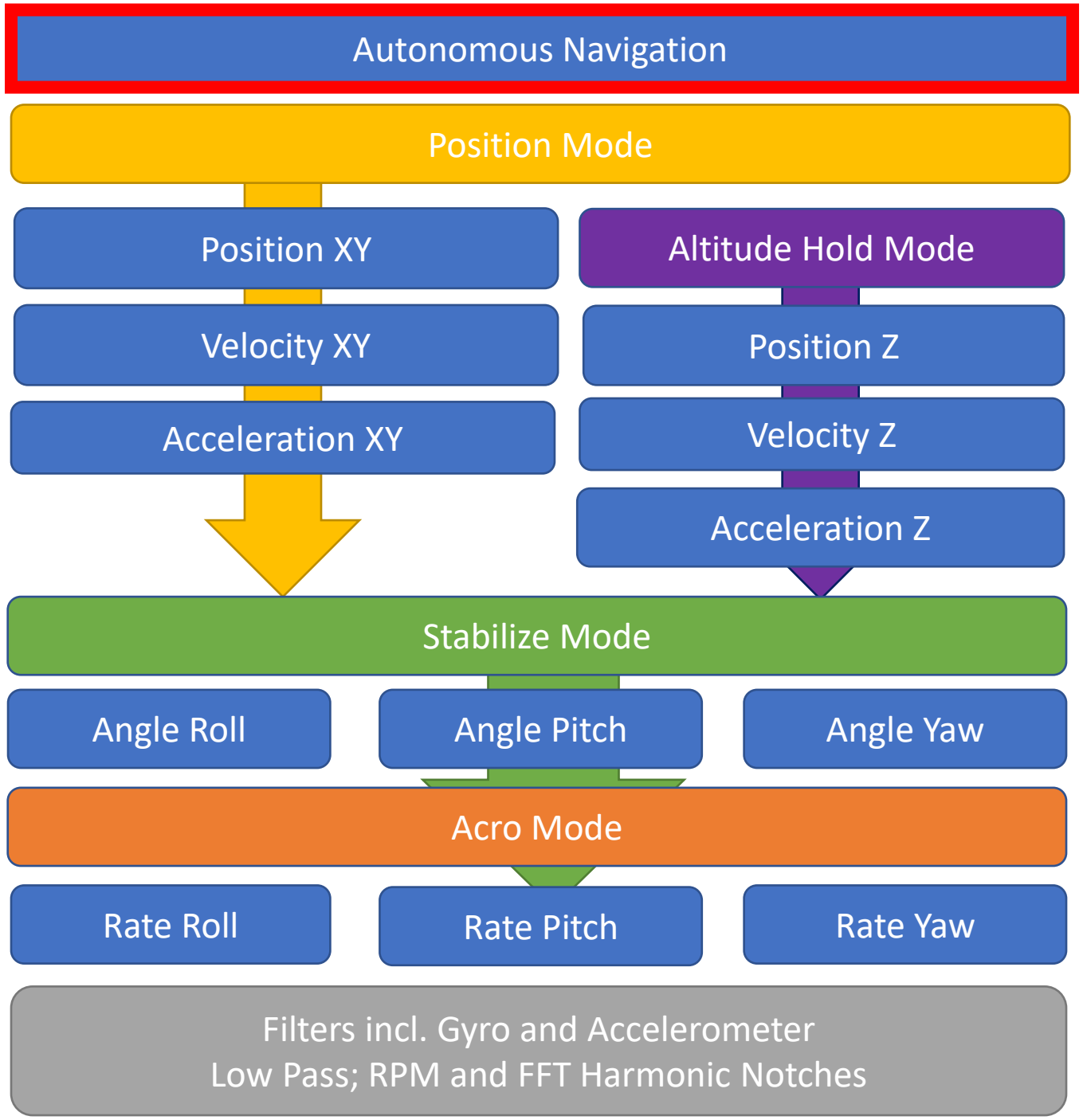

Autonomous Navigation
Position Mode
Position XY
Velocity XY
Acceleration XY
Altitude Hold Mode
Position Z
Velocity Z
Acceleration Z
Stabilize Mode
Angle Roll
Angle Pitch
Angle Yaw
Acro Mode
Rate Roll
Rate Pitch
Rate Yaw
Filters incl. Gyro and Accelerometer
Low Pass; RPM and FFT Harmonic Notches

Flight Behaviour Parameters – Position

Parameter	
MPC_ACC_HOR	Horizontal acceleration in autonomous and position modes
MPC_HOLD_MAX_XY	Maximum horizontal velocity below which position control is enabled (rather than velocity control)
MPC_JERK_MAX	Maximum horizontal and vertical jerk in Position and Altitude Mode
MPC_XY_VEL_MAX	Maximum horizontal velocity in Position and Autonomous modes

Position Mode Parameters

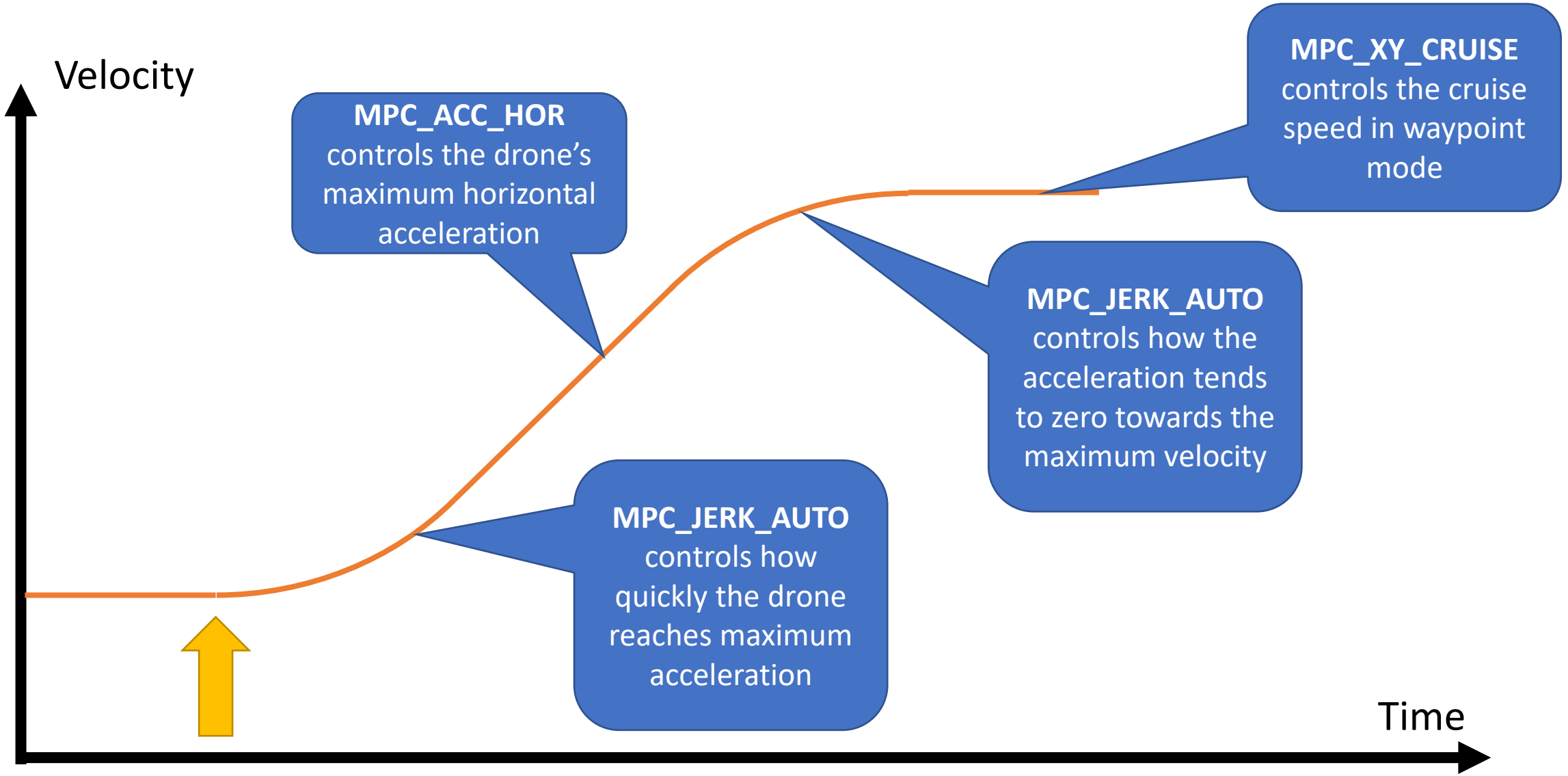



Autonomous Navigation
Position Mode
Position XY
Velocity XY
Acceleration XY
Altitude Hold Mode
Position Z
Velocity Z
Acceleration Z
Stabilize Mode
Angle Roll
Angle Pitch
Angle Yaw
Acro Mode
Rate Roll
Rate Pitch
Rate Yaw
Filters incl. Gyro and Accelerometer
Low Pass; RPM and FFT Harmonic Notches

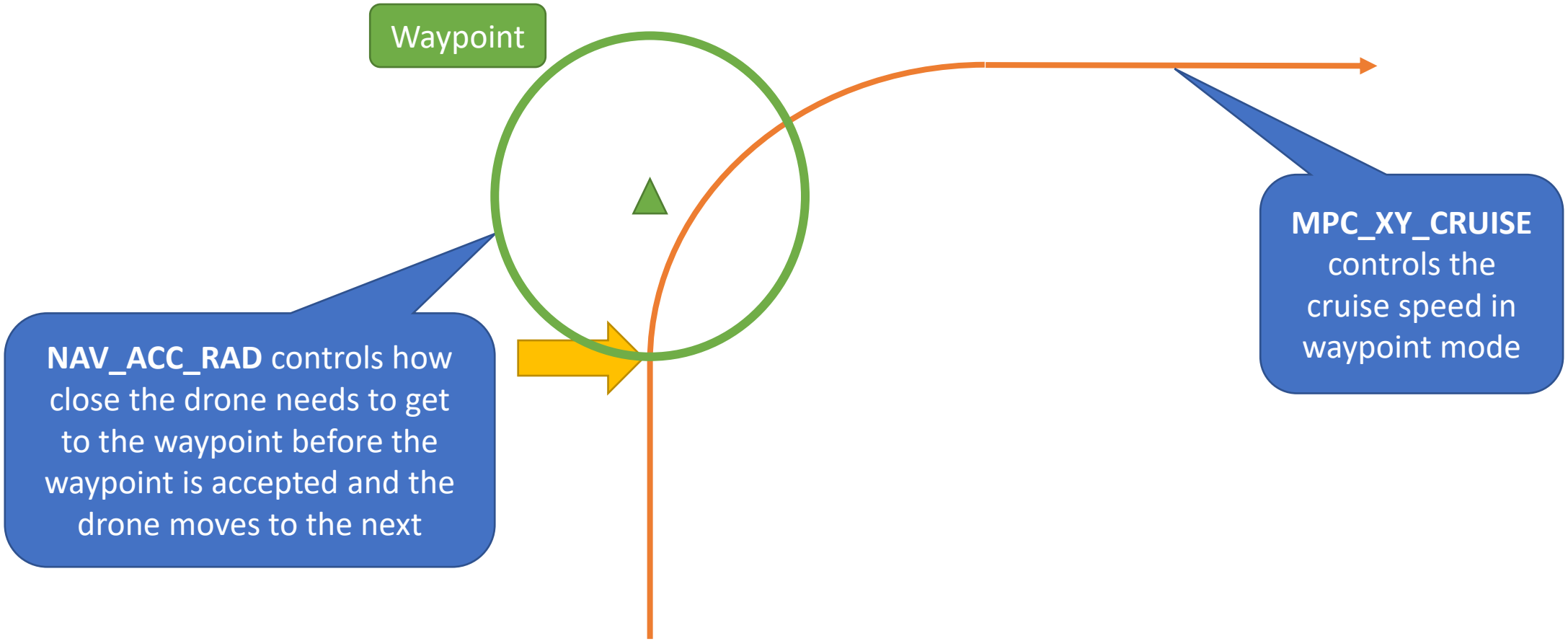
Flight Behaviour Parameters – Autonomous

Parameter	
MPC_JERK_AUTO	Jerk limit for Autonomous modes
MPC_XY_CRUISE	Cruise speed in Autonomous mode
MPC_YAWRAUTO_ACC	Maximum yaw acceleration in autonomous mode
MPC_YAWRAUTO_MAX	Maximum rotation rate in autonomous mode
MPC_YAW_MODE	Towards waypoint: point towards the next waypoint Towards home: always point towards take-off point Away from home: always point away from take-off point (easy for manual control) Along trajectory: point in the direction of travel Towards waypoint (yaw first): Yaw **before** moving towards the next waypoint Yaw fixed: Do not change yaw (useful when using a payload with independent yaw)
MPC_Z_V_AUTO_UP **MPC_Z_V_AUTO_DN**	Ascent velocity in autonomous mode Descent velocity in autonomous mode

Autonomous Horizontal Parameters



Autonomous Horizontal Parameters



Autonomous Vertical Parameters

